

Monza | Indo-MIM | BG Snap Assembly

Vendor: Indo-MIM
Part Description: BG Snap
Part number: 613-31681,613-38625,613-38626
Build: -
Process: Snap Assembly
Machine model number: -

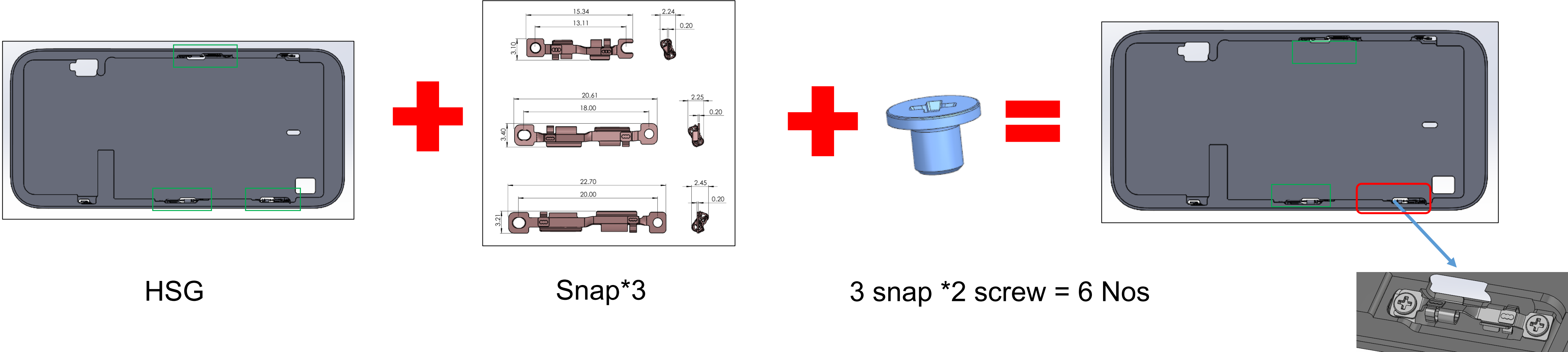
Revision	Date	Summary	Page
V1.0	03-12-2025	Initial Version	/
V1.1	25-12-2025	Update the OEE breakdown & tray & details	
V1.2	09-01-2026	Update the OEE breakdown & Standardization requirement	
V1.3	12-02-2026	Update the OEE breakdown & Standardization & Process Improvement	
V1.4	18-03-2026	Update on Linear conveyor system (Baeyal – Golytec) & LMI Snapshot sensor	

Monza | Snap Assy: DFM Check List

No.	Item	Check	DRI
1	RFP	Y	
2	Operational summary (OEE, C/T)	Y	
3	Layout w/dimension and Bom	Y	
4	Process Flow &Software flow	Y	
5	Machine maintenance & HMI	Y	
6	Key BOM List	Y	
7	Process Details	Y	
8	Fixture	Y	
9	Station Detail	Y	
10	Material feeding	Y	
11	Standardization (UI/Error code/OEE/Trace/PDCA/Fixture system/Component/EHS)	N	
12	Compatibility	N	
13	Retrofit	N	
14	Assembly, Calibration and Validation Plan	N	

RFP

Monza | Snap Assy: RFP and LQ Timeline



Assembly Method : Assemble Snap to HSG, meeting OEE requirement.

Assembly Requirement : Screw no missing, no floating, no tilting, no strip; product no damage.

Process Yield : 95% (machine in good condition, all consumable parts within service life, material deficient not included)

Monza | Snap Assy: RFP and LQ Timeline

L/T	SAT +IQ/OQ	PQ	MQ
Manufacturing and Assy lead time	Machine/ Fixture IQC/FAI, Machine Setup/Calibration/Dry run	CPK/ yield	OEE bring up
56 days	46 days	14 days	14 days

<u>S.No</u>	Process Description	LT (WK)
1	Std parts / Processing parts Procurement	7
2	Mechanical Assembly	2
3	Control Testing	1
4	Debug & Fine Tuning	2
5	Calibration(cameras & robots)	1
6	Manual sequence check	0.5
7	Auto sequence	1.5
8	Dry Run (Without part)	1
	Total WKs	16

Machine Overview

OEE Summary

Availability	93.04%
Performance	97.03%
1st Pass Yield	96.84%
Overall OEE	87.42%
UPH	629

OEE Detail Breakdown



C/T Breakdown				
Item	Station	QTY	C/T(s)	Headcount
1	Loading machine	1	3.5	(0.5)
2	Snap Assy & Screwing machine	3	5	/
3	Flipping Station	1	4	/
5	Unloading machine	1	3.5	(0.5)

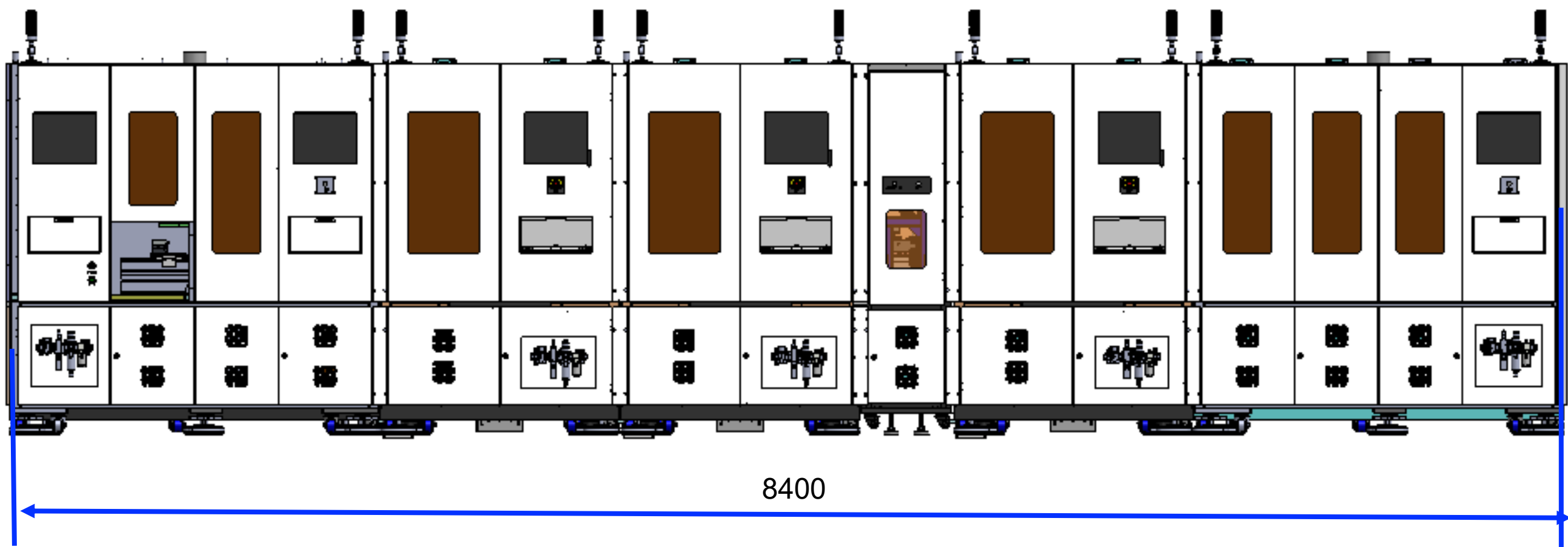
Line C/T 5s/pcs
TTL HC (1)

C/T Breakdown

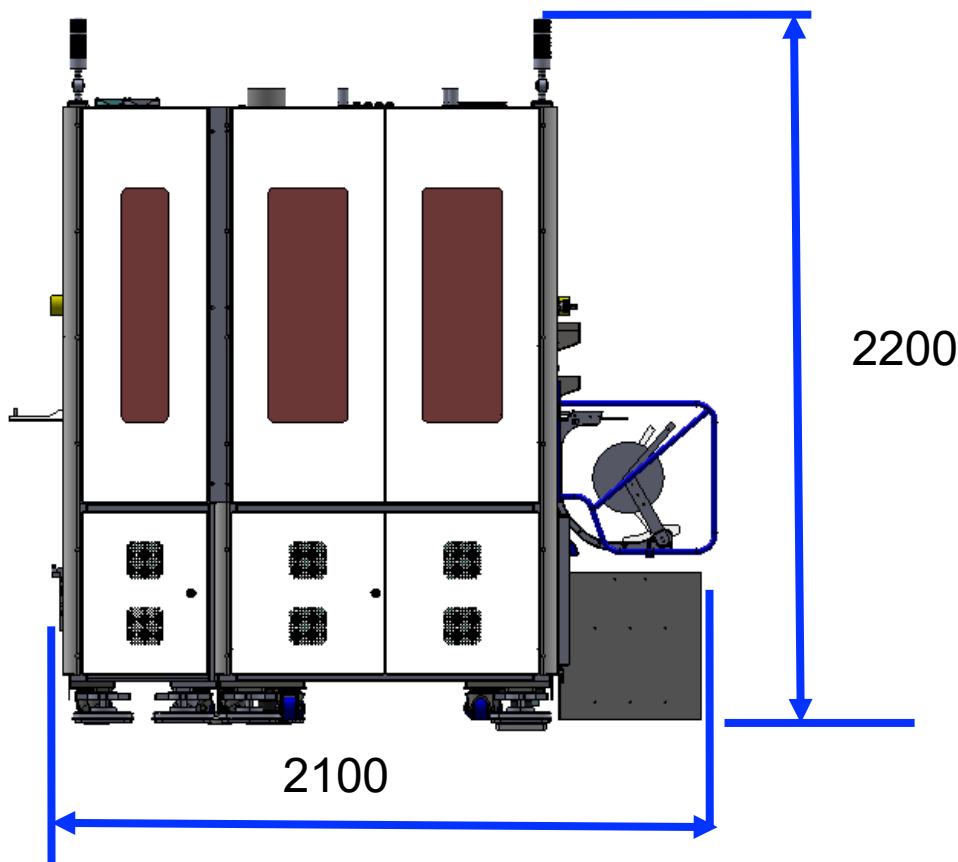


Monza | Snap Assy : Machine Overview –Layout

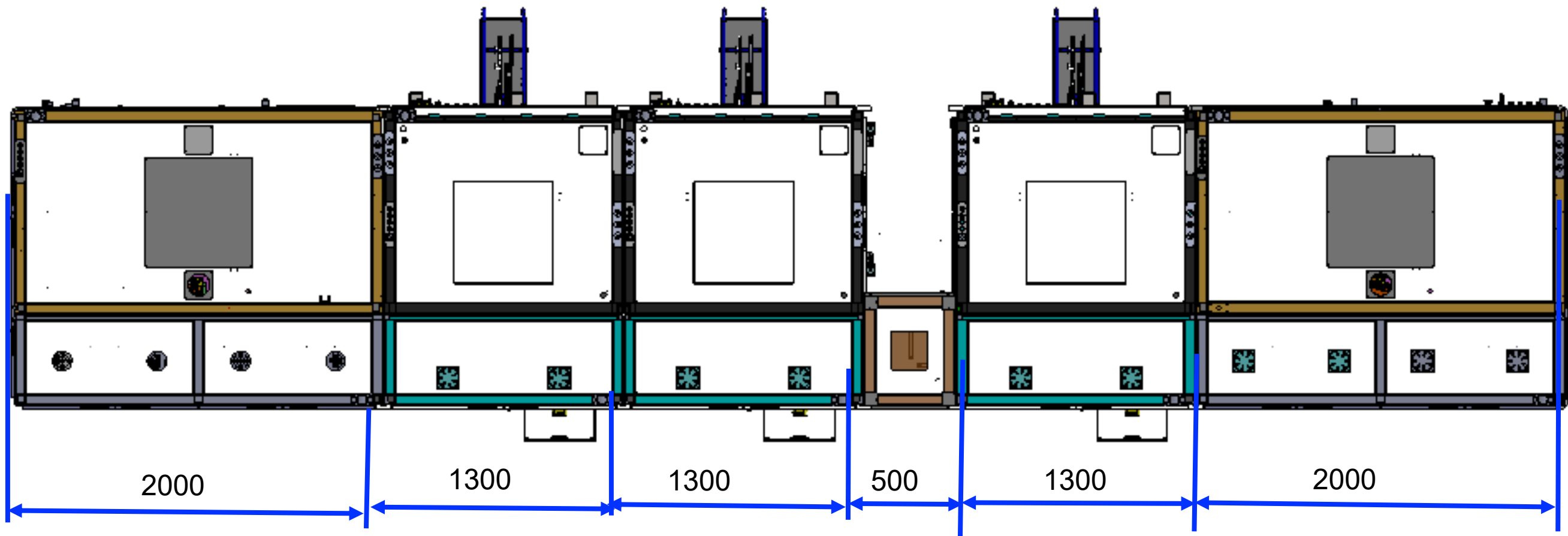
Front View



Side View

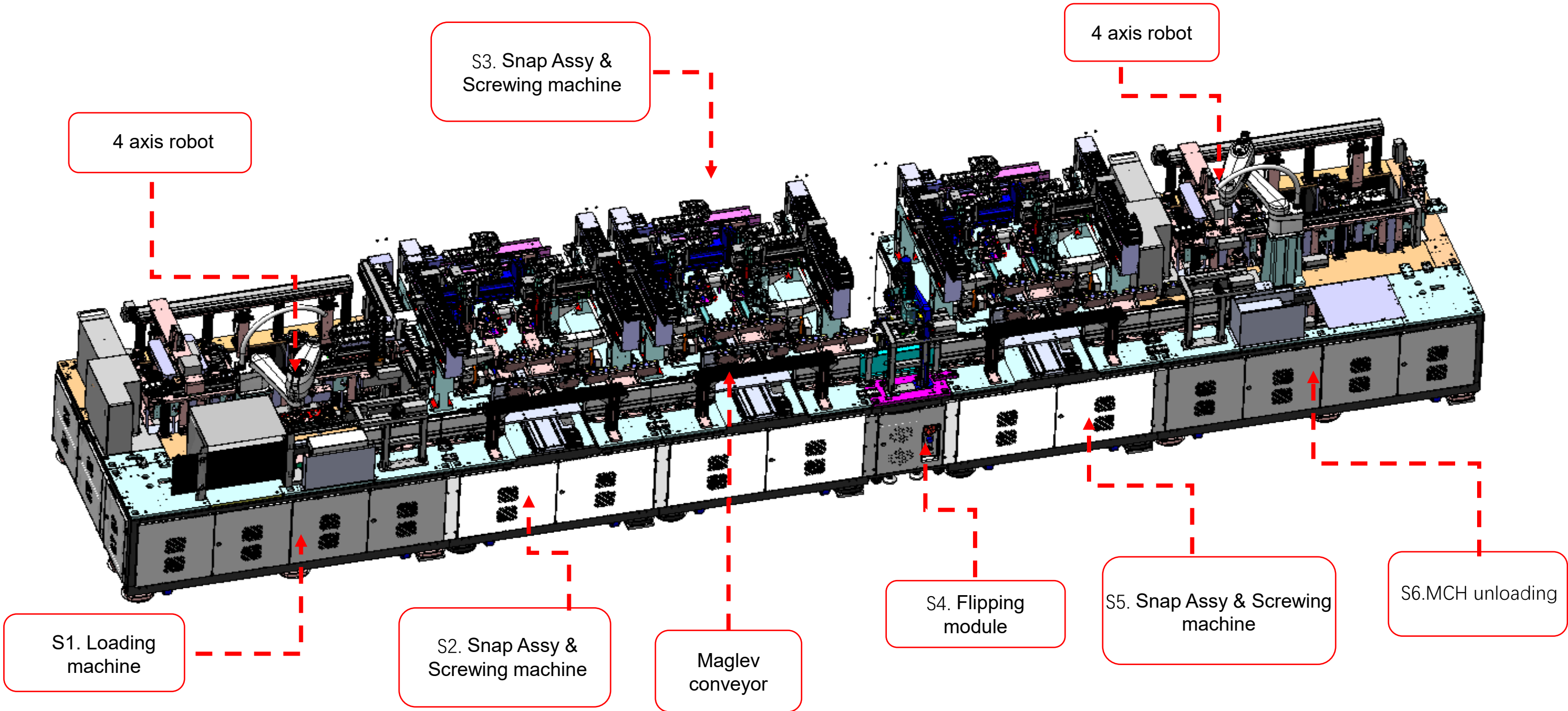


Top View



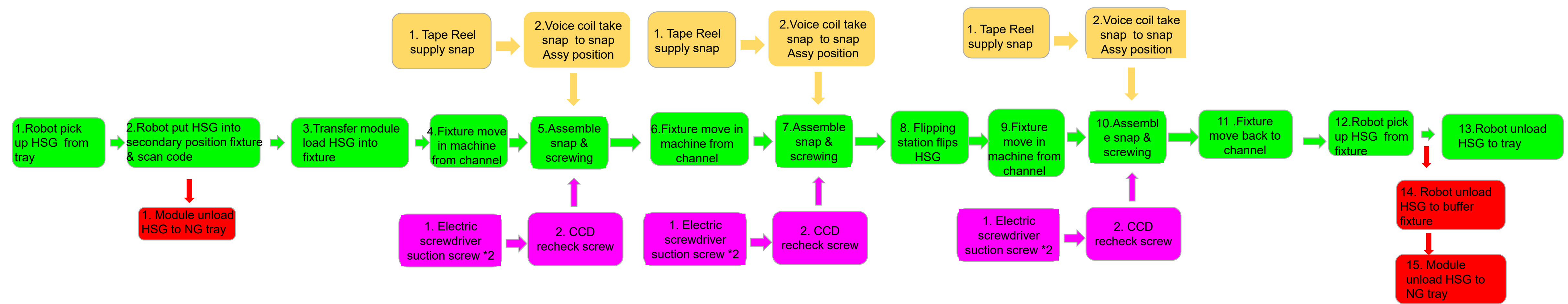
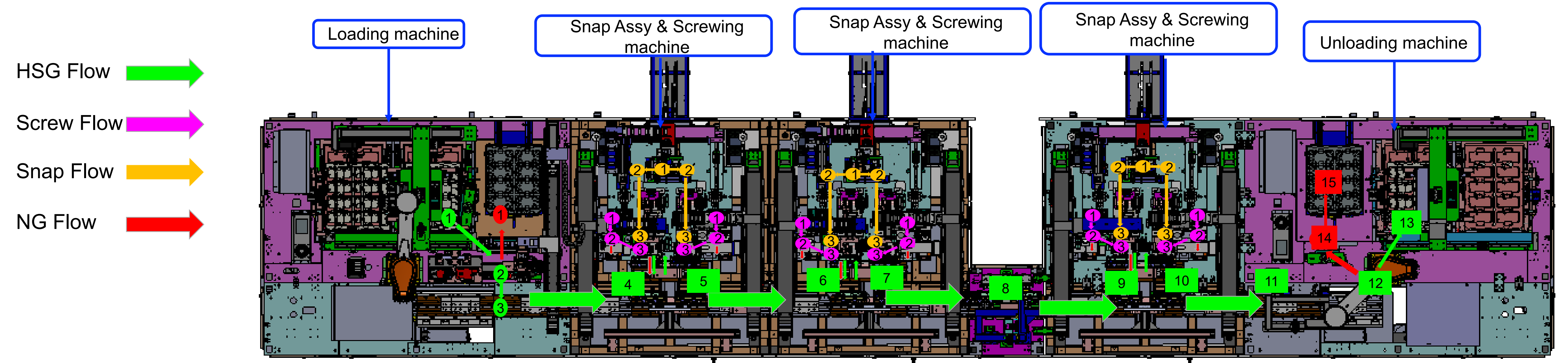
- 1.Voltage:220V
- 2.Power: 54KW
- 3.Air: 0.5~0.7Mpa
- 4.Labor: /
- 5. Machine weight:/

Monza | Snap Assy: Machine Overview– Station BOM

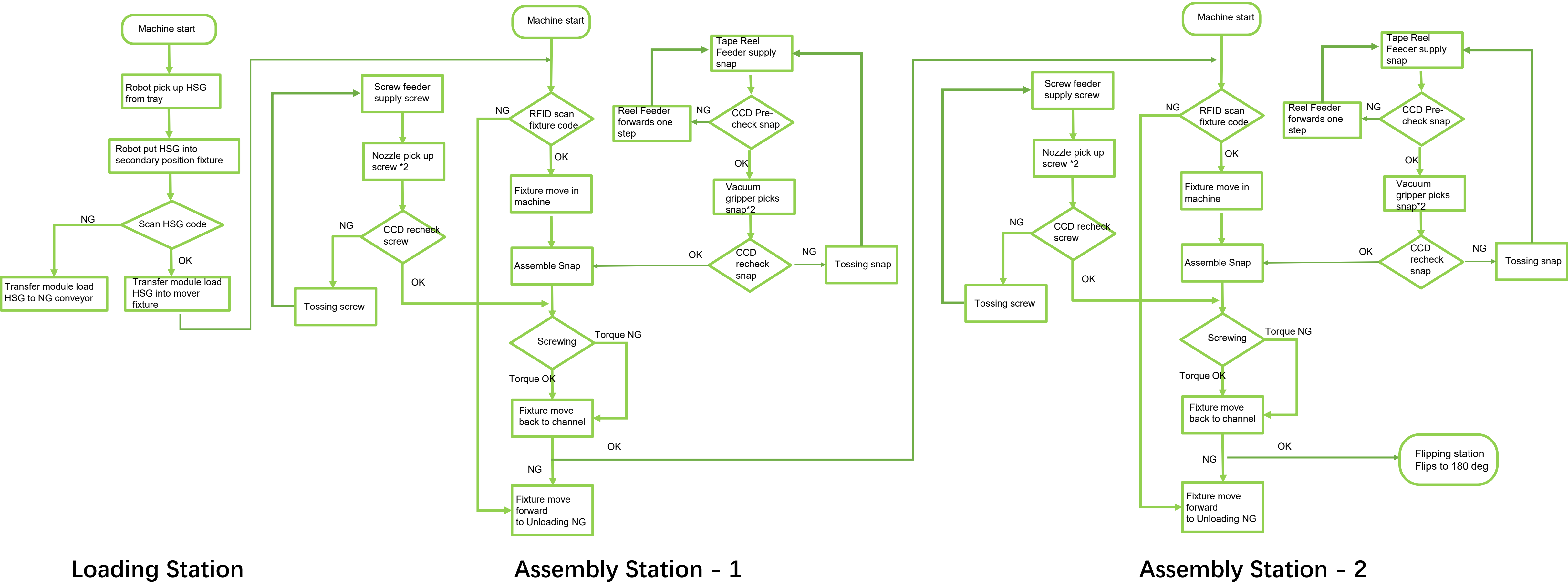


#	Module	Module Qty	Key parts	Key part QTY (TTL)
S1	Loading machine	1	4 axis robot	1
			X axis module	4
			Y axis module	1
			Z axis module	3
S2 & S3	Snap Assy & Screwing machine	2	X axis module	6
			Y axis module	10
			Z axis module	6
			Screw feeder	4
			Smart Screwdriver	8
			Tape Reel Feeder	4
S4	Flipping module	1	Z axis module	1
S5	Snap Assy & Screwing machine	1	X axis module	3
			Y axis module	5
			Z axis module	3
			Screw feeder	2
			Smart Screwdriver	4
			Tape Reel Feeder	2
S6	Unloading machine	1	4 axis robot	1
			X axis module	1
			Z axis module	2

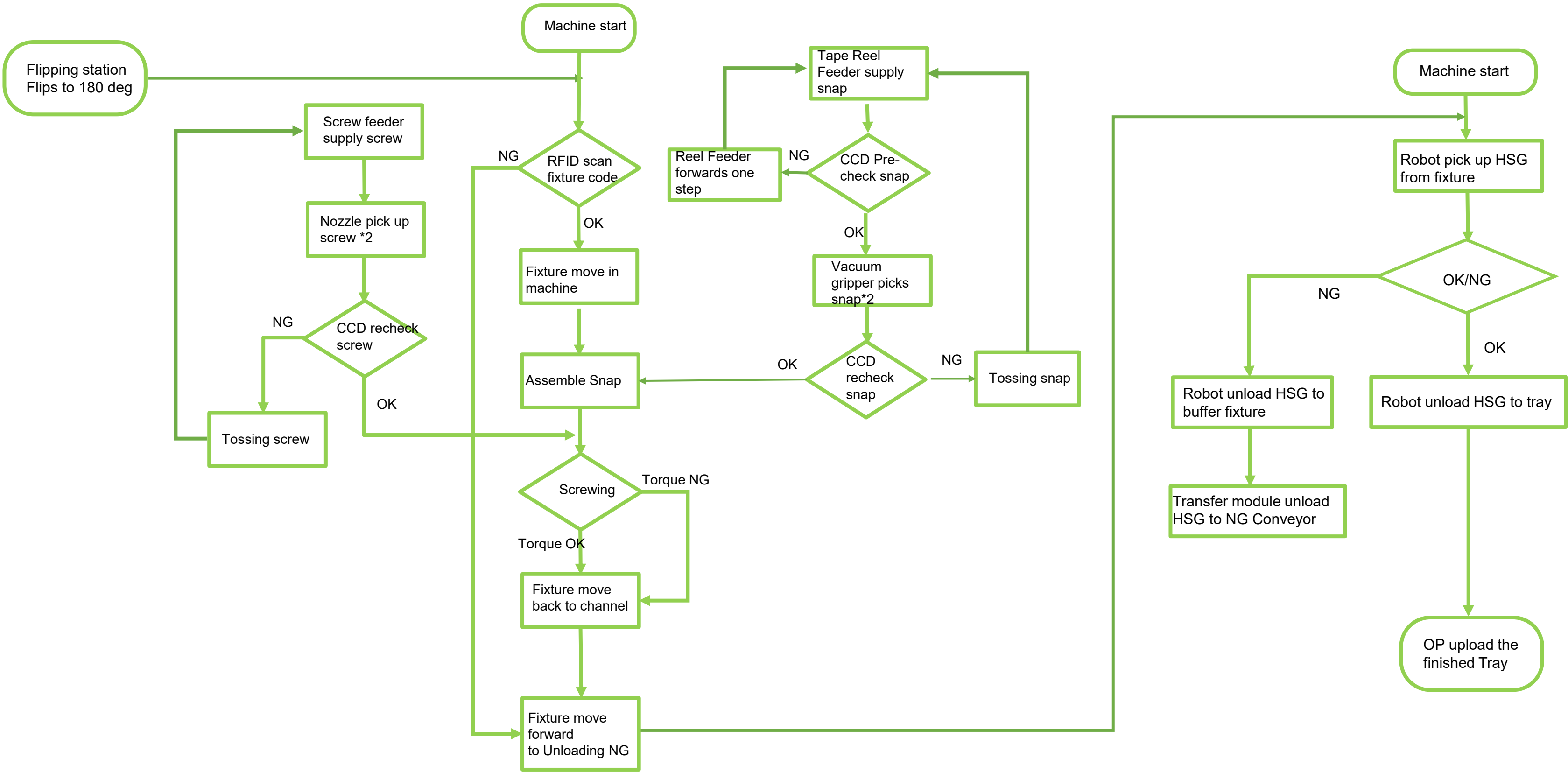
Monza | Snap Assy: Machine Overview - Process Flow & SW Flowchart



Monza | Snap Assy: Assy Flow Chart



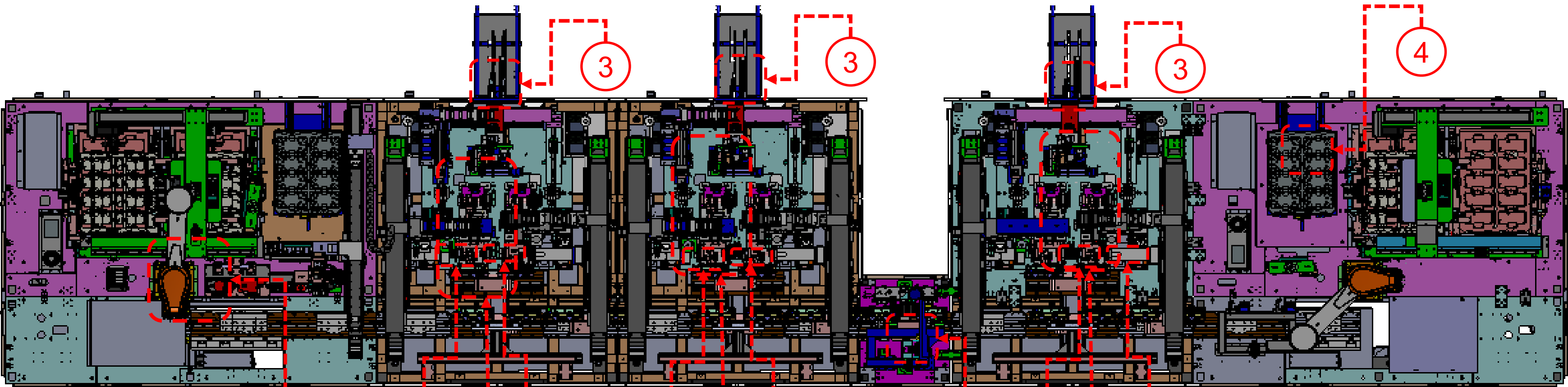
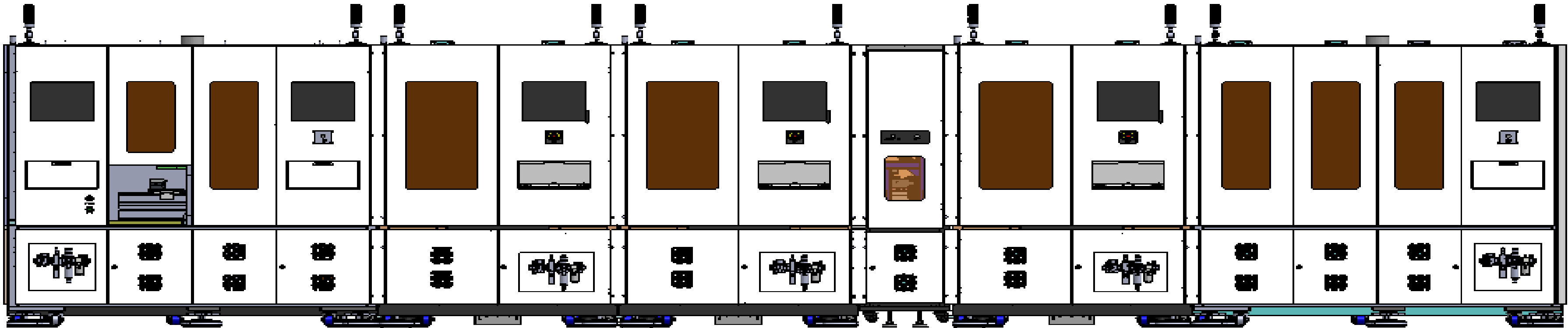
Monza | Snap Assy: Assy Flow Chart



Assembly Station - 3

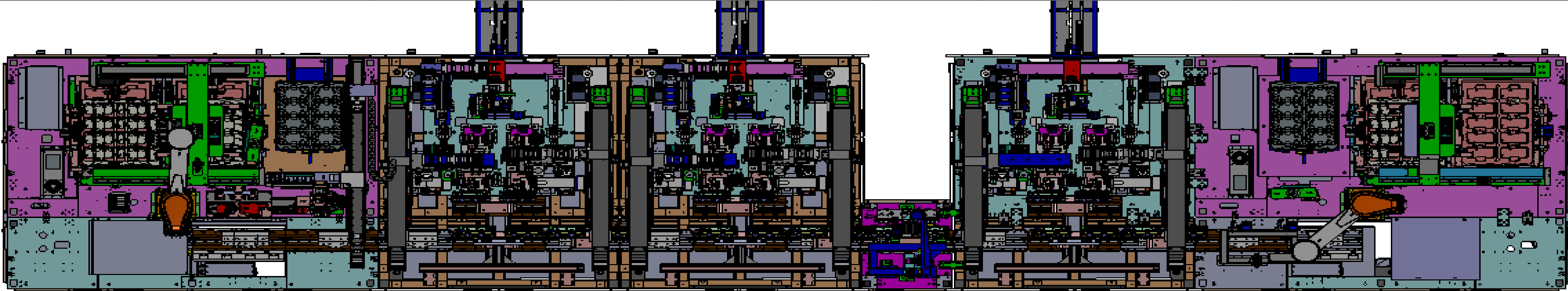
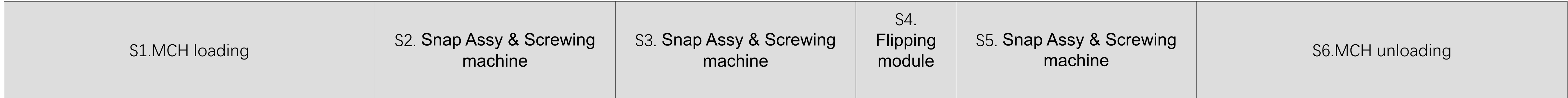
Unloading Station

Monza | Snap Assy: Machine Overview - Tossing



No.	1	2	3	4	5
Tossing item	HSG	Screw	Snap	HSG	Fixture
Tossing reason	• Scanning NG	• CCD recheck NG	• CCD recheck NG	• Torque NG	• Fixture maintenance

Monza | Snap Assy: Machine Overview - Traceability



S1.Loading machine
Scanner 1#
RFID 1#
HSG Barcode
Link HSG Barcode with Fixture ID
Ok or NG HSG Tossing

S2. Snap Assy & screwing
RFID 2#
Fixture code
Link HSG Barcode with Fixture ID
Ok or NG HSG Tossing

S3. Snap Assy & screwing
RFID 3#
Fixture code
Link HSG Barcode with Fixture ID
Ok or NG HSG Tossing

S4. Flipping Station
RFID 4#
Fixture code
Link HSG Barcode with Fixture ID
Ok or NG HSG Tossing

S5. Snap Assy & screwing
RFID 5#
Fixture code
Link HSG Barcode with Fixture ID
Ok or NG HSG Tossing

S6.Unloading machine
RFID 6#
HSG Barcode
Link HSG Barcode with Fixture ID
Ok or NG HSG Tossing

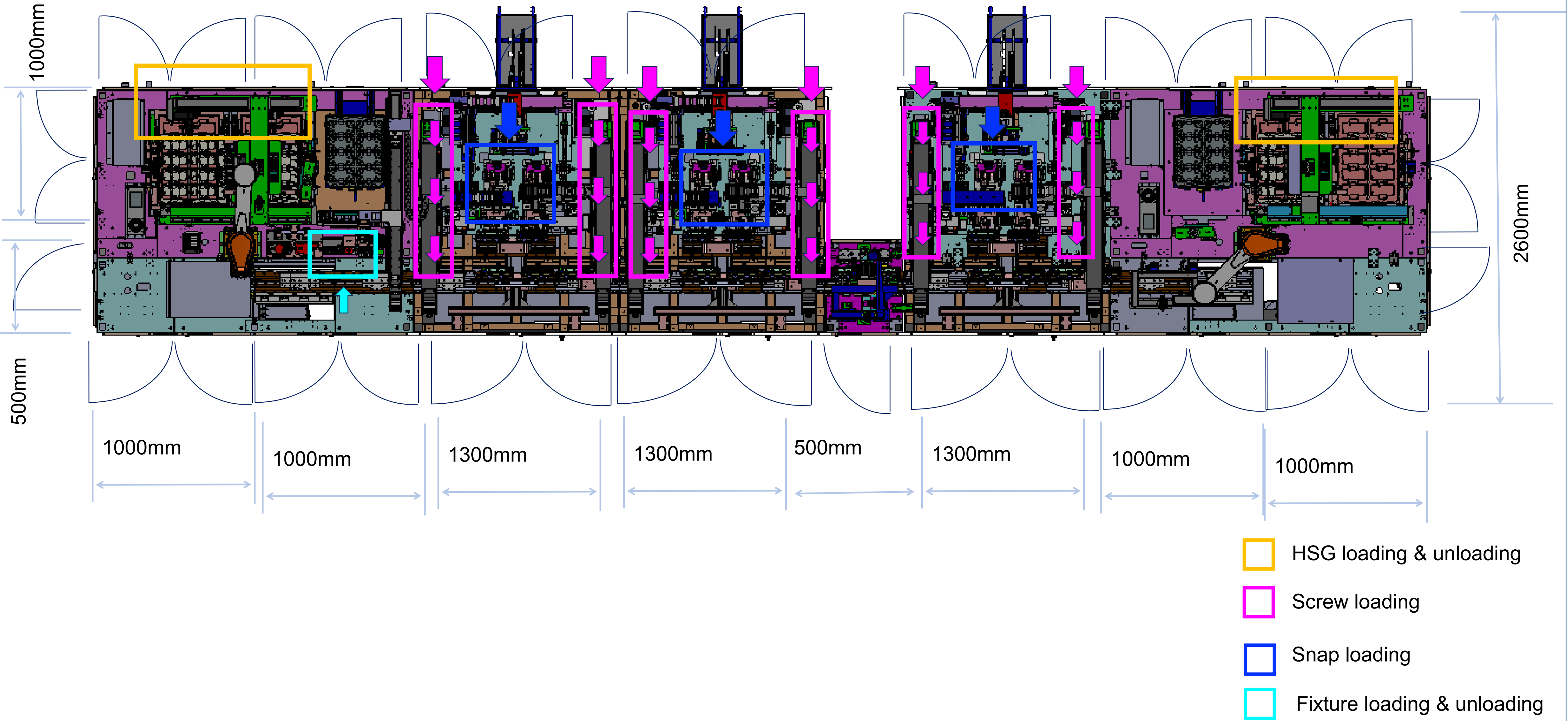
Scanner

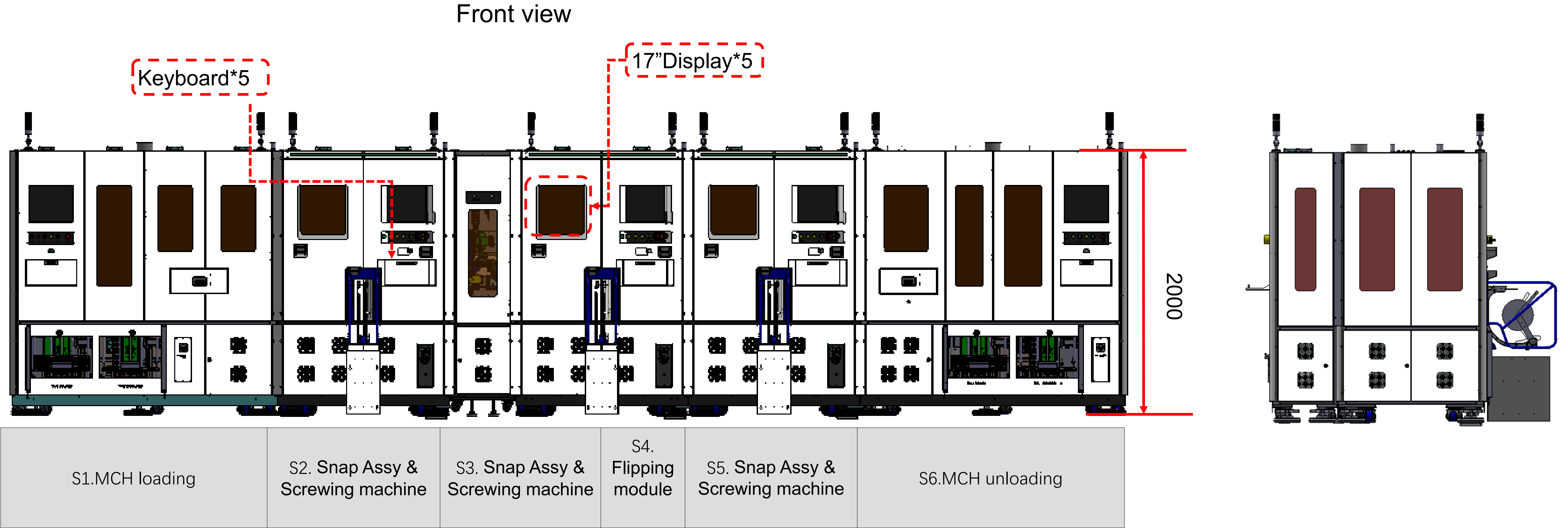
CCD for scanner

RFID Reader

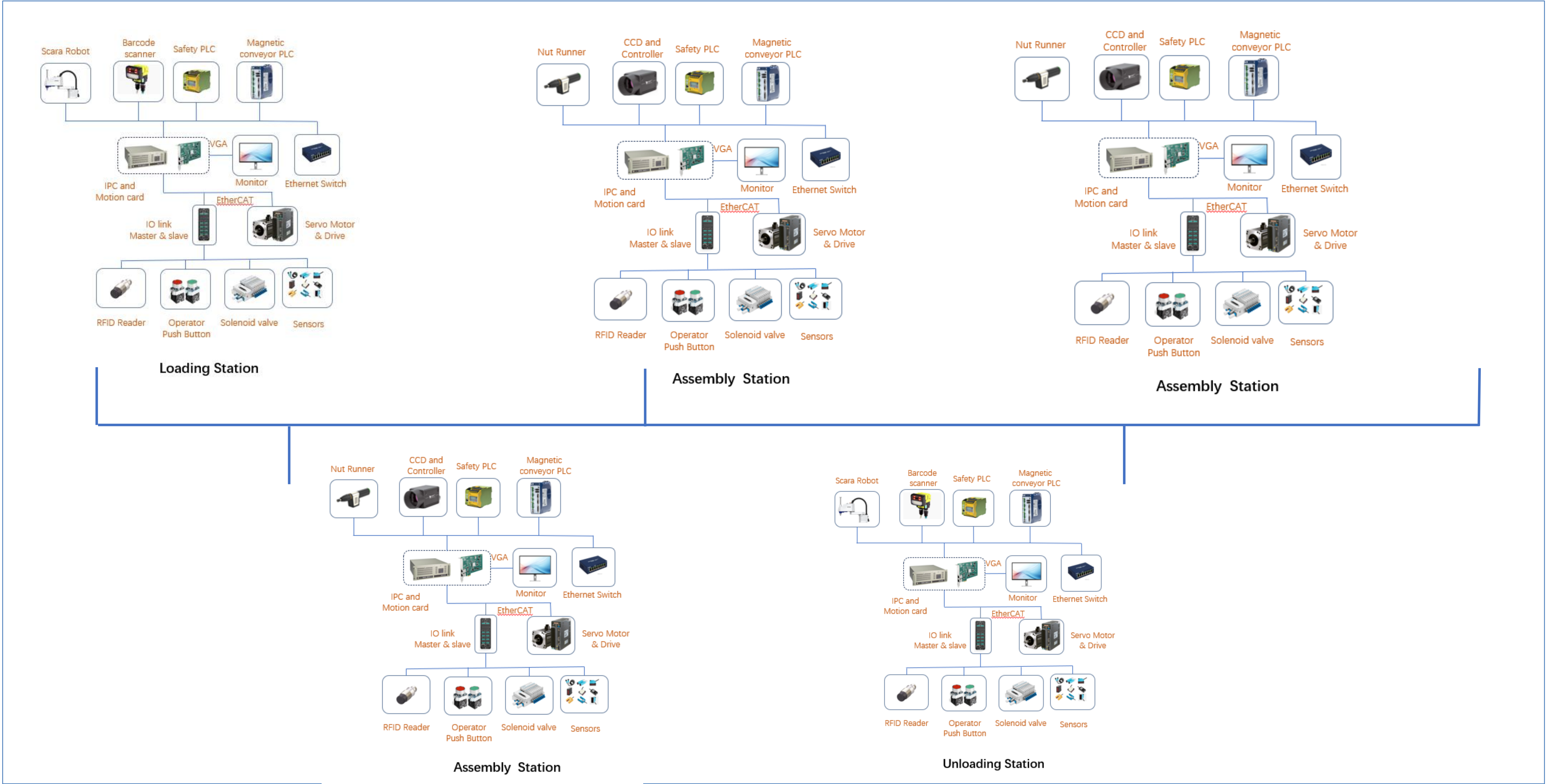
Station # & Name	
Scanner	RFID
Read object	Read object
Task description	
Judgement	

Monza | Snap Assy: Machine Overview - Maintenance Space

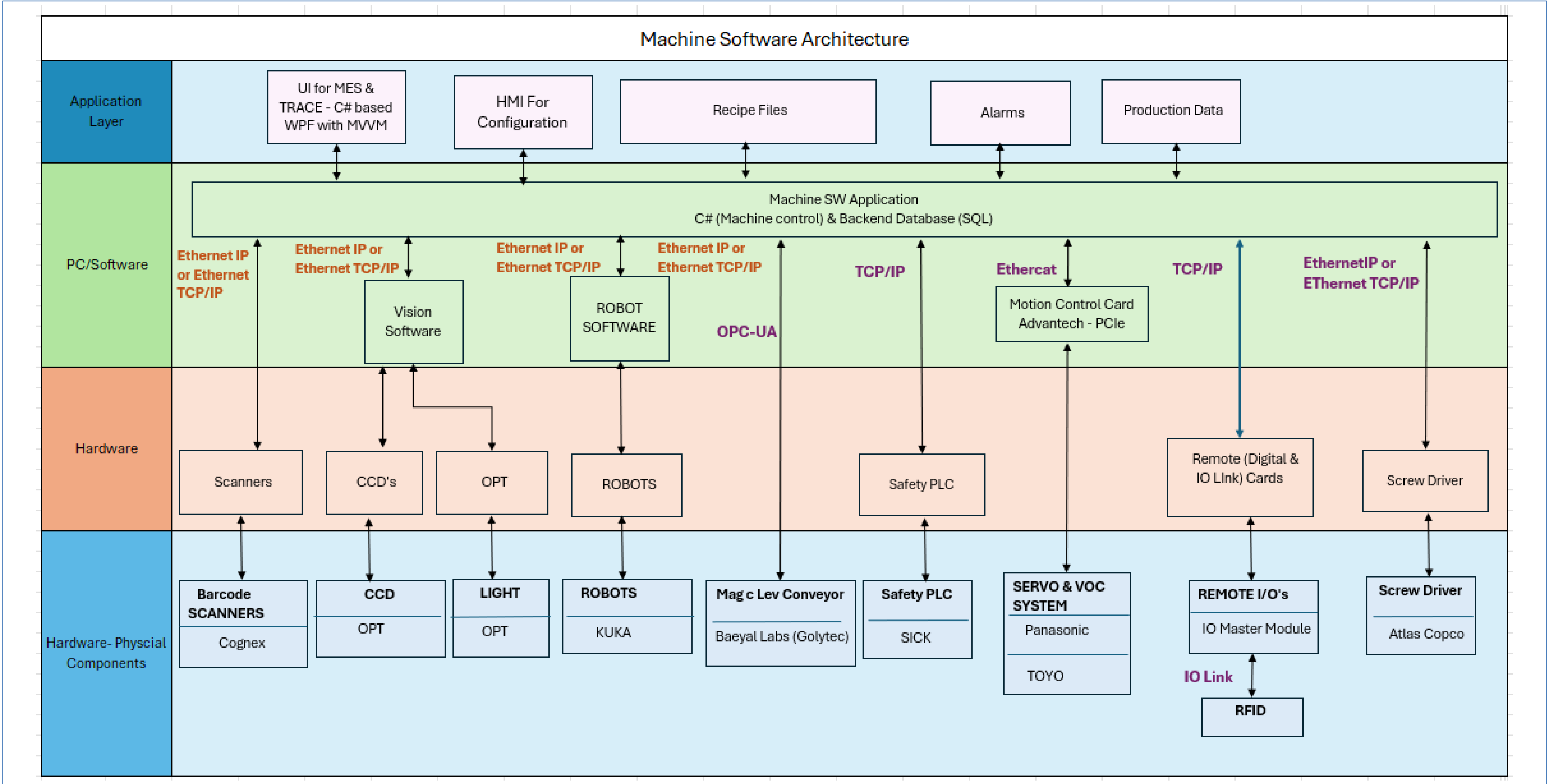




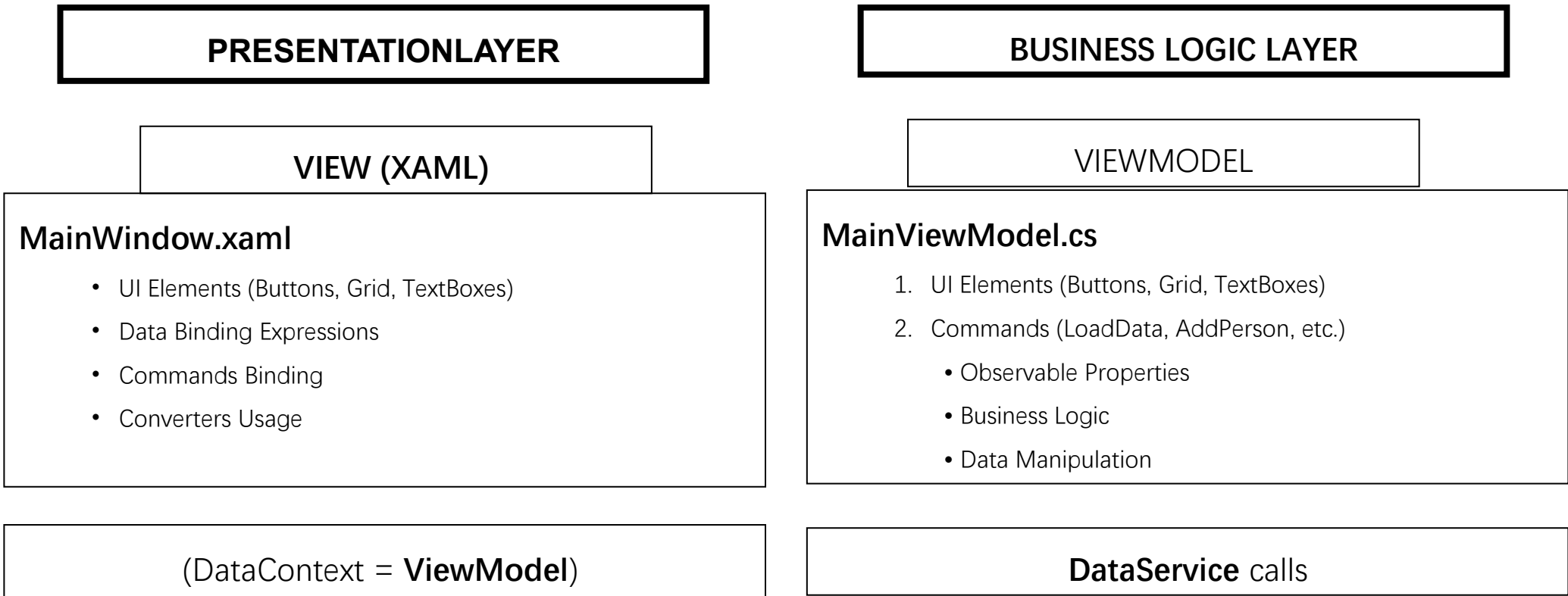
Monza | Snap Assy: Control system



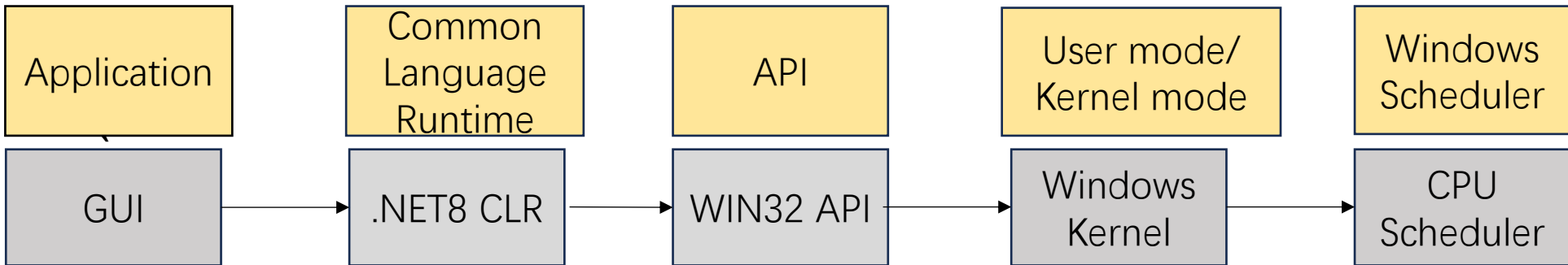
Monza | Snap Assy: Control system



1. MAIN APPLICATION
 - ProjectName.exe (Main WPF application)
2. CORE MODULES
 - PJName.Actuation (Actuator control)
 - PJName.FSM (Finite State Machine)
 - PJName.Link (Communication links)
 - MachineState (State management)
3. SERVICE LAYER (Abstraction)
 - [Service]Interface (Contracts/interfaces)
 - [Service] (Implementations)
 - Examples:
 - DataBaseService + DataBaseServiceInterface
 - HttpService + HttpServiceInterface
 - MotionCardServiceInterface + MotionCardServiceAdvMotAPI
 - Database ServiceLogservice
 - Scannerservice + ScanerserviceInterface
 - TCPservice + TCPserviceInterface
4. EXTERNAL INTEGRATIONS
 - AdvMotAPI (Motor/actuator API)
 - ScannerServiceCognex (Cognex scanner)



Interaction Flow



Aspect	.NET 8 Runtime	Windows Kernel
Thread & Process	Manages managed threads and tasks.	Schedules threads and handles CPU cores.
Memory	Uses GC for objects.	Allocates virtual memory and stack/heap.
I/O	Provides high-level I/O APIs.	Performs actual device and disk/network access.
Timers & Events	Runs managed timers and events.	Handles kernel timers and signals threads.
CPU Usage	Sets thread priority; relies on kernel.	Controls CPU scheduling and priorities.

Threading & Task Management

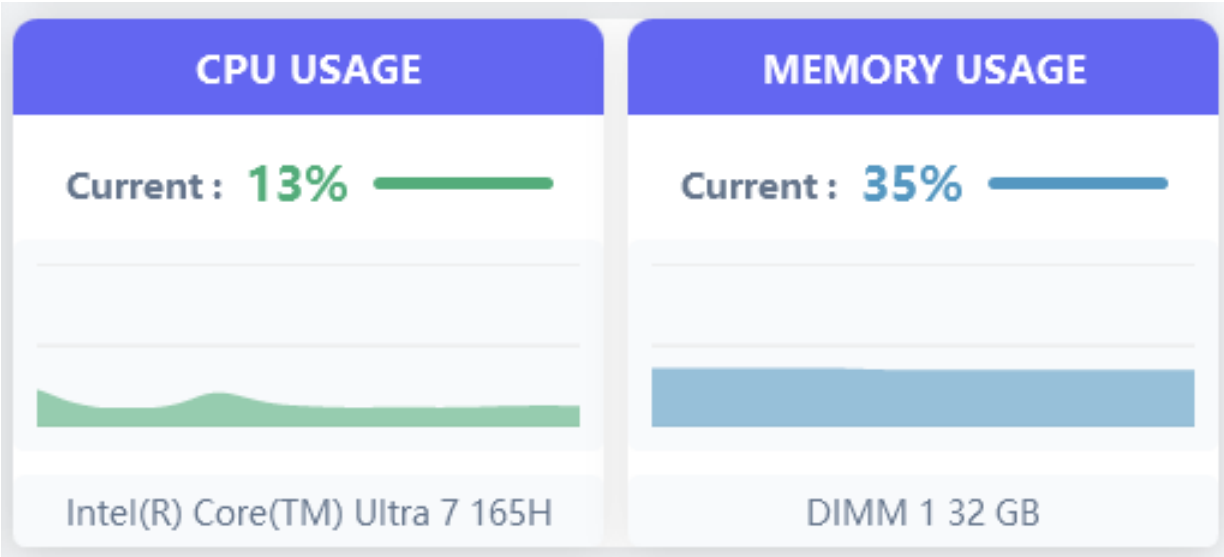
Multi-threaded Execution & Task Management

- **UI Thread:** Main thread handles user interaction and screen updates.
- **Background Worker Threads:** Run robot control, data logging, and I/O tasks independently to keep UI responsive.
- **Task Parallel Library (TPL):** Enables parallel execution of tasks without manual thread creation, improving CPU usage.
- **ThreadPool:** Reusable threads for short-lived tasks and timers; reduces thread creation overhead.
- **Timers:** Execute periodic operations via ThreadPool; high-frequency timers increase CPU load.
- **Garbage Collector Threads:** Dedicated background threads clean unused memory; frequent GC can spike CPU.

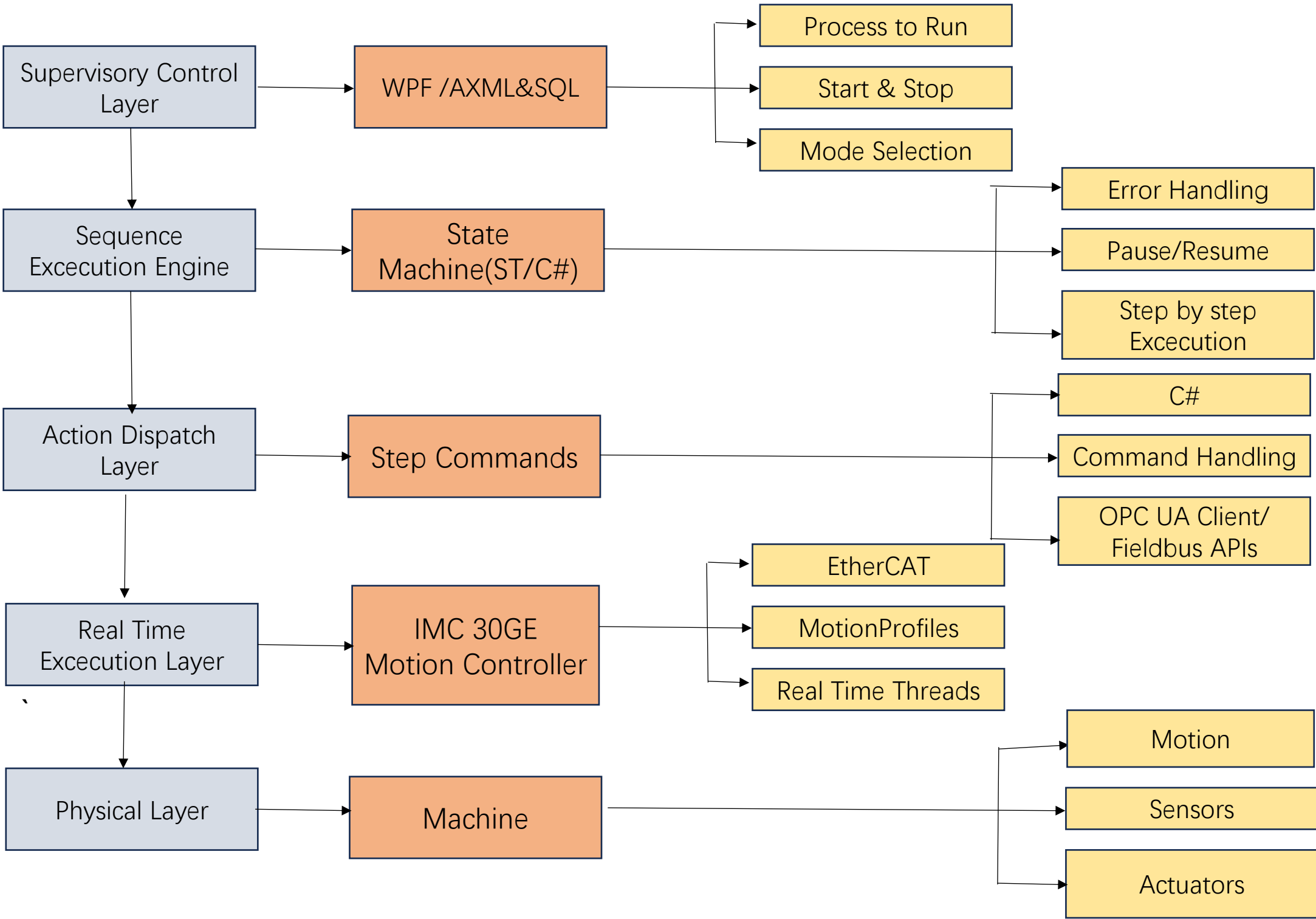
Component	Thread Type	Purpose
UI	Main thread	User interaction
Robot Control	Background Task	Motion & status execution
Data Logging	Background Task	File/DB logging
Timers	ThreadPool	Periodic execution
Garbage Collector	Background Thread	Memory cleanup

CPU Usage Control & Optimization

- **ProjectName.exe** is a background application or service running on your system.
- It is currently using **very low CPU (~13%)** and about **440 MB of memory**, which is normal.
- The process shows **35% in 32GB disk memory usage, network, or GPU activity**, which is normal.



SEQUENCE OF AUTOMATION FLOW



Design Time Configuration

- Design-Time Configuration refers to **predefined machine and system parameters**
- Stored persistently in **XAML configuration files**
- Created and maintained **outside runtime execution**
- Used dynamically by runtime logic without recompilation

Role of XAML Configuration file

- XAML acts as the **single source of truth**
- Structured and human-readable
- Editable using **VS Code**
- Supports schema validation
- Version-controlled for traceability

Parameter Configuration

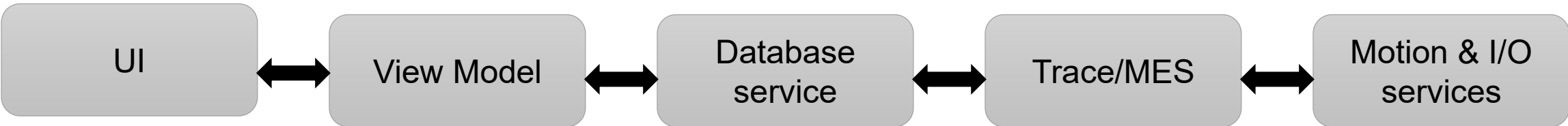
➤ SQL-Based Parameter Configuration Architecture

- The system uses a centralized SQL database (MySQL) for parameter configuration.
- Configuration data is accessed through MVVM-based services.
- Prism framework enables loose coupling and modular architecture.
- Supports Trace, MES, and MES OEE Dashboard, PDCA parameter segregation.

➤ SQL Database Structure for Parameters

- Parameters Table**
 - Defines Station Name, SN No, Parameter Name, data type, unit, and module.
- Parameter Values Table**
 - Stores actual parameter values per machine process data.

MVVM Parameter Handling Flow

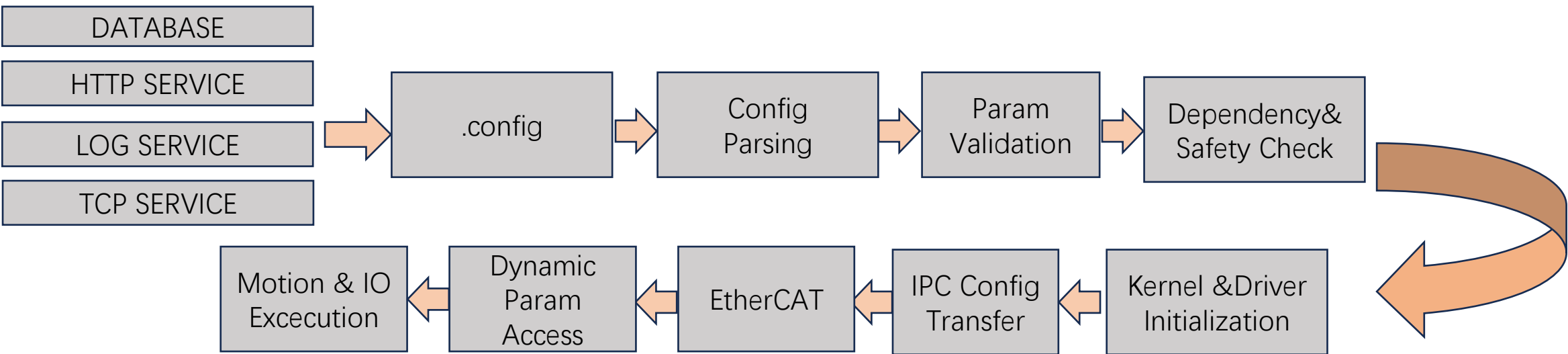


- UI binds parameter values using WPF data binding.
- View Model loads and validates parameters.
- Database Service performs SQL operations.
- Trace & MES logic consumes validated data.
- Motion and IO services execute control logic.

Prism Framework

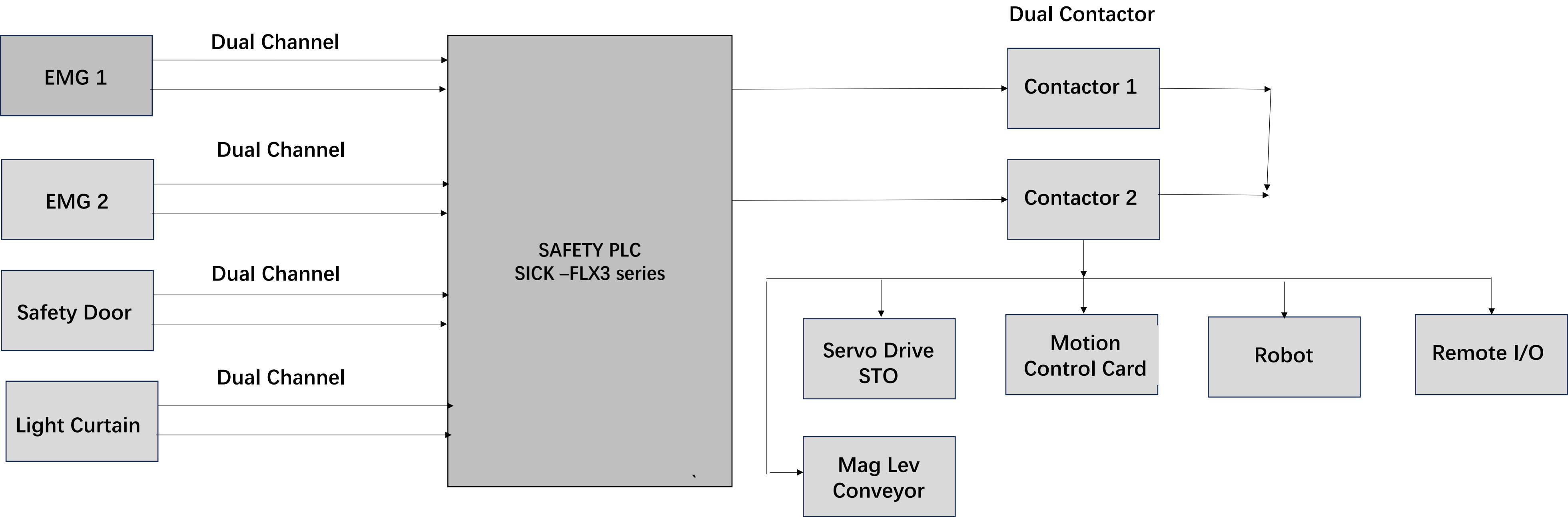
- Dependency Injection for services
- Event Aggregator for parameter update notifications
- Command binding for Save / Load actions
- Region management for dynamic UI screens

Configuration parameter Process



Configuration parameter File view

- All configuration parameters are stored in external files (e.g., Database.config, HTTPService.config, Logservice.config, TCPservice.config).



Note: All Safety Devices complies under safety standards.

Safety Door – Ple | Cat4 | SIL3 | ISO13849-1

Light Curtain – Cat4 | Ple | SIL3 | ISO13849-1, maintain safety calculation Regulation ISO13855

Robot Safety Standard – ISO10218-1

Servo Drive – With STO

Monza | Snap Assy: Machine Overview - BOM & Warranty

Key Components										
Item	Machine	Module	Function	Key parts	Brand	Model	Q'ty	Warranty	New/Reuse	Consumable
S1	Loading machine	Material Storage module	Transfer tray	Y axis	Toyo	GTH8-L20-600-BC-P40-C4P5	1	12M	New	N
				cylinder	SMC	CDUJB12-15DM-F8PL	2	12M	New	N
				cylinder	SMC	MXQ20-75AS-M9PL	1	12M	New	N
				cylinder	SMC	MGPM16-20Z-M9PL	6	12M	New	N
			Clamping of material tray	cylinder	SMC	MXQ8-20A-M9PL	2	12M	New	N
				cylinder	FESTO	MGPM16-20Z-M9PL	6	12M	New	N
				cylinder	FESTO	CDUJB16-20DM	4	12M	New	N
		Flipping Gantry module	Load HSG	X axis	Toyo	GTH8-L20-1000-BM-P40-C4	1	12M	New	N
				Y axis	HIWIN	KK60D10P-300A1-F0	1	12M	New	N
				Z axis	HIWIN	KK60D10P-300A1-F0	1	12M	New	N
				Cylinder	SMC	CDUJB10-30DM-F8PL2	2	12M	New	N
				Rotating cylinder	CHELIC	RTL18-8G2	2	12M	New	N
		Robot	Pick up HSG	4-axis Robot	Kuka	KR 13 R850 Z200-2 CS	1	12M	New	N
		Control system	IPC	IPC	ADVANTECH	IPC 7132	1	12M	New	N
			Axis Control	Motion Control card	ADVANTECH	PCIE-1203-16A	1	12M	New	N
			Remote I/O Module	I/O Link Master			3	12M	New	N
			Remote I/O Module	I/O Link Slave			16	12M	New	N
			Link fixture	RFID	PEPPERL+FUCHS	IQT1-18GM-IO-V1	2	12M	New	N
		Security Configuration	Safety Door Latch	Safety Door Latch	Keyence	GS-51P5	16	12M	New	N
			Safety Grating	Safety Grating	Keyence	GL-VM39FP_N_1	2	12M	New	N

Monza | Snap Assy: Machine Overview - BOM & Warranty

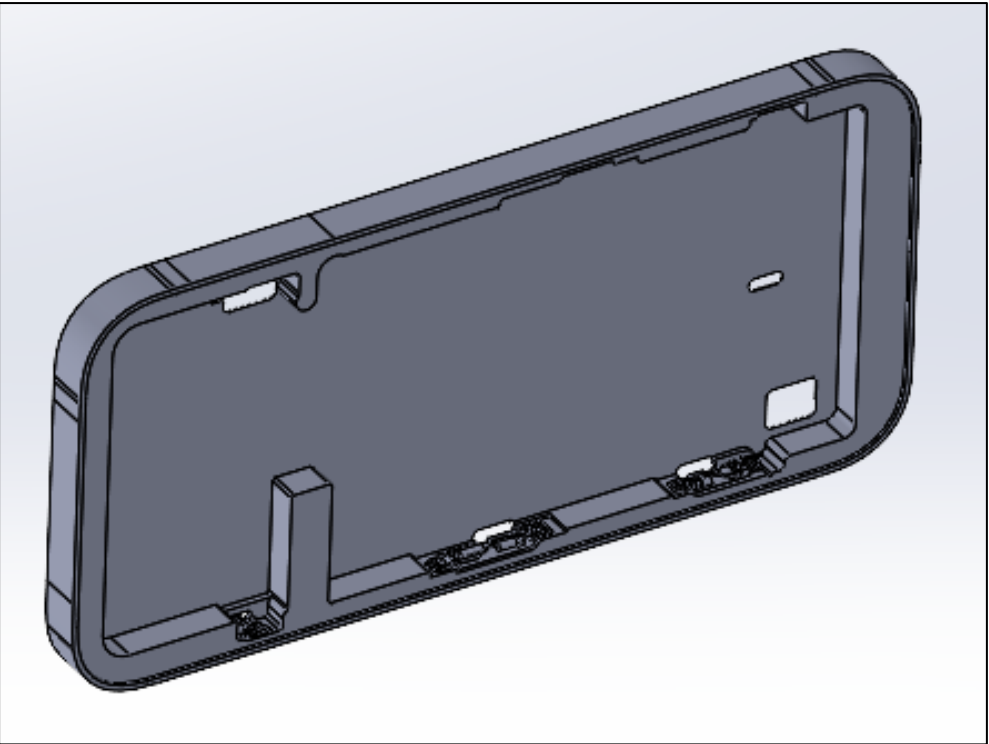
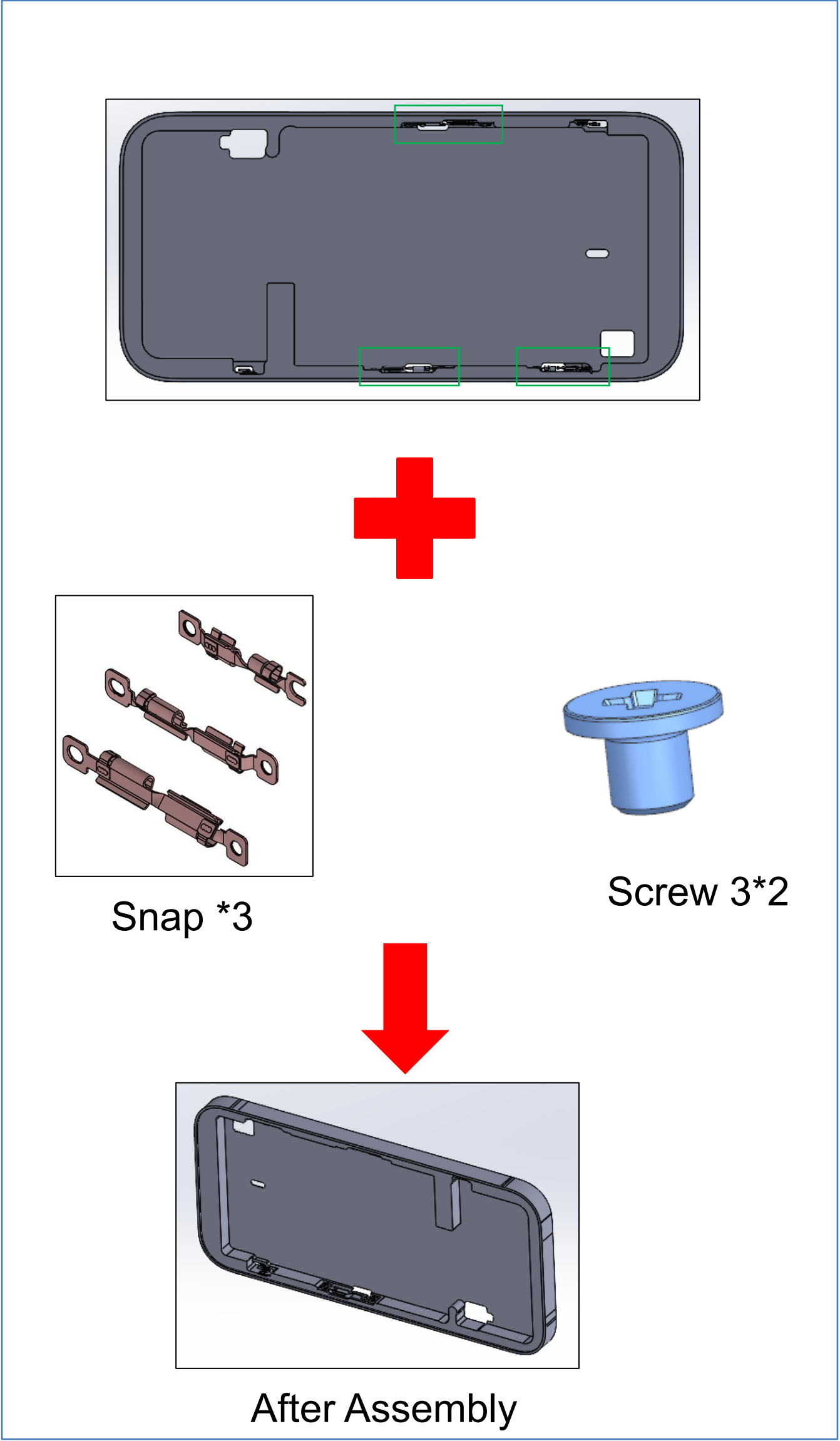
Key Components										
Item	Machine	Module	Function	Key parts	Brand	Model	Q'ty	Warranty	New/Reuse	Consumable
S2,S3, S5	Snap Assy & Screwing machine	Screwing module	Screwing	Smart Screwdriver	Atlas	QMC-21-10-HM4	4	12M	New	N
				Smart Screwdriver Controller	Atlas	MTF6000	4	12M	New	N
				Screw Feeding Device	Deprag	DEPRAG Mini Screw Feeder_189x103x156	2	12M	New	N
				Y axis	HIWIN	KK86D20P-740A1-F0C	2	12M	New	N
				X axis	HIWIN	KK86D10C-440A1-F0 KK	2	12M	New	N
				Z axis	HIWIN	KK60D10C-300A1-F0	2	12M	New	N
				cylinder	SMC	MXQ12-10B-M9PVL	2	12M	New	N
		Snap loading module	Snap loading	X axis	HIWIN	KK60D10C-500A1-F0	1	12M	New	N
				Y axis	HIWIN	KK60D10C-300A1-F0	1	12M	New	N
				Voice coil	Toyo	VLR2565	2	12M	New	N
		Snap feeding module	Snap feeding	Tape Reel Feeder	YIG	Tape Reel Feeder	2	12M	New	N
			Transfer snap	Y axis	Toyo	GTH5-L10-300-BL (R) -Y10-C3	2	12M	New	N
			Assemble snap	cylinder	SMC	MSQAZAE	2	12M	New	N
		Lower CCD	Recheck & correct screw	CCD	OPT	SCI-ZXCM2500-GMP11-003	2	12M	New	N
		Control system	IPC	IPC	ADVANTECH	IPC 7132	1	12M	New	N
			Axis Control	Motion Control card	ADVANTECH	PCIE-1203-32A	1	12M	New	N
			Remote I/O Module	I/O Link Master			3	12M	New	N
			Remote I/O Module	I/O Link Slave			12	12M	New	N
			Link fixture	RFID	PEPPERL+FUCHS	IQT1-18GM-IO-V1	2	12M	New	N

Monza | Snap Assy: Machine Overview - BOM & Warranty

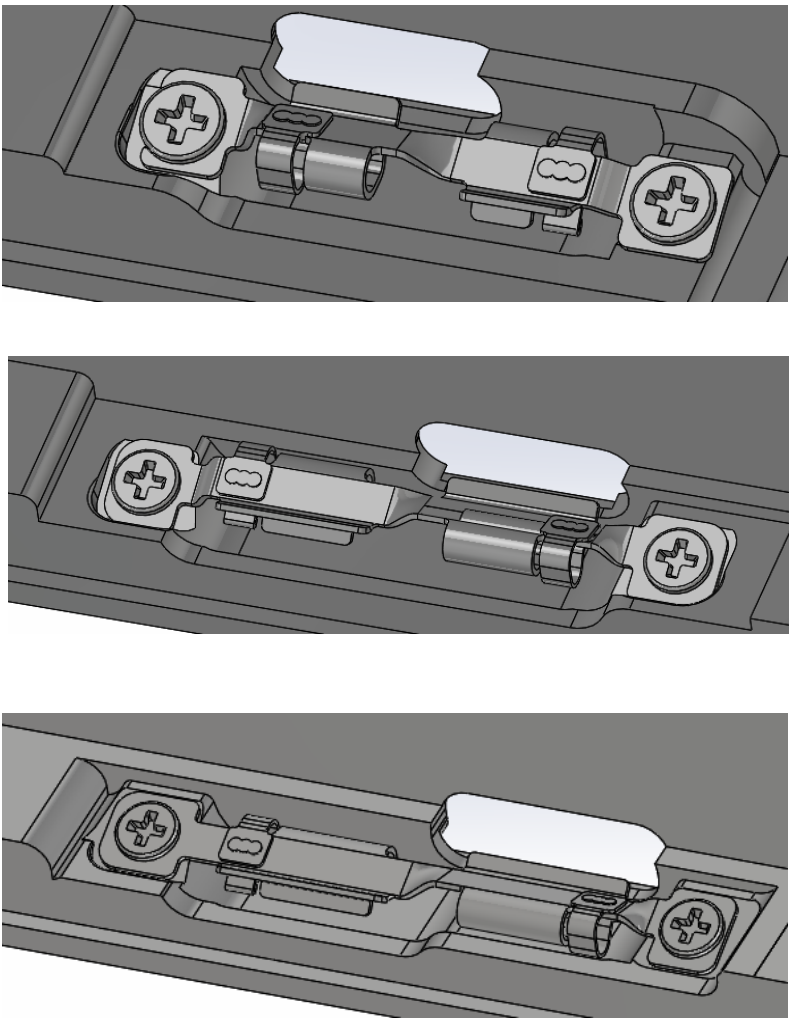
Key Components										
Item	location	Module	Function	Key parts	Brand	Model	Q'ty	Warranty	New/Reuse	Consumable
S6	Unloading machine	Storage module	Transfer tray	Y axis	Toyo	GTH8-L20-600-BC-P40-C4P5	1	12M	New	N
				cylinder	SMC	CDUJB12-15DM-F8PL	2	12M	New	N
				cylinder	SMC	MXQ20-75AS-M9PL	1	12M	New	N
				cylinder	SMC	MGPM16-20Z-M9PL	6	12M	New	N
			Clamping of material tray	cylinder	SMC	MXQ8-20A-M9PL	2	12M	New	N
				cylinder	FESTO	MGPM16-20Z-M9PL	6	12M	New	N
				cylinder	FESTO	CDUJB16-20DM	4	12M	New	N
		Robot	Pick up HSG	4-axis Robot	Kuka	KR 13 R850 Z200-2 CS	1	12M	New	N
				Rotating cylinder	CHELIC	RTL18-8G	2	12M	New	N
		Control system	IPC	IPC	ADVANTECH	IPC 7132	1	12M	New	N
			Axis Control	Motion Control card	ADVANTECH	PCIE-1203-16A	1	12M	New	N
			Remote I/O Module	I/O Link Master			3	12M	New	N
			Remote I/O Module	I/O Link Slave			12	12M	New	N
			Link fixture	RFID	PEPPERL+FUCHS	IQT1-18GM-IO-V1	2	12M	New	N
		Security Configuration	Safety Door Latch	Safety Door Latch	Keyence	GS-51P5	6	12M	New	N
			Safety Grating	Safety Grating	Keyence	GL-VM39FP_N_1	2	12M	New	N

Process Detail

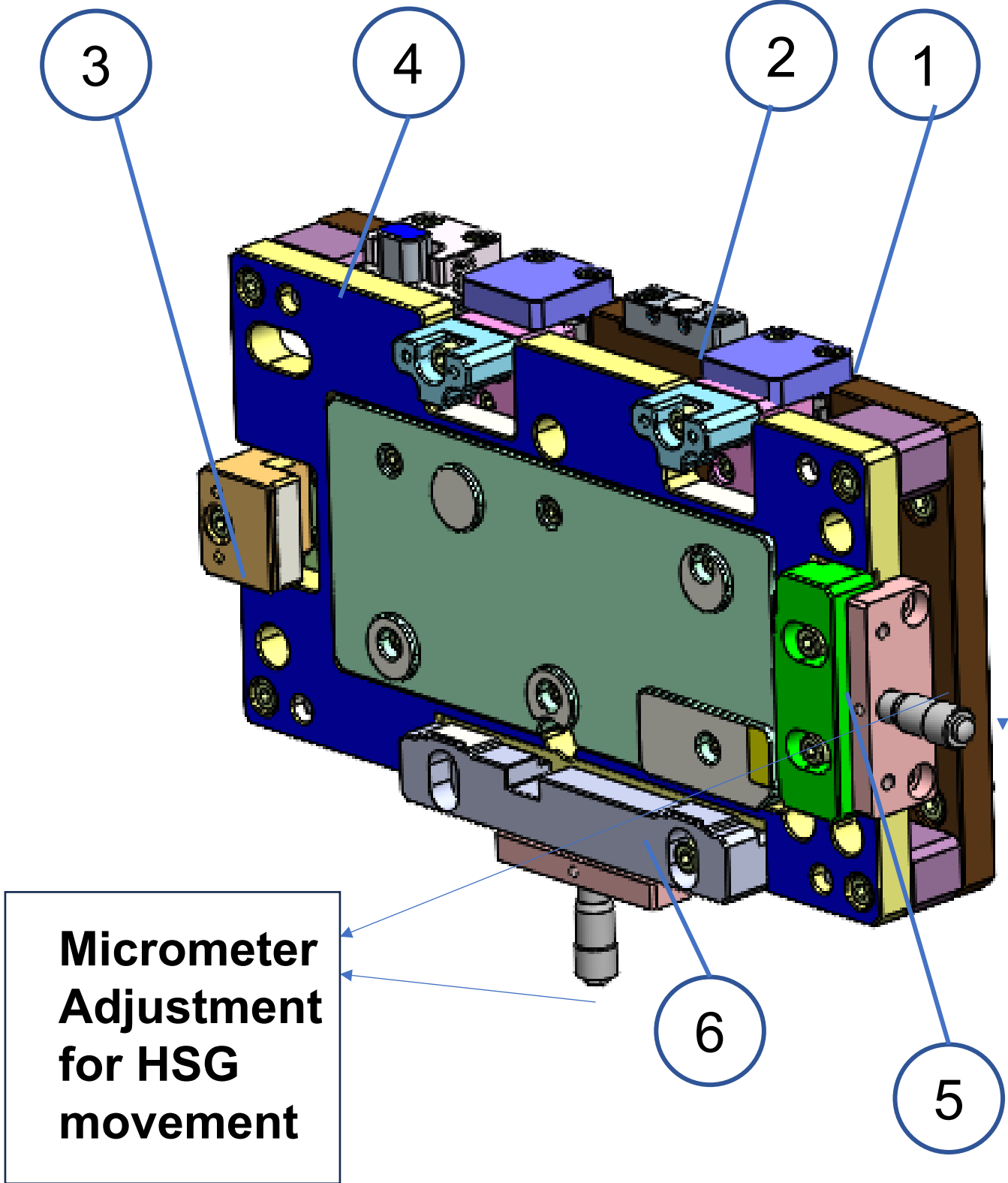
Monza | Snap Assy: Individual Process Details



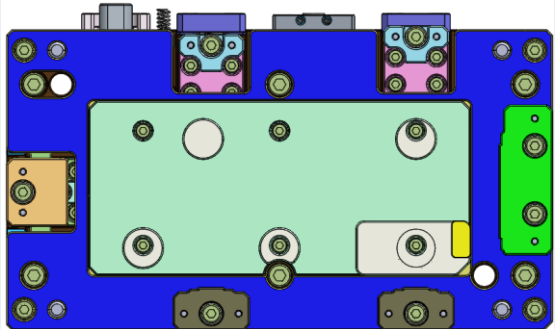
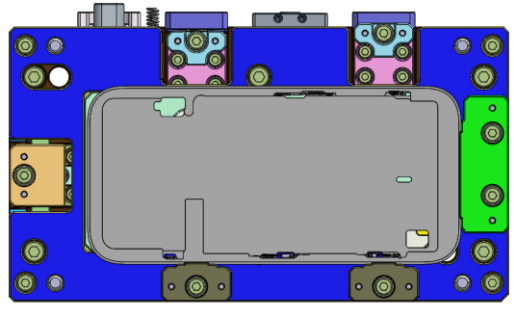
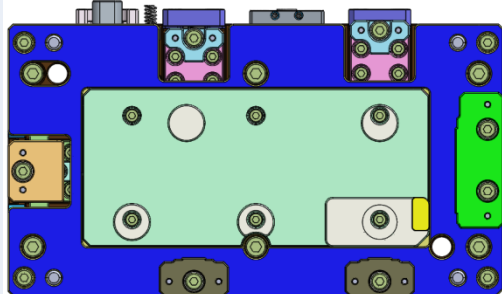
Screwing



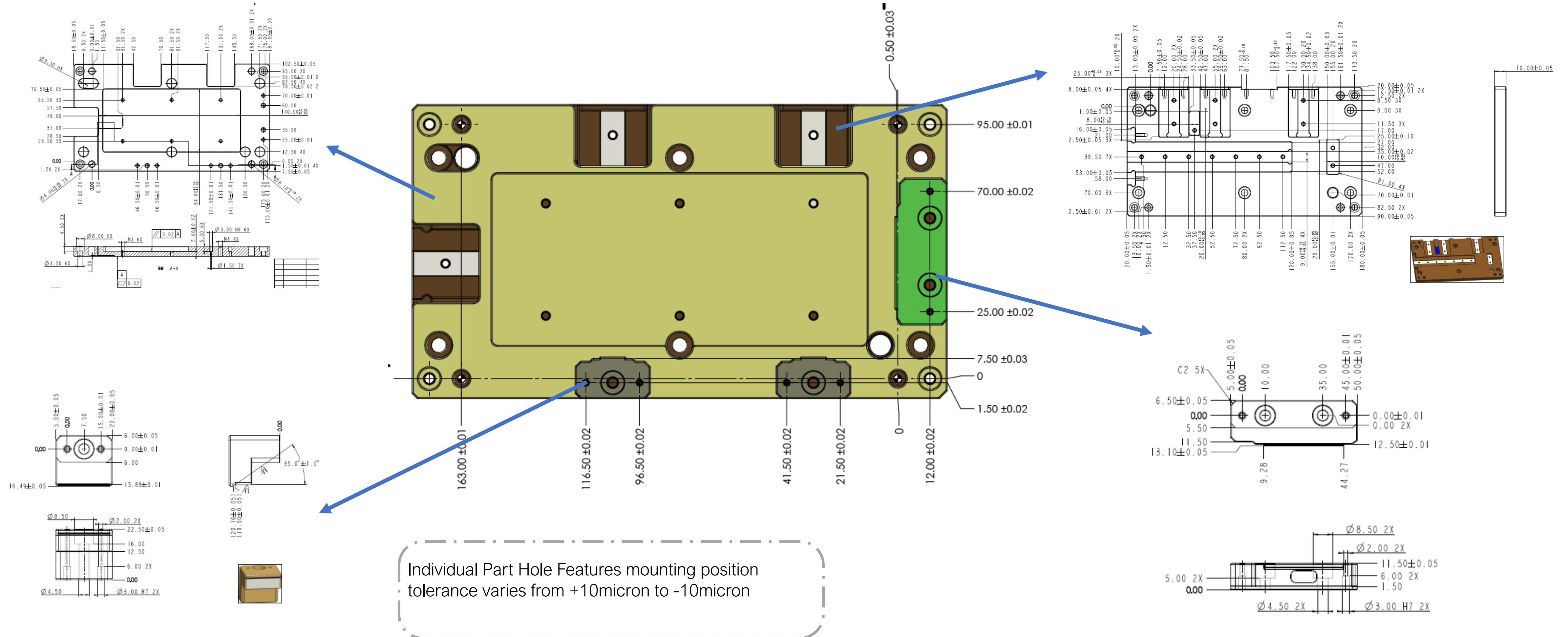
Fixture



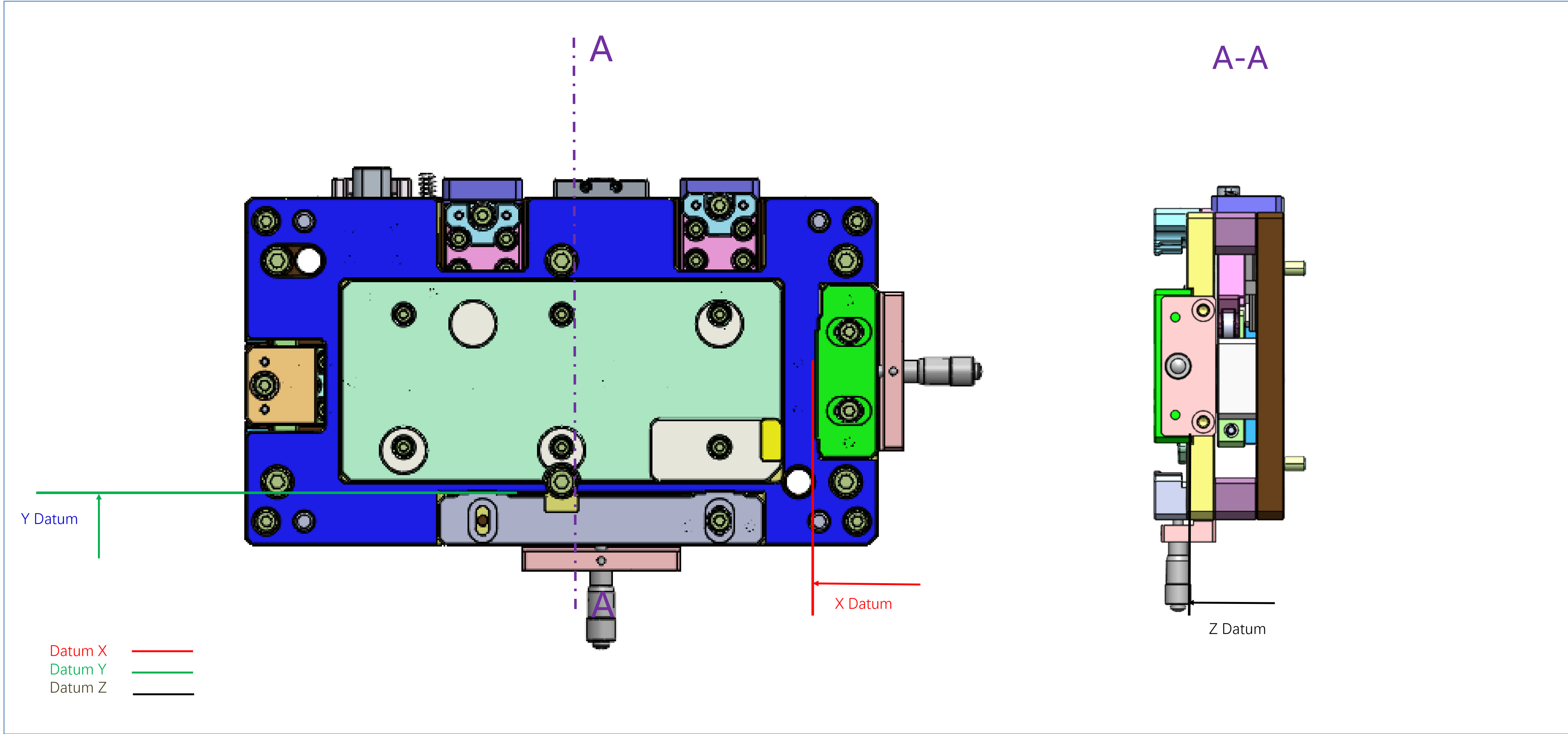
No.	Name.	Material
1	Long side push block	PPS
2	Product support plate	PPS
3	Short side push block	PPS
4	Unlock the block	S136
5	Short side datum block	PPS
6	Long side datum block	PPS

Actions	Step1	Step2	Step3
Diagram			
Description	Two sides unclamps	Put HSG in fixture and push clamps HSG for position	Unloading HSG
Spring Force		Long side push block:8N Short side push block:15N	

Monza | Snap Assy: Fixture – Stack up Tolerance



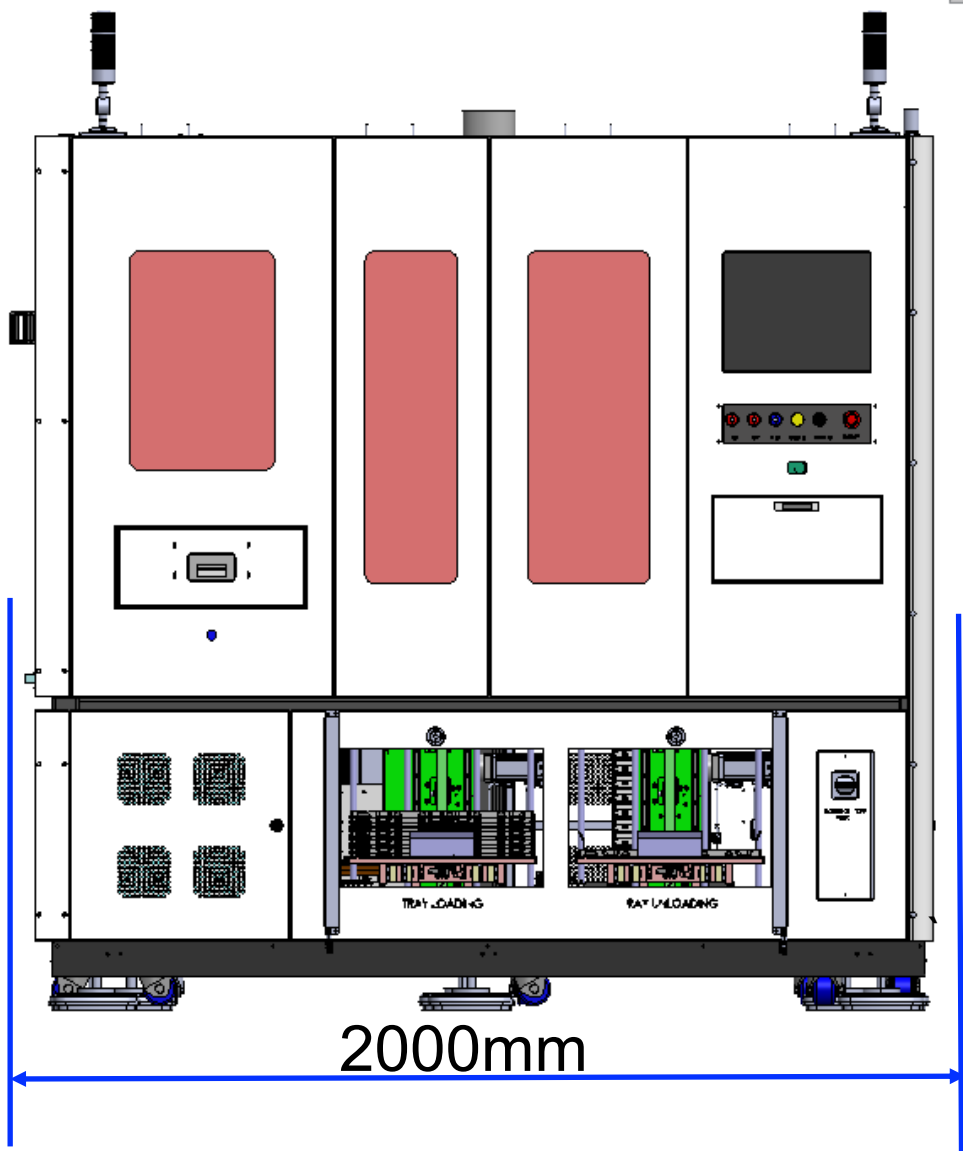
Datum - Fixture in Snap Assy & Screwing machine



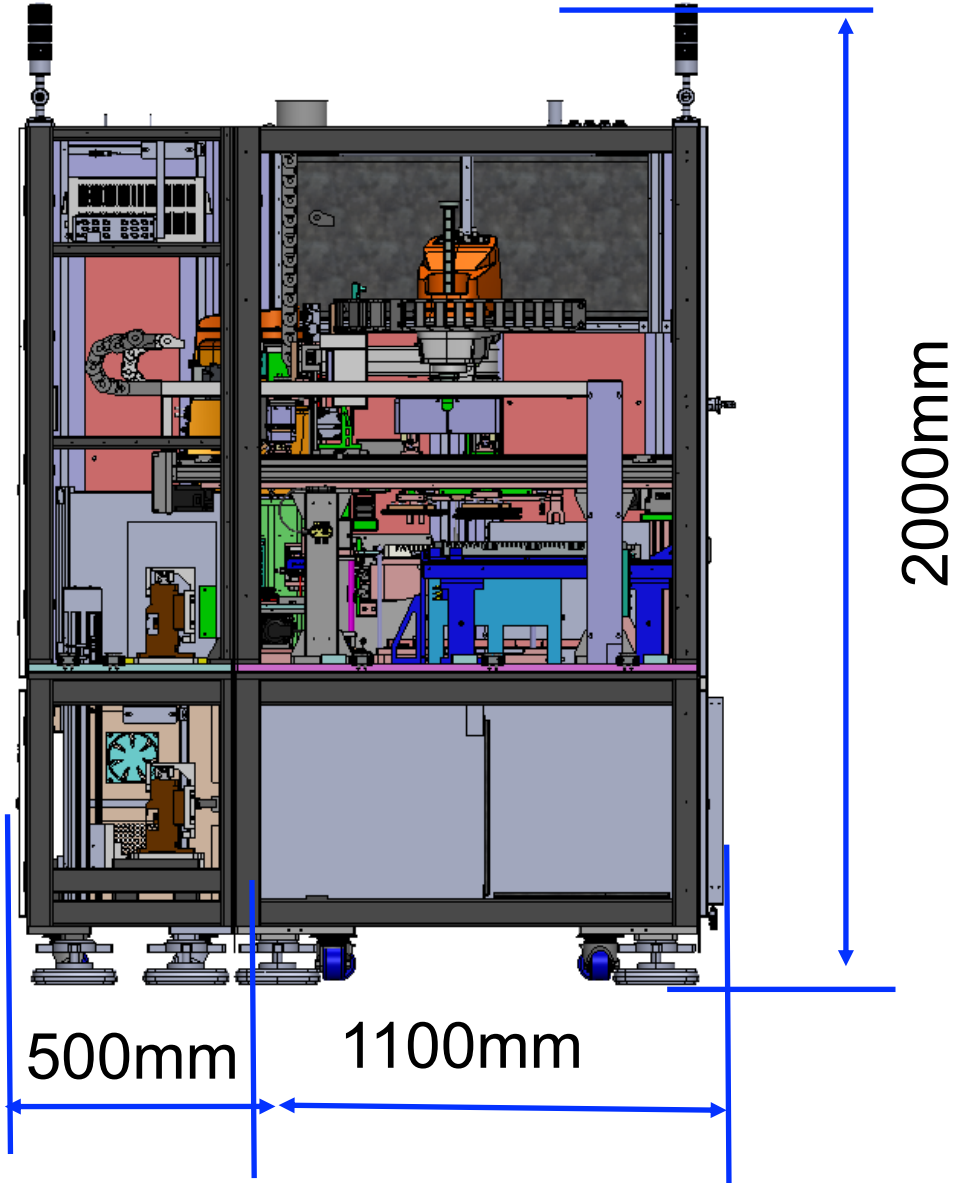
Station Details- Loading Machine

Monza | Snap Assy : Machine Overview –Layout

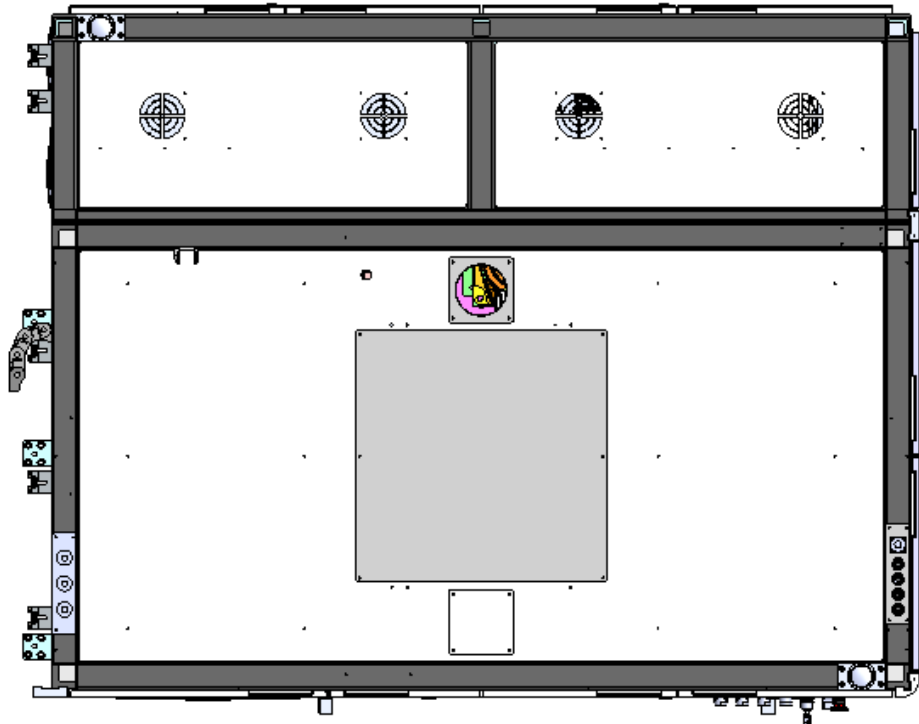
Front View



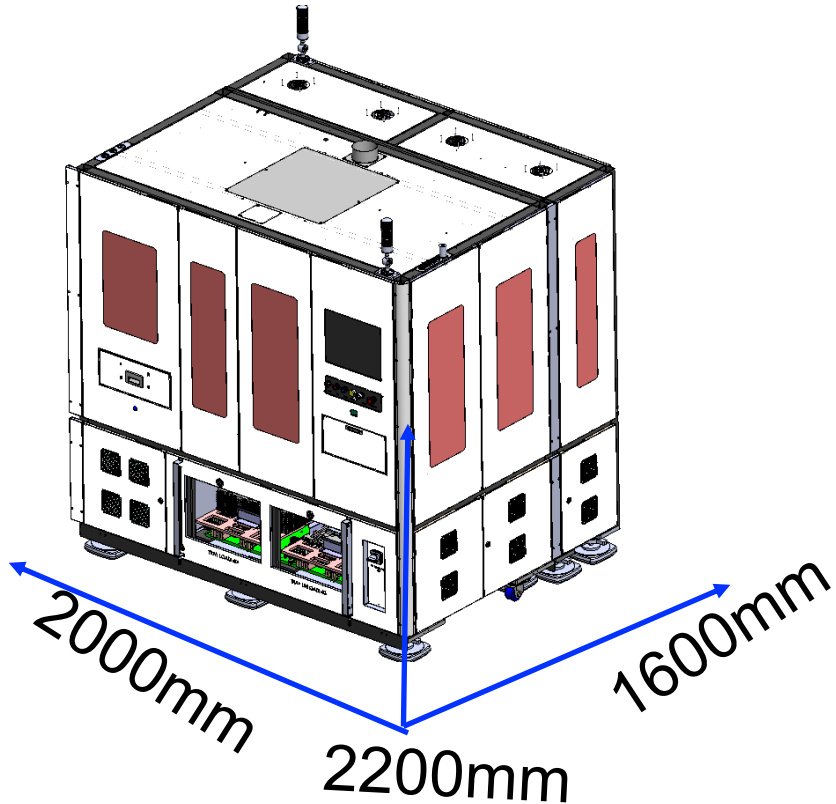
Side View



Top View

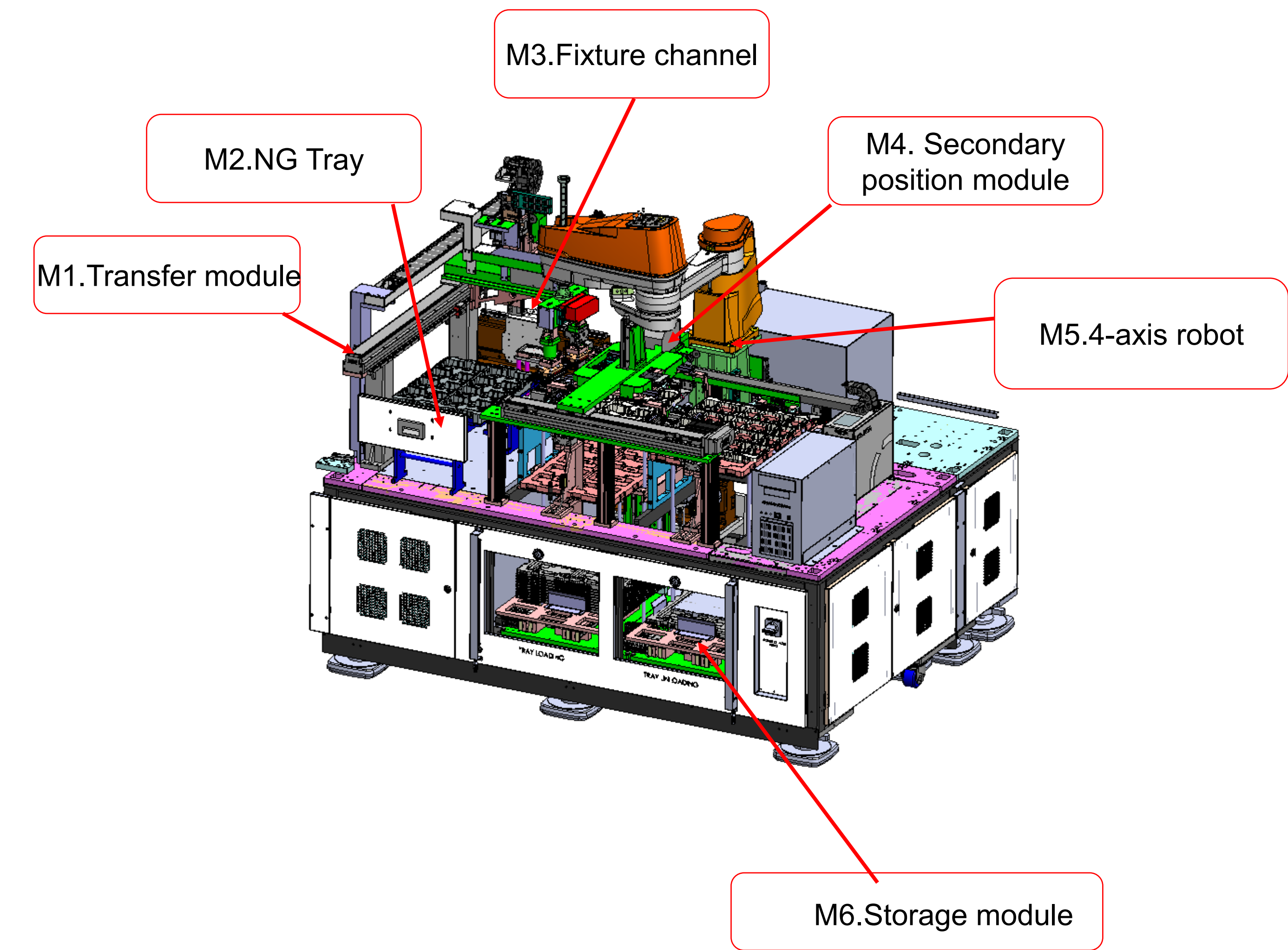


Axis View



- 1.Voltage:220V
- 2.Power: 10Kw
- 3.Air: 0.5~0.7Mpa
- 4.Labor:/
- 5. Machine Weight: 2000kg

Monza | Snap Assy: Machine Overview– Station BOM

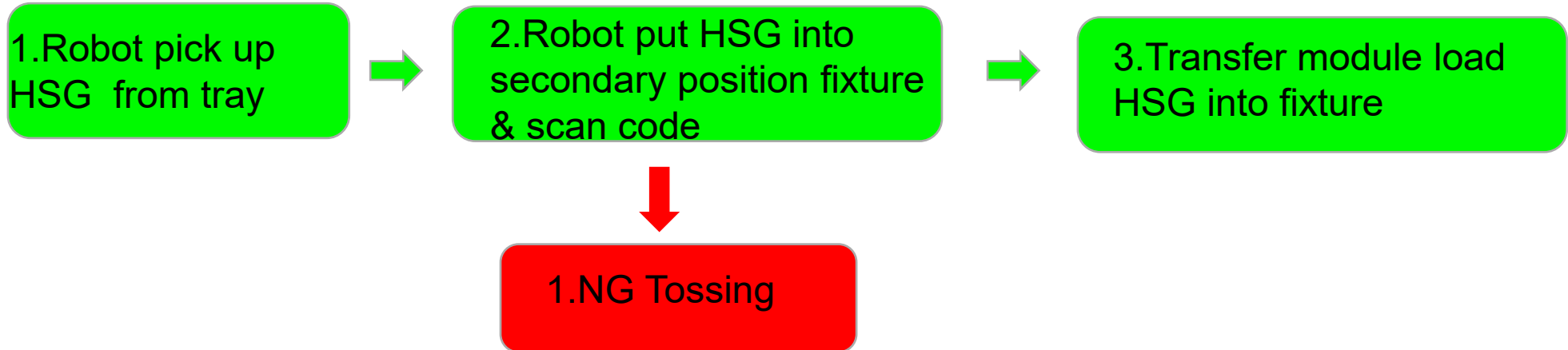
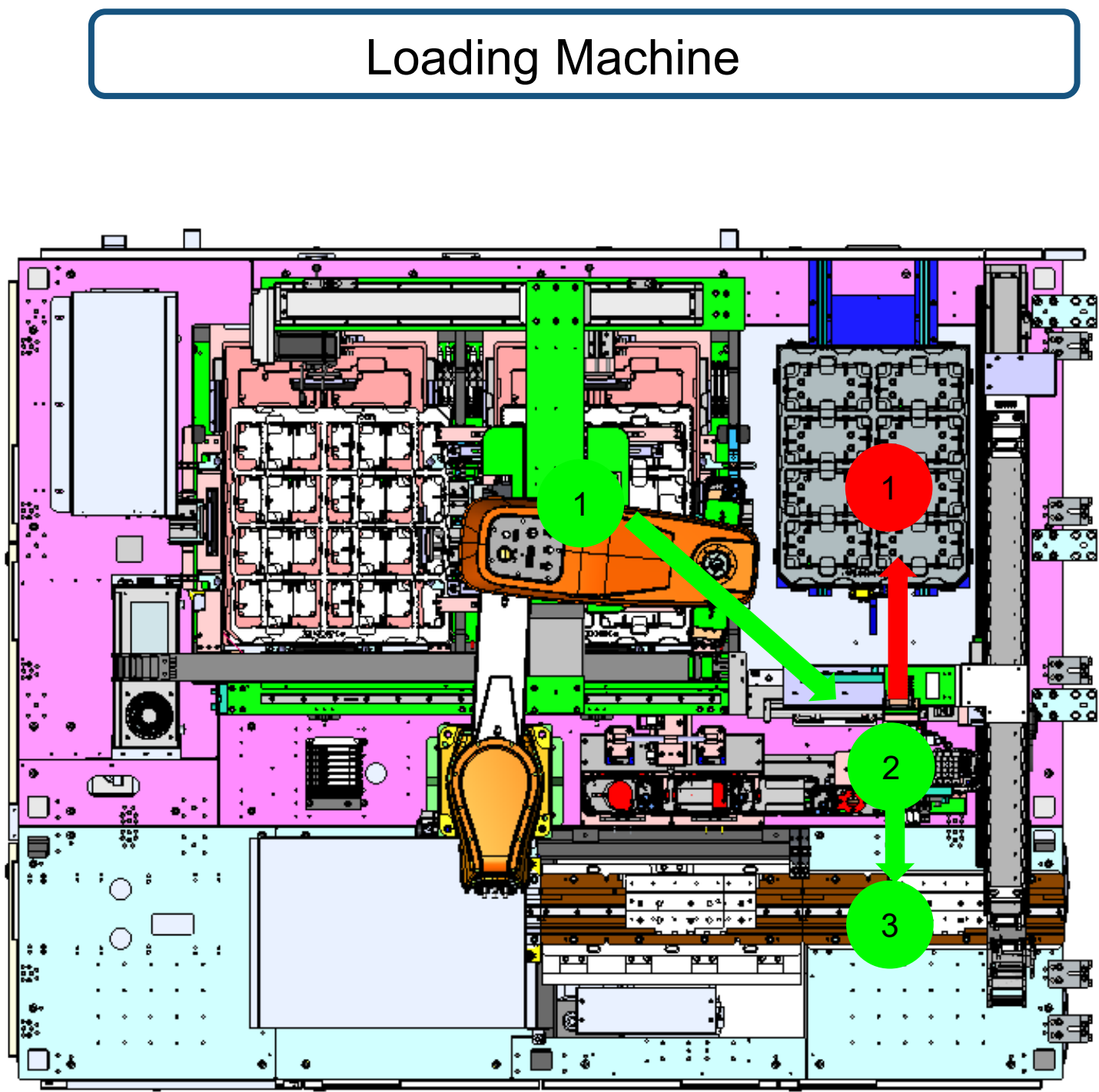


#	Module	Module Qty	Key parts	Key part QTY (TTL)
M1	Transfer module	1	X axis module	1
			Y axis module	1
			Z axis module	1
			Rotating cylinder	2
M2	NG Conveyor	1	/	/
M3	Fixture channel	1	channel	1
		1	RFID	1
M4	Secondary position module	1	/	/
M5	4-axis robot	1	Robot	2
M6	Storage module	2	Storage	1

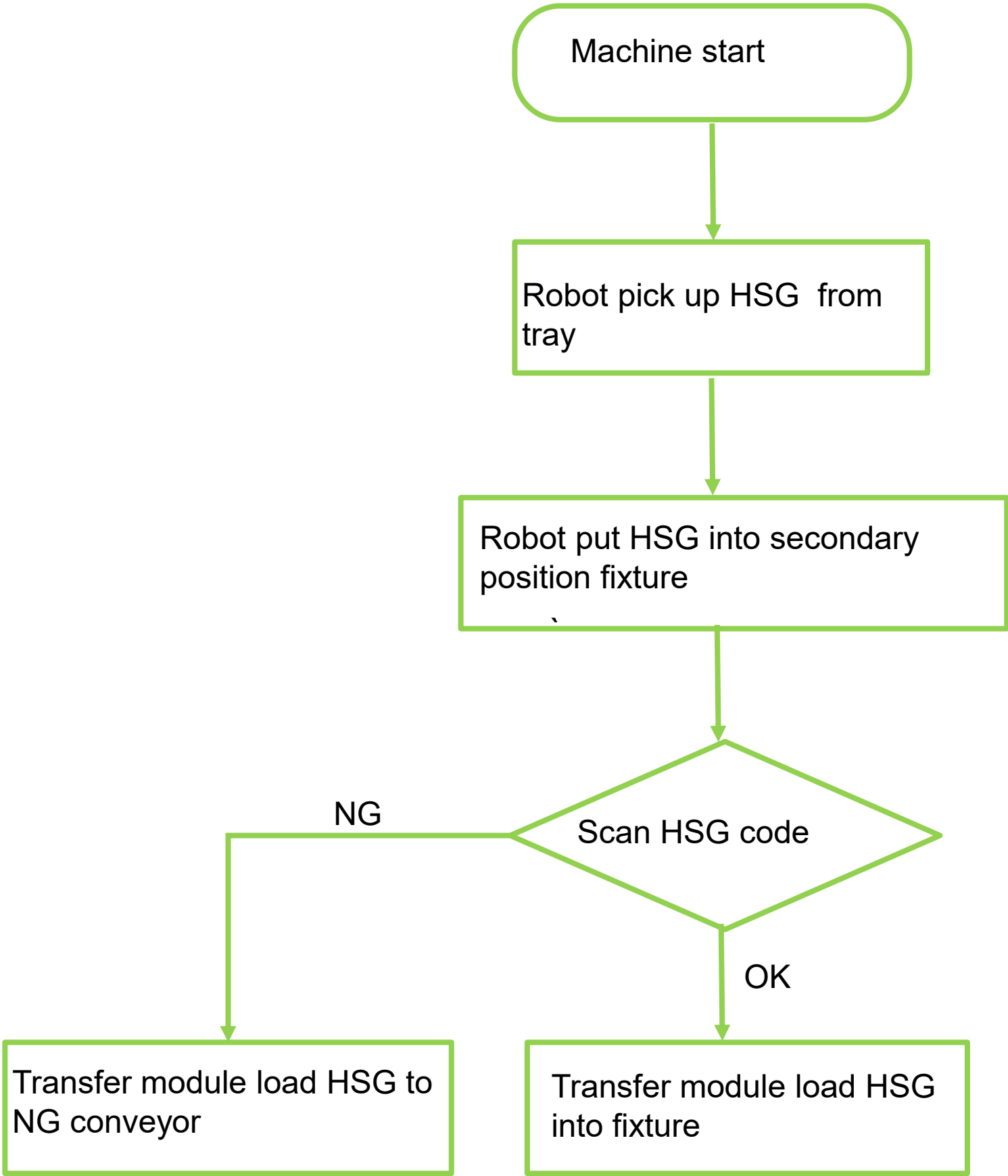
Monza | Snap Assy: Machine Overview - Process Flow & SW Flowchart

HSG Flow 

NG Flow 

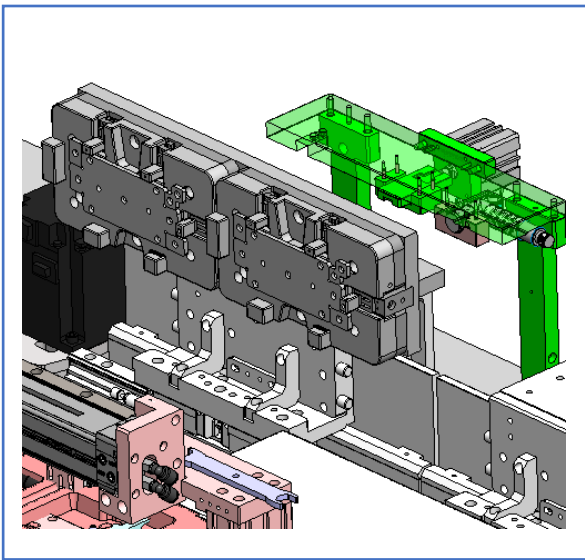


Monza | Snap Assy: Flow Chart

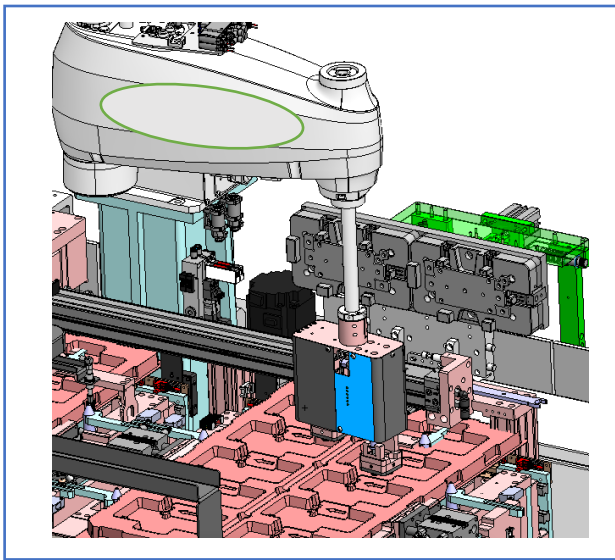
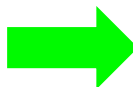


Monza | Snap Assy: Module Process Flow

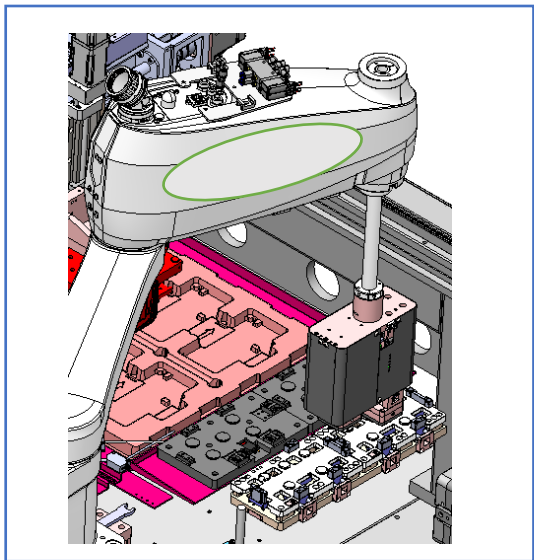
➡ HSG flow
➡ NG flow



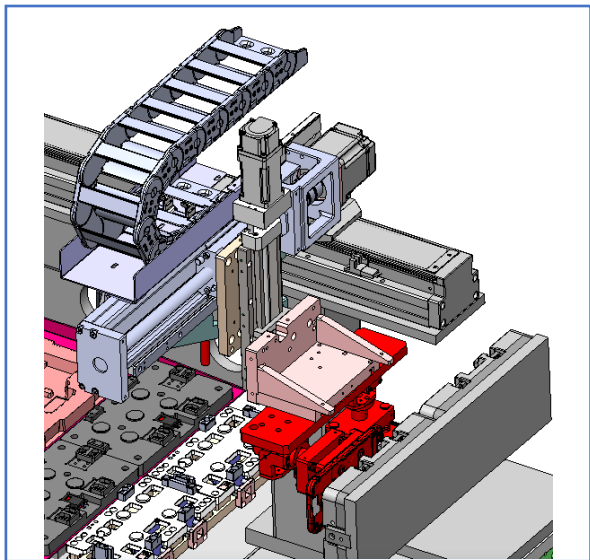
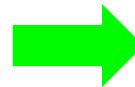
Fixture move to loading position



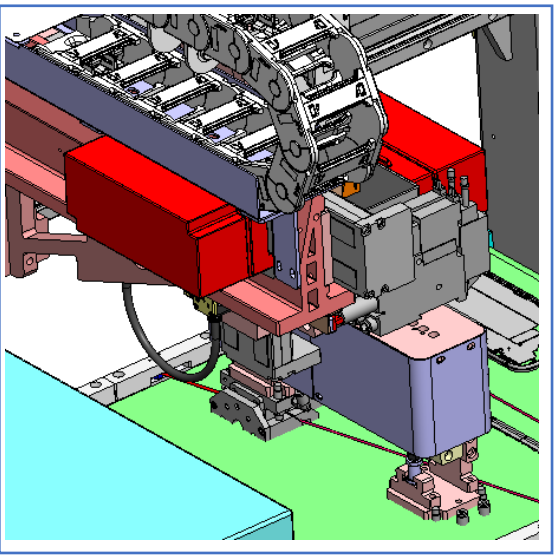
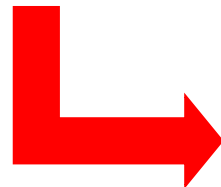
Robot pick up HSG from tray



Robot put HSG into secondary position fixture

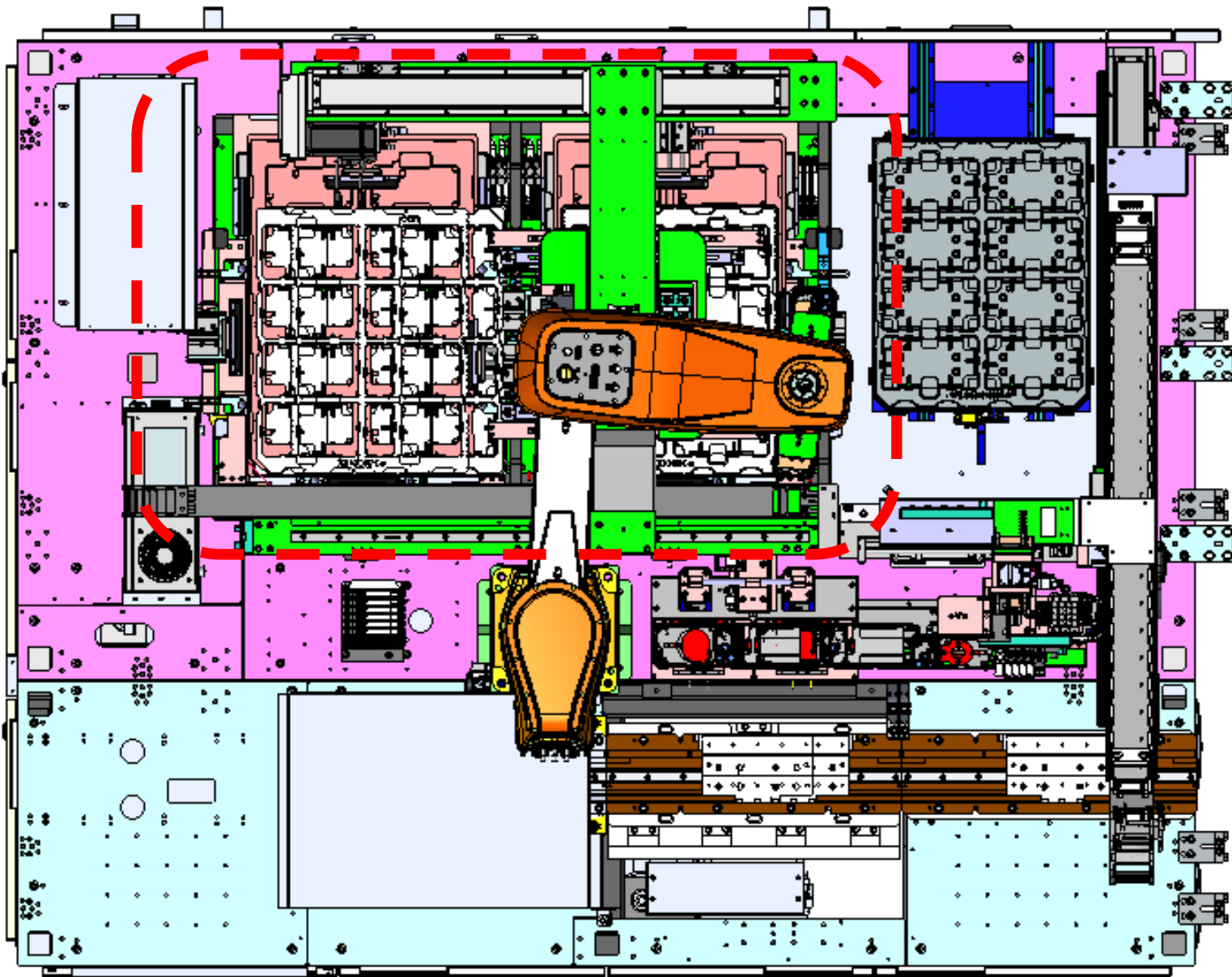


Transfer module load HSG into fixture

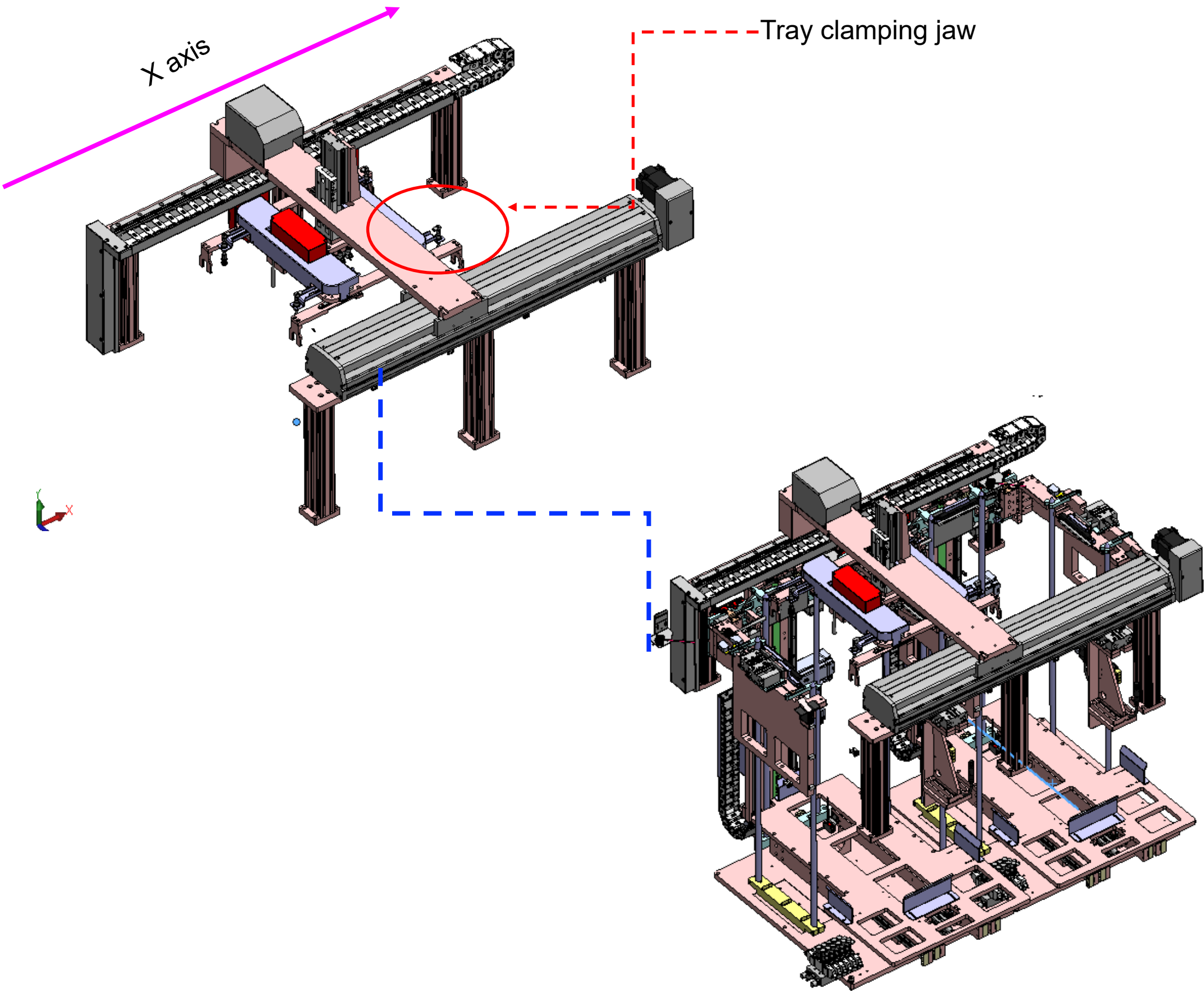


NG Tossing

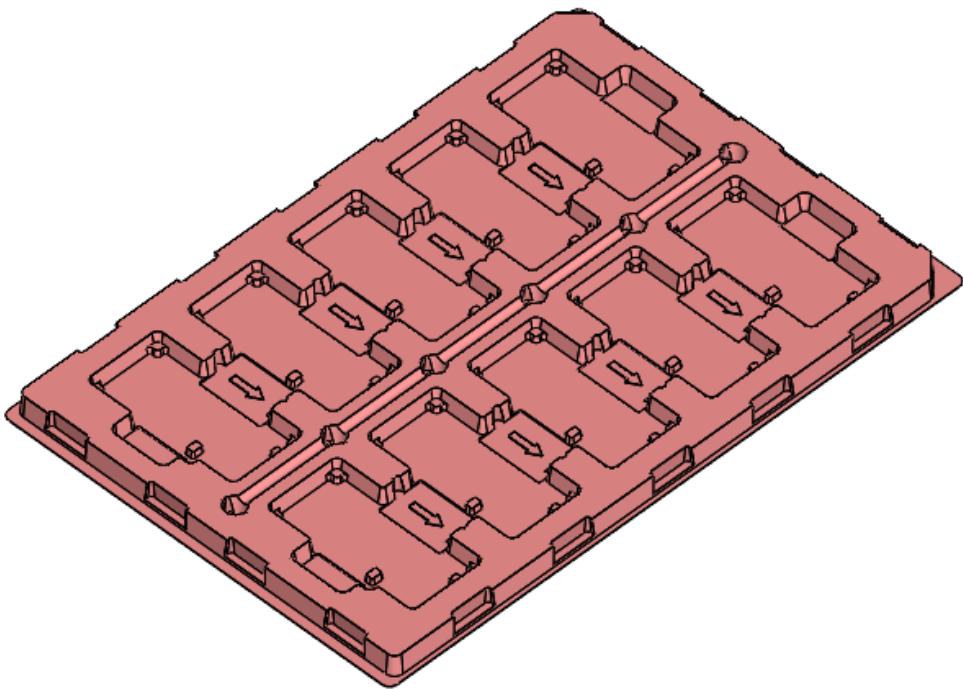
Monza | Snap Assy: Module Details – Storage Module



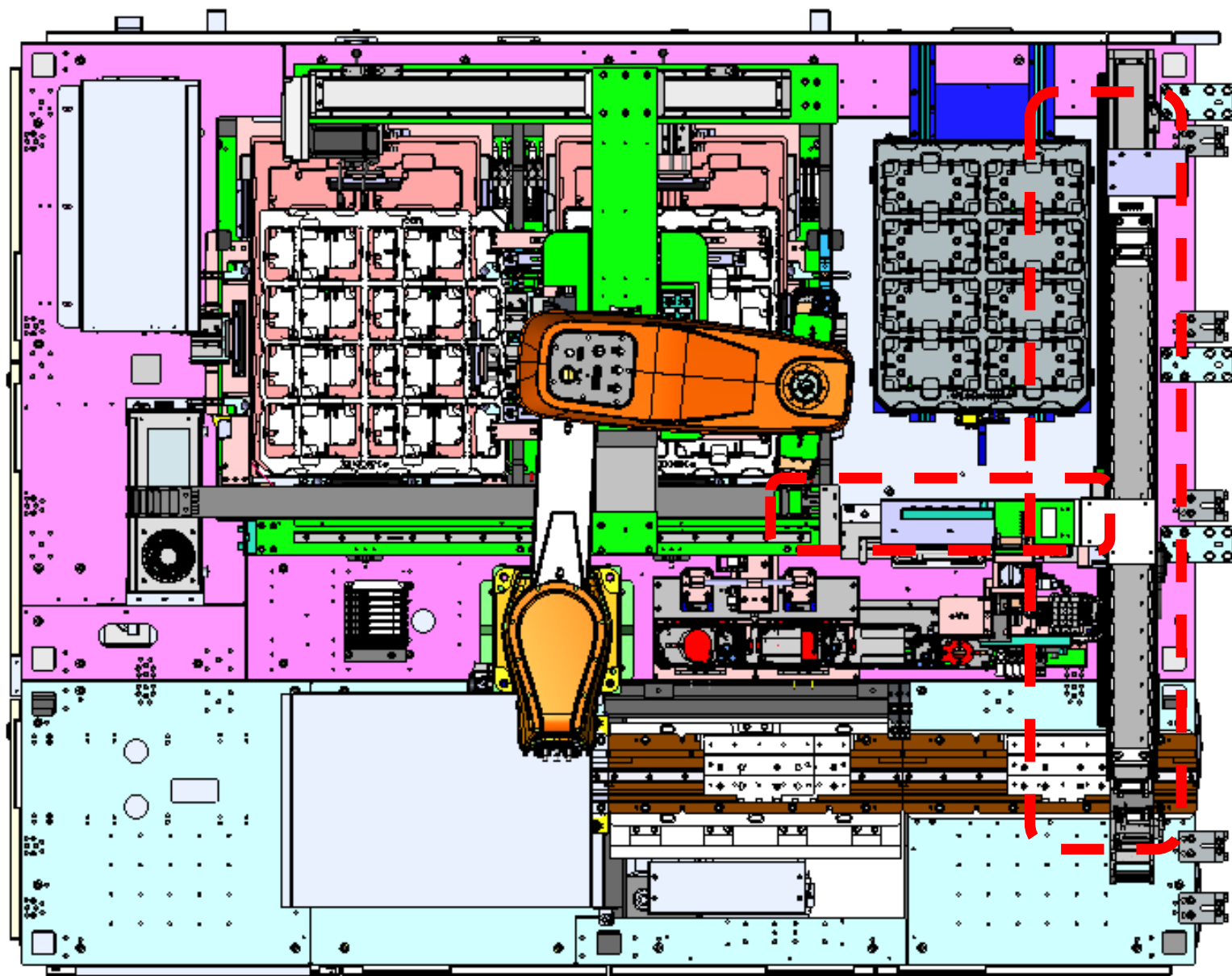
No.	Name	Model	Qty	Brand	Remark
1	X axis module	GTH8-L20-600-BC-P40-C4P5	1	Toyo	Stroke : 600mm Repeatability : $\pm 0.02\text{mm}$



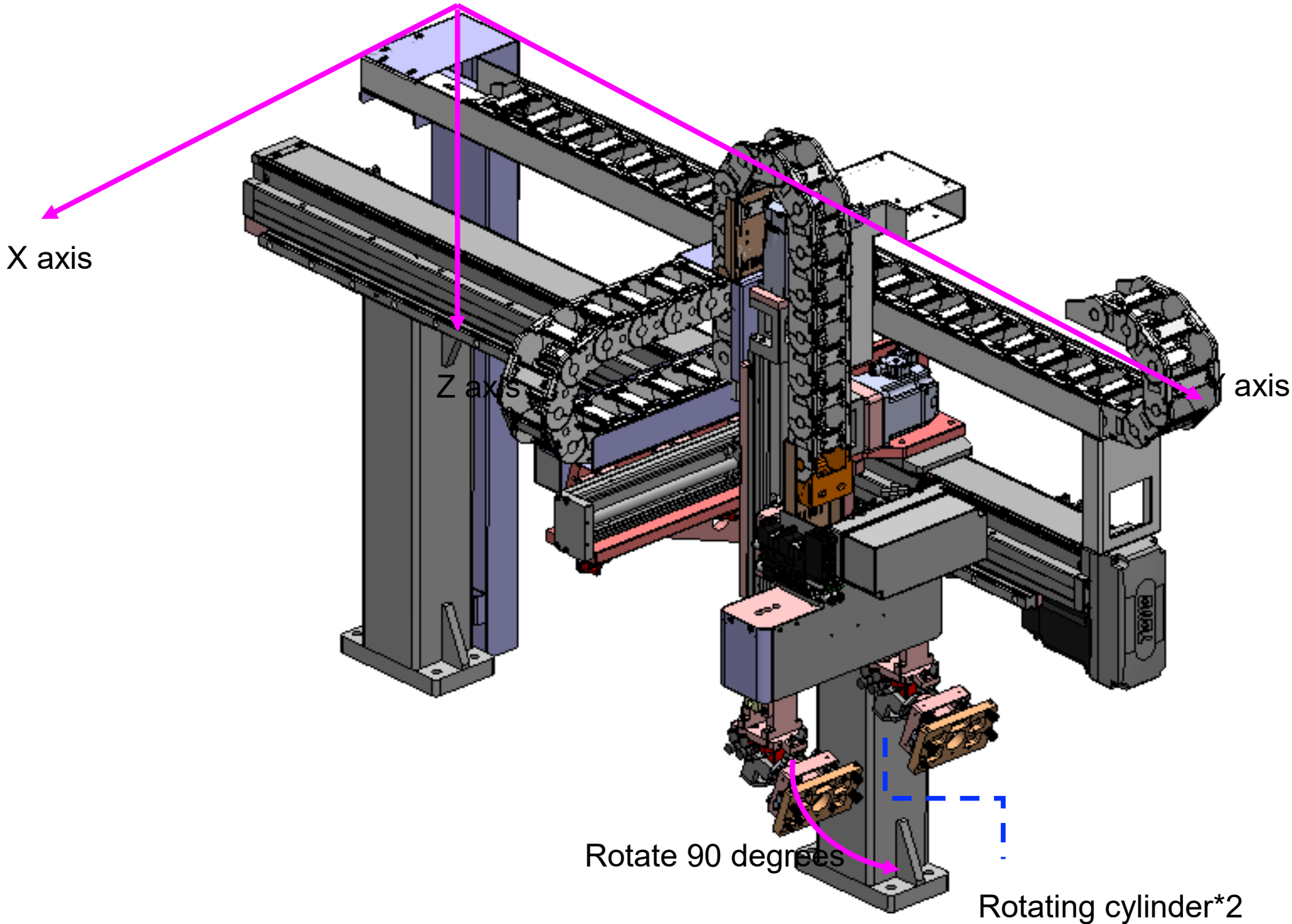
Maximum tray size: 580*390
Minimum tray size: 420*350



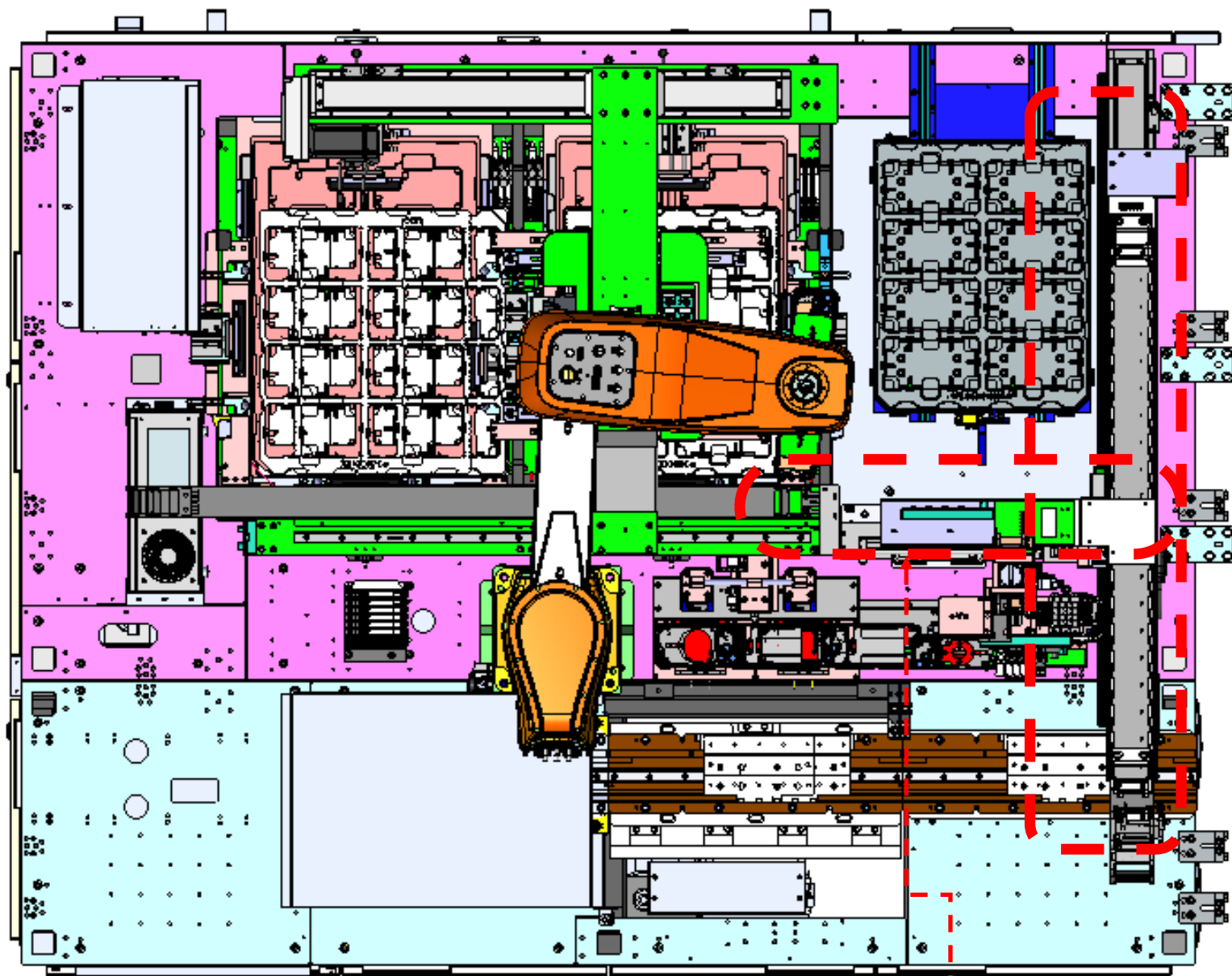
Monza | Snap Assy: Module Process Flow – Flipping module



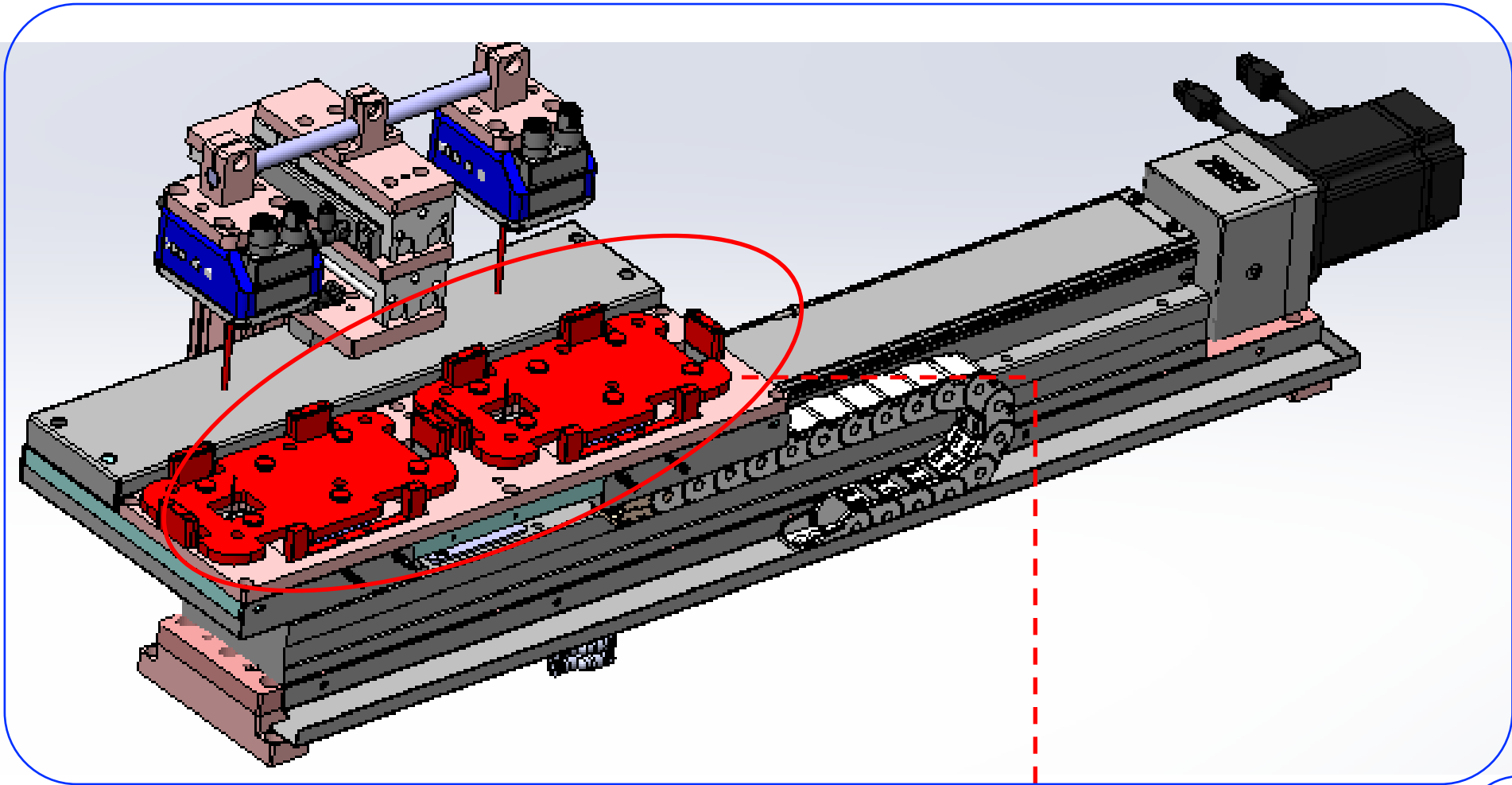
No.	Name	Model	Qty	Brand	Remark
1	X axis module	KK60D10P-400A1-F0	1	HIWIN	Stroke : 400mm Repeatability : $\pm 0.02\text{mm}$
2	Y axis module	GRH8-L20-950-BM-200W	1	Toyo	Stroke : 950mm Repeatability : $\pm 0.02\text{mm}$
3	Z axis module	KK60D10P-300A1-F0	1	HIWIN	Stroke : 300mm Repeatability : $\pm 0.02\text{mm}$



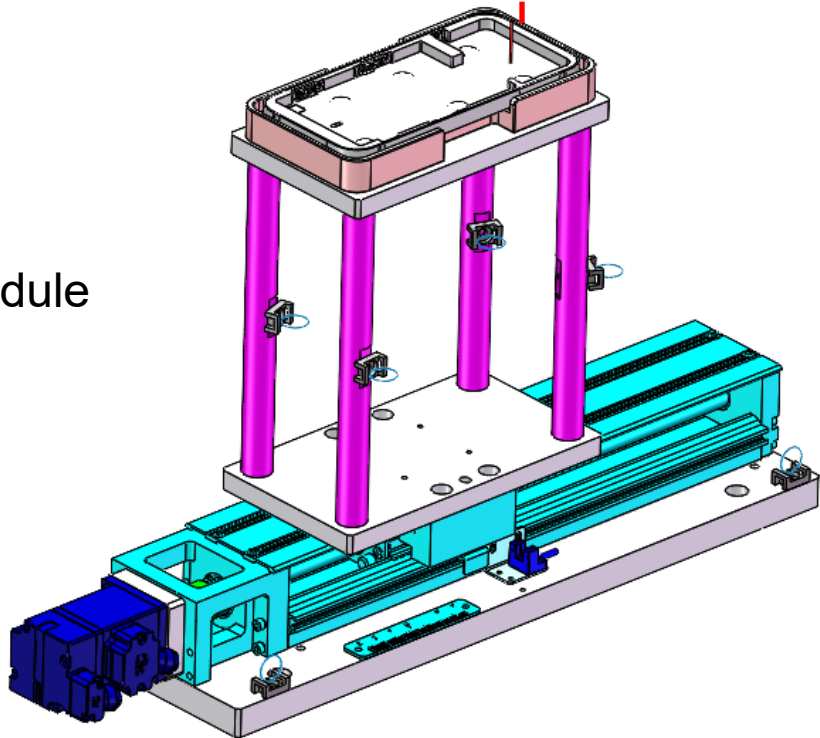
Monza | Snap Assy: Module Details - Secondary Position Module & Fixture Channel



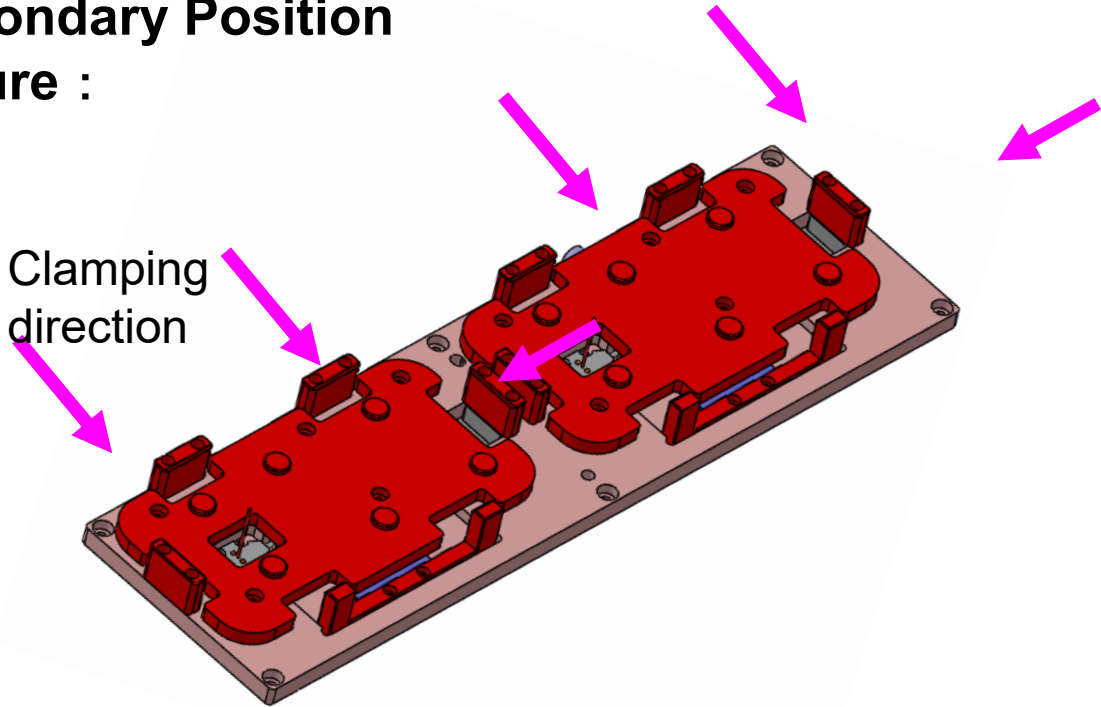
No.	Name	Model	Qty	Brand	Remark
1	X axis module	GTH8-L20-400-BC-P20-C4	1	TOYO	Stroke : 400mm Repeatability : $\pm 0.02\text{mm}$

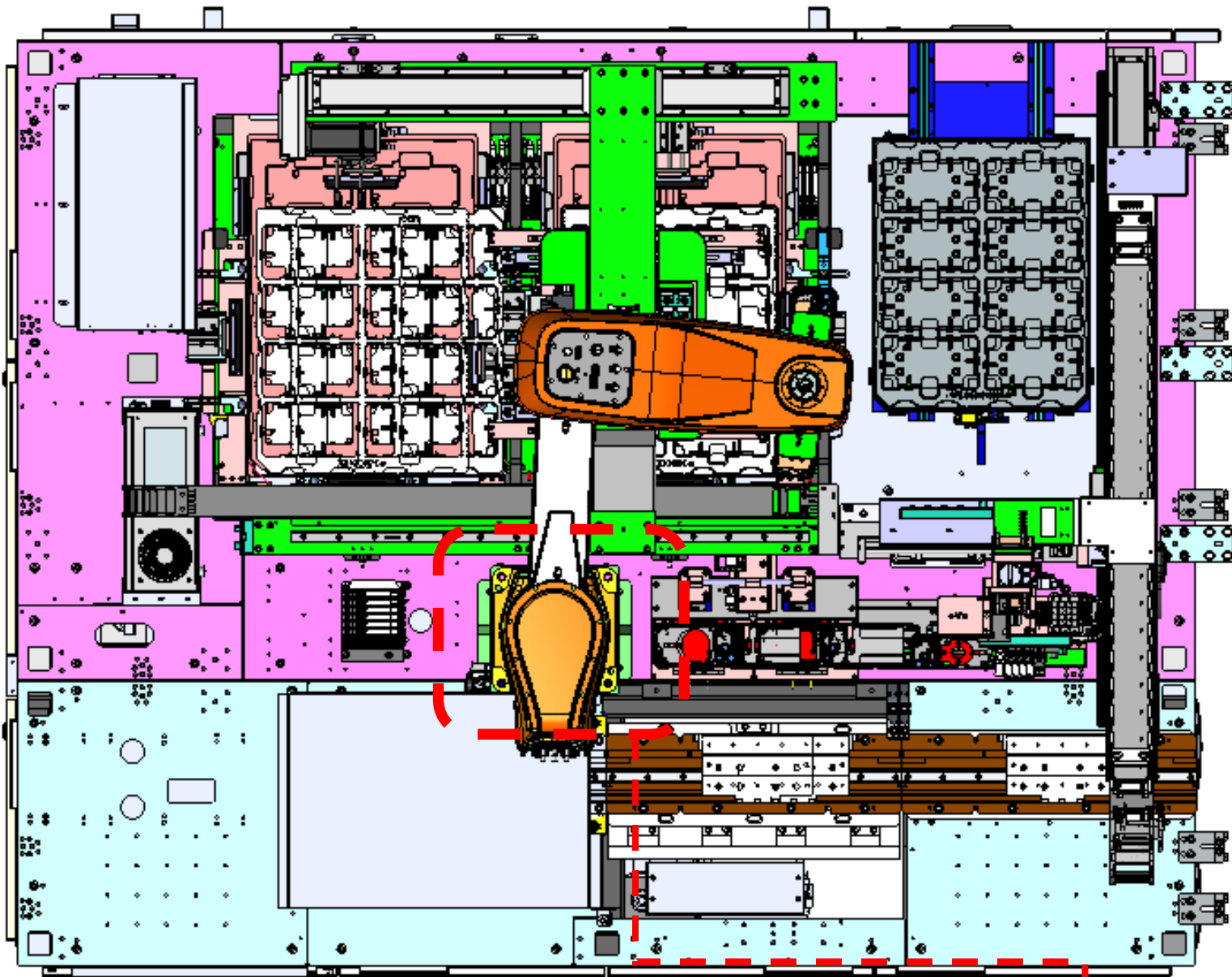


Buffer module



Secondary Position
fixture :





No.	Name	Model	Qty	Brand	Remark
1	4-axis robot	KR 13 R850 Z200-2 CS	1	KUKA	

Basic data

Designation	KR 13 R850 Z200-2 CS
Number of axes	4
Number of controlled axes	4
Volume of working envelope	0.397 m³
Pose repeatability XY	± 0.025 mm
Pose repeatability Z	± 0.01 mm
Pose repeatability R	± 0.01 °
Weight	approx. 46.5 kg
Rated payload	6 kg
Maximum payload	13 kg
Maximum reach	850 mm
Protection rating (IEC 60529)	IP20
Sound level	< 80 dB (A)
Mounting position	Floor
Footprint	210 mm x 210 mm

4.9.2 Axis data, KR 13 R850 Z200-2 CS

Axis data

Motion range	
Motion range, axis 1	±145 °
Motion range, axis 2	±145 °

95/233 | www.kuka.com

MA KR 13 SCARA-2 CS V1 | Issued: 21.02.2025

KR 13 SCARA-2 CS

Motion range, axis 3	-200 mm / 0 mm
Motion range, axis 4	±360 °
Speed with rated payload	
Axis velocity, axis 1	390 °/s
Axis velocity, axis 2	650 °/s
Axis velocity, axis 3	1.1 m/s
Axis velocity, axis 4	2700 °/s

The direction of motion and the arrangement of the individual axes may be noted from the following diagram.

Dimensions: mm

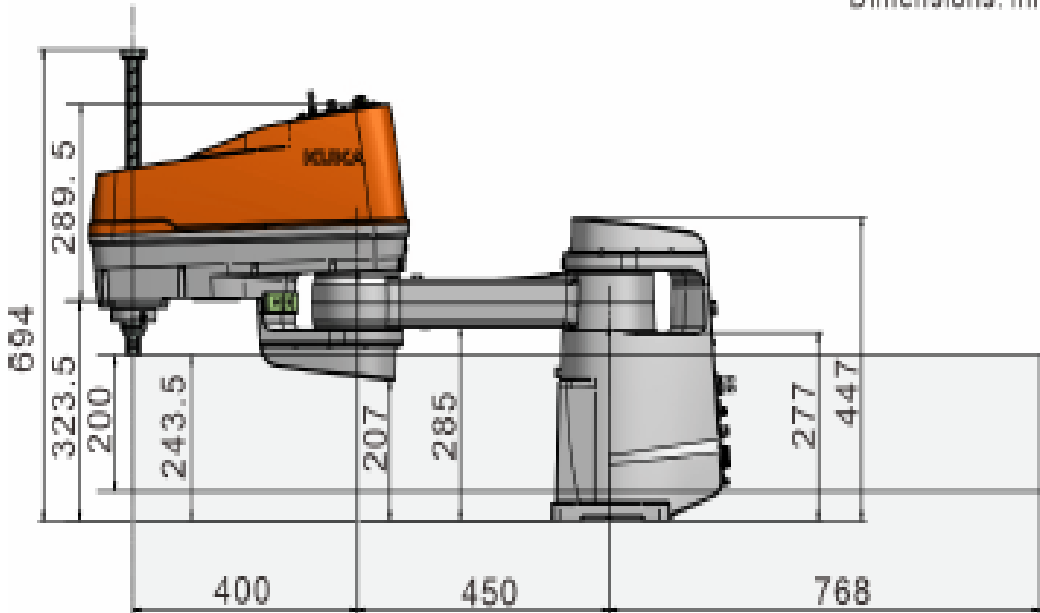


Fig. 4-65: KR 13 R850 Z200-2 CS, working envelope, side view

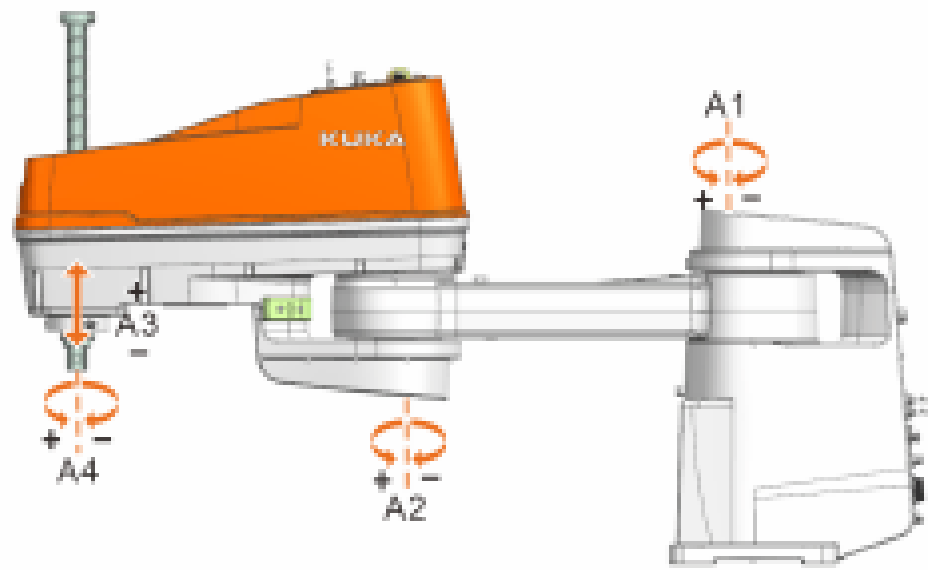


Fig. 4-64: Direction of manipulator axes

Technical data

MA KR 13 SCARA-2 CS V1 | Issued: 21.02.2025

www.kuka.com | 95/233

KR 13 SCARA-2 CS

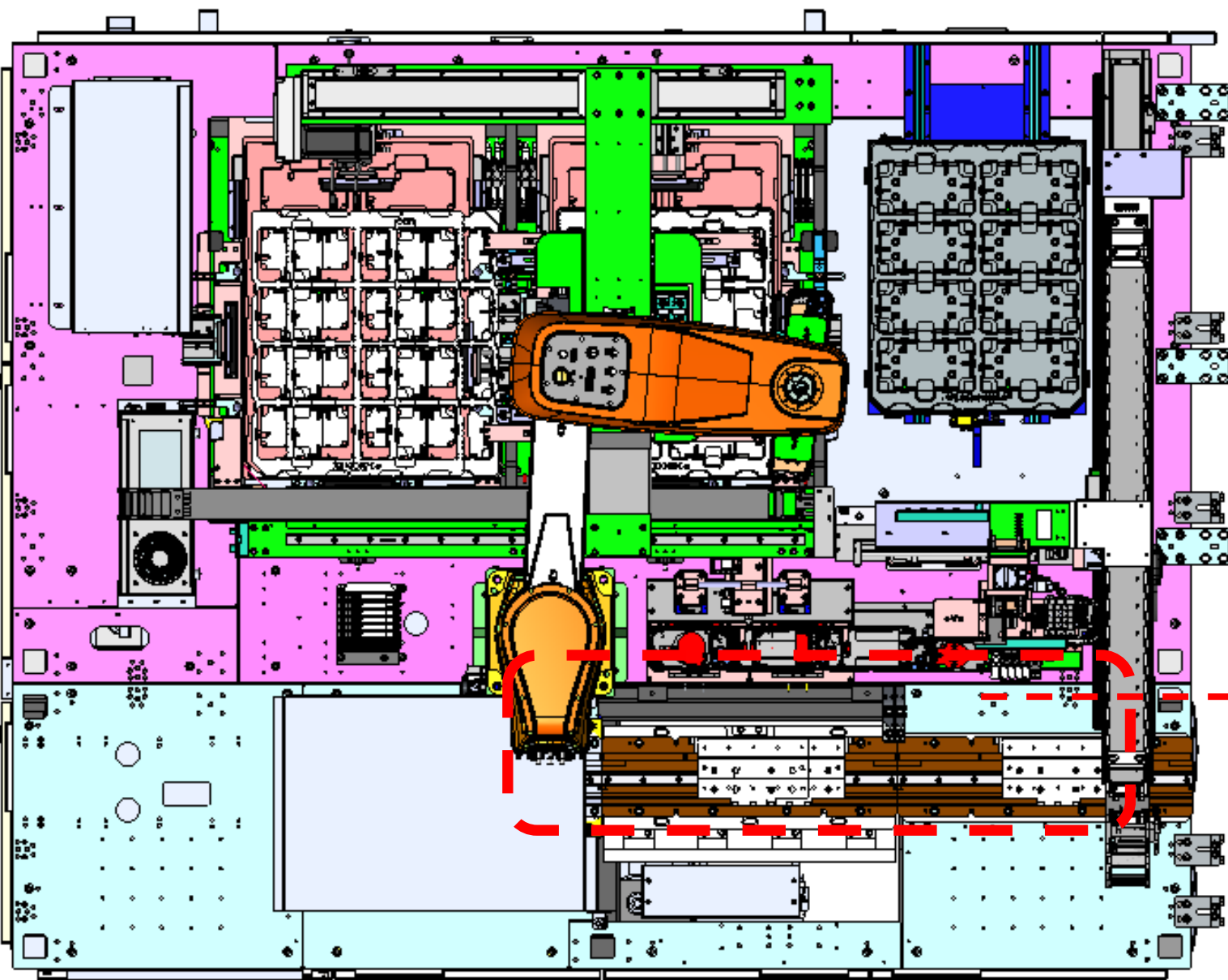
Designation	KR 13 R850 Z200-2 CS
Default color	KUKA Industrial Orange (RAL 2009); Signal white (RAL 9003)
Controller	KR CS Box-2
Time per cycle	0.44 s
Cleanroom class (ISO 14644-1)	-

Ambient conditions

Humidity class (EN 60204)	-
Classification of environmental conditions (EN 60721-3-3)	3K22
Ambient temperature	
During operation	0 °C to 45 °C (273 K to 318 K)
During storage/transportation	-25 °C to 60 °C (248 K to 333 K)

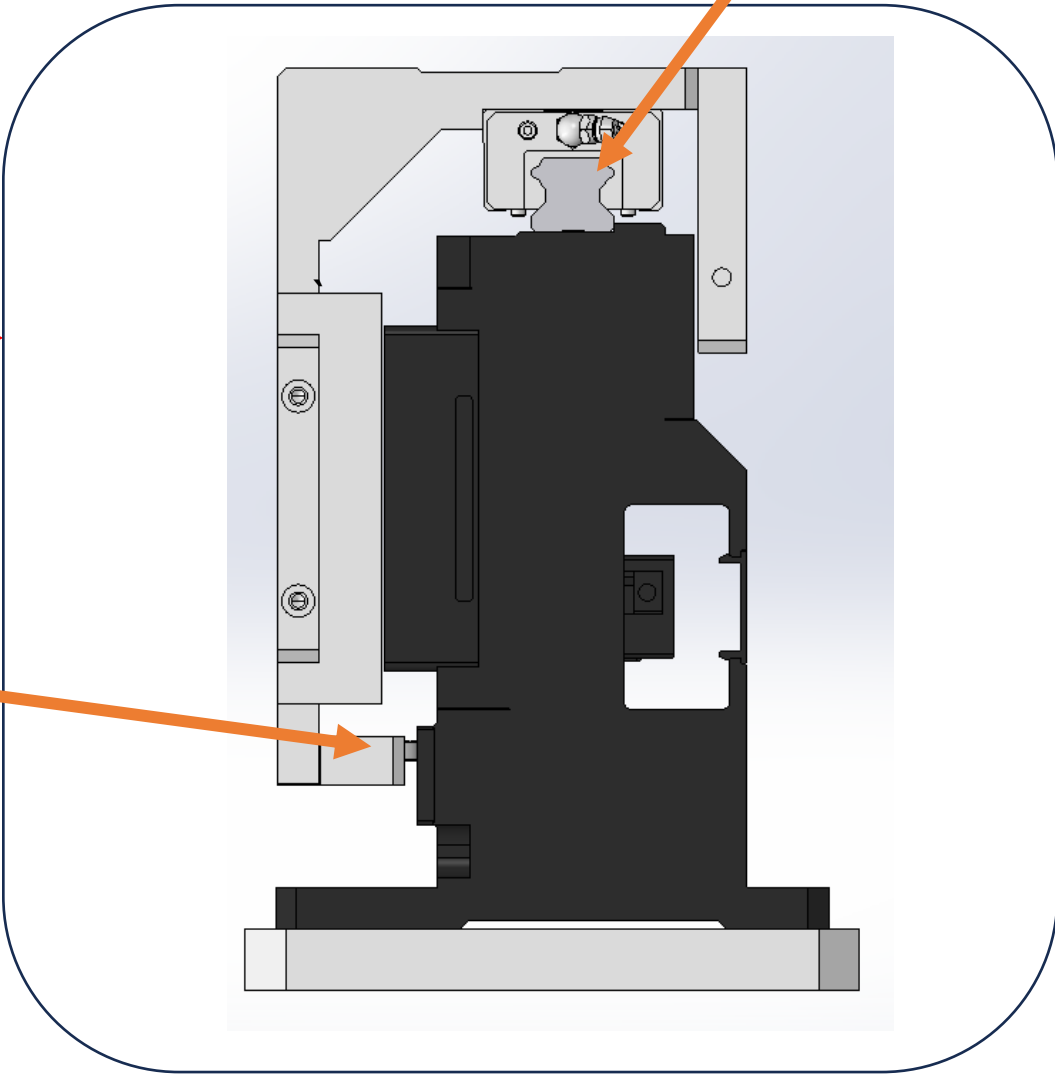
Note: When operating at low temperatures, it is necessary to reduce the speed and acceleration of the robot for preheating before normal use.

Monza | Snap Assy: Module Details – Fixture Channel

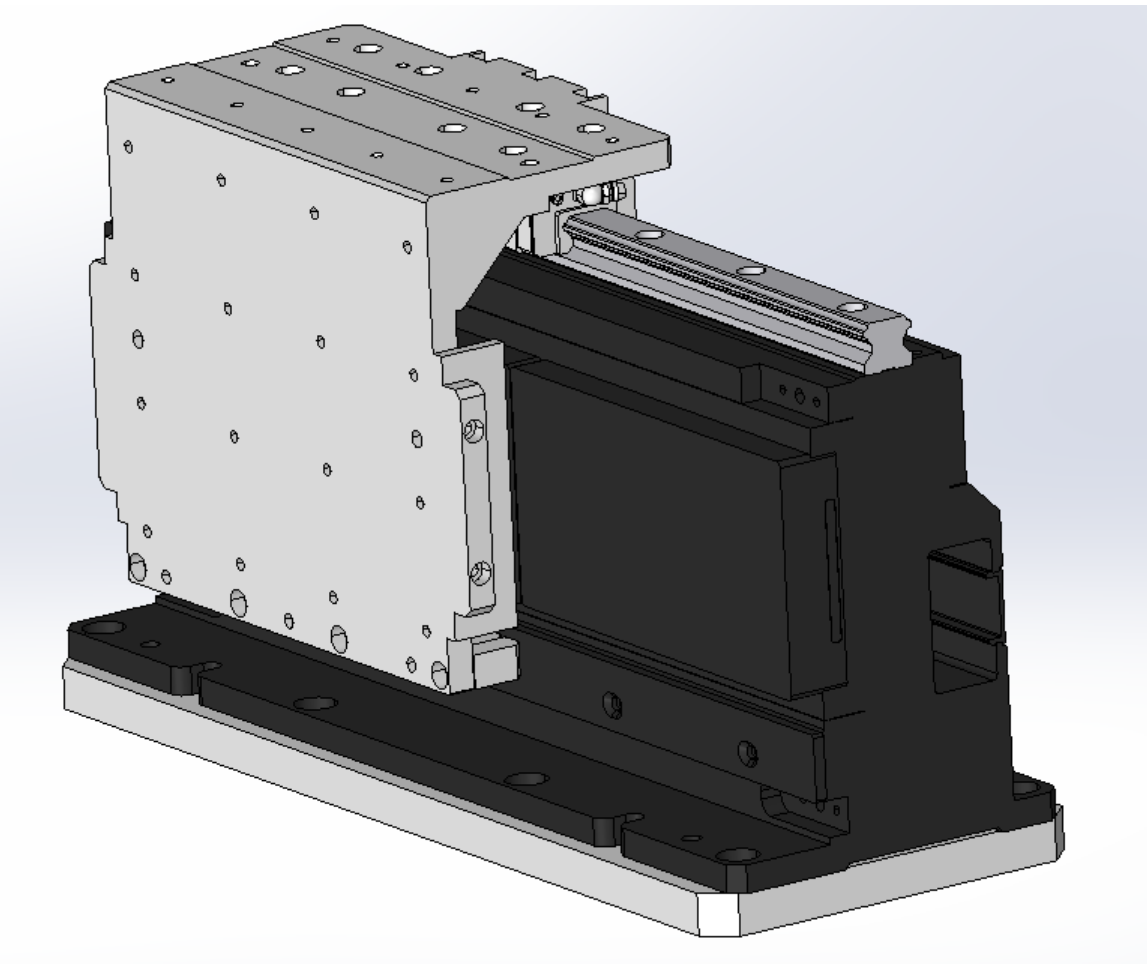


No.	Name	Model	Qty	Brand	Remark
1	Flow channel	Beckhoff	1	Beckhoff	Magnetic levitation channel

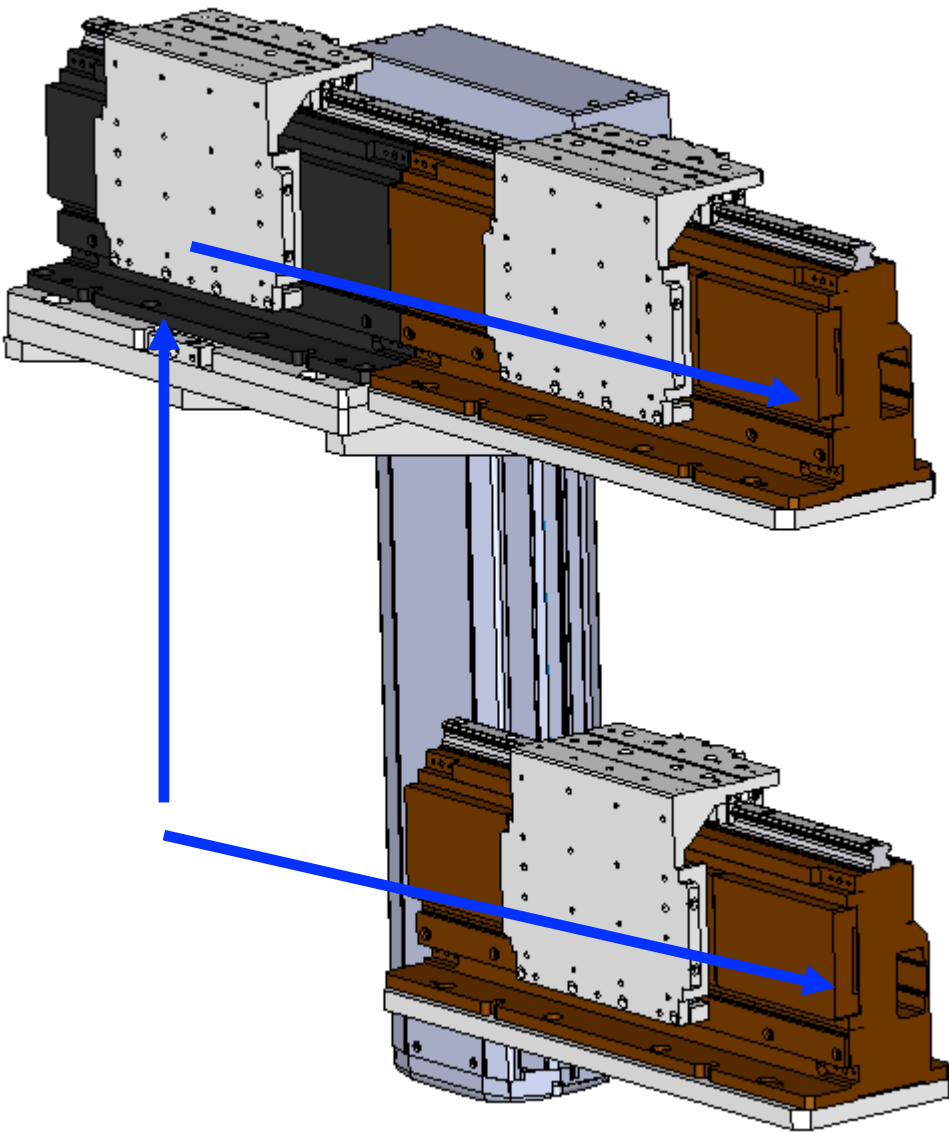
LM Rail with
LM Block



Roller Guide

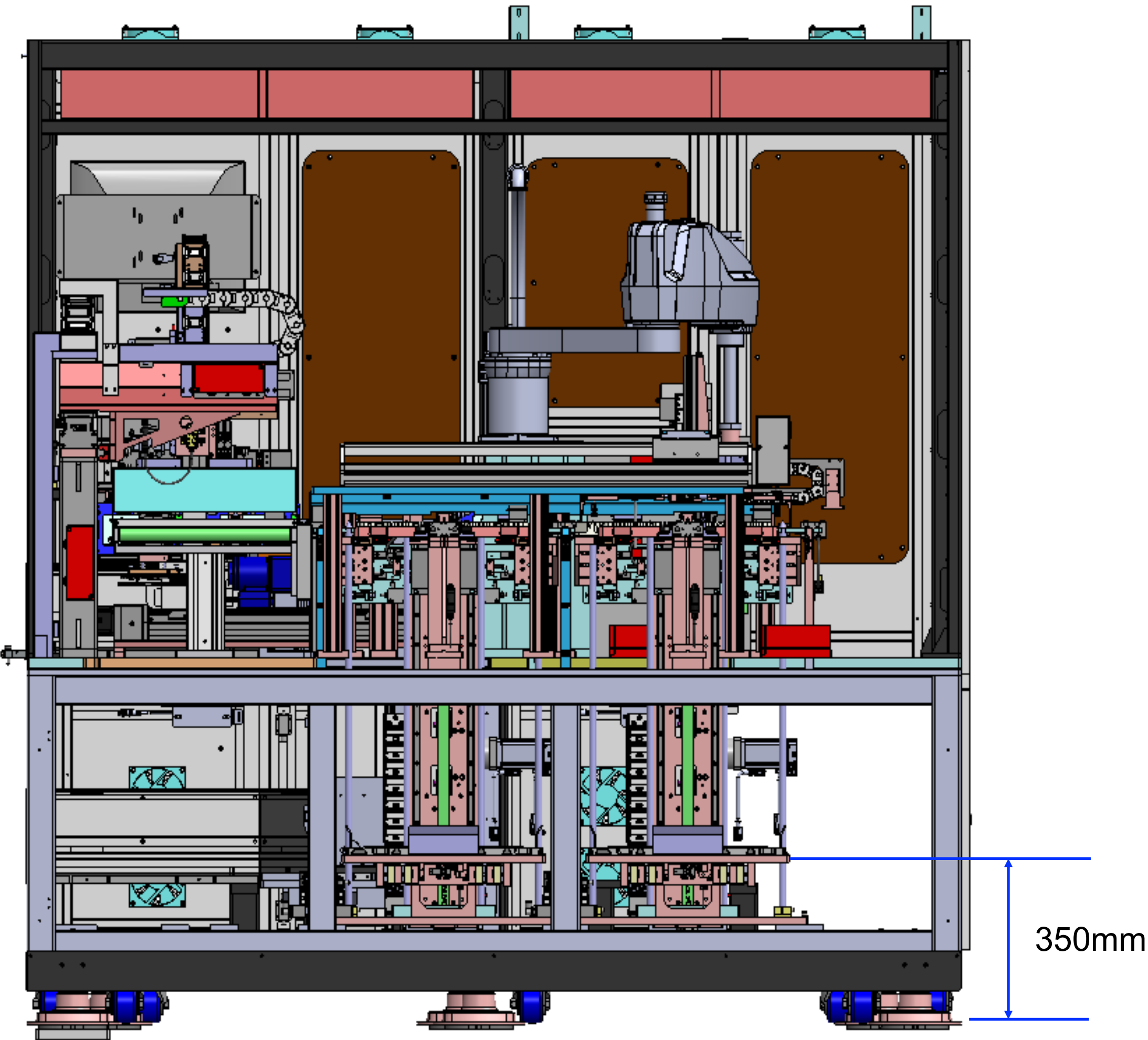
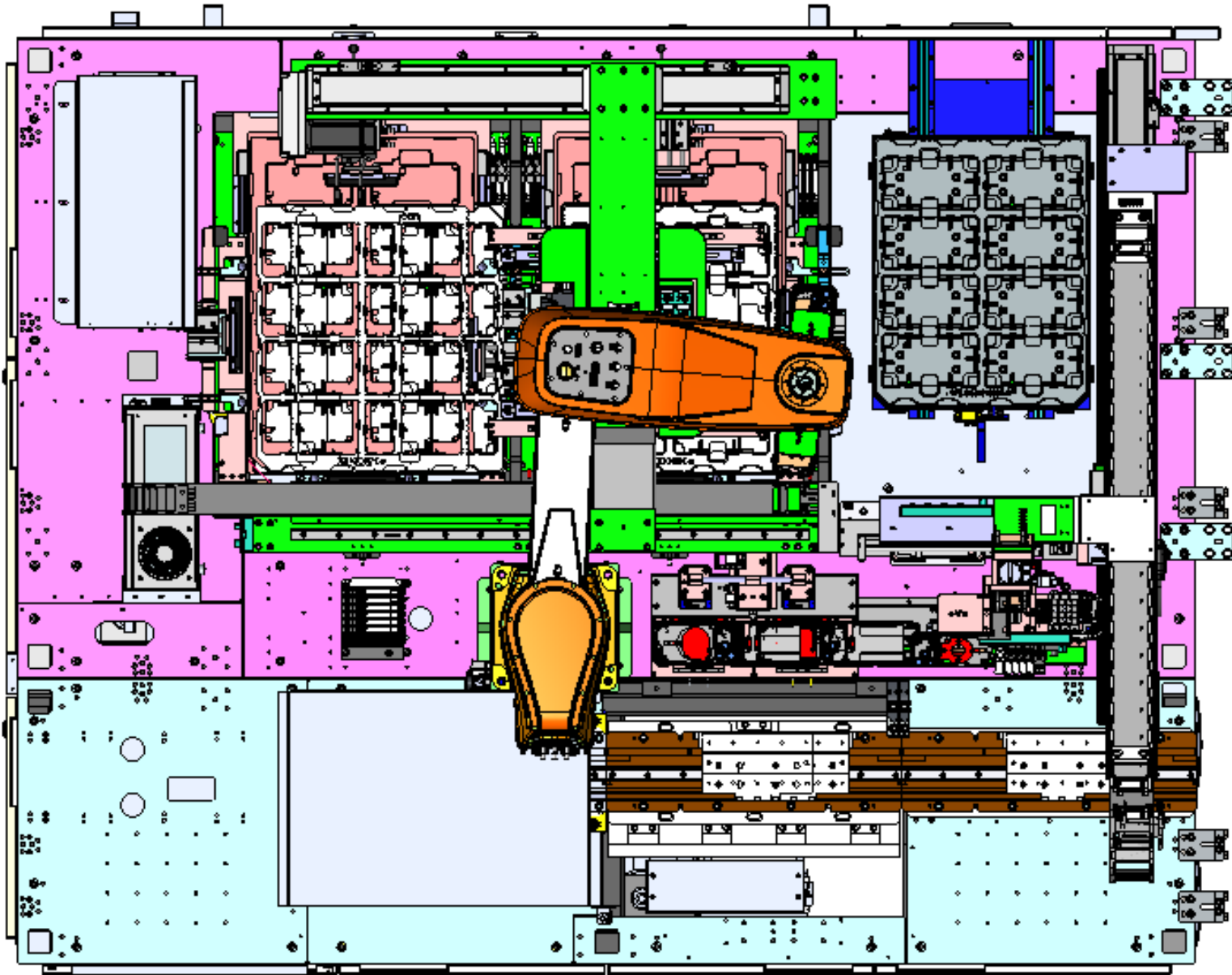


Flow channel :



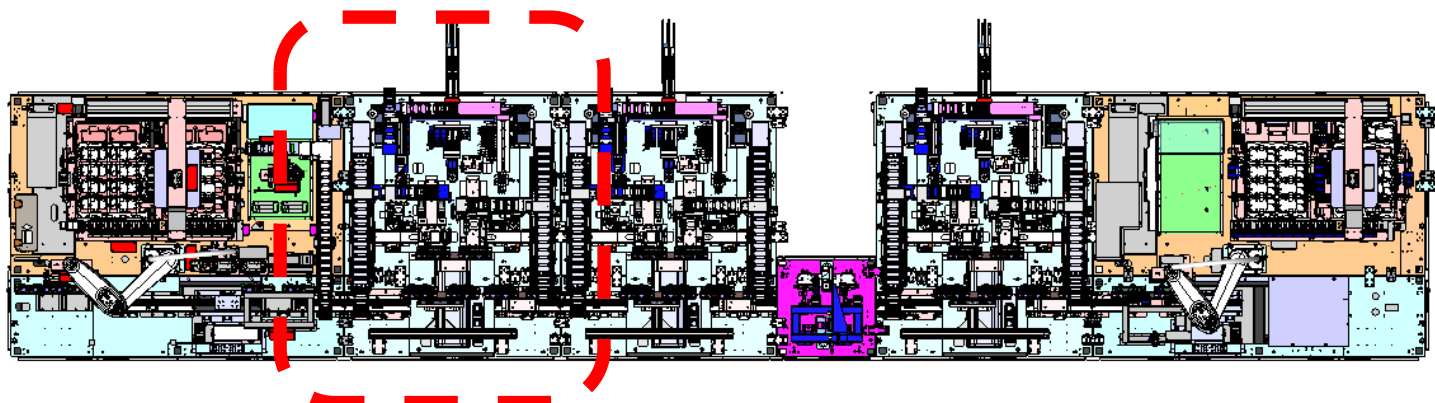
Switching between upper and lower layers

Monza | Snap Assy: Module Details – L/UL Tray Height details

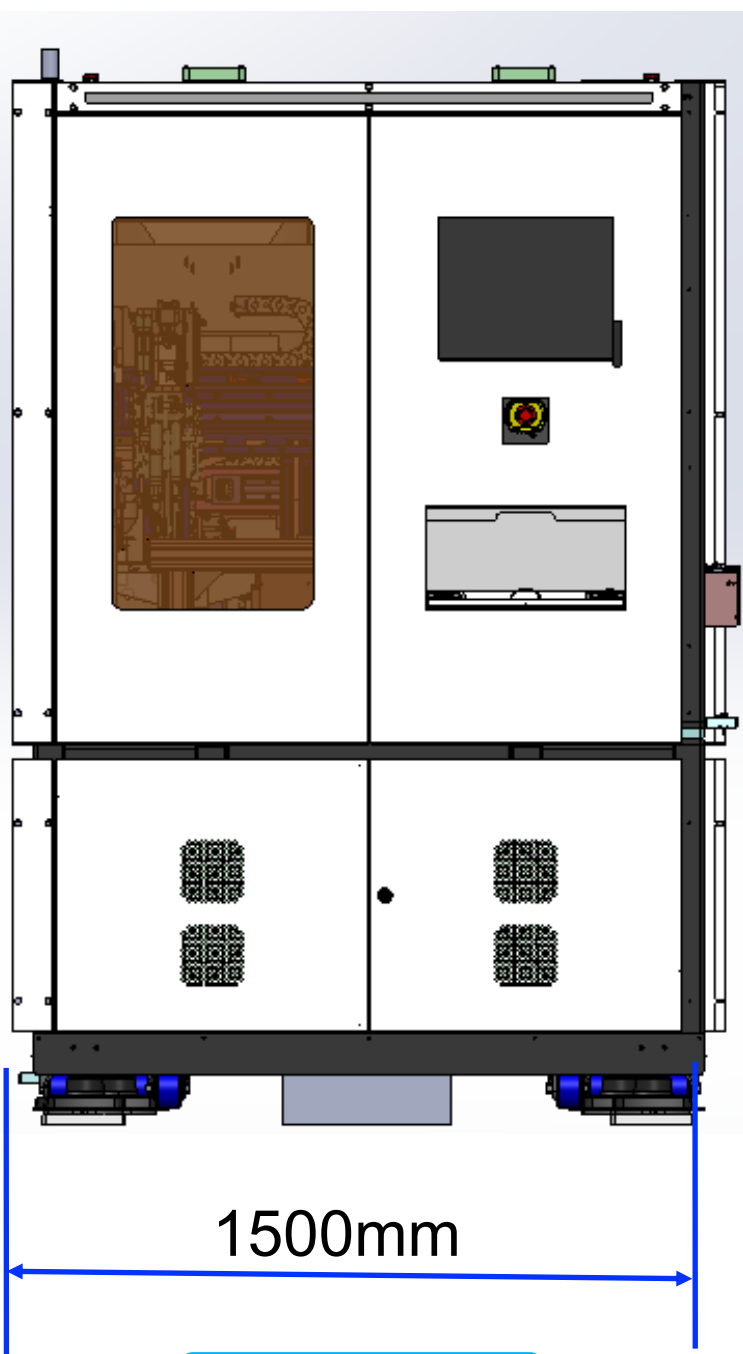


Station Details Snap Assy & Screwing Machine

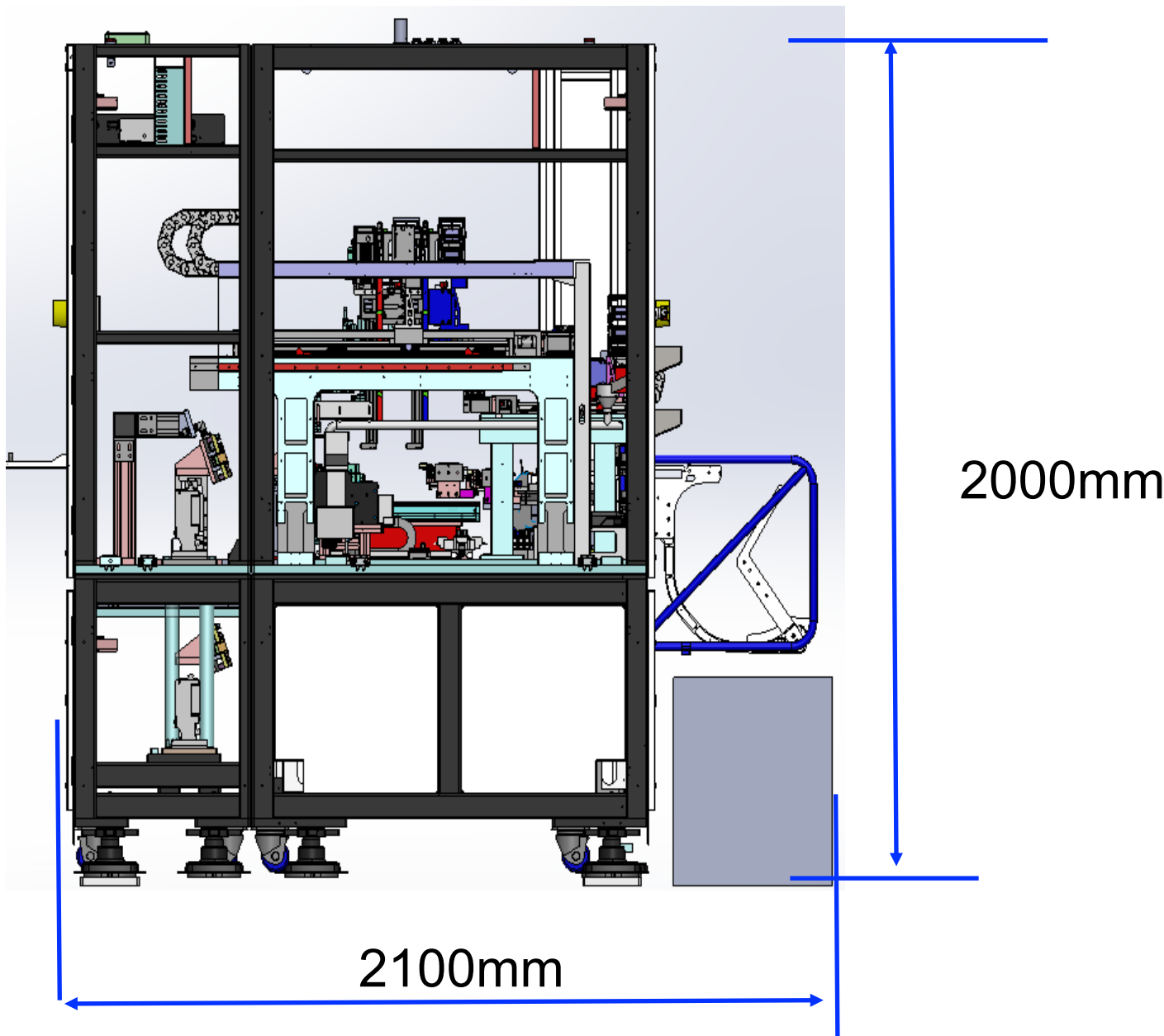
Monza | Snap Assy : Machine Overview –Layout



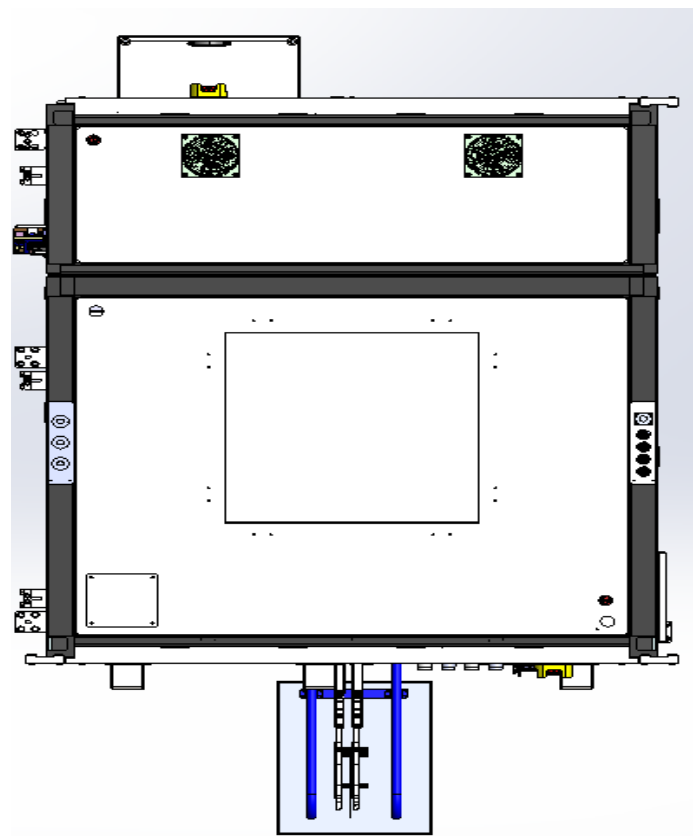
Front View



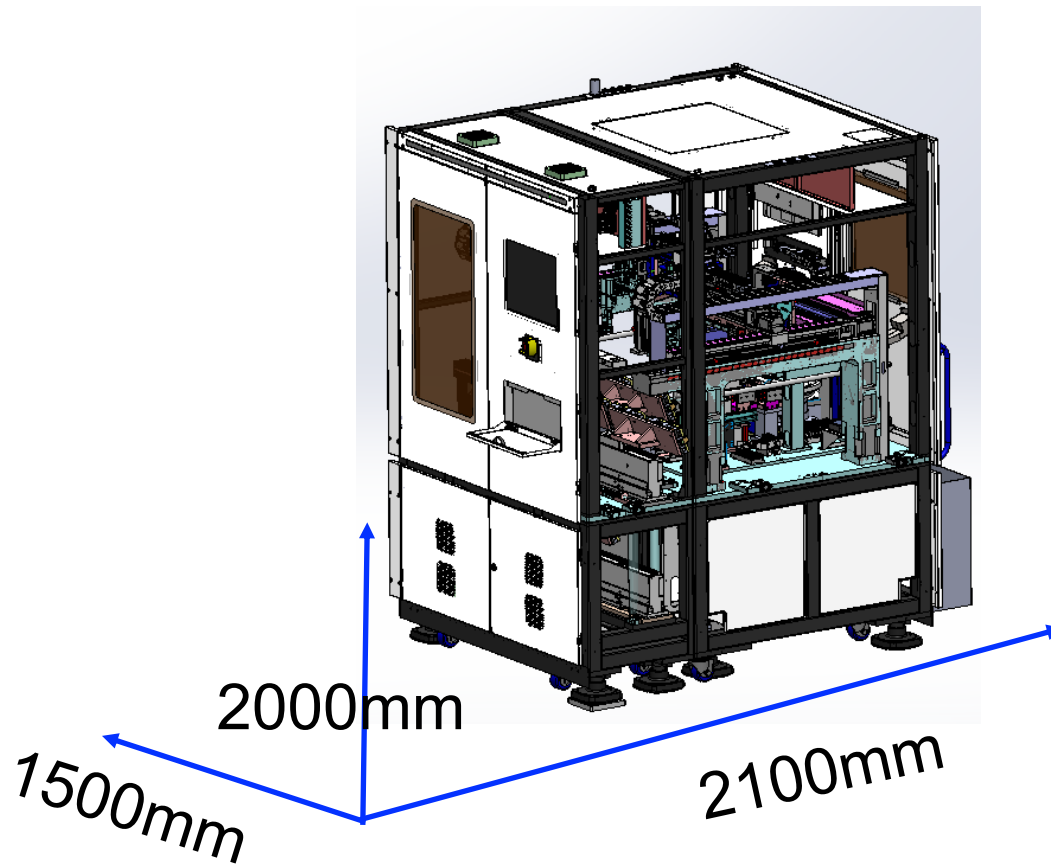
Side View



Top View

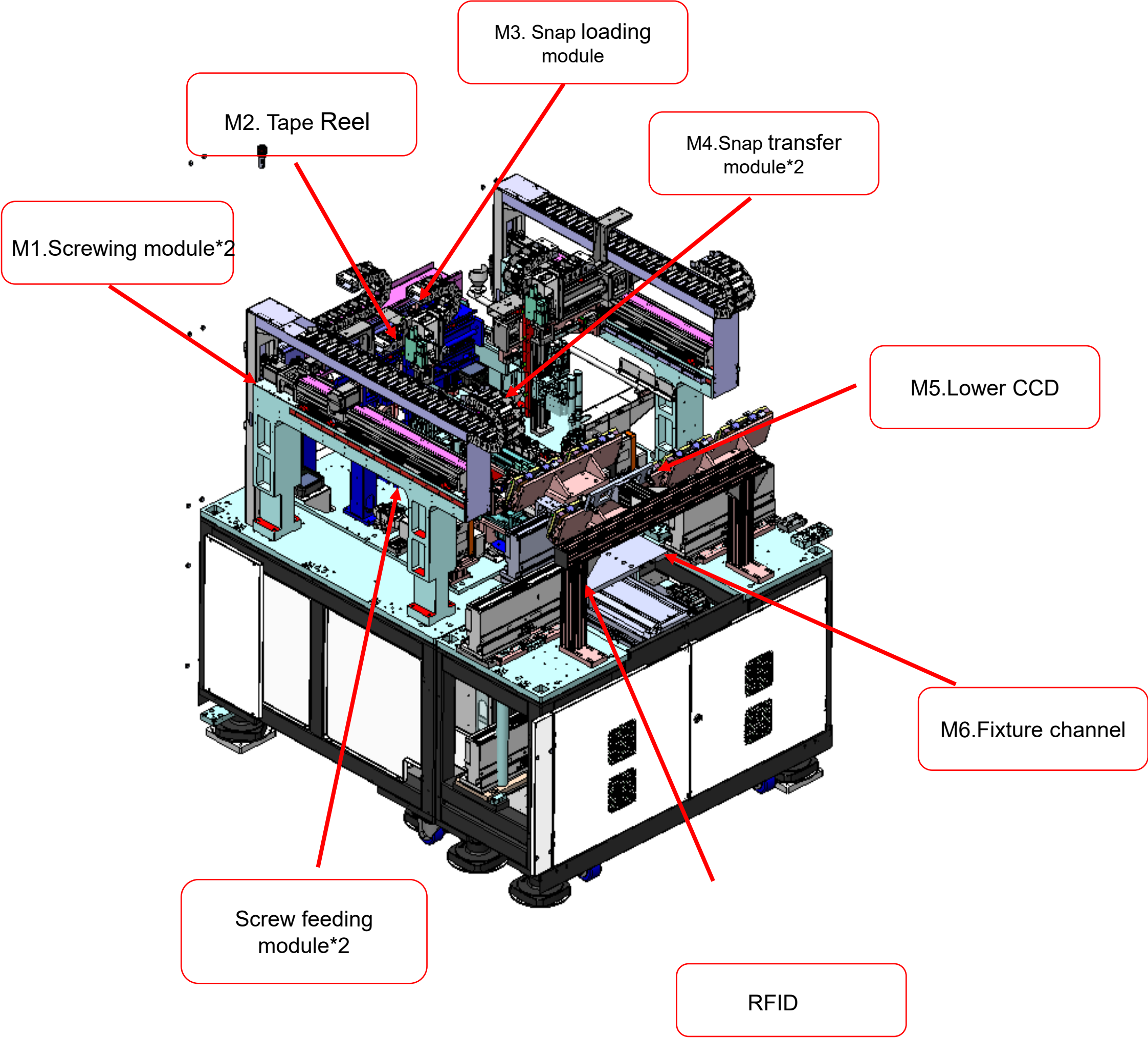


Axis View



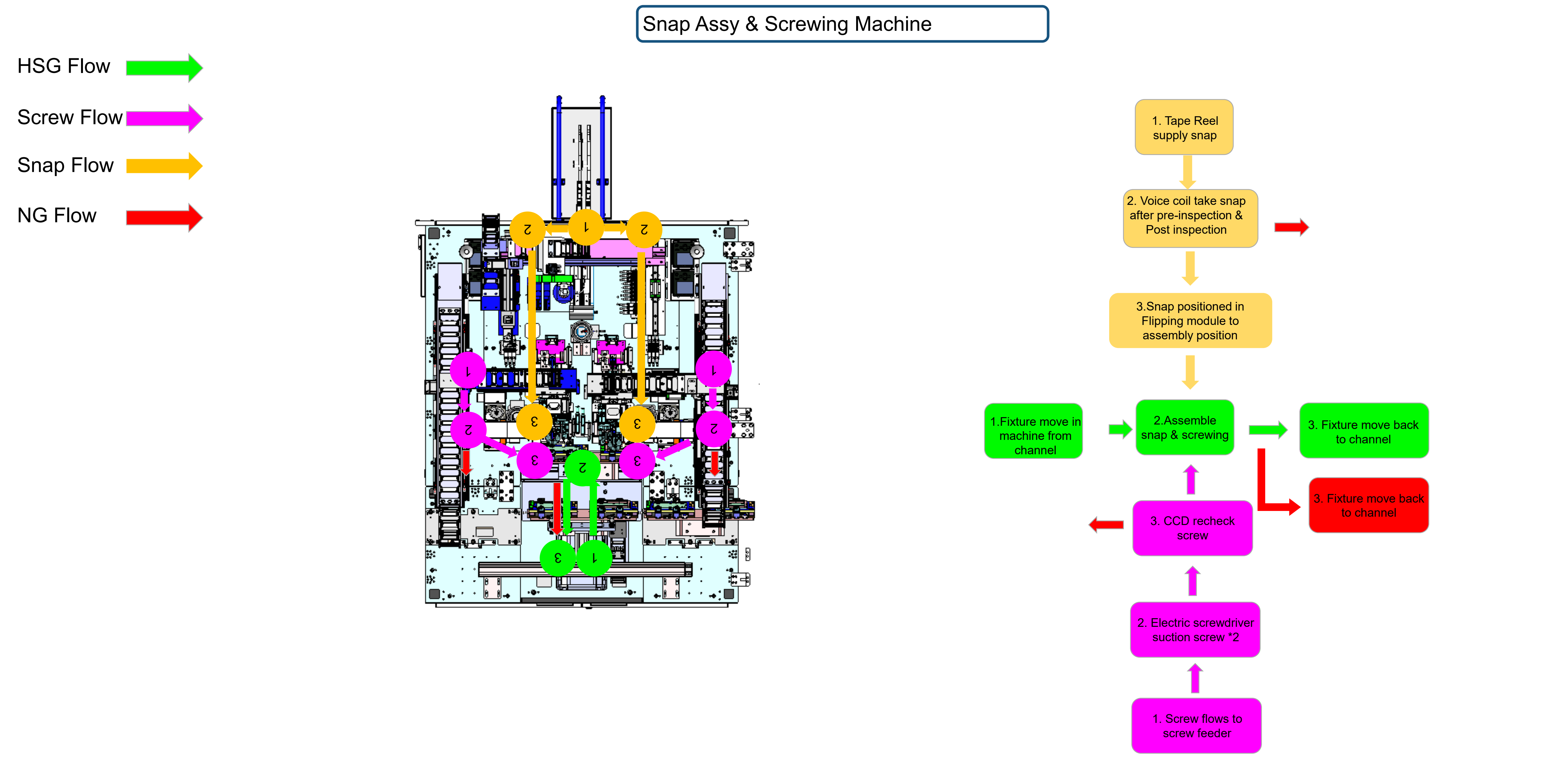
- 1.Voltage:220V
- 2.Power: 10.5Kw
- 3.Air: 0.5~0.7Mpa
- 4.Labor:/
- 5. Machine Weight: 2000kg

Monza | Snap Assy: Machine Overview– Station BOM

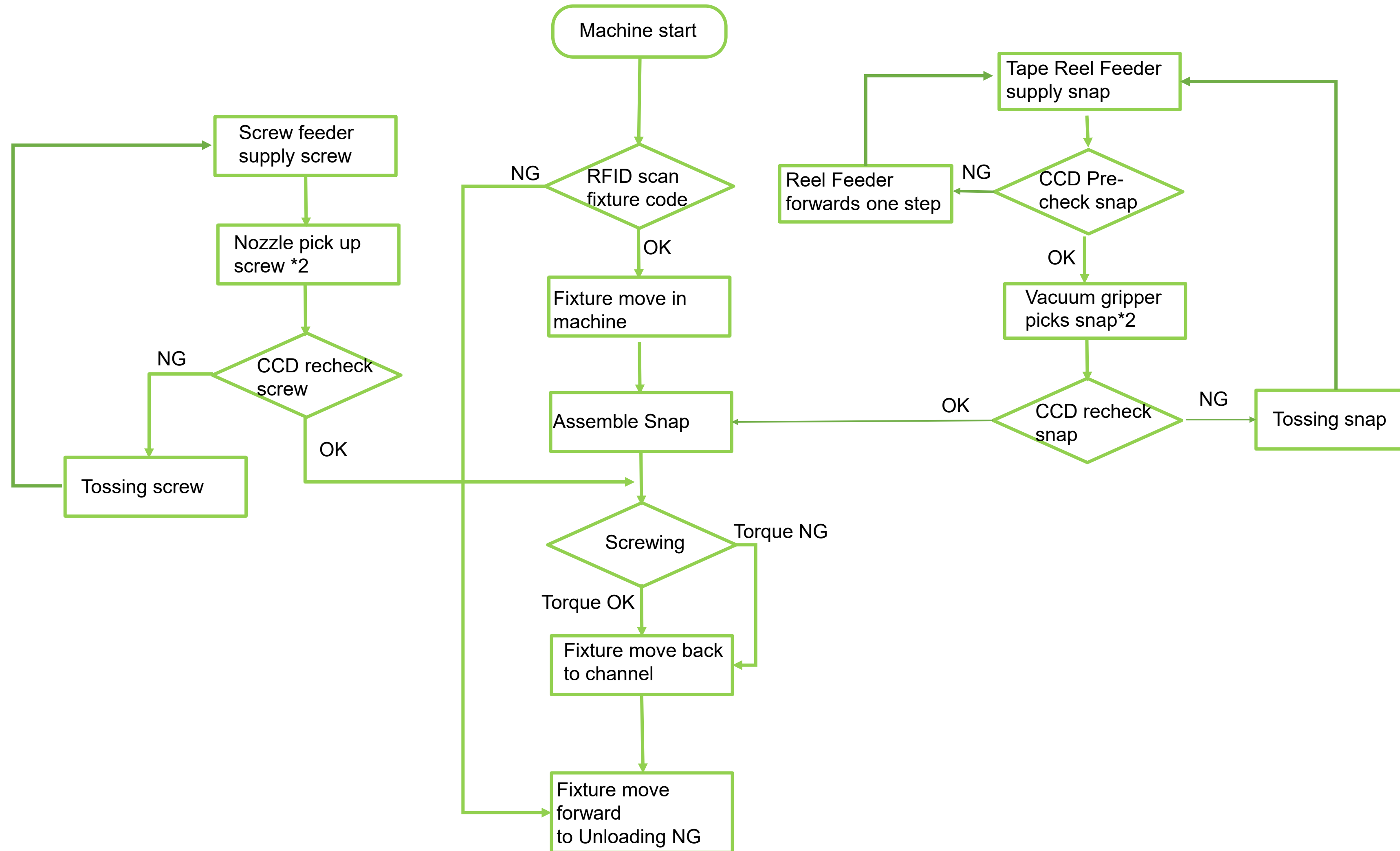


#	Module	Module Qty	Key parts	Key part QTY (TTL)
M1	Screwing module	2	X axis module	1
			Y axis module	1
			Z axis module	1
			Smart Screwdriver	4
M2	Tape Reel Feeder	2	Feeder	1
M3	Snap loading module	1	X axis module	1
			Y axis module	1
			Z axis module	1
M4	Snap transfer module	2	Y axis module	1
M5	Lower CCD	2	CCD	1
M6	Fixture channel	1	/	/
M7	Top CCD	1	CCD	1
M8	Lower CCD	1	CCD	1

Monza | Snap Assy: Machine Overview - Process Flow & SW Flowchart

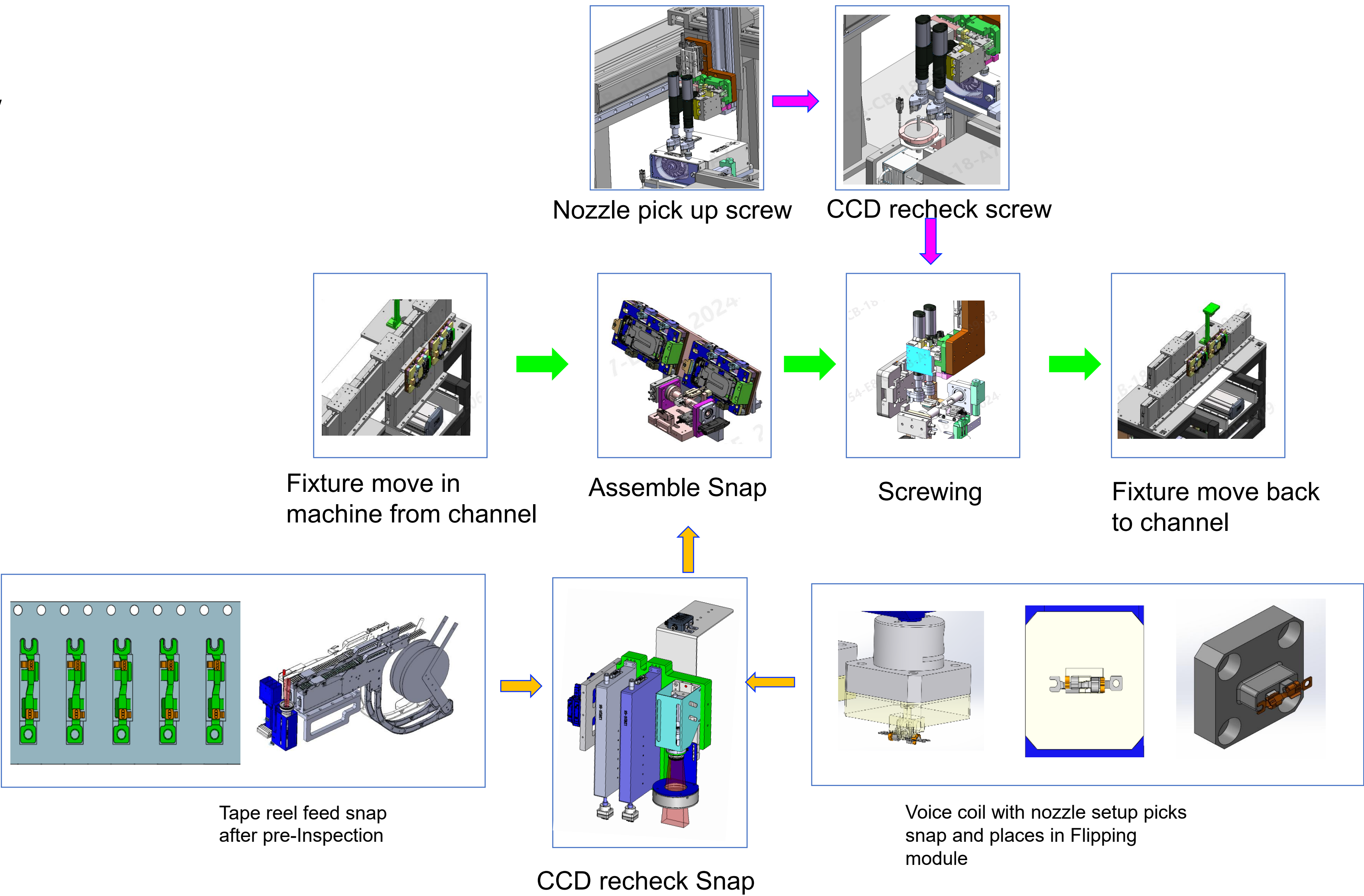


Monza | Snap Assy: Flow Chart



Monza | Snap Assy: Module Process Flow

- ➡ HSG flow
- ➡ Snap flow
- ➡ Screw flow

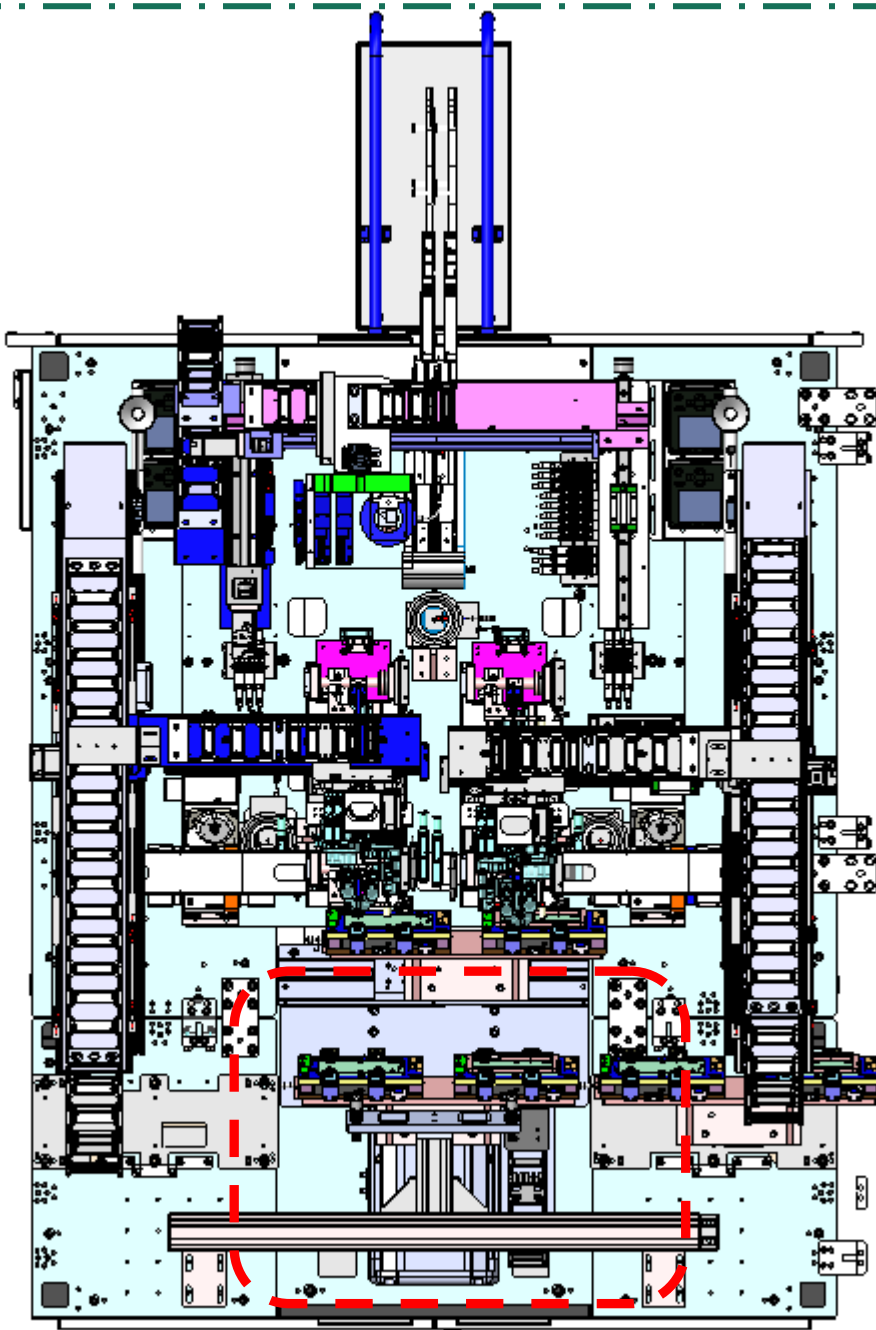


Tolerance Analysis / Engineering Analysis:

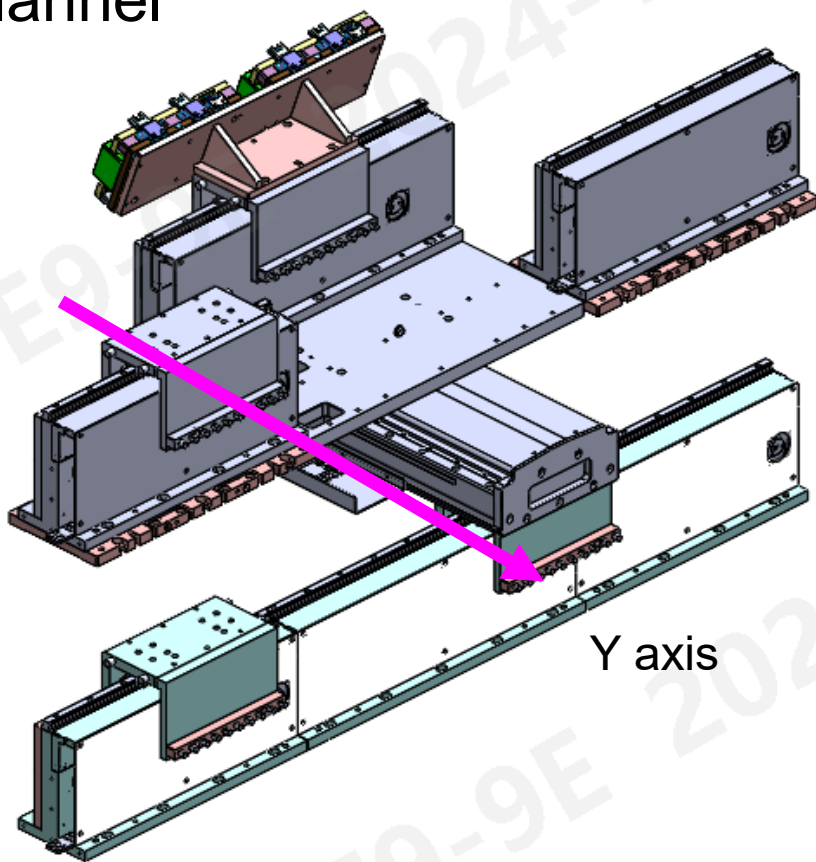
Description	Nominal	Upper	Lower
X Axis	0.000	0.020	0.020
Y Axis	0.000	0.020	0.020
Z Axis	0.000	0.020	0.020
Summary		(+)Tol	(-)Tol
Worst Case		0.060	0.060
RSS		0.03464	0.03464

$$\text{SQRT}[0.02^2+0.02^2+0.02^2]=0.03464$$

Monza | Snap Assy: Module Details – Flow Channel

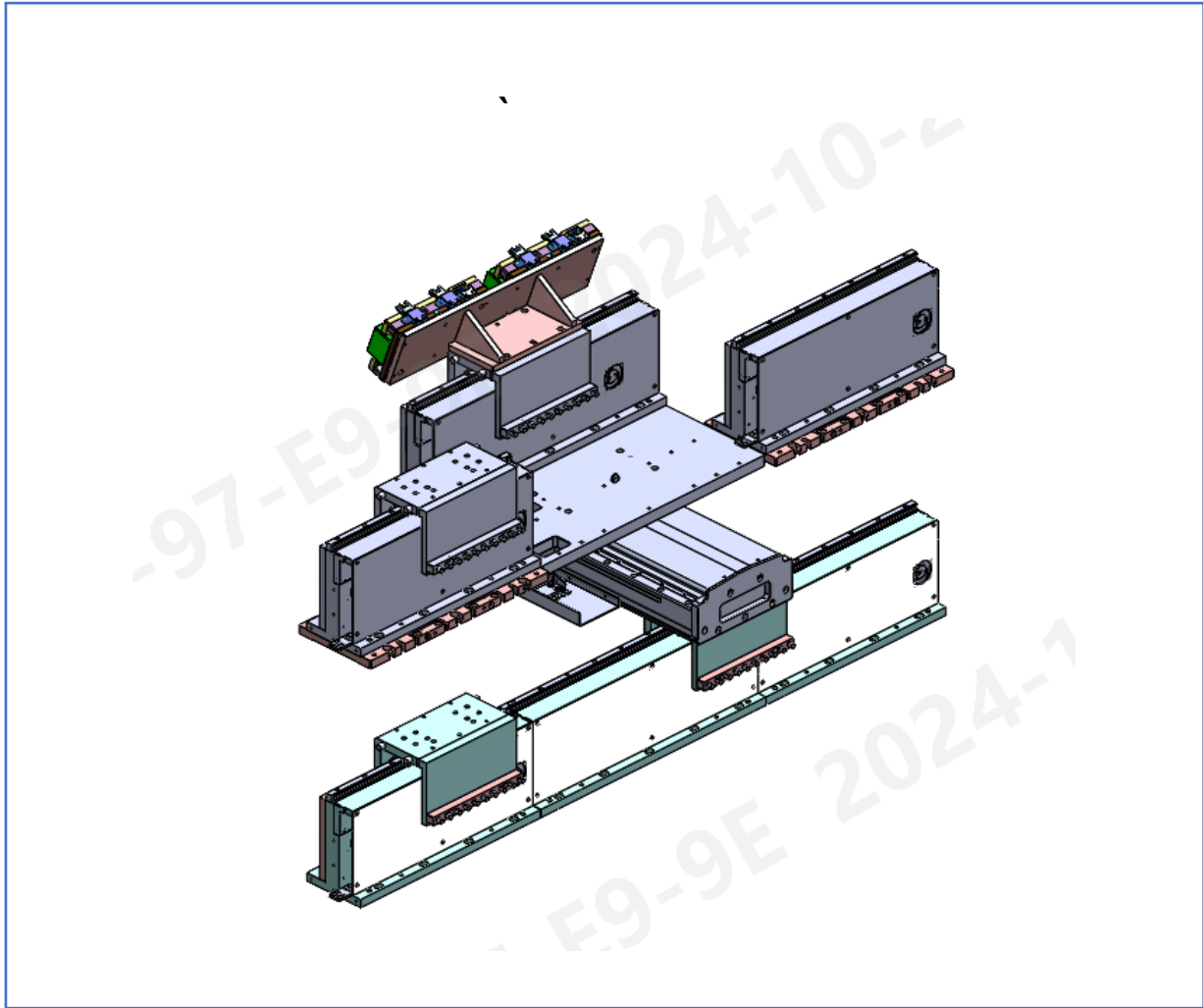


Flow channel

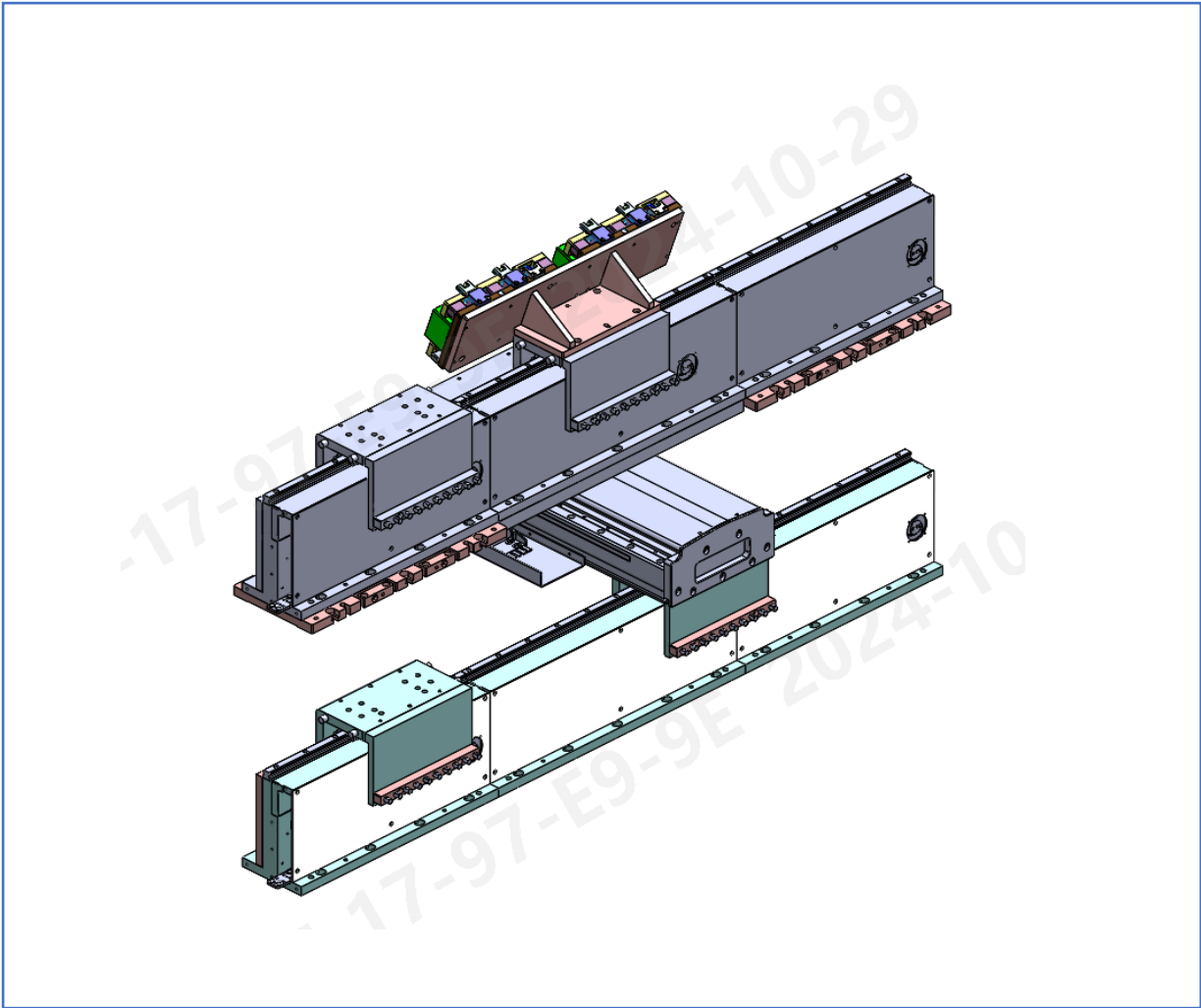
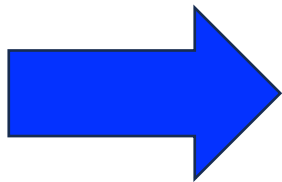


Y axis

No.	Name	Model	Qty	Brand	Remark
1	Y axis module	YTD	1	-	Stroke : 250mm Repeatability : $\pm 0.02\text{mm}$
2	Flow channel	Beckhoff	1	Beckhoff	Magnetic levitation channel

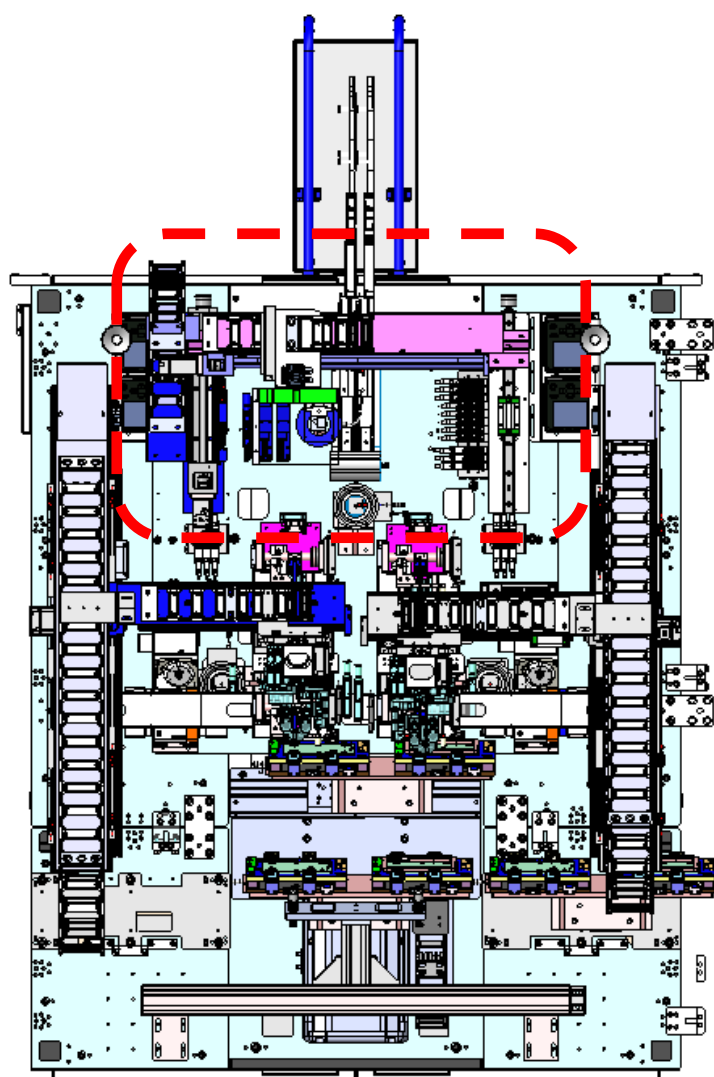


Fixture move in machine from channel

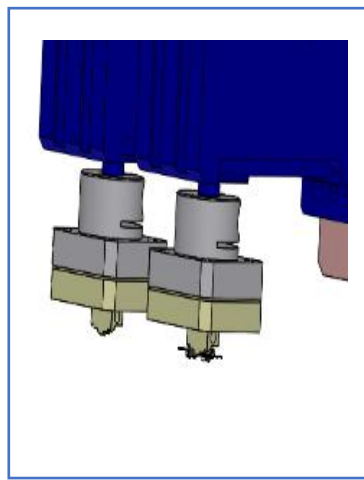
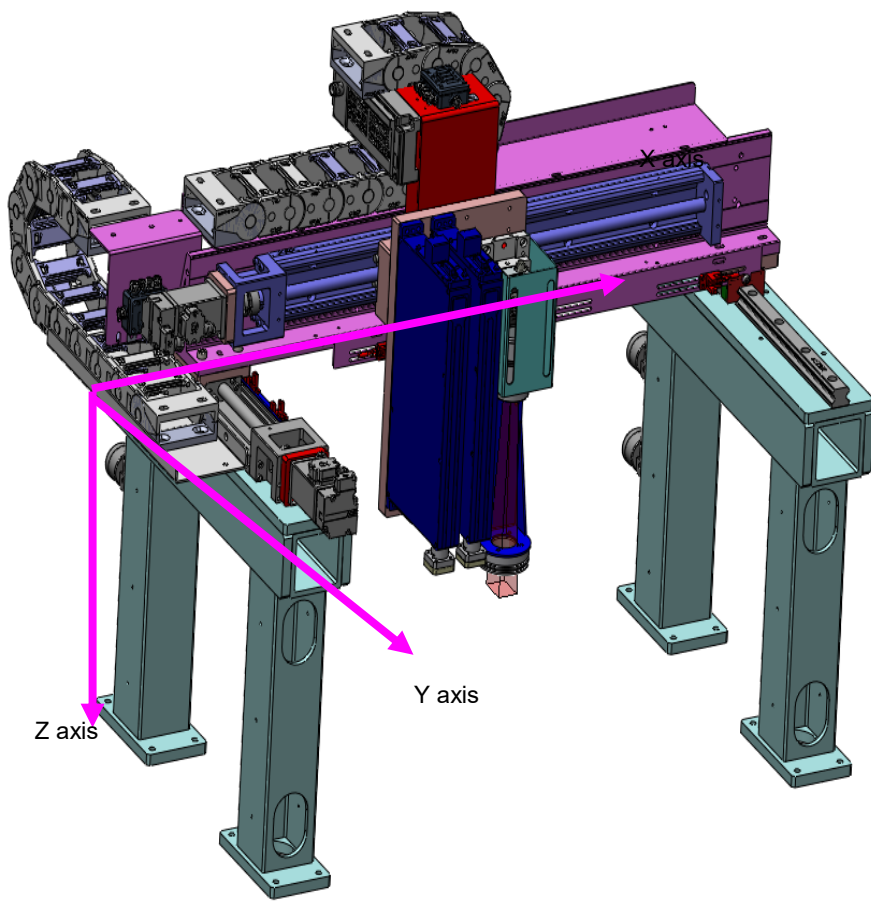
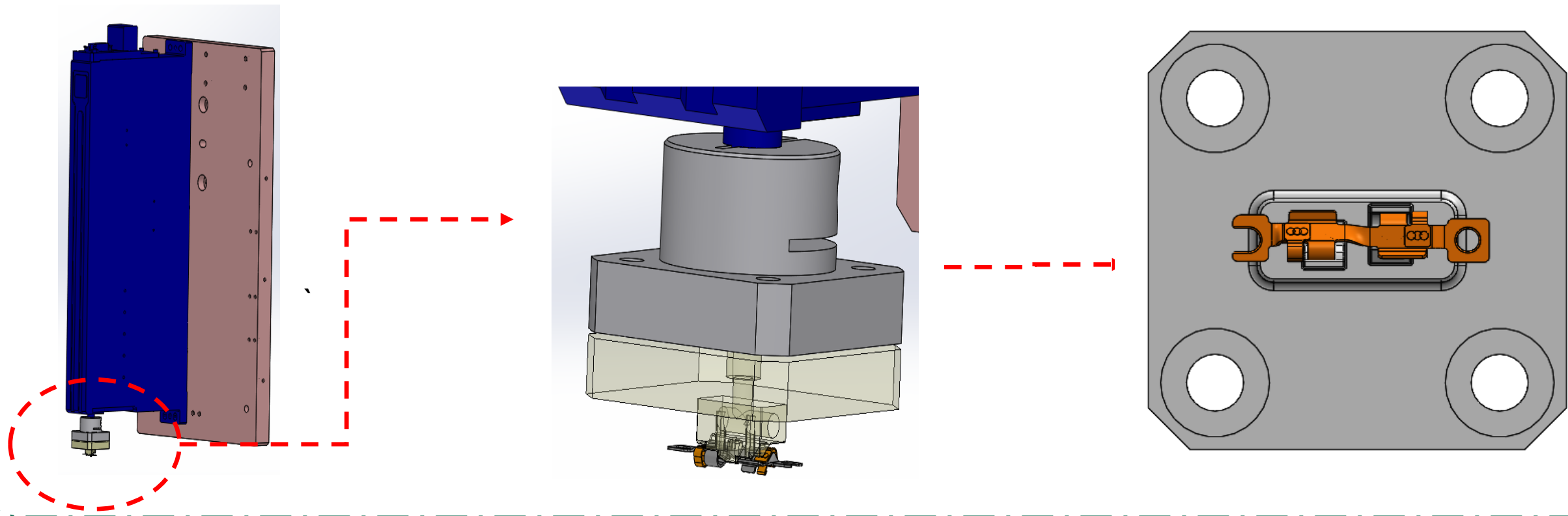


Fixture move back to channel

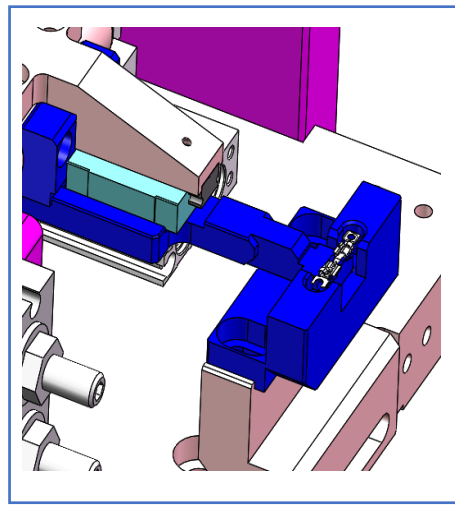
Monza | Snap Assy: Module Details – Snap Gantry



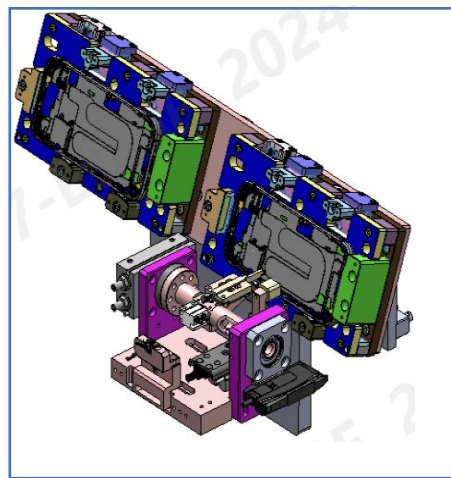
No.	Name	Model	Qty	Brand	Remark
1	X axis module	KK60D10C-400A1-F0	1	HIWIN	Stroke : 400mm Repeatability : $\pm 0.02\text{mm}$
2	Y axis module	KK60-300A1-F0	1	HIWIN	Stroke : 300mm Repeatability : $\pm 0.02\text{mm}$
3	Z axis module	VLR25-S65-A10-05-SE-N-A001	1	TOYO	Stroke : 65mm Repeatability : $\pm 0.02\text{mm}$



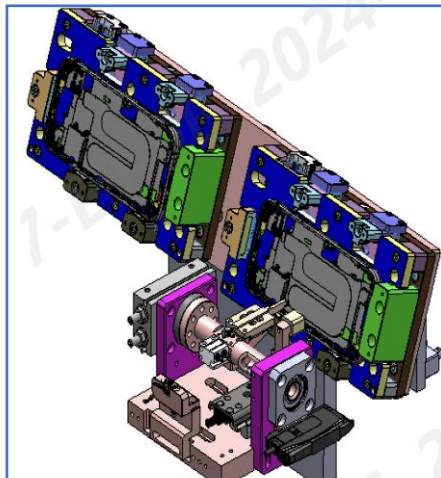
Voice coil motor pick up Snap



Put Snap on Flipping module

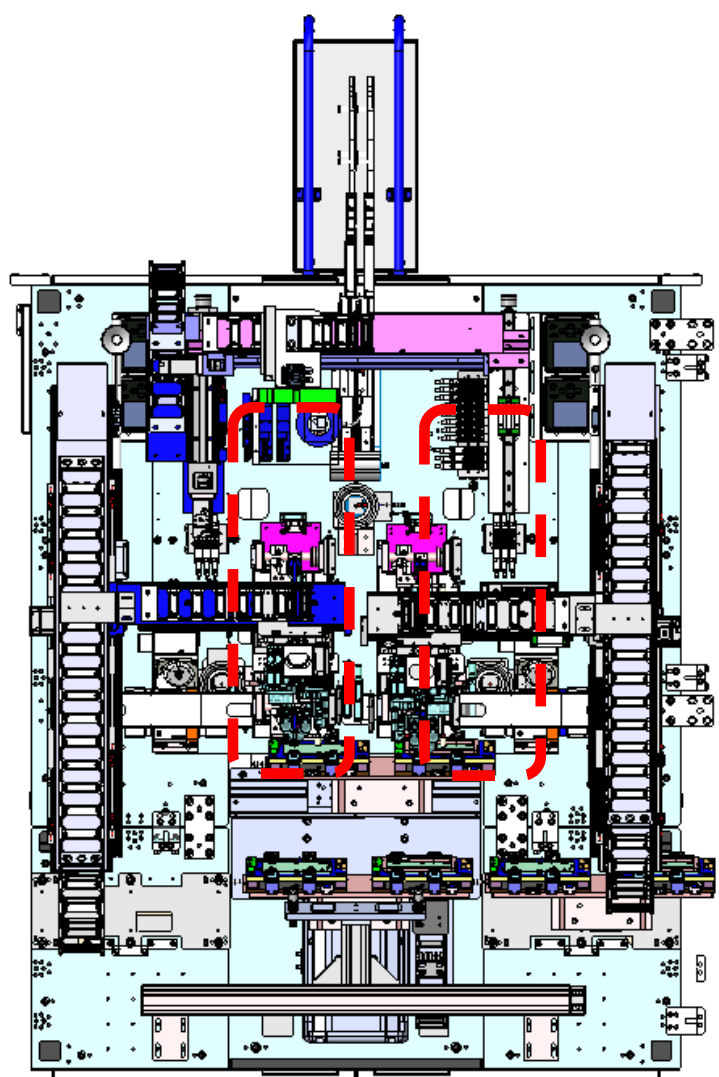


Assemble Snap

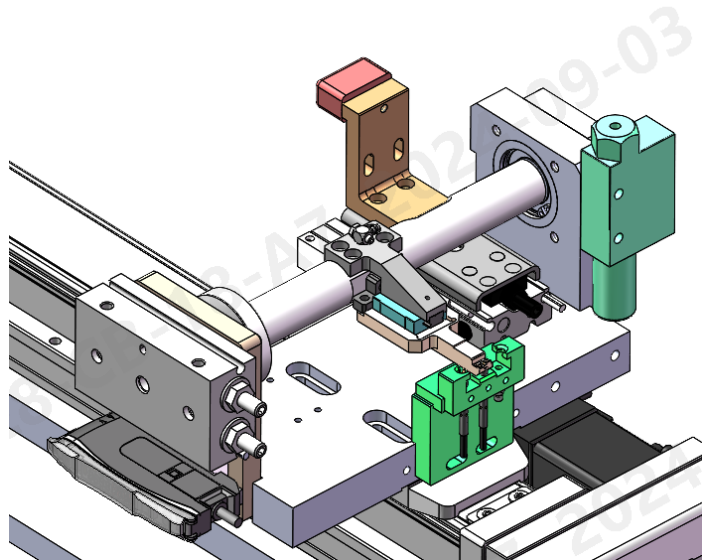


Screwing

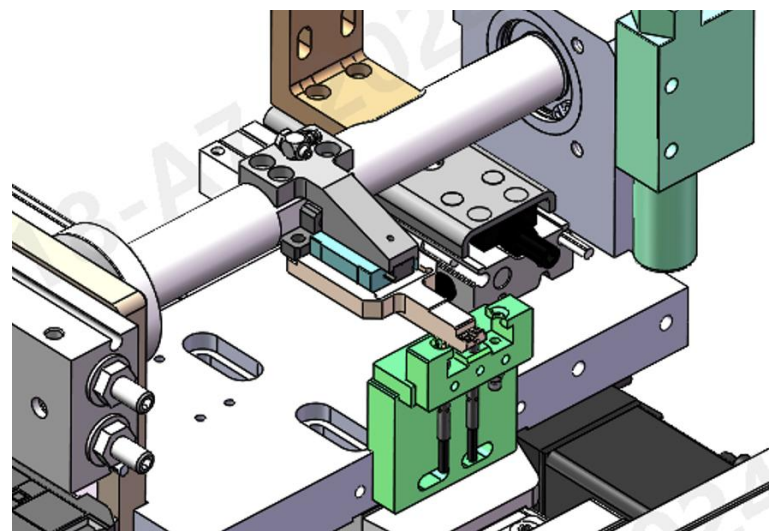
Monza | Snap Assy: Module Details –Snap Transfer Module



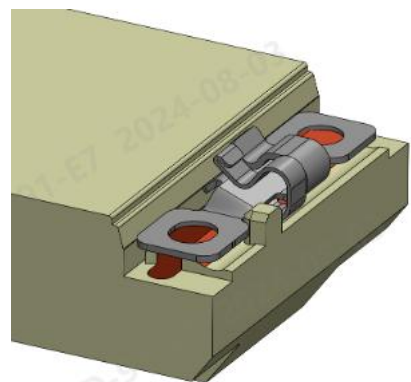
No.	Name	Model	Qty	Brand	Remark
1	Y axis module	GTH5-L10-300-BL(R)-Y10-C3	2	Toyo	Stroke : 300mm Repeatability : ±0.02mm



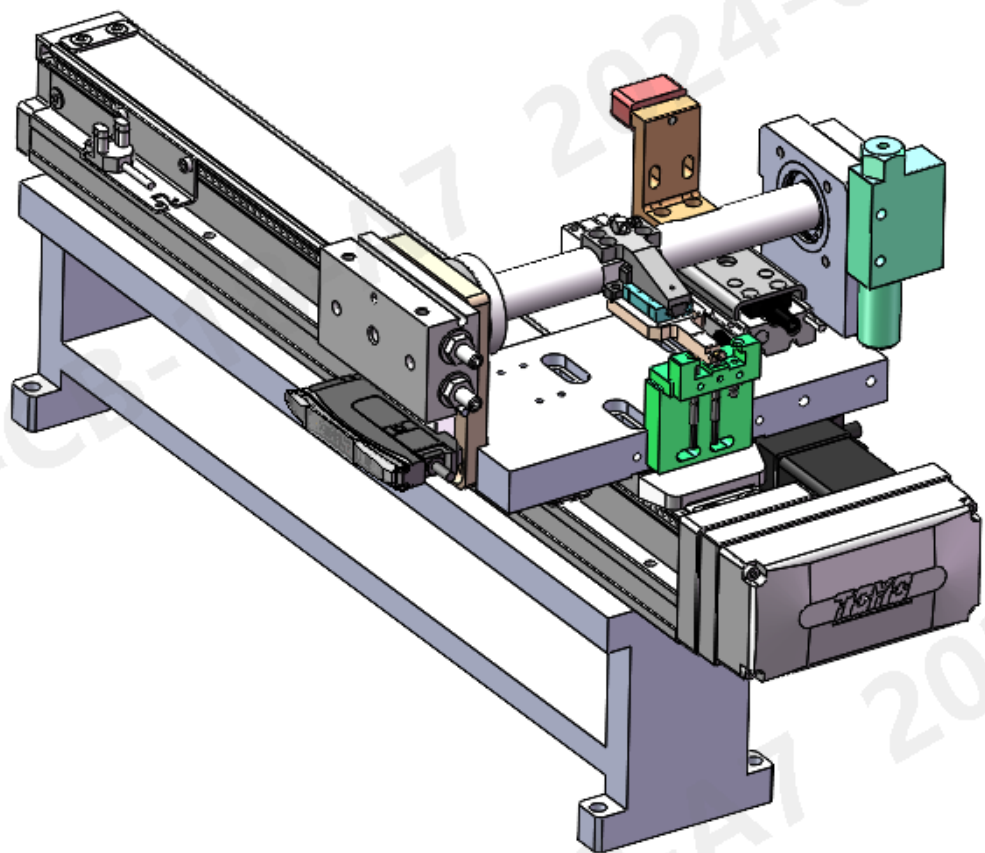
Snap Pickup using Voc method



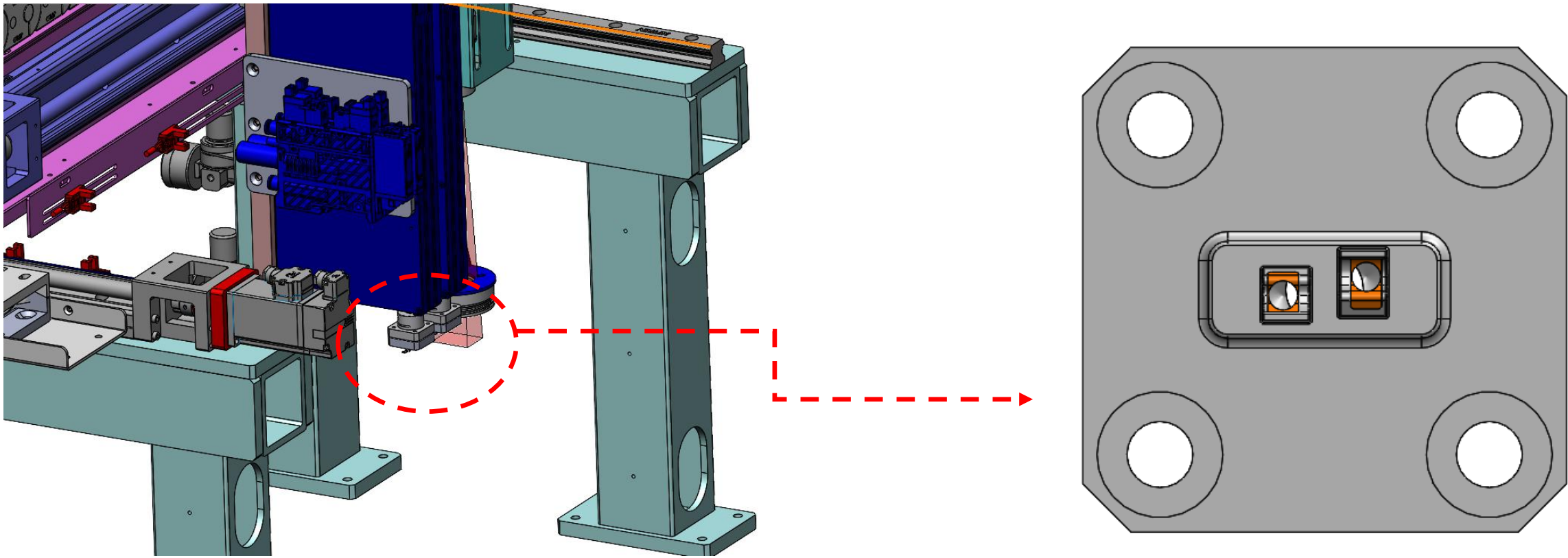
Flipper transfer module



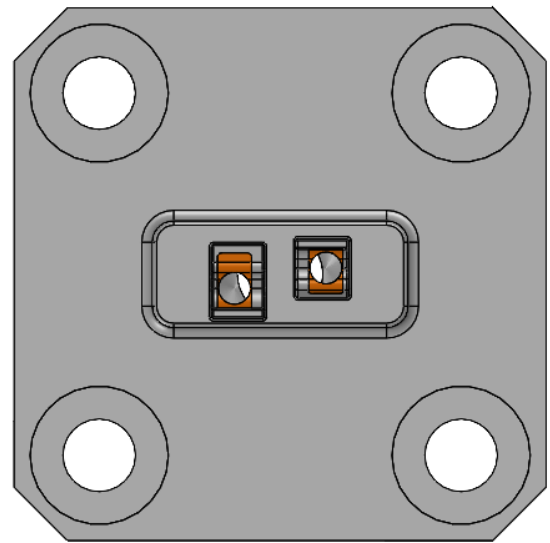
Snap nozzle
(different sizes for Snap)



Monza | Snap Assy: Gripper 1 Vacuum Area



Pitch – 13 mm

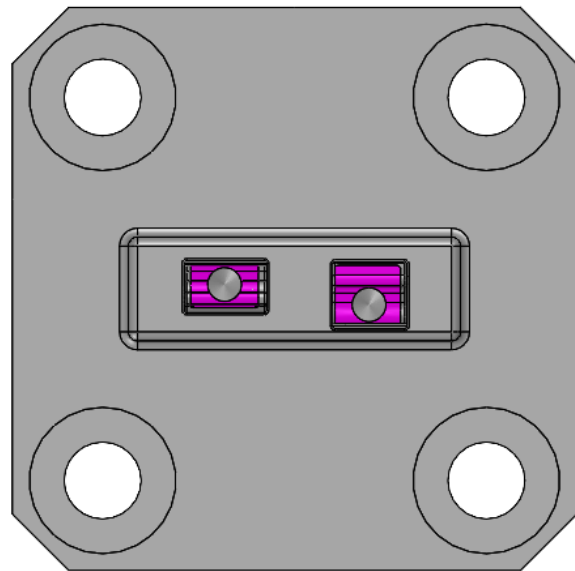


Nozzle hole area $S = S1 + S2$
 $= \pi r^2 + \pi r^2$ (i.e dia 1.5mm)
 $S = 3.535 \text{ mm}^2$

Suction Force
 $F = P \cdot S$
 $= 0.05 \text{ Mpa} \cdot 3.535 \text{ mm}^2 = 0.177 \text{ N}$
 $= 0.0177 \text{ kg}$

SNAP weight 0.000081kg, Suction force greater than spring weight, suction force ok

Pitch – 18 mm

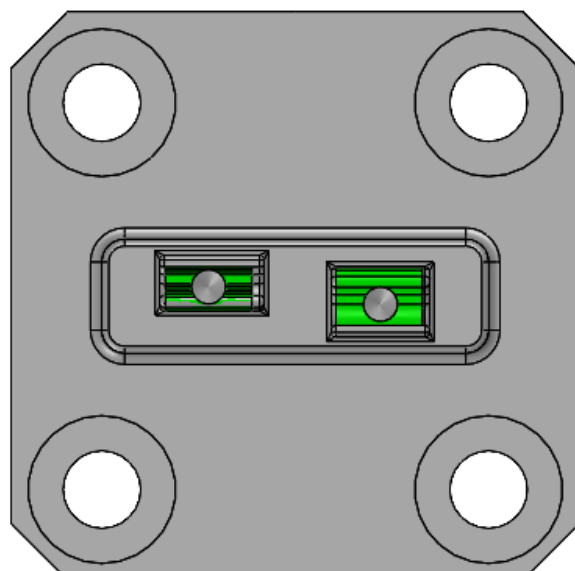


Nozzle hole area $S = S1 + S2$
 $= \pi r^2 + \pi r^2$ (i.e dia 1.5mm)
 $S = 3.535 \text{ mm}^2$

Suction Force
 $F = P \cdot S$
 $= 0.05 \text{ Mpa} \cdot 3.535 \text{ mm}^2 = 0.177 \text{ N}$
 $= 0.0177 \text{ kg}$

SNAP weight 0.000123kg, Suction force greater than spring weight, suction force ok

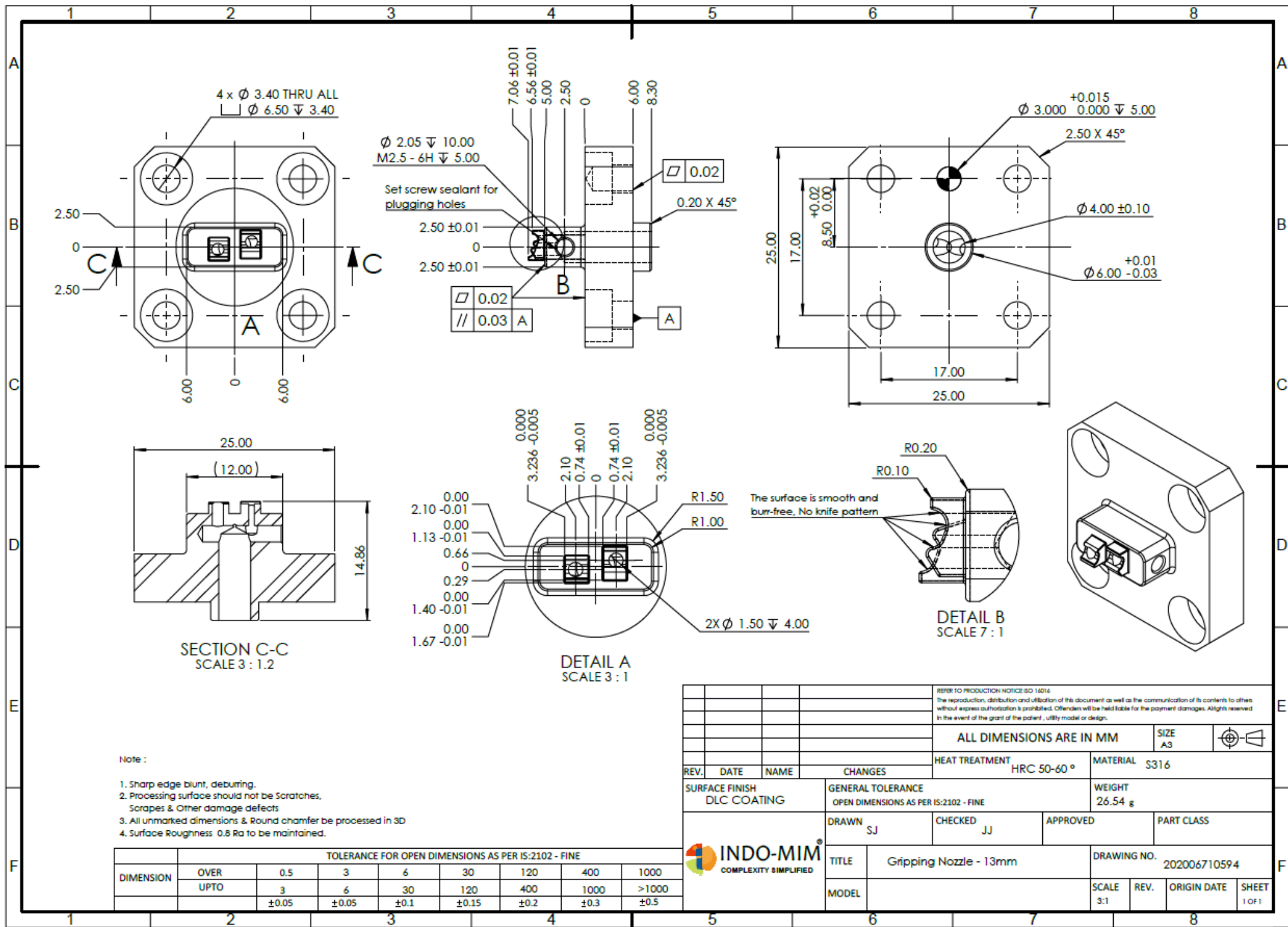
Pitch – 20 mm



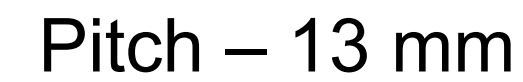
Nozzle hole area $S = S1 + S2$
 $= \pi r^2 + \pi r^2$ (i.e dia 1.5mm)
 $S = 3.535 \text{ mm}^2$

Suction Force
 $F = P \cdot S$
 $= 0.05 \text{ Mpa} \cdot 3.535 \text{ mm}^2 = 0.177 \text{ N}$
 $= 0.0177 \text{ kg}$

SNAP weight 0.000123kg, Suction force greater than spring weight, suction force ok



GENERAL / EXTERNAL

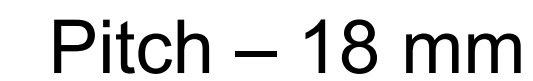


$$S = 2.0498 \text{ mm}^2$$

Suction Force

$$\begin{aligned} F &= P \cdot S \\ &= 0.05 \text{ Mpa} \cdot 2.0498 \text{ mm}^2 = 0.1025 \text{ N} \\ &= 0.01025 \text{ kg} \end{aligned}$$

SNAP weight 0.000081kg, Suction force greater than spring weight, suction force ok

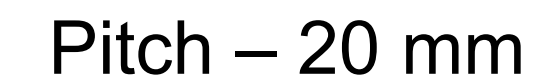


$$S = 3.0867 \text{ mm}^2$$

Suction Force

$$\begin{aligned} F &= P \cdot S \\ &= 0.05 \text{ Mpa} \cdot 3.0867 \text{ mm}^2 = 0.1543 \text{ N} \\ &= 0.01543 \text{ kg} \end{aligned}$$

SNAP weight 0.000123kg, Suction force greater than spring weight, suction force ok.



$$S = 2.9067 \text{ mm}^2$$

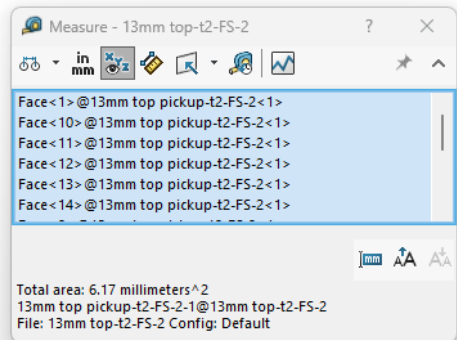
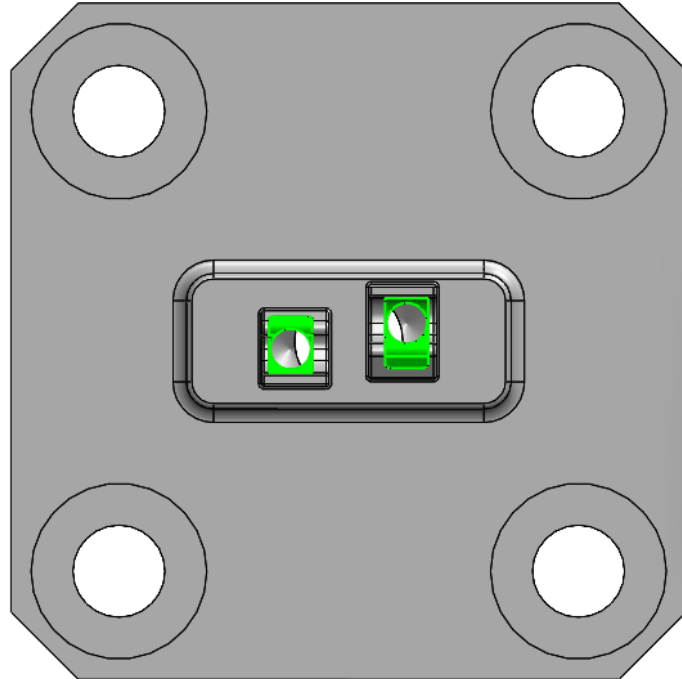
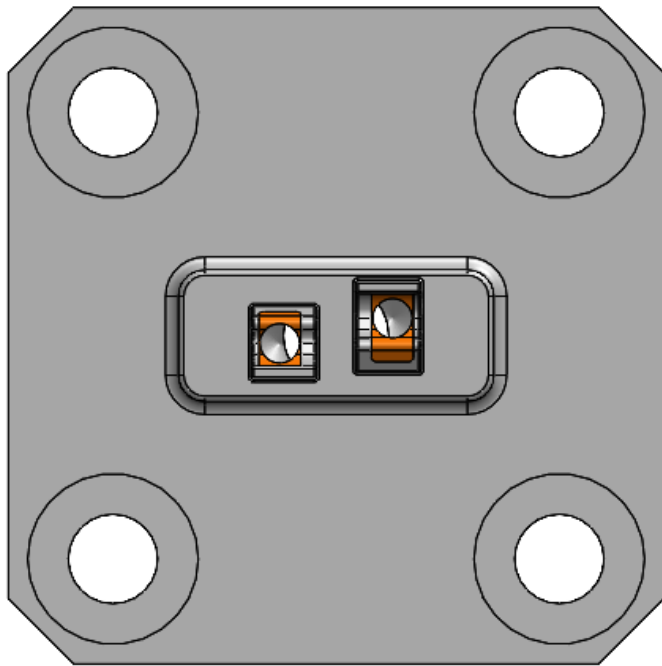
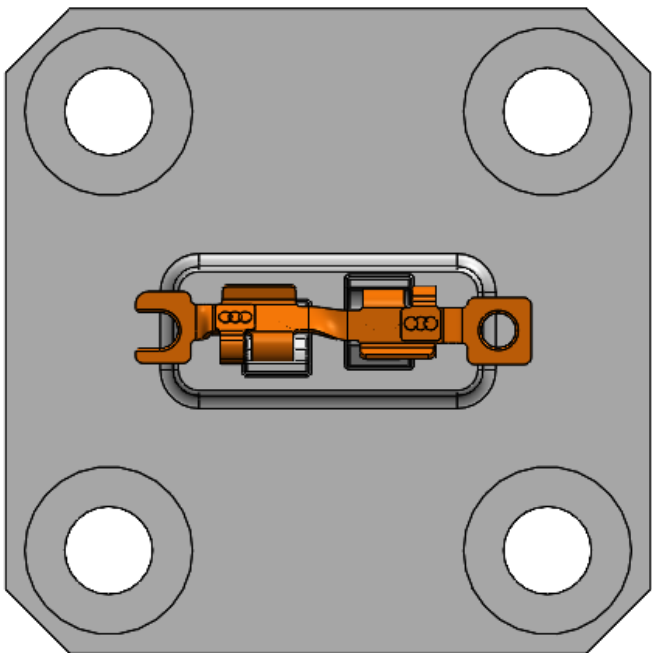
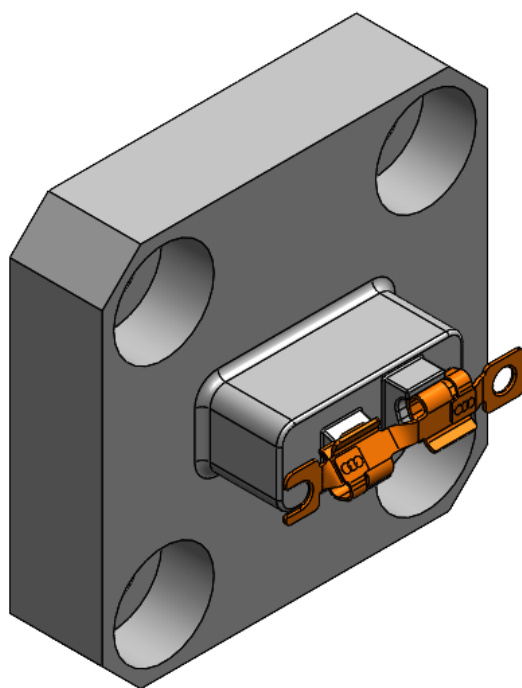
Suction Force

$$\begin{aligned} F &= P \cdot S \\ &= 0.05 \text{ Mpa} \cdot 2.9067 \text{ mm}^2 = 0.1453 \text{ N} \\ &= 0.01453 \text{ kg} \end{aligned}$$

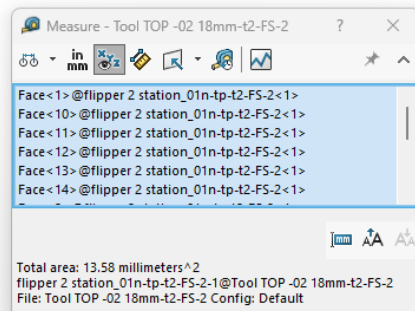
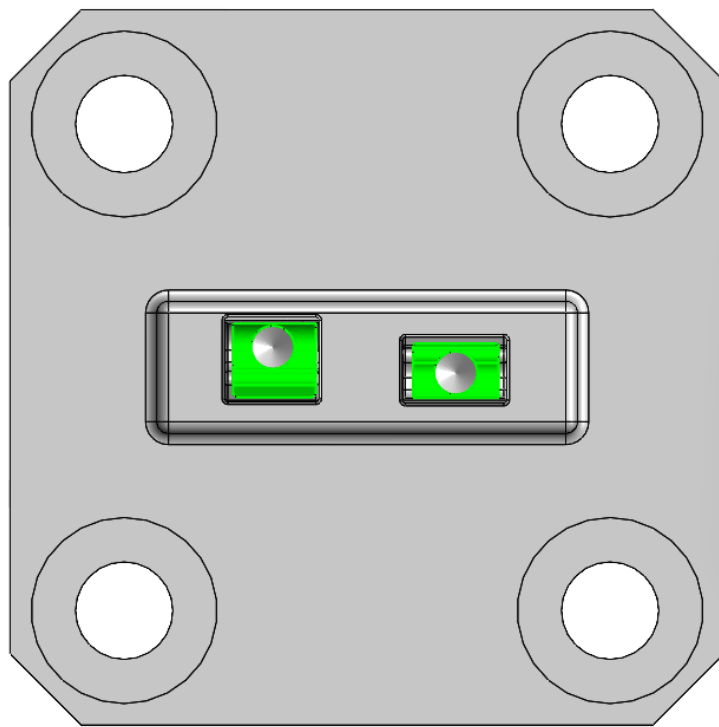
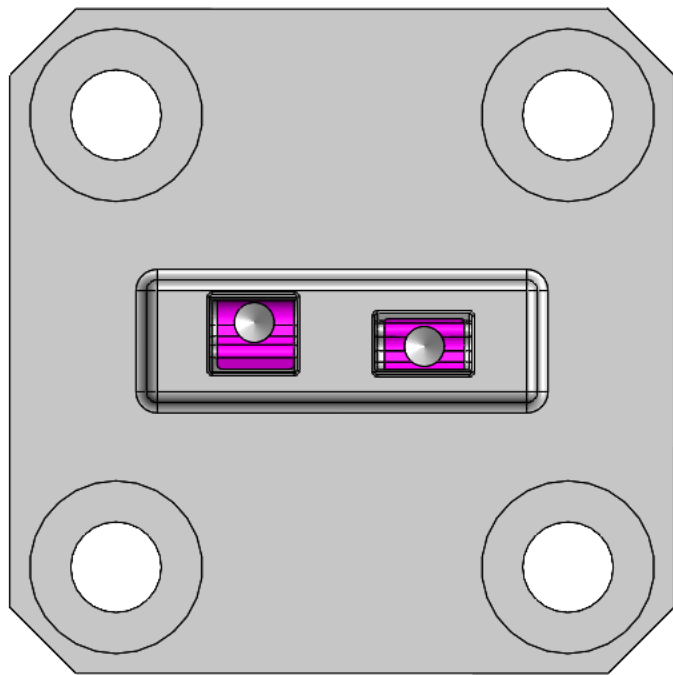
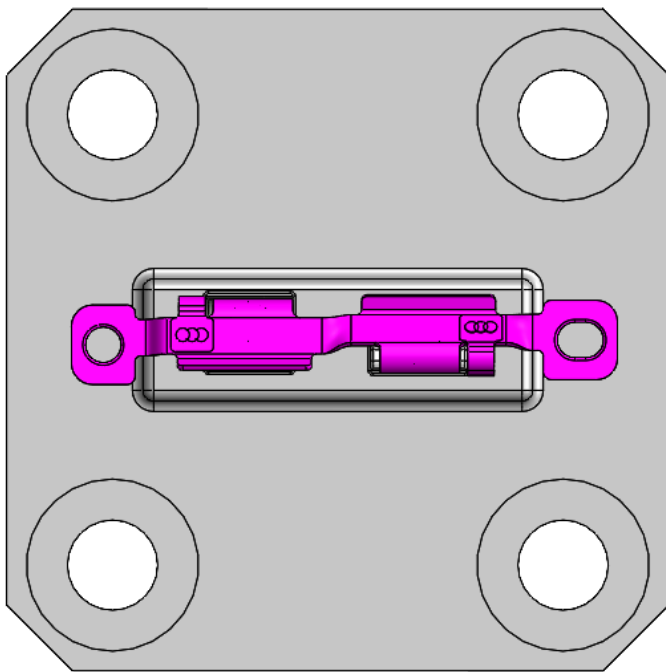
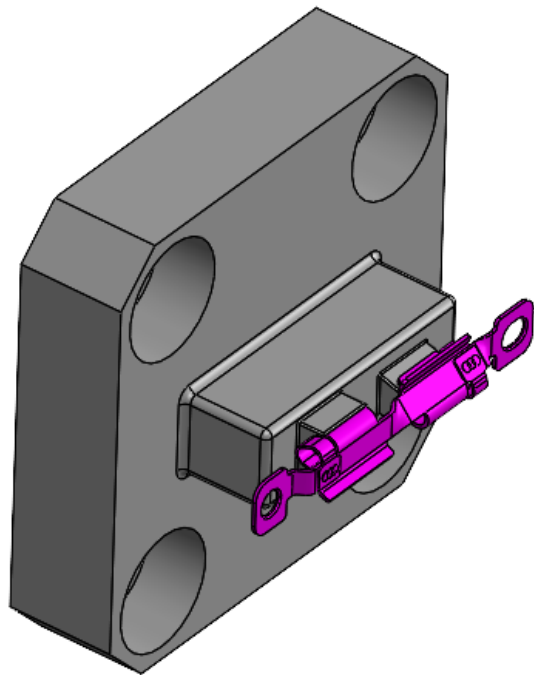
SNAP weight 0.000123kg, Suction force greater than spring weight, suction force ok.

Monza | Snap Assy: Gripper 1 Total contact area

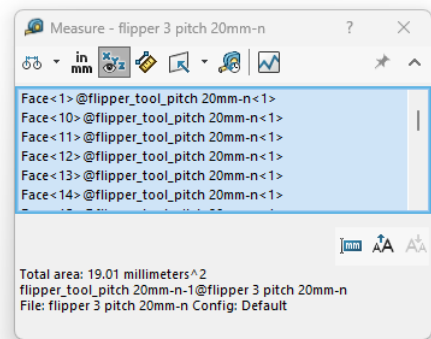
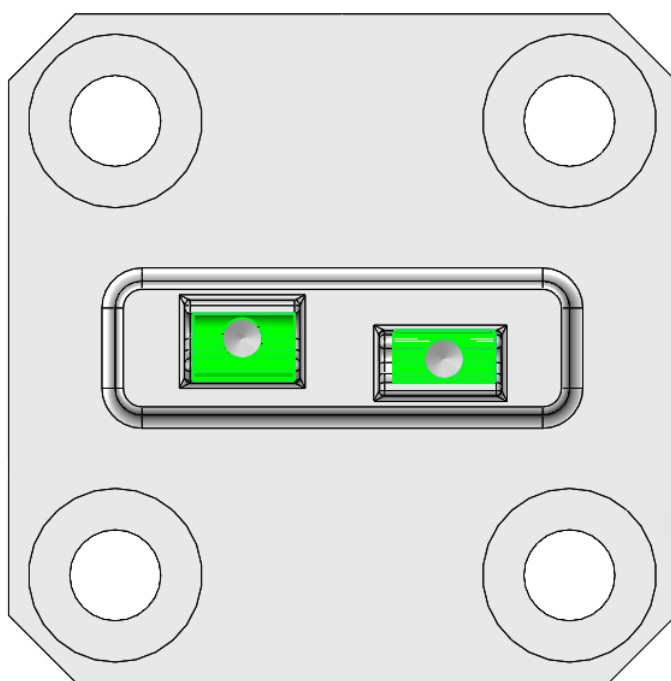
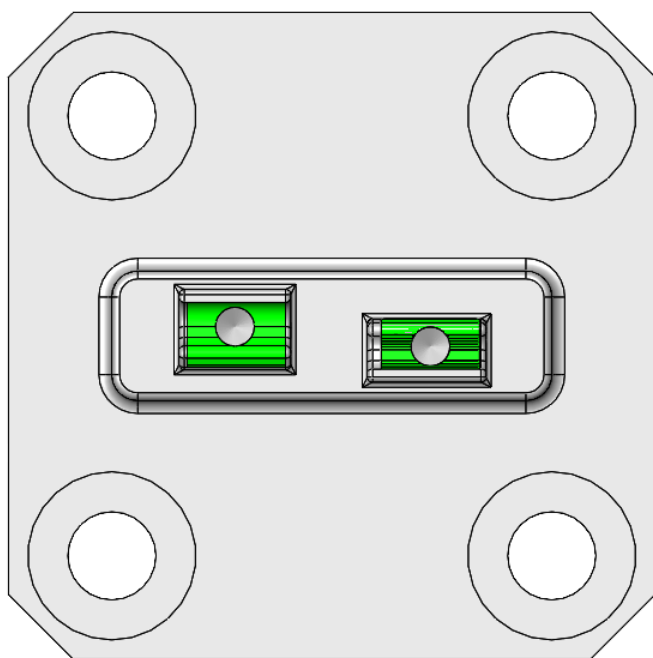
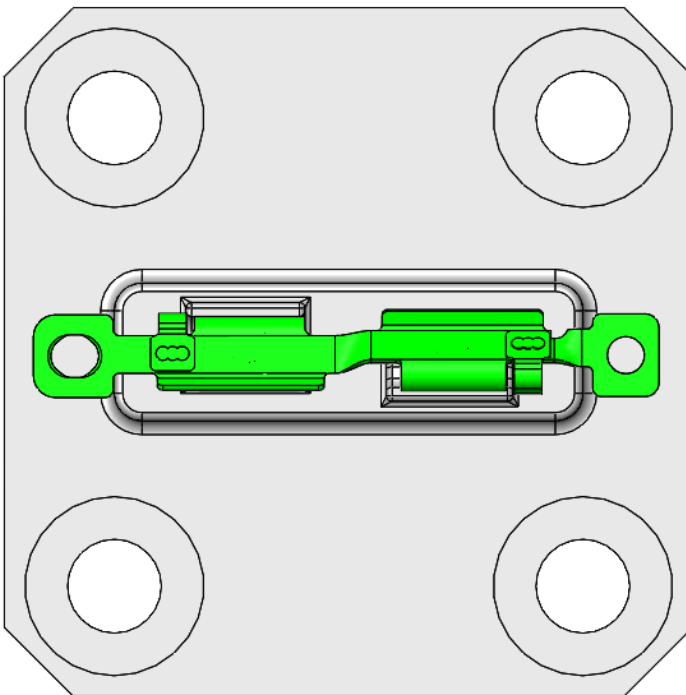
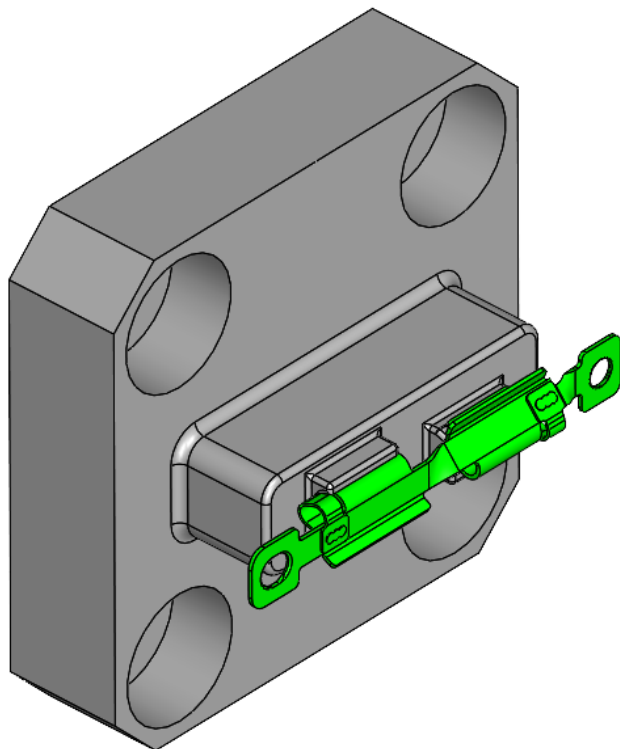
Pick-up Tool 613-31681
(AZH767) Pitch 13mm



Pick-up Tool 613-38625
(AUV687) Pitch 18mm

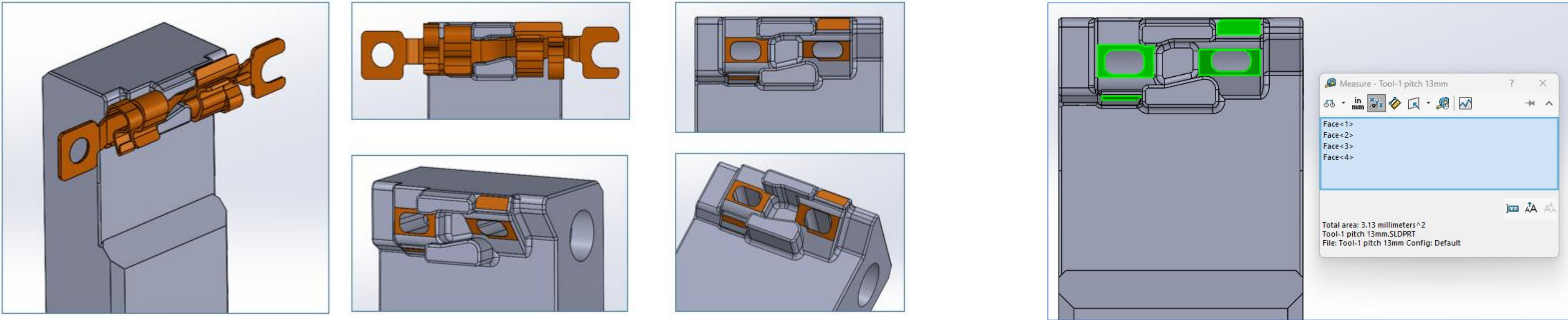


Tool 613-38626
(AUV688) Pitch 20mm

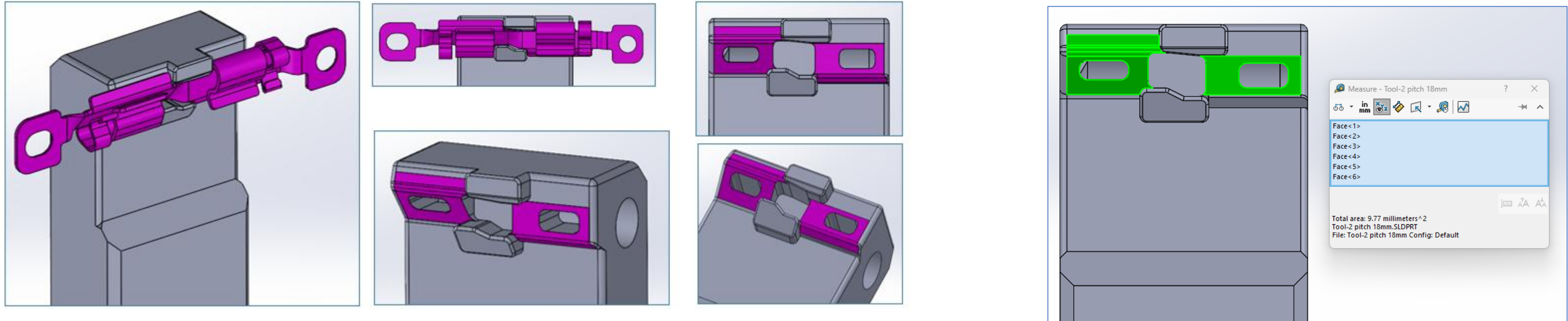


Monza | Snap Assy: Gripper 2 Total contact area

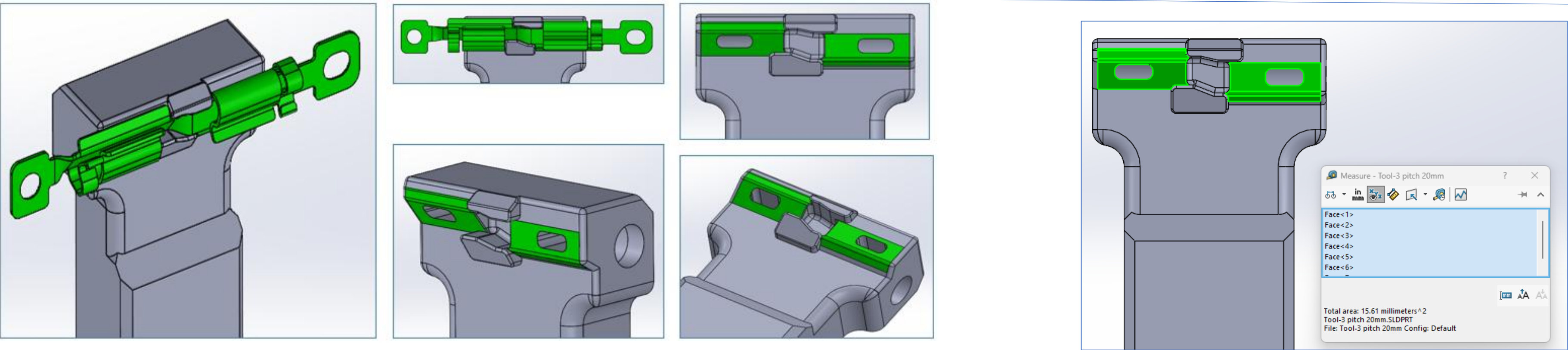
Pick-up Tool 613-31681
(AZH767) Pitch 13mm



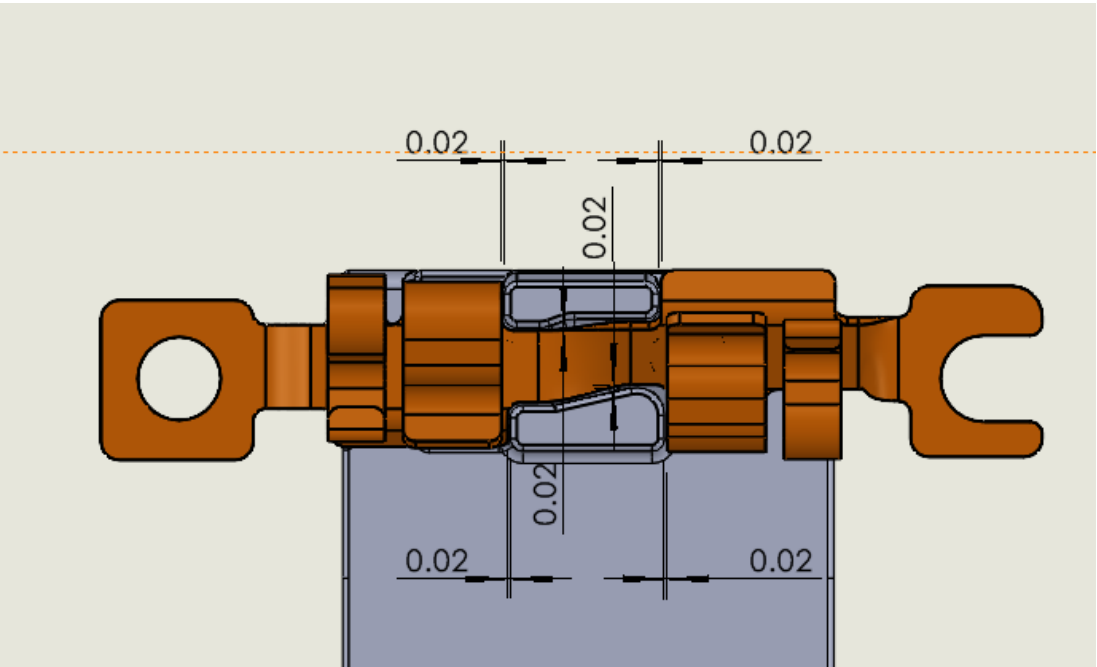
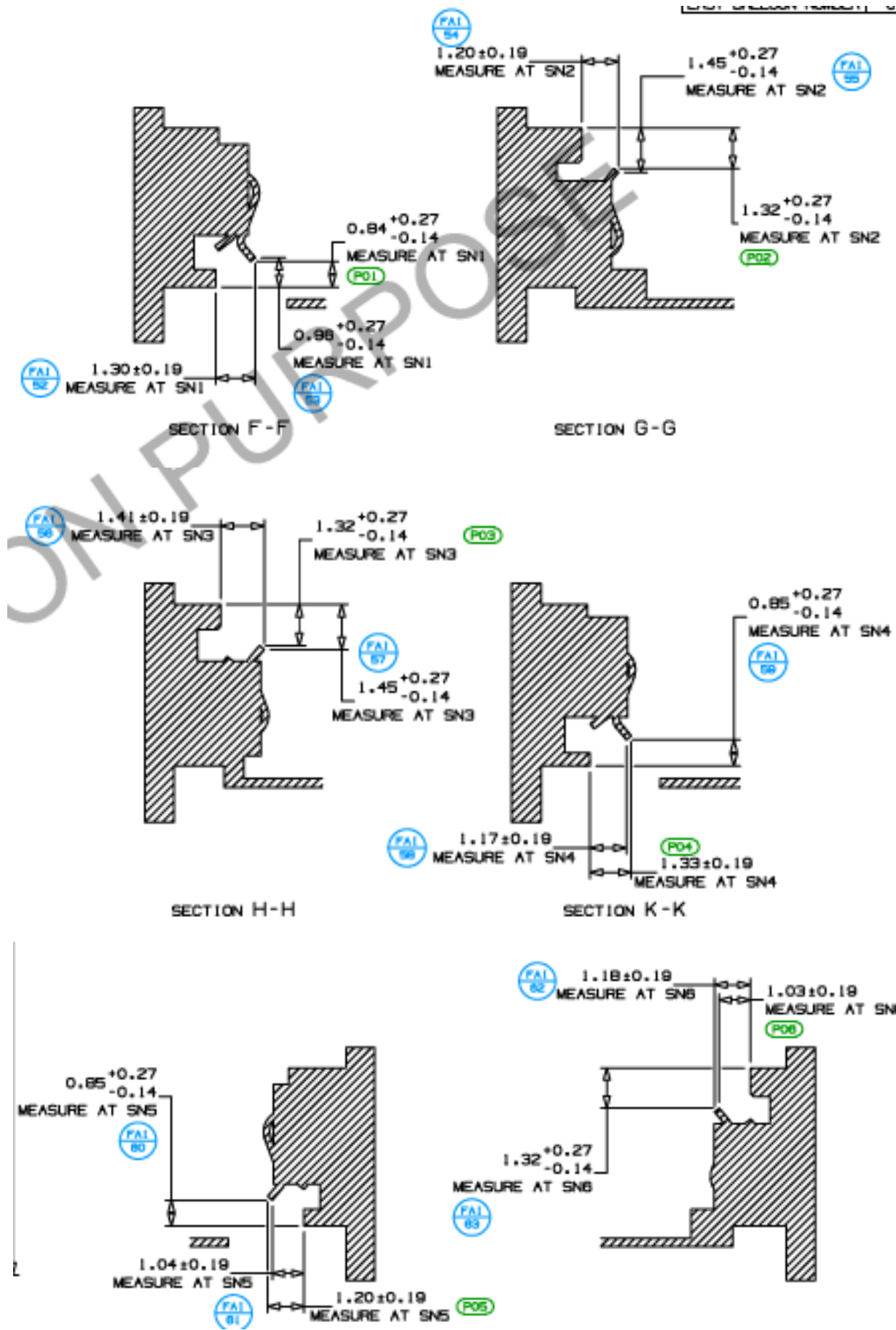
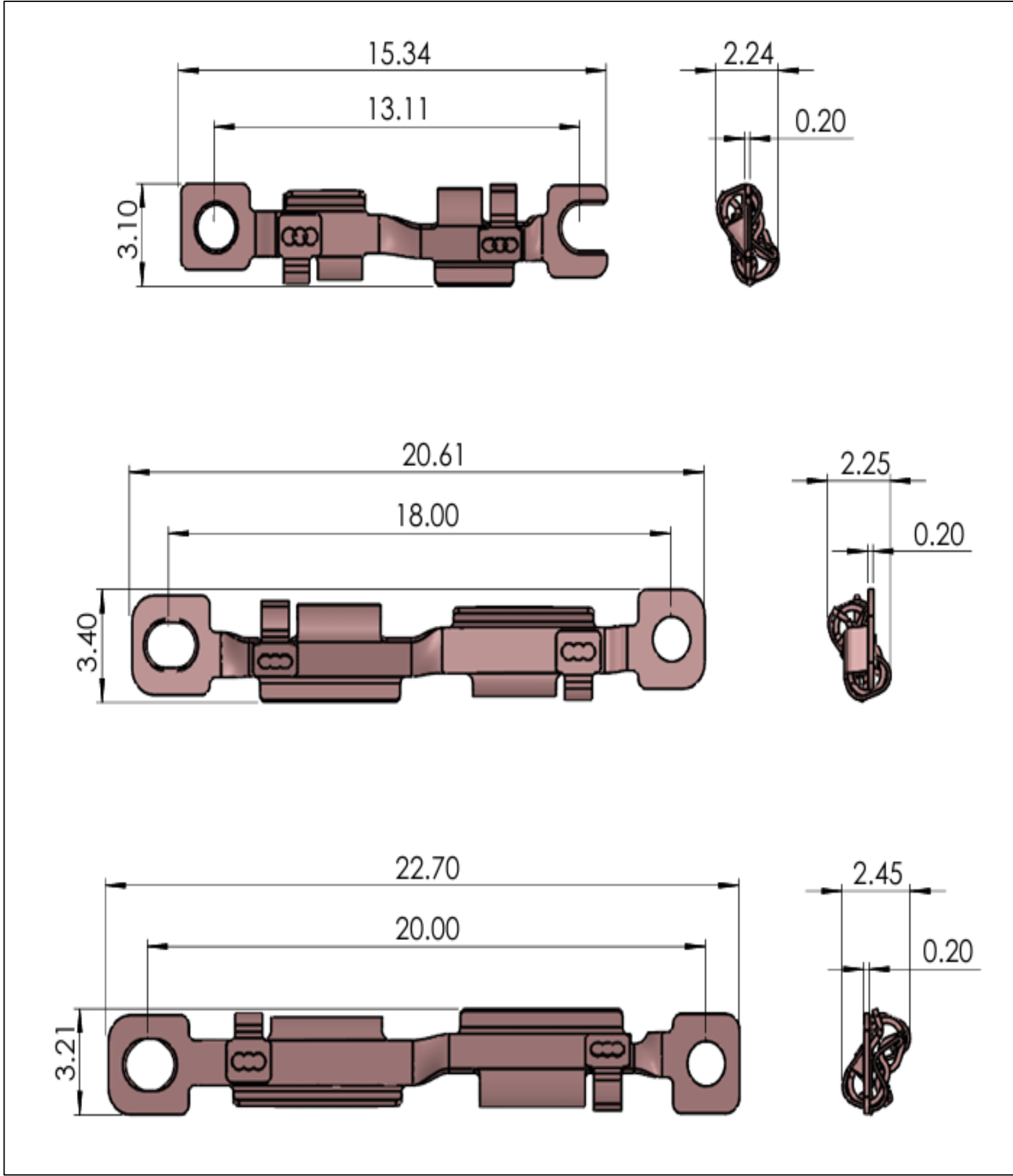
Pick-up Tool 613-38625
(AUV687) Pitch 18mm



Tool 613-38626
(AUV688) Pitch 20mm



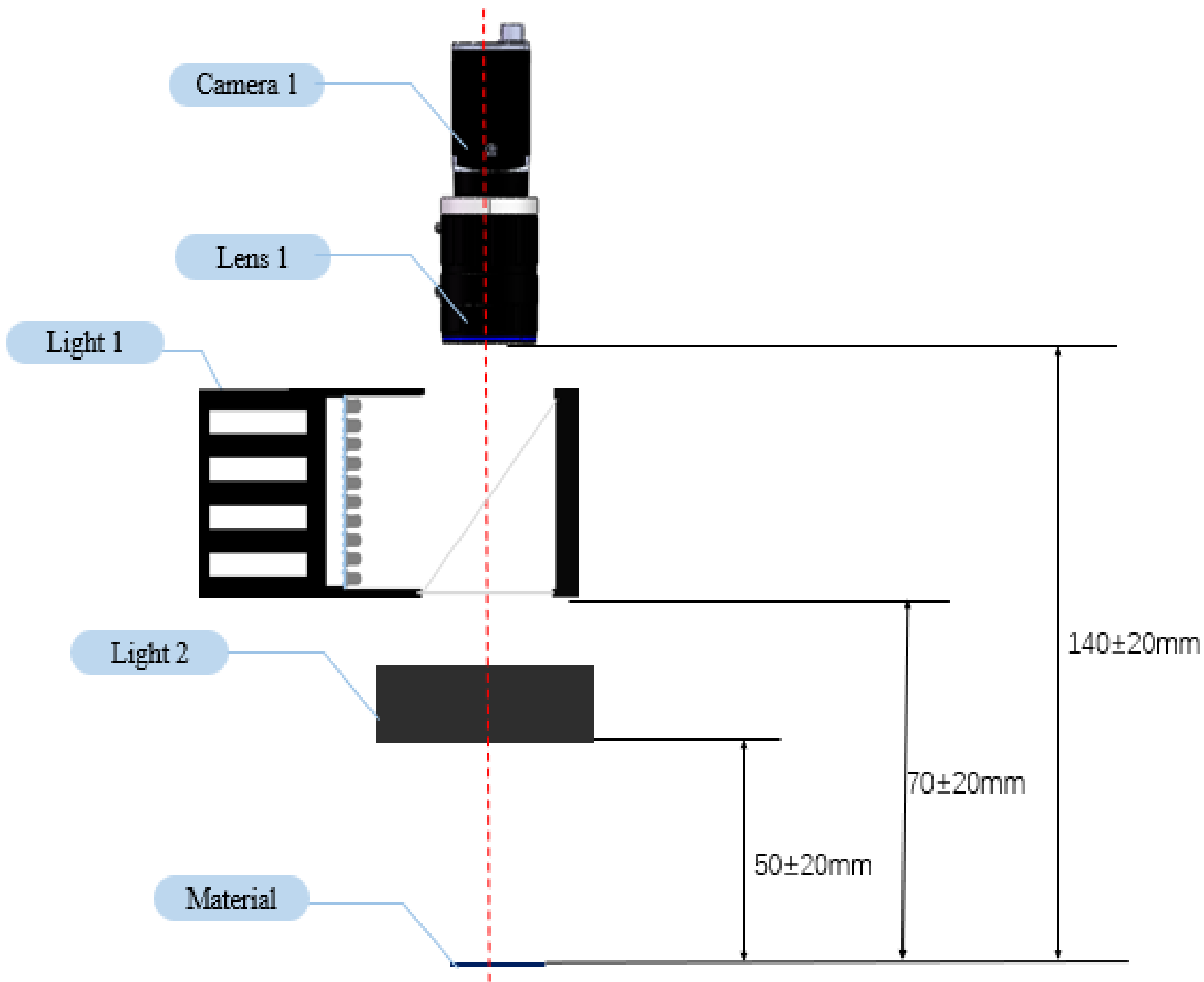
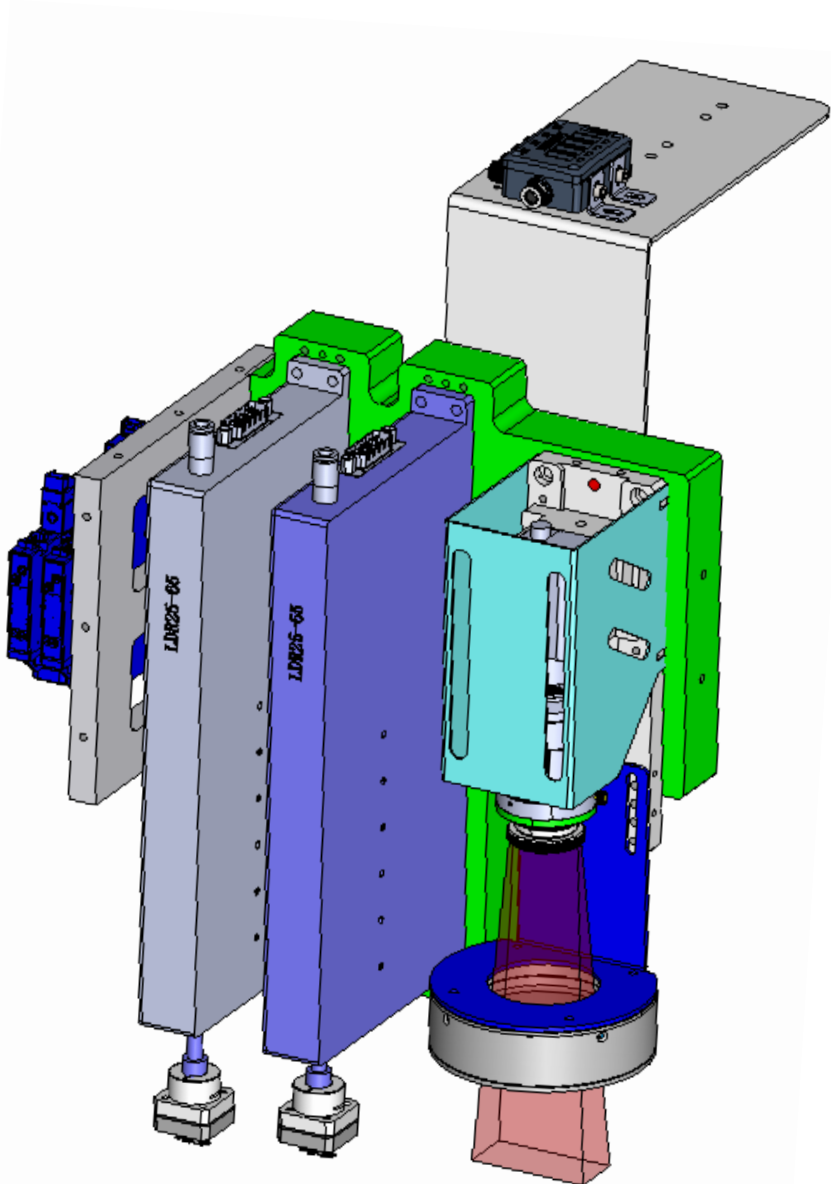
Monza | Snap Assy: Snap Stack-Up Tool Vs Snap



Snap Alignment tolerance Varies from +270 micron to -140micron as mentioned.,

Clearance between the Snap And EOAT tool in XY Alignment is Kept at +20 micron and -20 micron which sufficiently satisfies Stack-Up

Monza | Snap Assy: Process Improvement – Snap Picking CCD



Schematic diagram of setup (Front View)

$$\text{Spatial resolution (mm/pixel)} = \frac{\text{Field of View (mm)}}{\text{Number of pixels in that direction}}$$

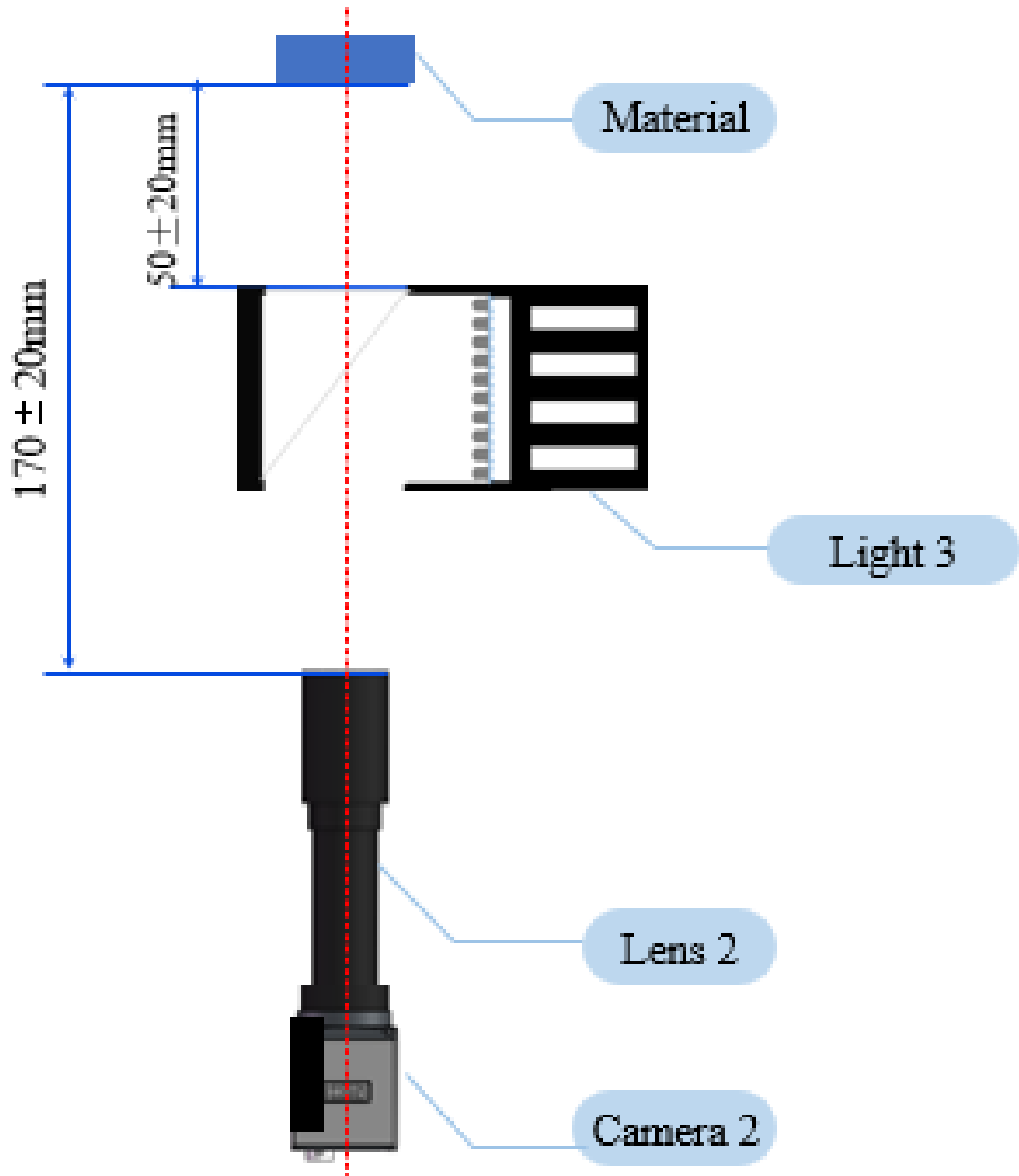
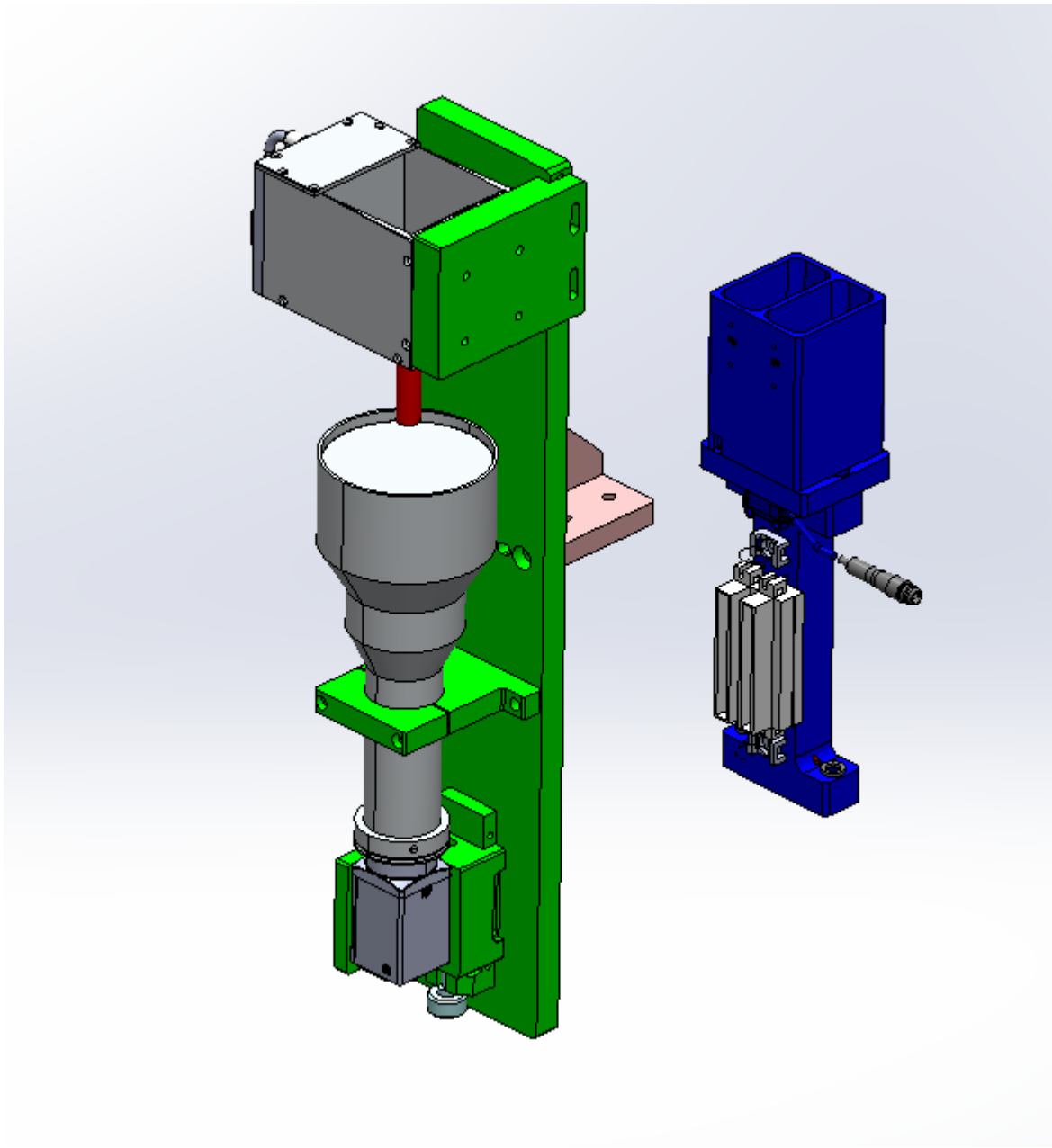
$$\text{Sub-pixel unit} = \frac{1}{10} \text{ pixel}$$

$$\text{Edge precision} = \text{Spatial resolution} \times \text{Sub-pixel unit}$$

$$\text{Single-edge Accuracy (mm)} \approx \text{Spatial resolution} \times \text{sub-pixel precision}$$

$$\text{Repeatability (mm)} \approx 2 \times \text{Single-edge accuracy}$$

Station		Quantity
CAM1	ZXCM1200-GLP117-01-T	1
Lens1	Fixed Focal Length Lens	
Lens1 Model	AC3514-10M-T	1
Working Distance	140±20mm	
FOV	30mm * 22.63mm	
Spatial Resolution	0.00746mm/pixel	
Aperture Value	F5.6	
Extension Ring	JQ5MM-T	1
Light Source1	Co-axial Light	
Light Model 1	CTD60-B-T	1
Light Source2	Blue Ring Light	
Light Model 2	RLA960-B-T	1



Schematic diagram of setup (Front View)

Spatial resolution (mm/pixel) = $\frac{\text{Field of View (mm)}}{\text{Number of pixels in that direction}}$

Sub-pixel unit = $\frac{1}{10}$ pixel

Edge precision = Spatial resolution × Sub-pixel unit

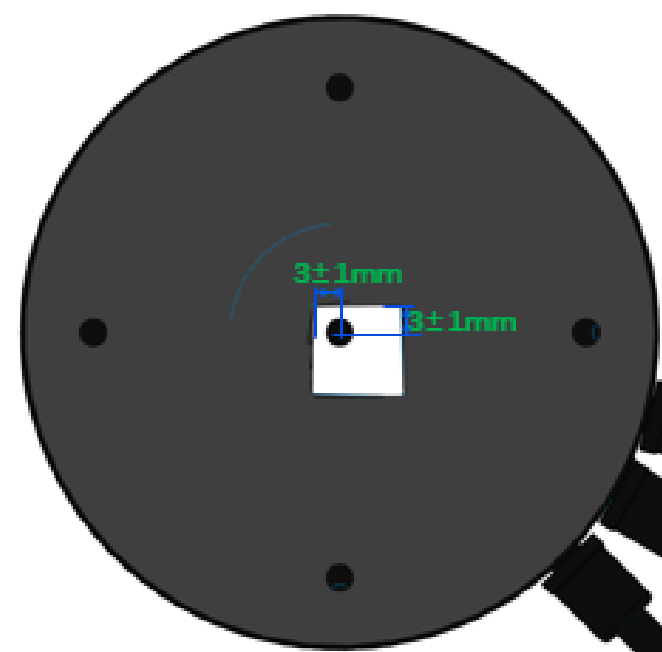
Single-edge Accuracy (mm) ≈ Spatial resolution × sub-pixel precision

Repeatability (mm) ≈ 2 × Single-edge accuracy

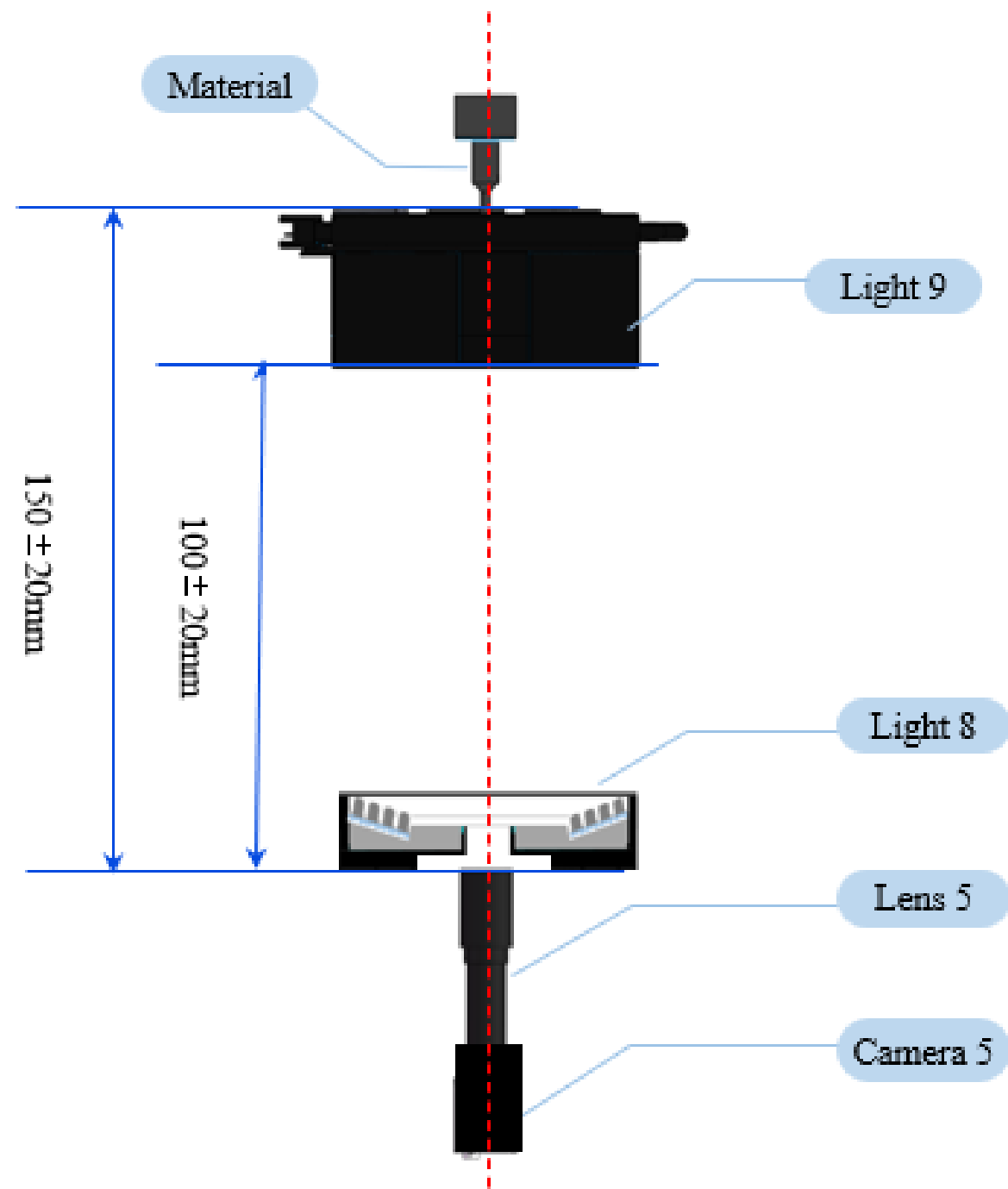
Station		Quantity
Cam 2	ZXCM1200-GMP117-10-T	1
Lens 2	Telecentric lens	
Lens Model 2	ADW0470-170-01-T	1
Working Distance	170 ± 20mm	
FOV	30mm*22mm	
Spatial Resolution	0.0074mm/pixel	
Light Source 3	White Coaxial Light	
Light Model 3	CTD60-V21-W-T	1

Monza | Snap Assy: Process Improvement –Screw Lower CCD with Lens prism

The square opening of the light source needs to be level with the FOV and requires fine-tuning.



schematic diagram of Light source top view



Schematic diagram of setup Front View

$$\text{Spatial resolution (mm/pixel)} = \frac{\text{Field of View (mm)}}{\text{Number of pixels in that direction}}$$

$$\text{Sub-pixel unit} = \frac{1}{10} \text{ pixel}$$

$$\text{Edge precision} = \text{Spatial resolution} \times \text{Sub-pixel unit}$$

$$\text{Single-edge Accuracy (mm)} \approx \text{Spatial resolution} \times \text{sub-pixel precision}$$

$$\text{Repeatability (mm)} \approx 2 \times \text{Single-edge accuracy}$$

Station		Quantity
Cam 8	ZXCM600-GLP118-01-T	1
Lens 8	Telecentric lens	
Lens Model 8	ADT03-150-01-T	1
Working Distance	150 ± 20mm	
FOV	24.58mm*16.38mm	
Spatial Resolution	0.008mm/pixel	
Light Source 8	Blue Ring Light	
Light Model 8	RLA730-B-T	1
Light Source 9&10	Combined Light source	
Light Model 9&10	ALXS7090-B-T	1

OPT-CDF06-05IC184B

▼ Machine Vision

Features

With a built-in light source and CH05 series lenses of different apertures, it achieves integrated shooting, enabling multi-angle shooting of the inner wall and top of the workplace. The high resolution can detect details of any objects. Different apertures affect the depth of field of the workplace without affecting the field of view.

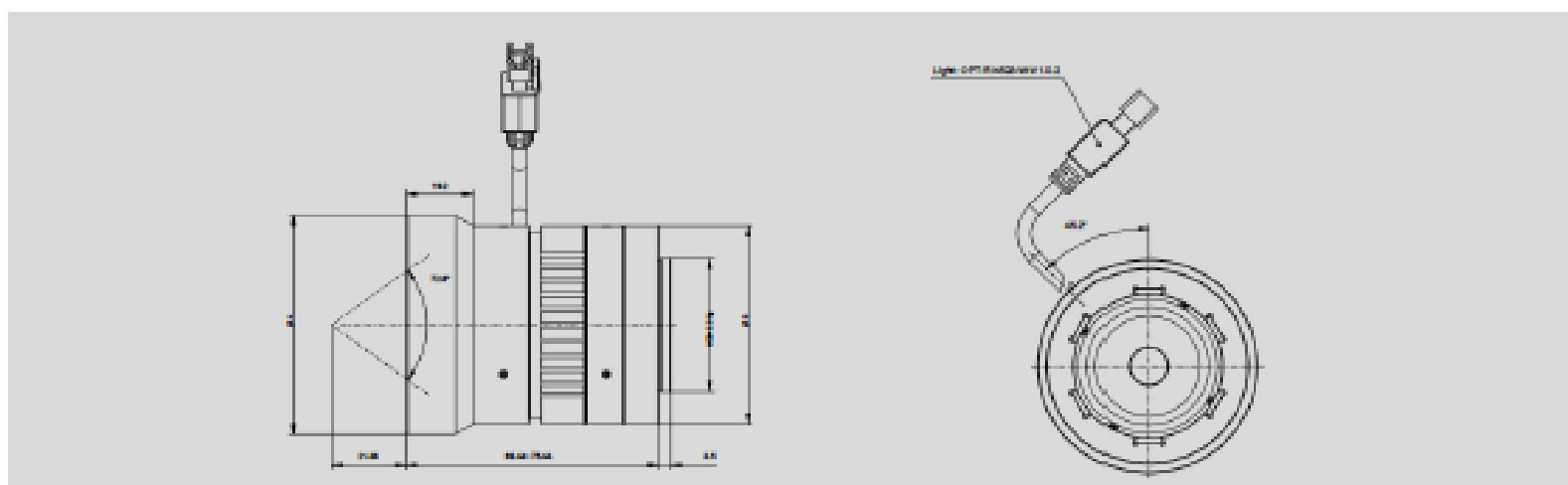


Product Parameters

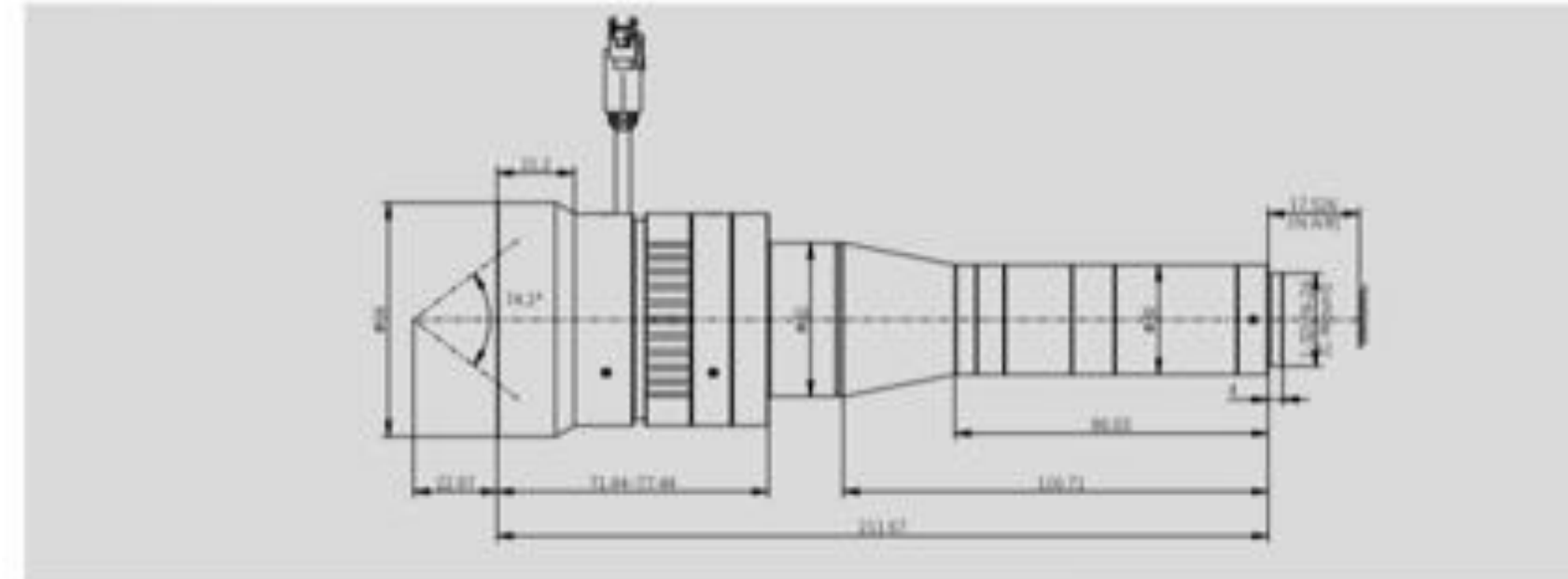
Camera compatibility (mm)	16(1.1")	Built-in light source parameters	
Image circle (mm)	Ø10.0	model	OPT-RI4523-W-V1.0.2
Working distance (mm)	20.0-26.0	color	white
aperture	F6.5-F16	Voltage	24V
Interface	C-Mount	power	2.16W
Dimensions (mm)	Ø84*(208.64~214.64)	Interface	SMR-02V-B_2PIN
Weight (kg)	0.50		

Inspecting workplace dimensions		
Maximum outer diameter (mm)	Maximum height (mm)	Illustrate
9.0	±6.5	When the outer diameter of the workplace is less than 9.0 mm, the maximum height within the field of view is 6.0 mm.

Module size

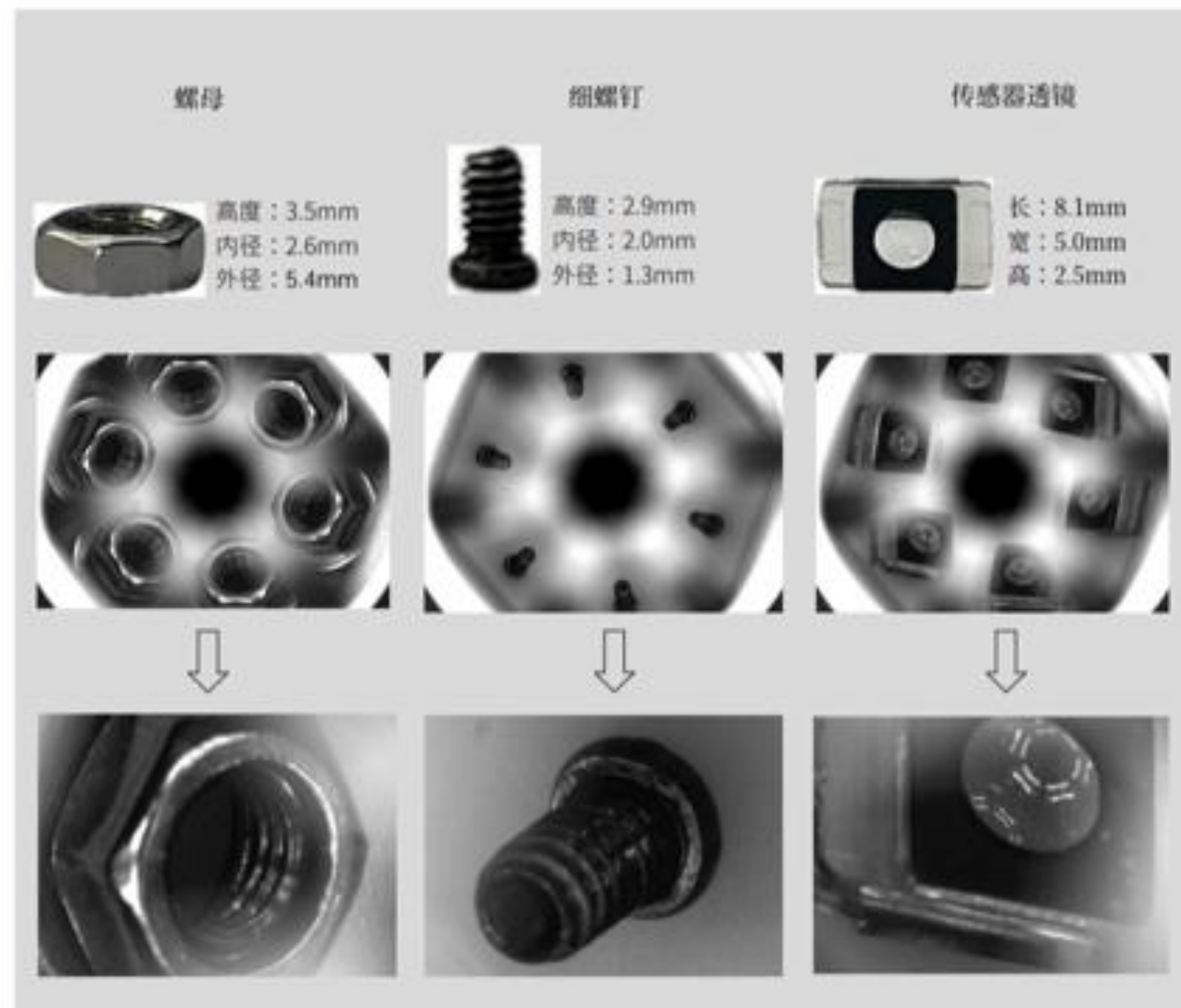


組合尺寸



应用案例

适用于检测小尺寸物体的侧面特征(腔体内表面特征),例如检测药物容器、塑料盖、螺钉、电子器件等



Function

1.Screw recheck

Description

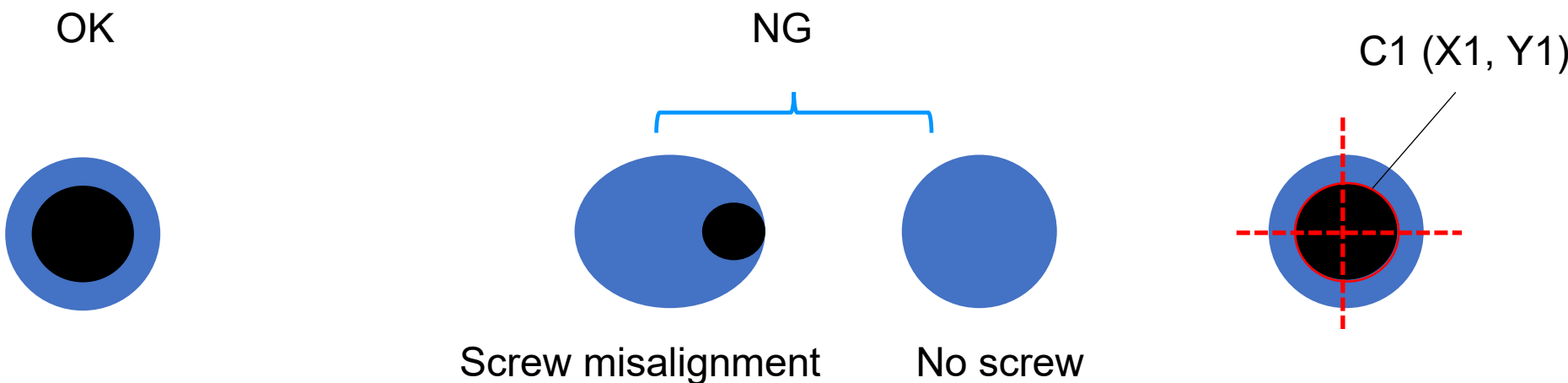
1. Screw recheck

By grasping the area of the screw, determine whether it is misalignment.
Take the area at the correct time as the reference.

2. Screw correction

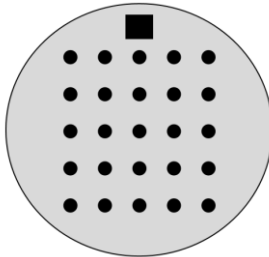
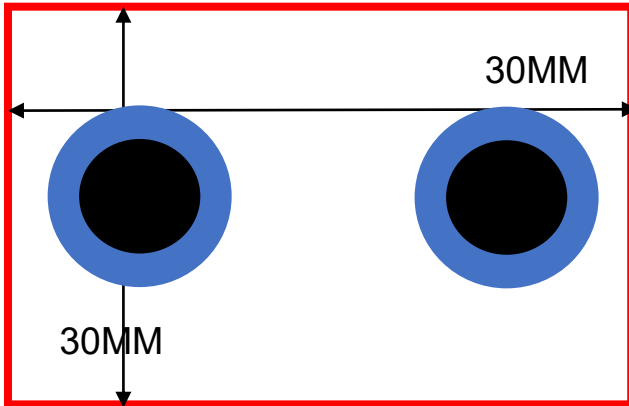
2.1 Grab the center C1 (X1, Y1) of the screw as the positioning reference.

2.2 Pixel accuracy: 0.005mm/pixel



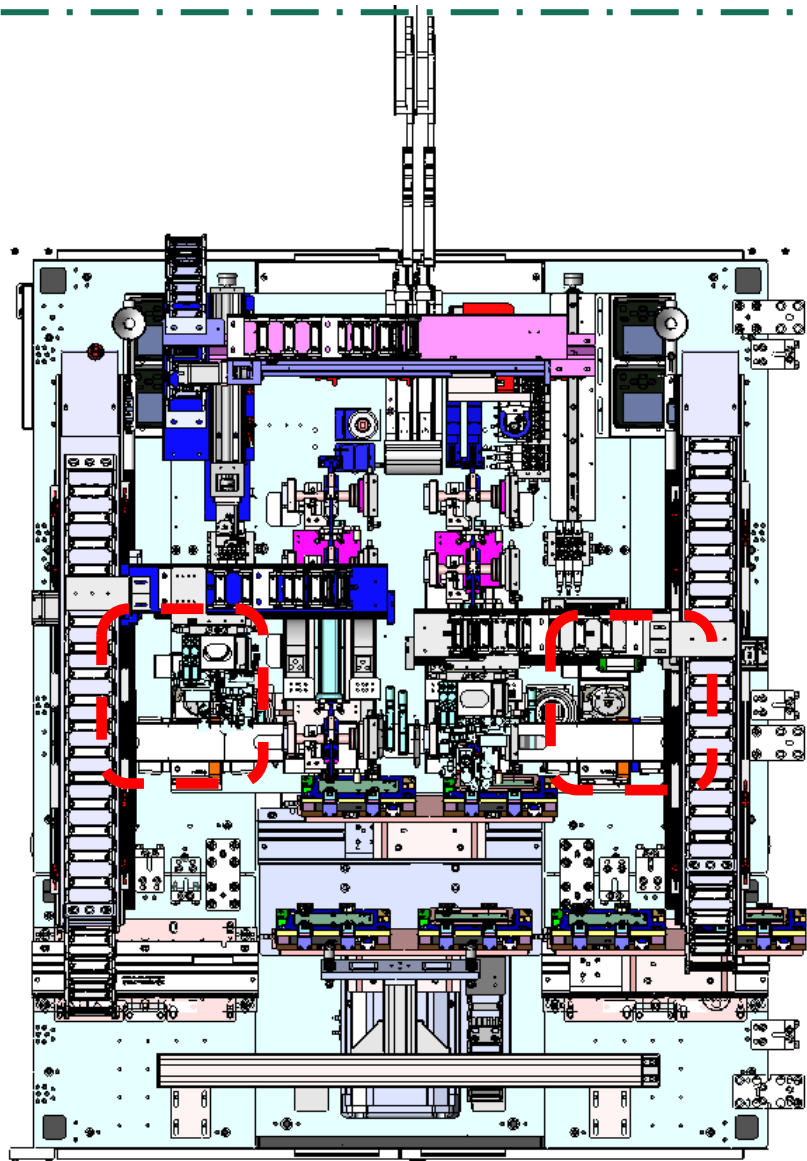
Criteria

Material in FOV

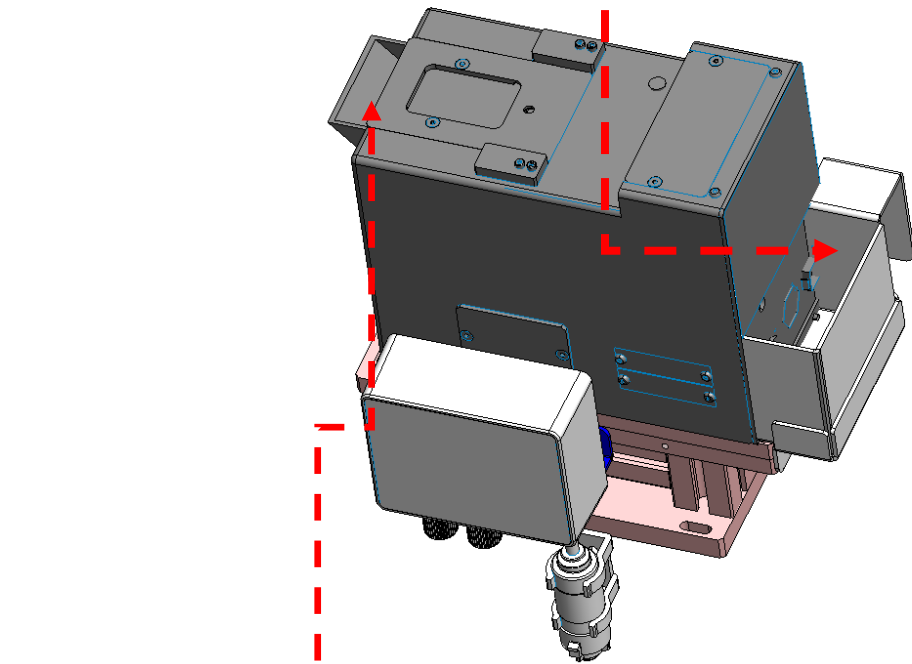


Calibration plate

Monza | Snap Assy: Module Details – Screw Feeding Module

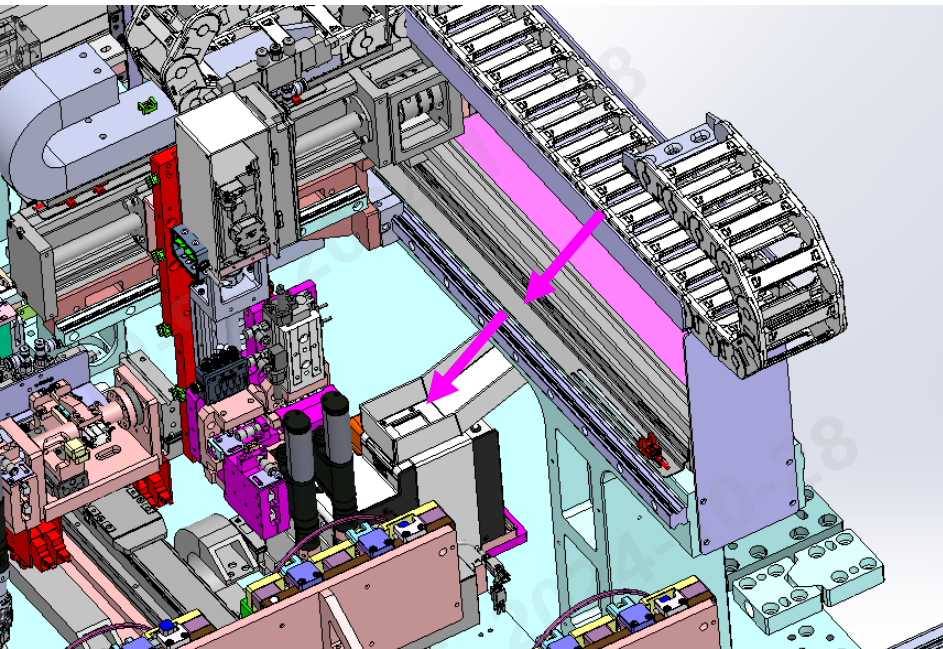
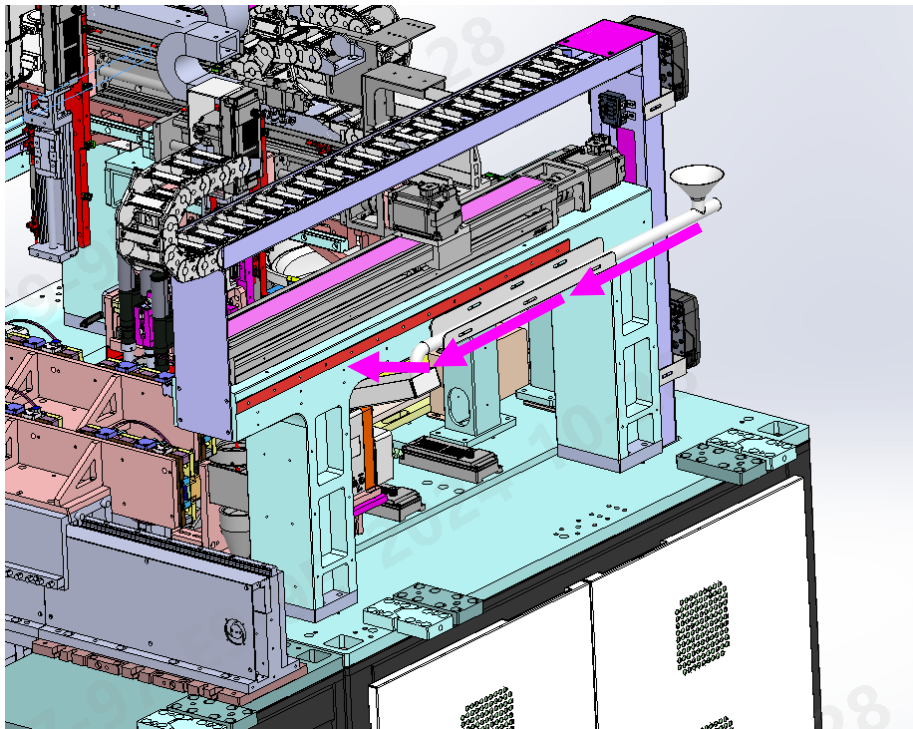


Tossing screw

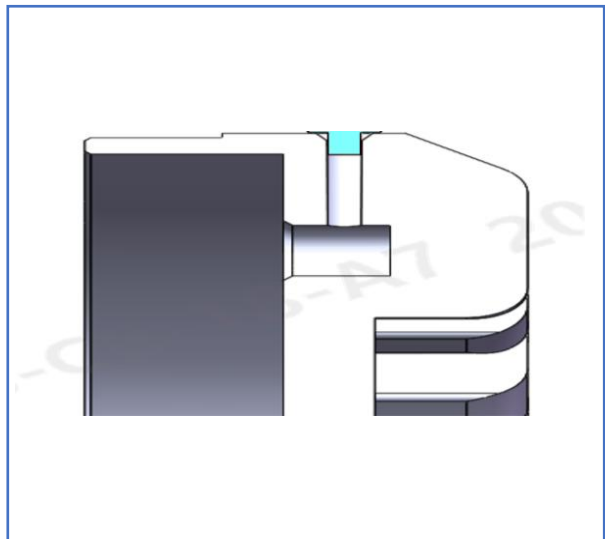


Screw feed position

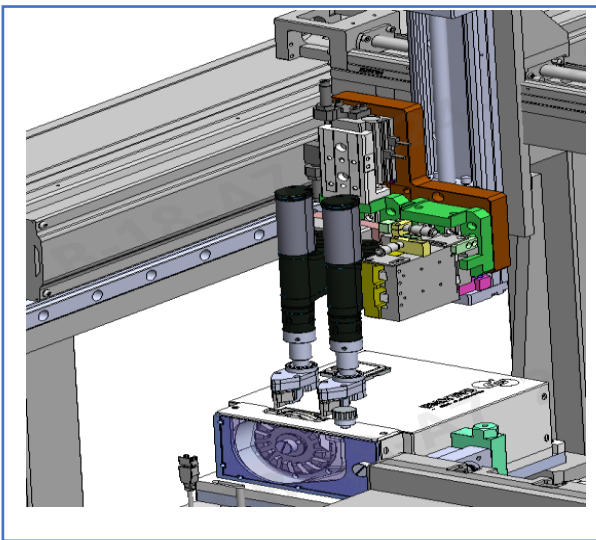
No.	Name	Model	Qty	Brand	Remark
1	Screw feeder	DEPRAG Mini Screw Feeder_189x103x156	2	DEPRAG	/



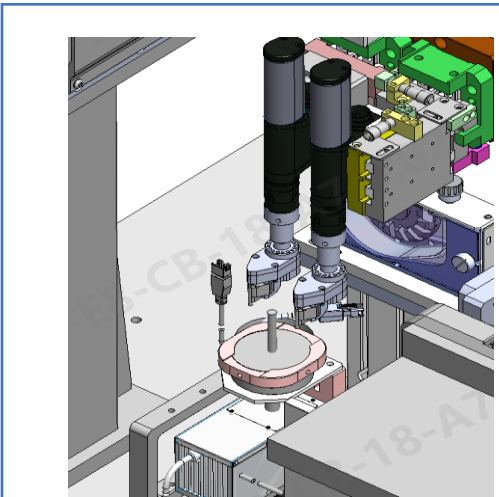
Supply screw path



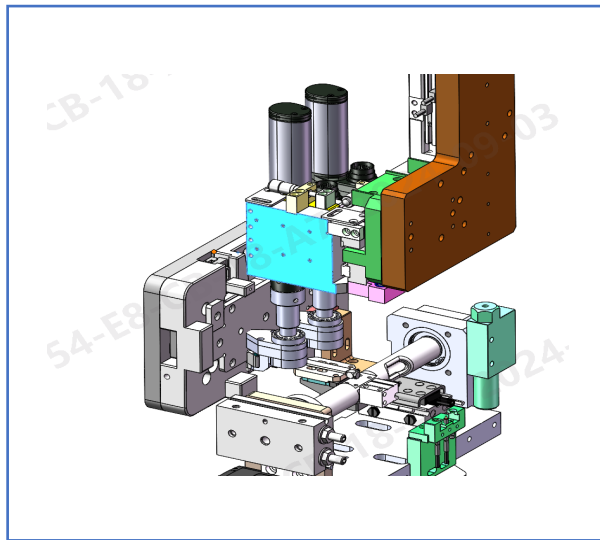
Screw feeder supply Screw



Nozzle pick up screw

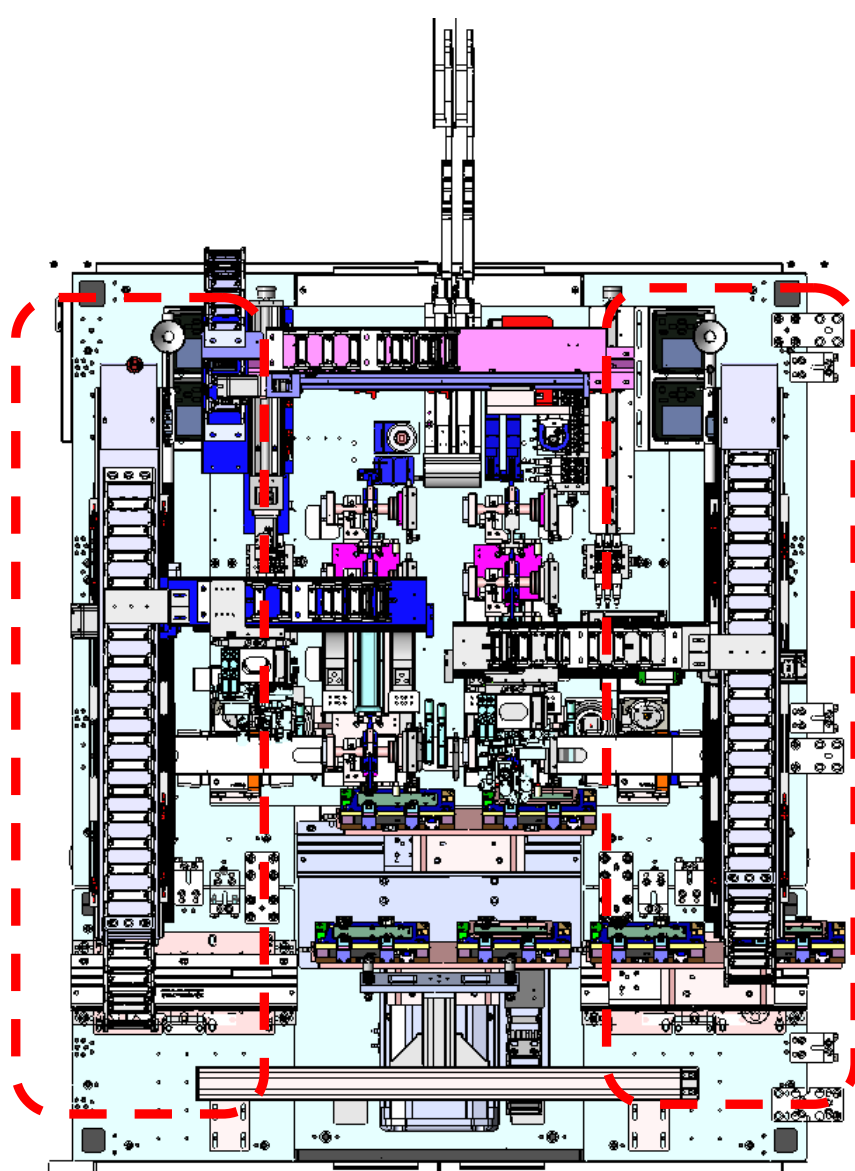


CCD Recheck screw

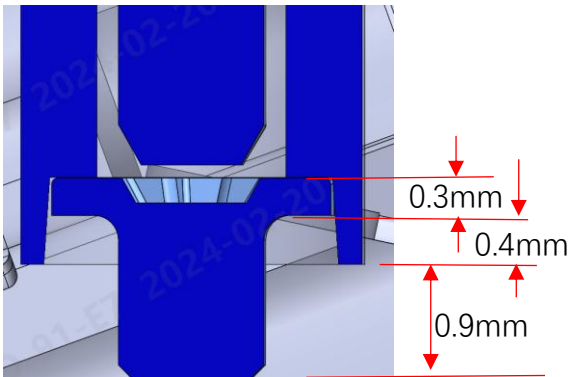
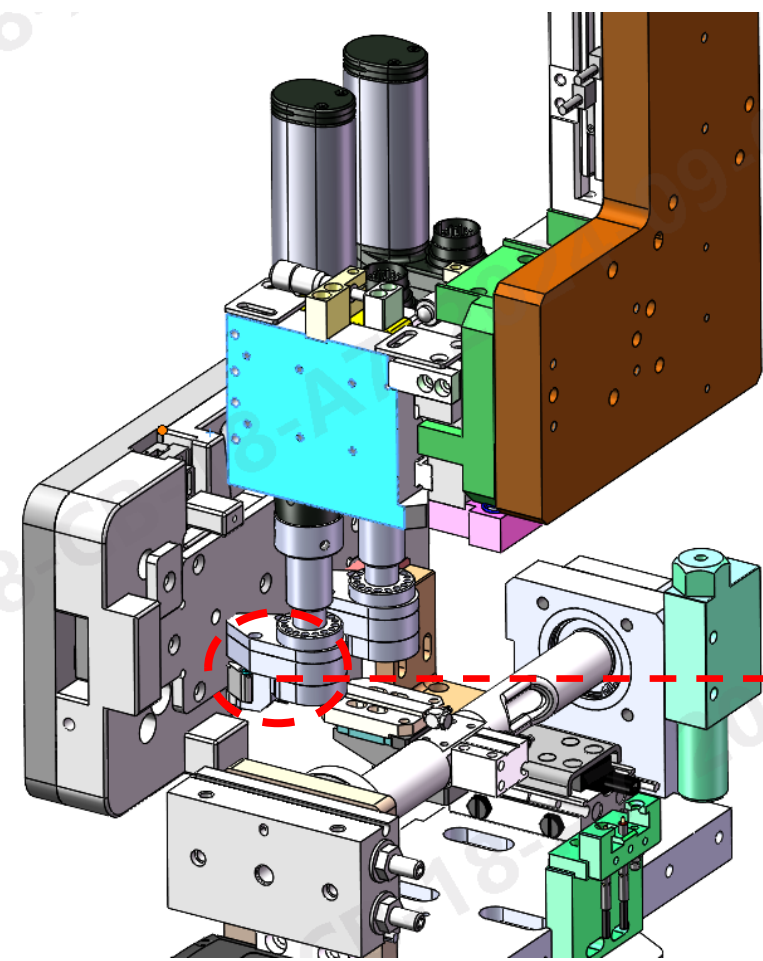
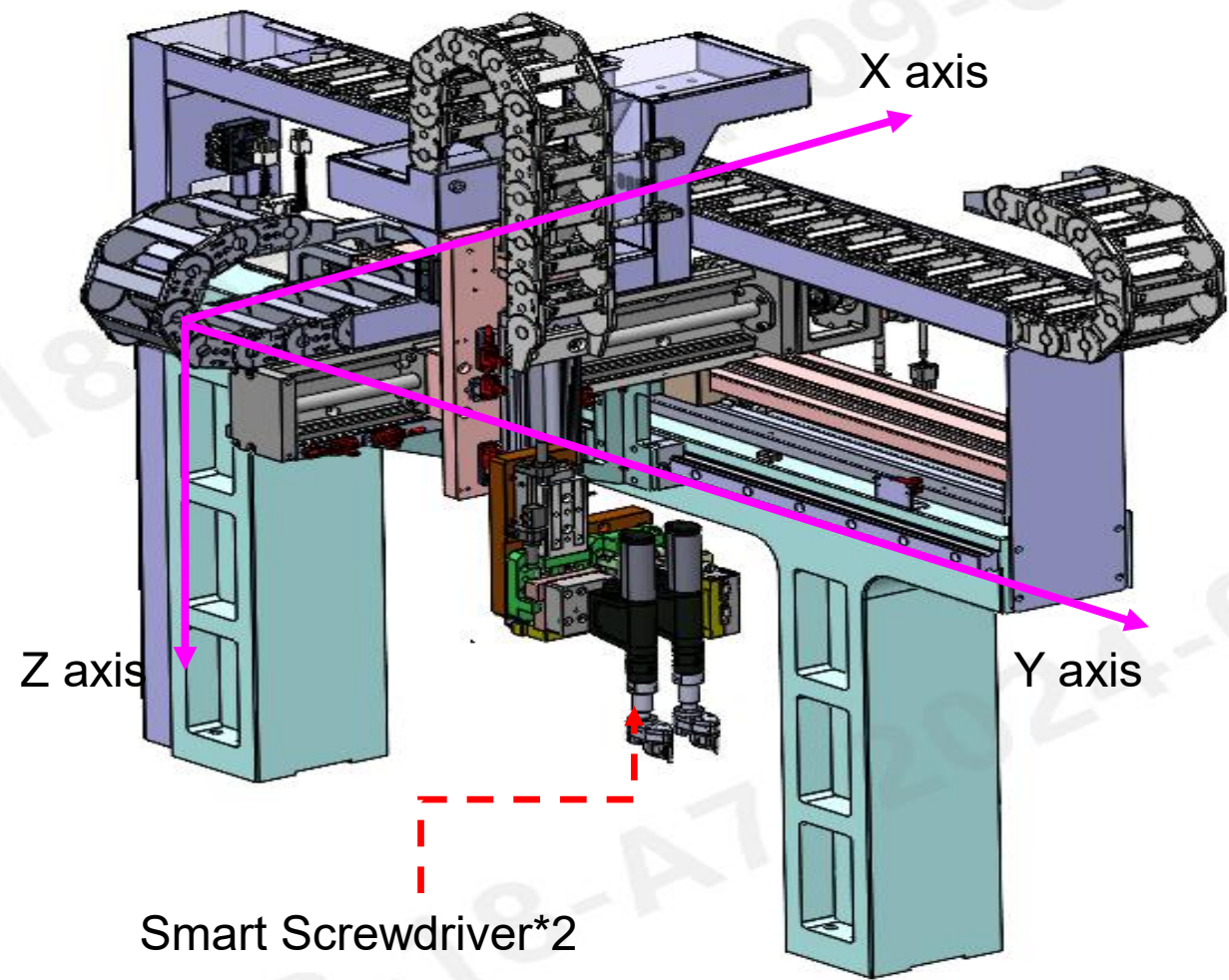


Screwing

Monza | Snap Assy: Module Details – Screwing Module



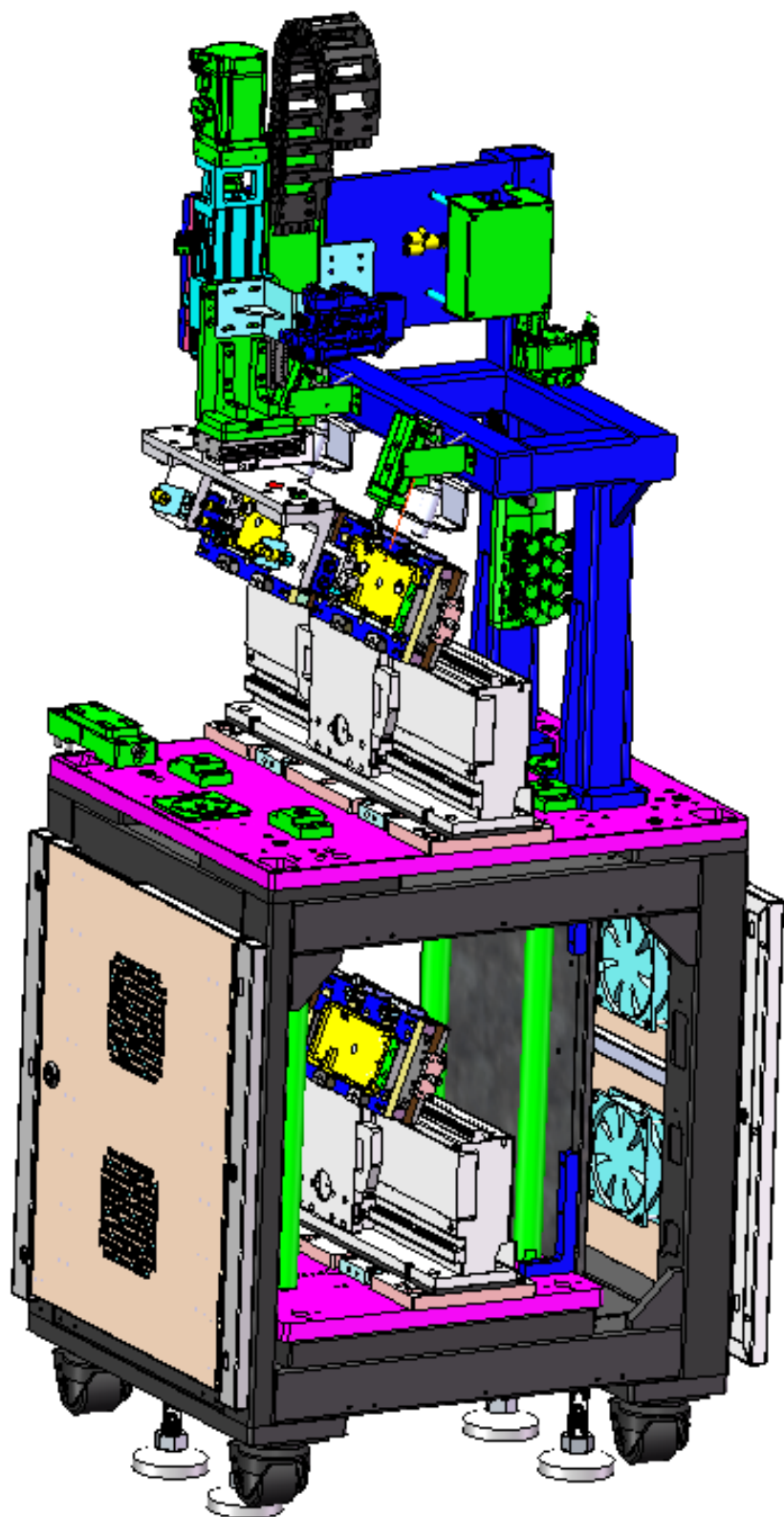
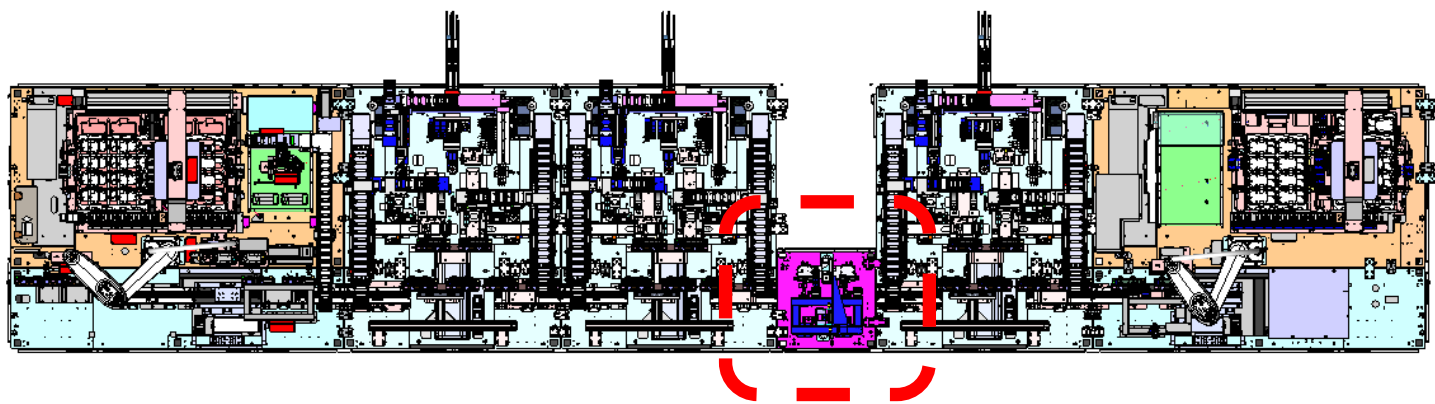
No.	Name	Model	Qty	Brand	Remark
1	X axis module	KK86D20P-740A1-F0	2	HIWIN	Stroke : 740mm Repeatability : $\pm 0.02\text{mm}$
2	Y axis module	KK86D20C-340A1-F0	2	HIWIN	Stroke : 340mm Repeatability : $\pm 0.02\text{mm}$
3	Z axis module	KK60-300A1-F0	2	HIWIN	Stroke : 300mm Repeatability : $\pm 0.02\text{mm}$
4	Smart Screwdriver	QMC-21-10-HM4	4	Atlas	/



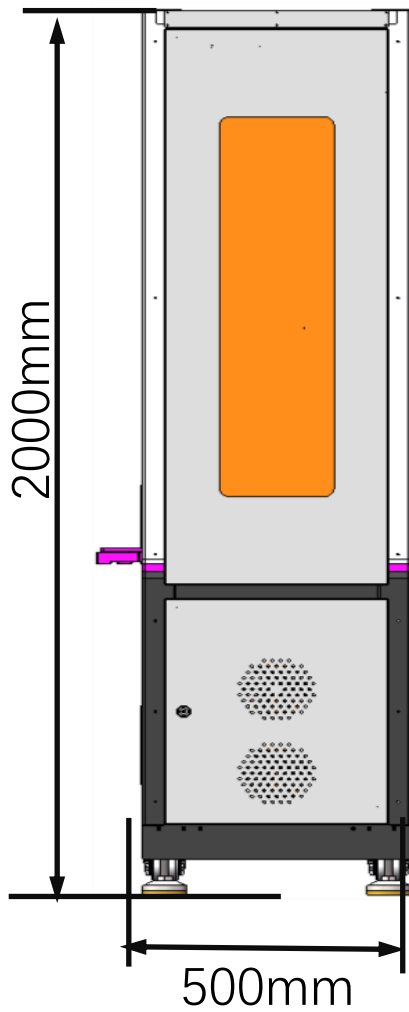
Sectional view of Nozzle & screw

Station Details- Flipping Station

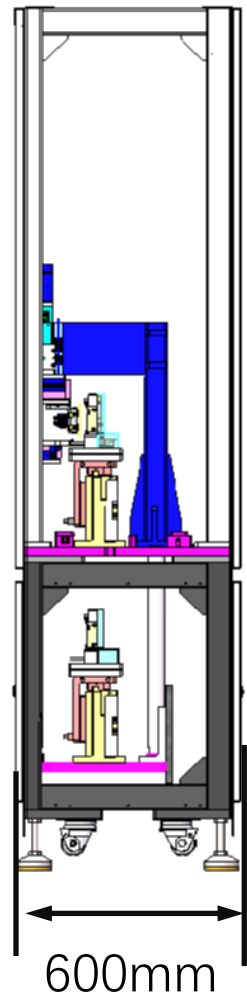
Monza | Snap Assy : Machine Overview –Layout



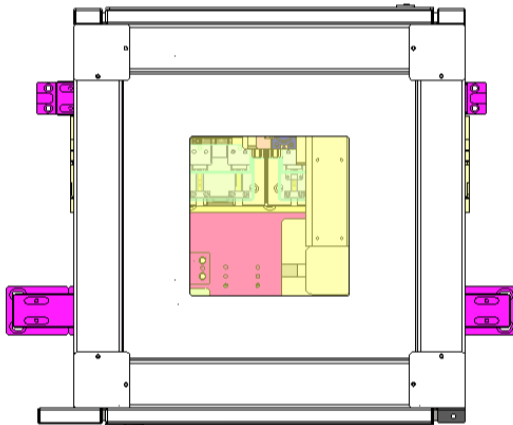
Front View



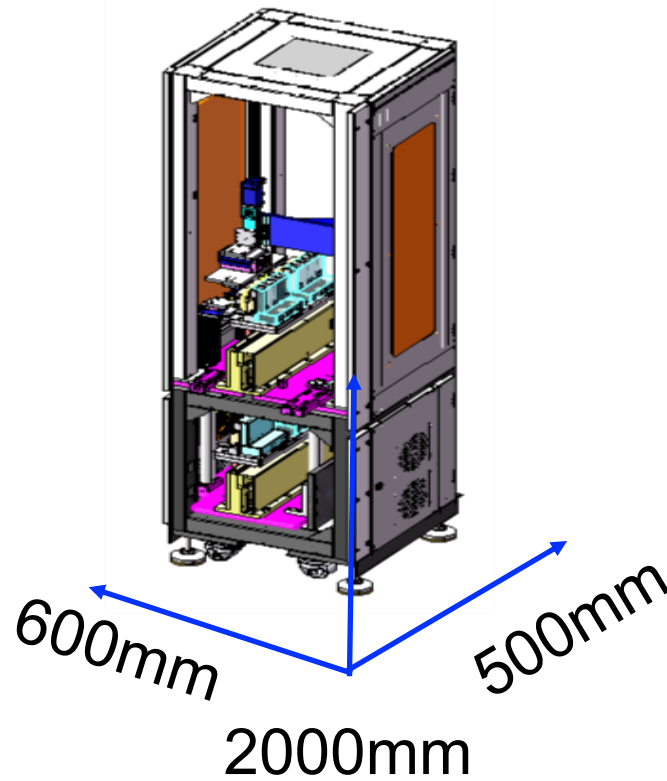
Side View



Top View

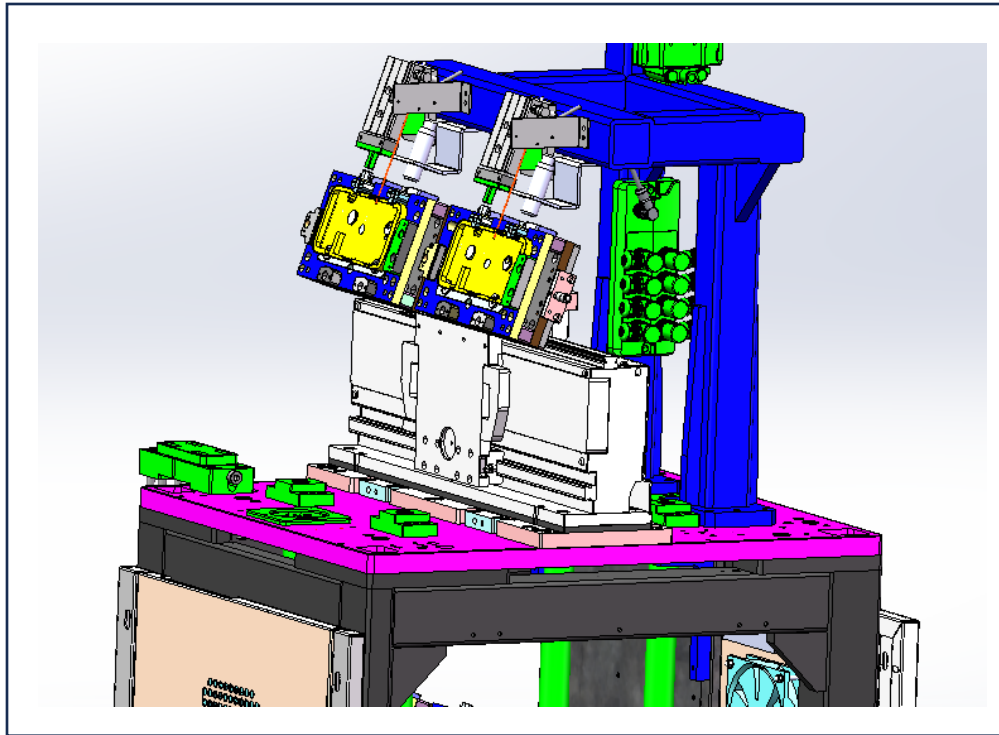


Axis View

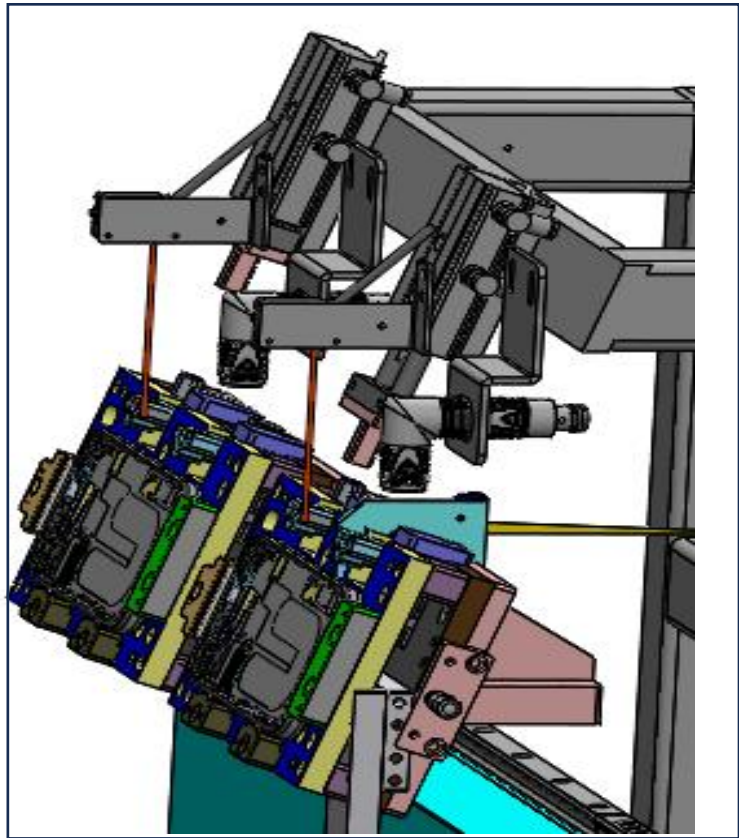


- 1. Power supply: 220V
- 2. Work rate: 2KW
- 3. Air source: 0.5~0.7Mpa
- 4. Personal ratio: 0
- 5. Estimated weight: 500kg

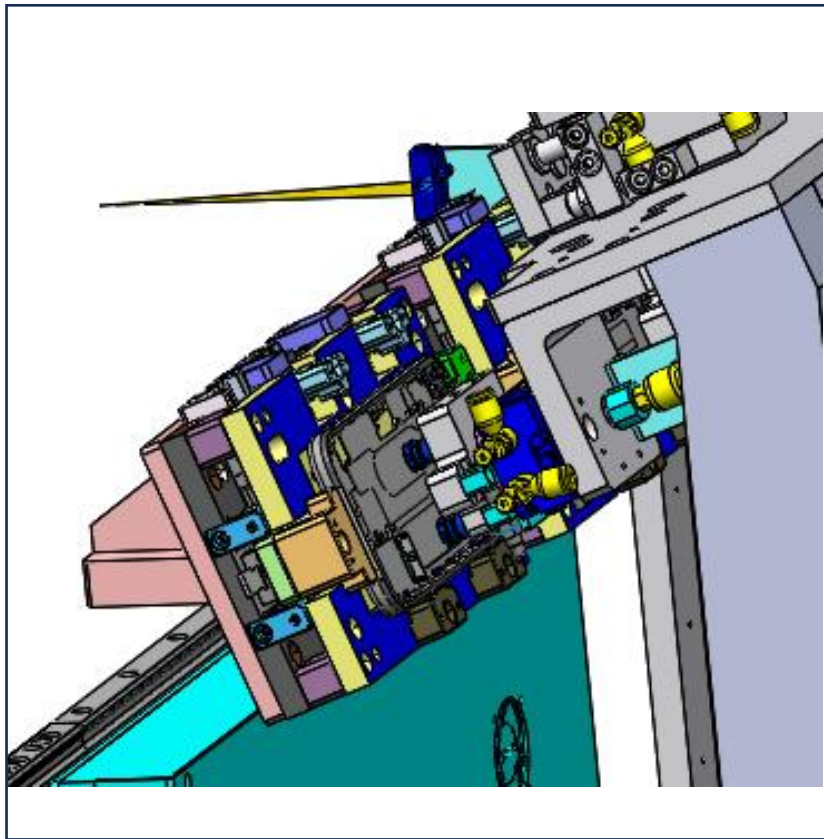
Monza | Snap Assy: Module Process Flow



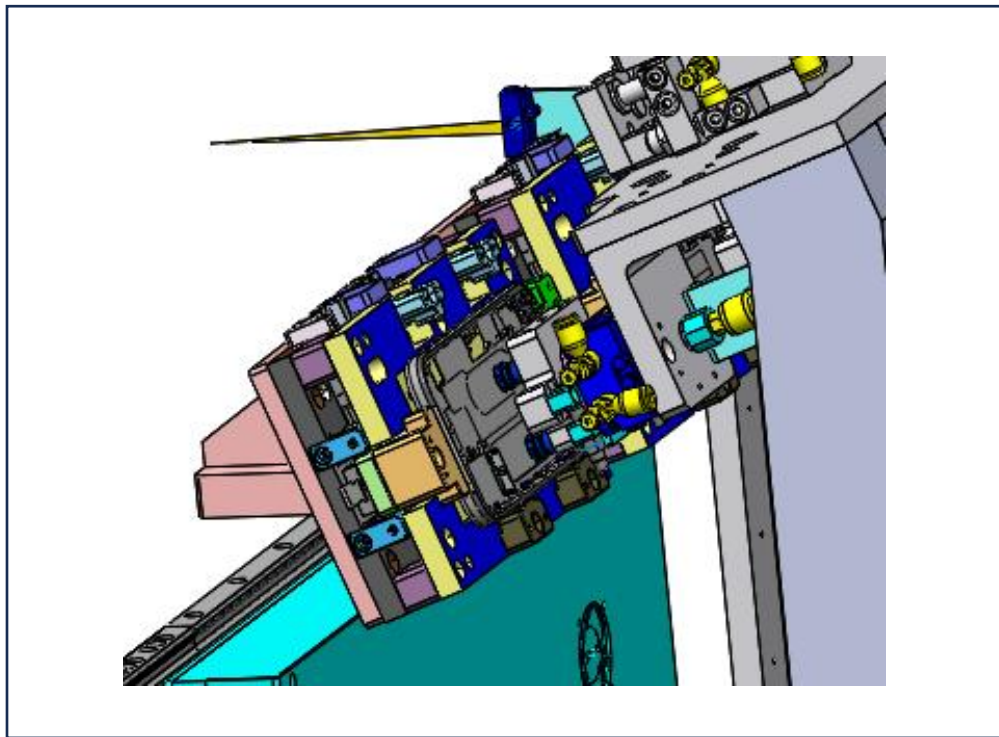
1. Fixture flow to flipping position



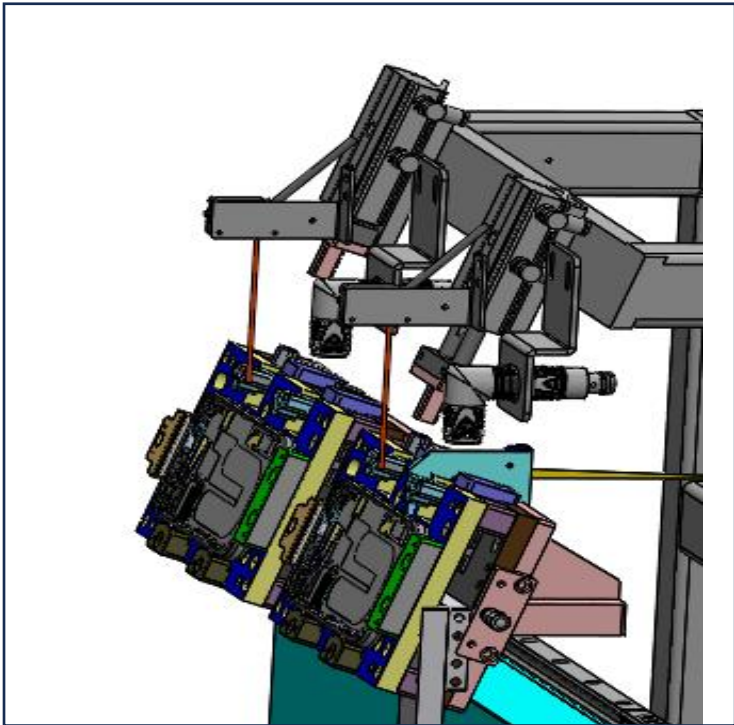
2. Unlocking cylinder extends to unlock fixture



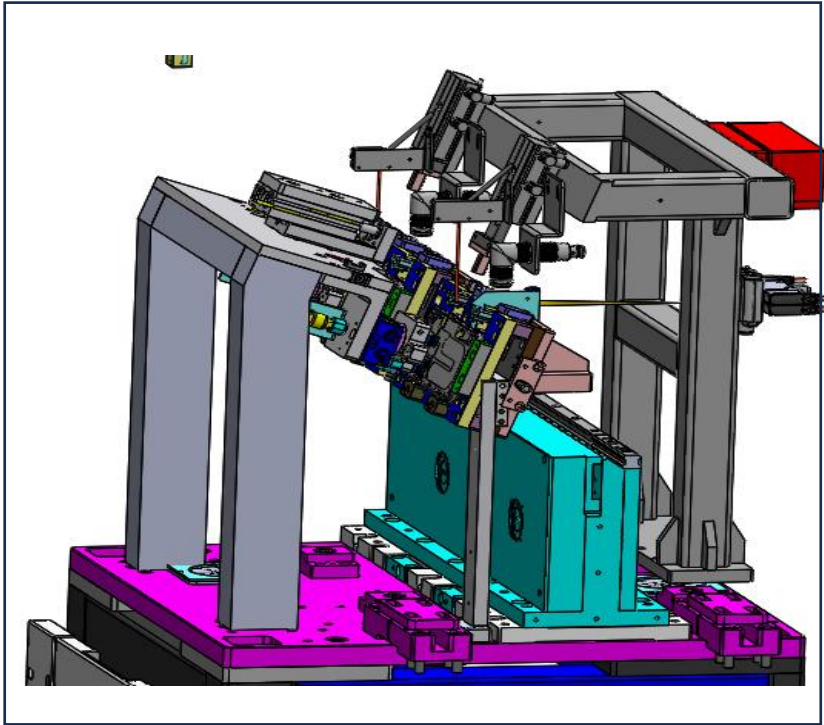
3. Flip module suction MCH



4. Flipping fixture 180°



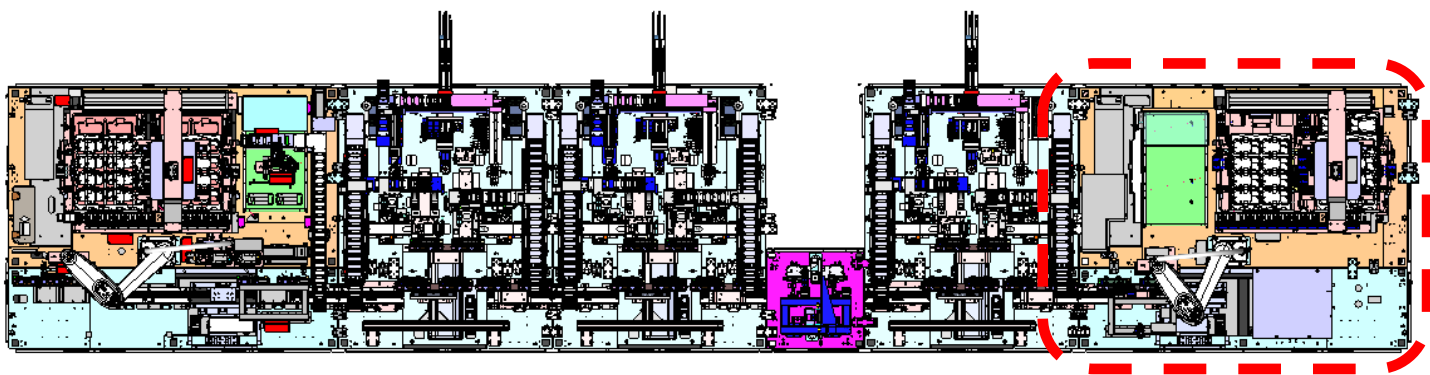
5. Place the flipped MCH into the fixture



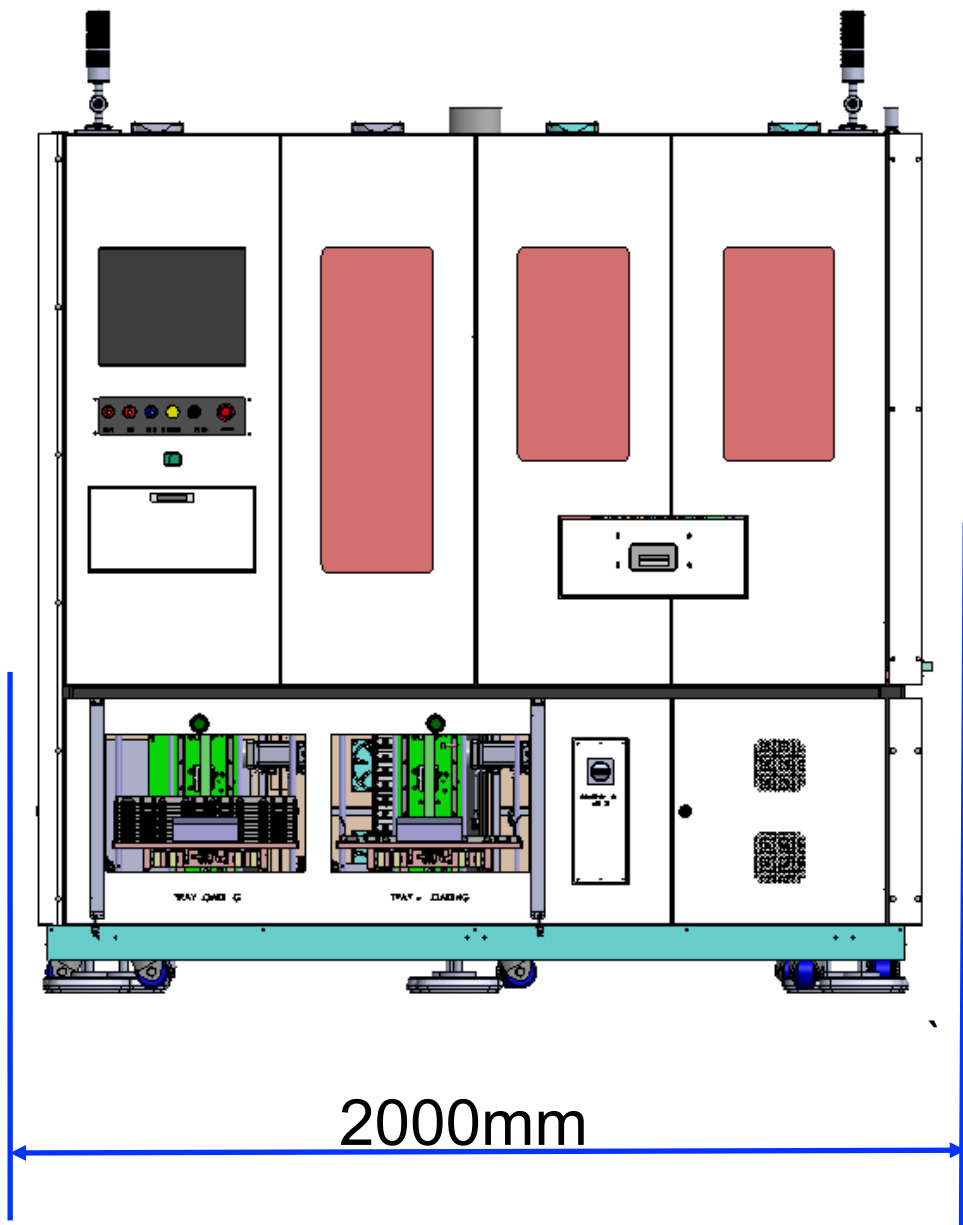
6. Flip module and unlock module back to initial position

Station Details- Unloading Machine

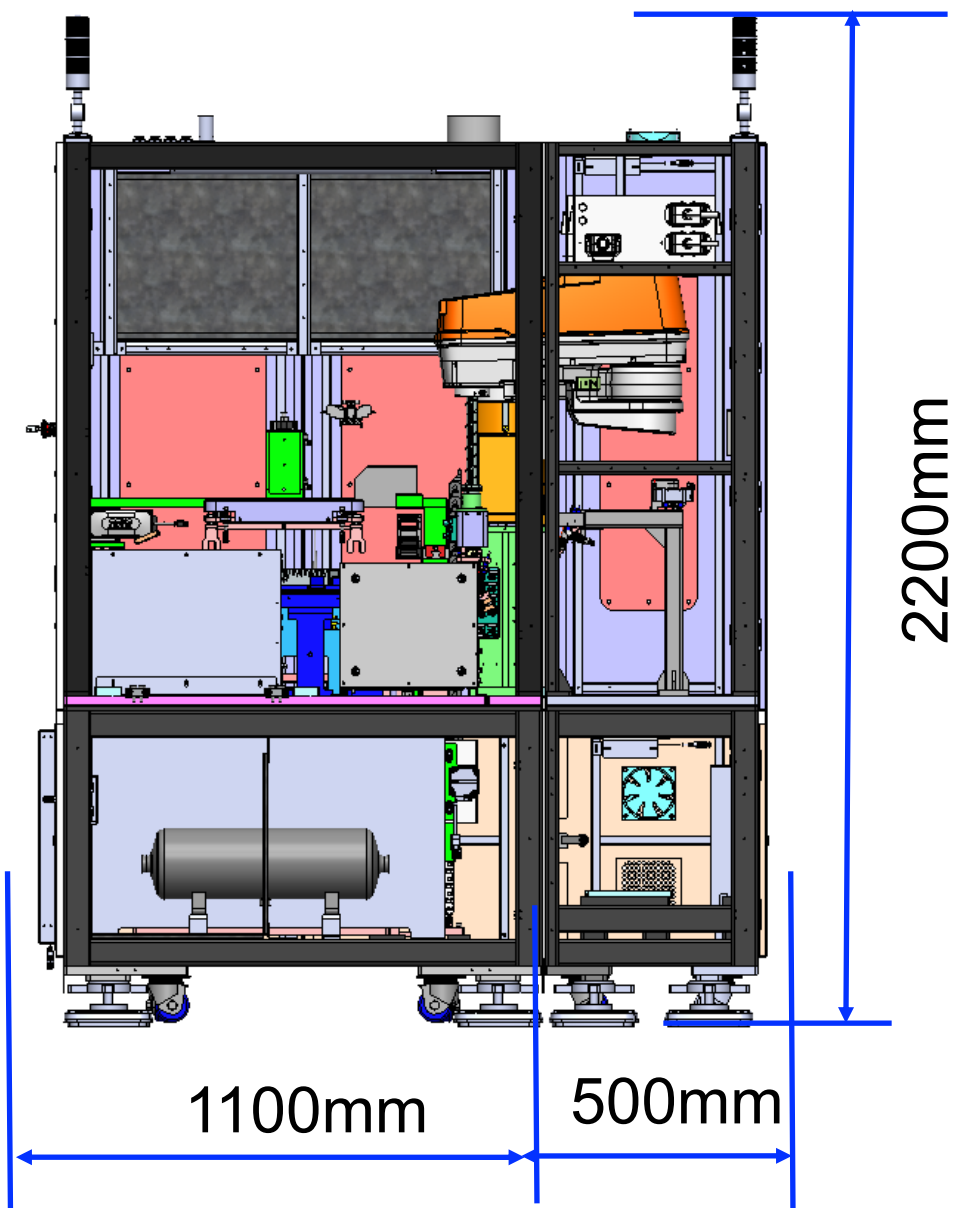
Monza | Snap Assy : Machine Overview –Layout



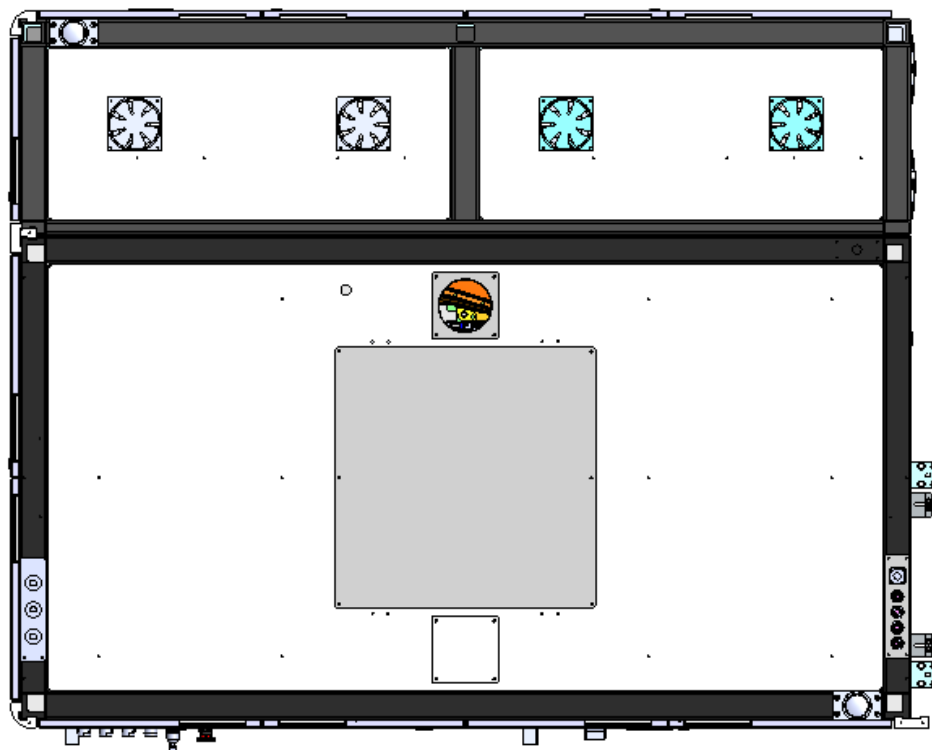
Front View



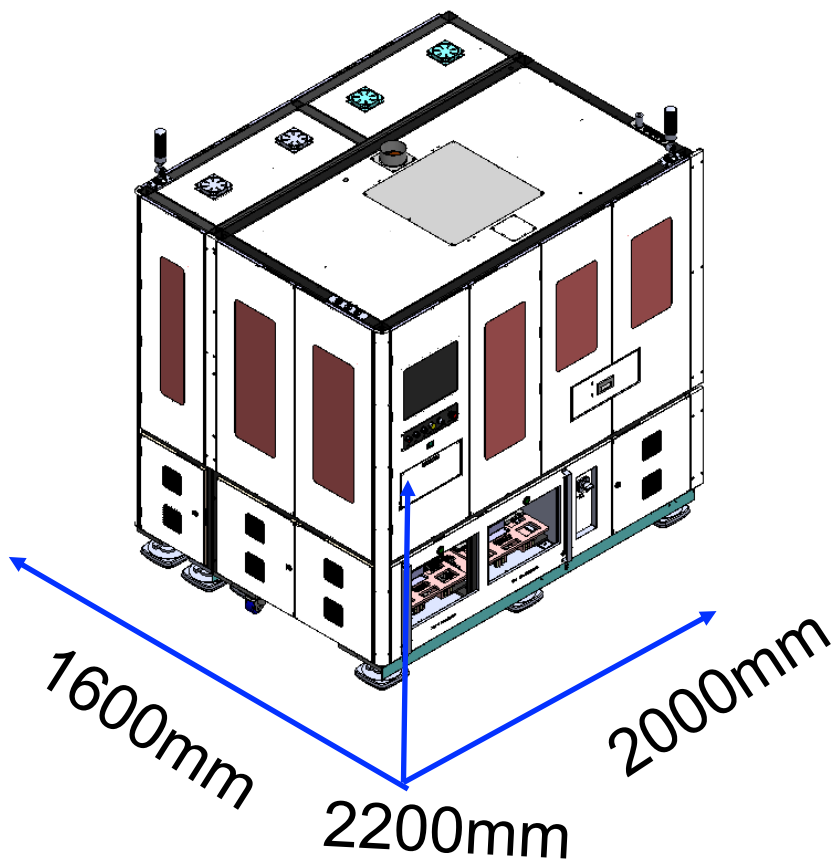
Side View



Top View

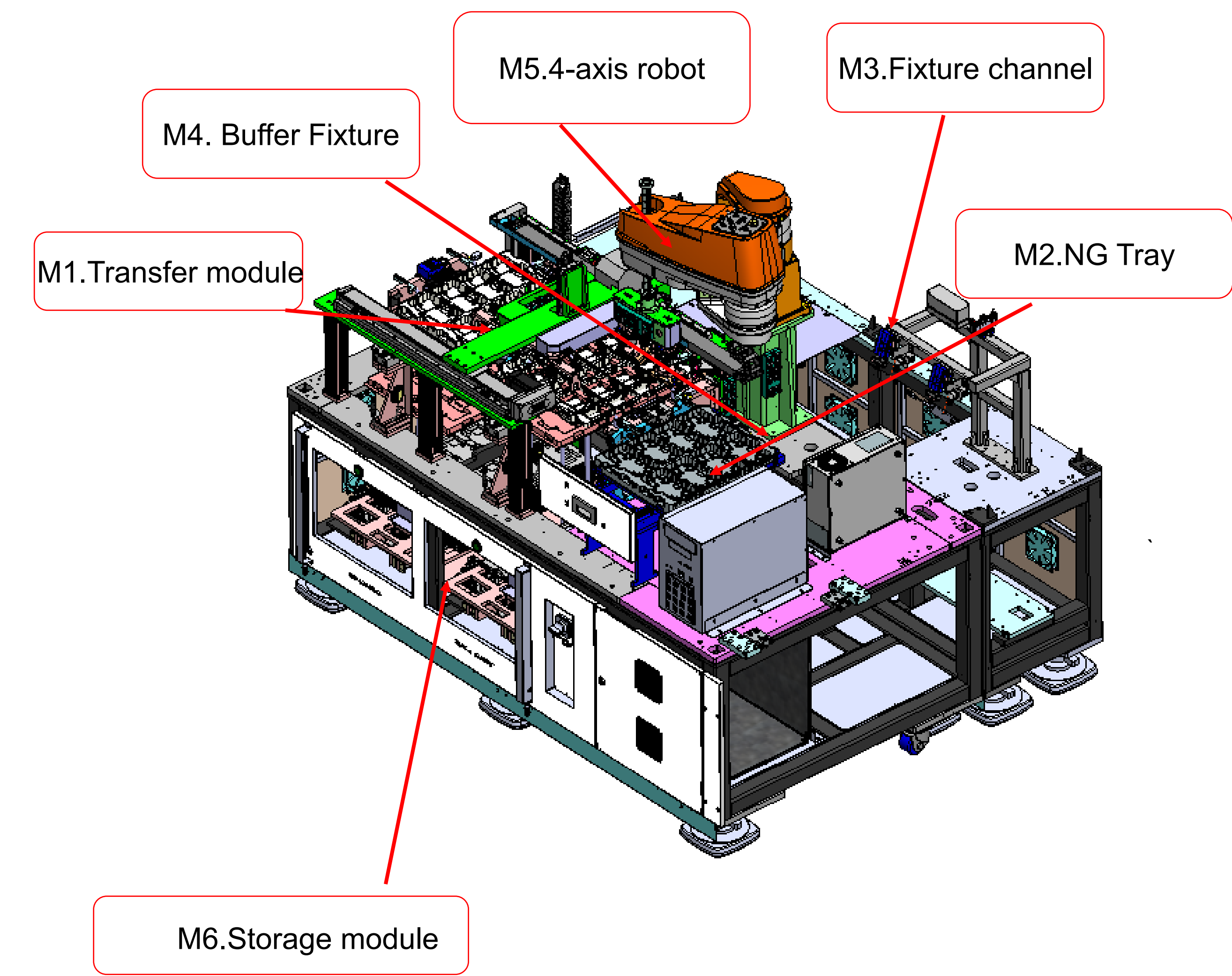


Axis View



- 1.Voltage:220V
- 2.Power: 10Kw
- 3.Air: 0.5~0.7Mpa
- 4.Labor:/
- 5. Machine Weight: 2000kg

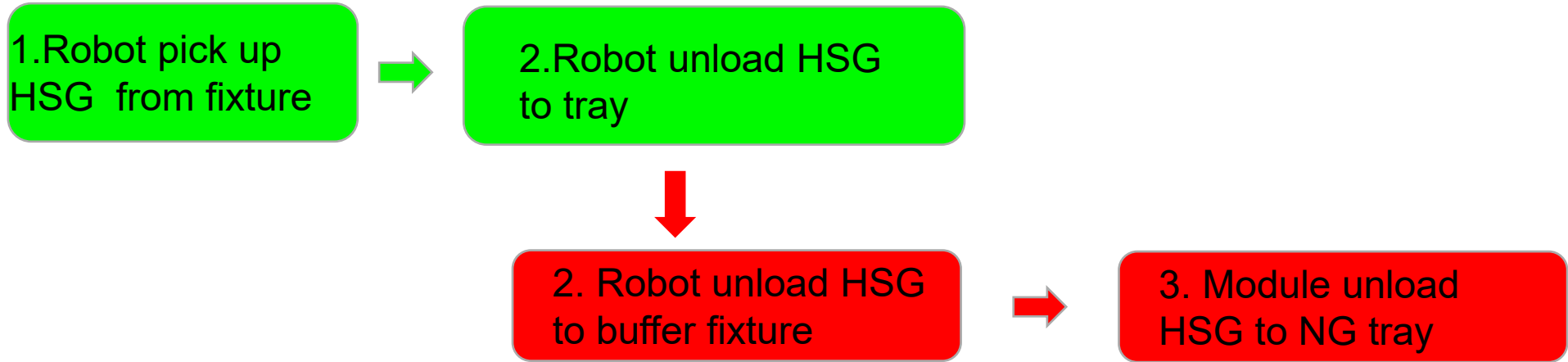
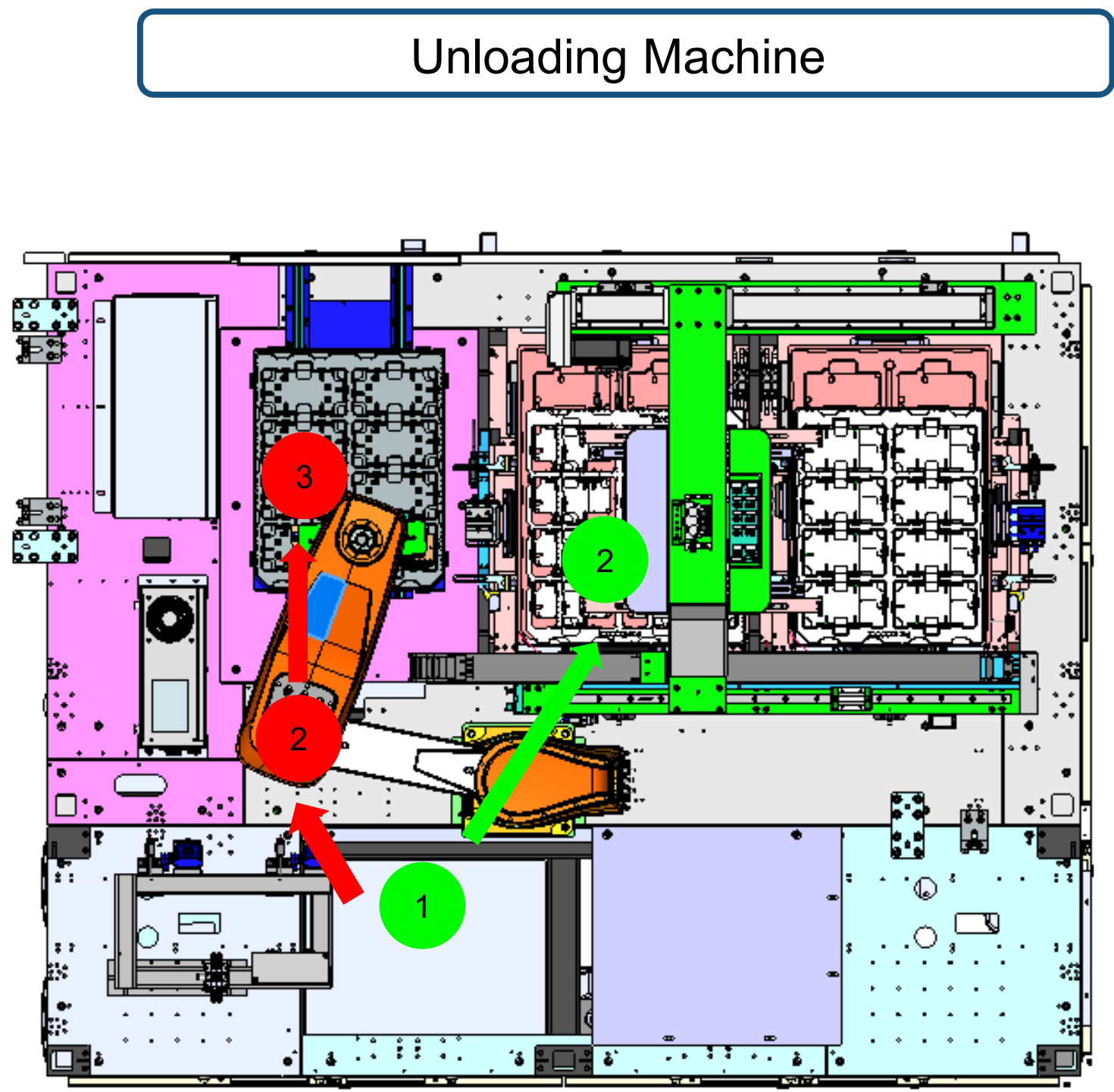
Monza | Snap Assy: Machine Overview– Station BOM



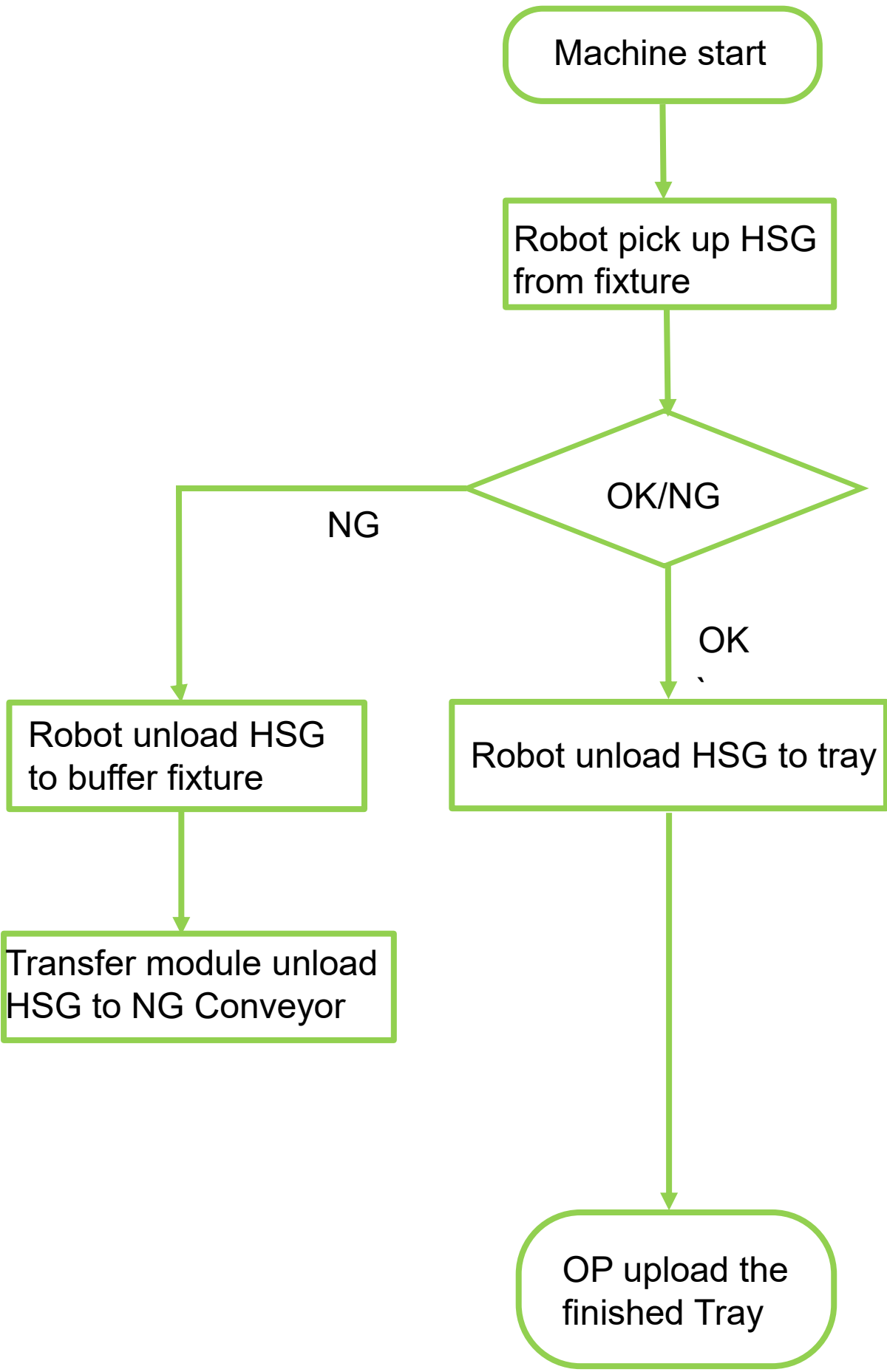
#	Module	Module Qty	Key parts	Key part QTY (TTL)
M1	Transfer module	1	X axis module	1
			Y axis module	1
			Z axis module	1
			Rotating cylinder	2
M2	NG tray	1	/	/
M3	Fixture channel	1	channel	1
		1	RFID	1
M4	Buffer fixture	1	/	/
M5	4-axis robot	1	Robot	2
M6	Storage module	3	Storage	1

HSG Flow 

NG Flow 

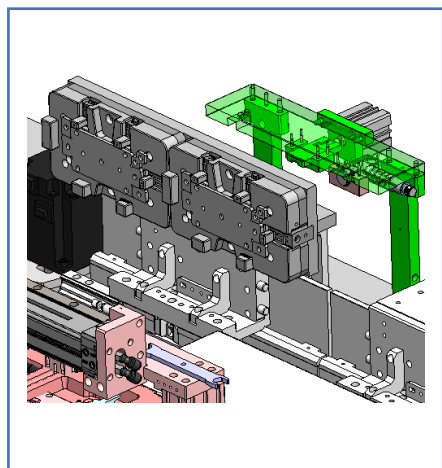


Monza | Snap Assy: Flow Chart

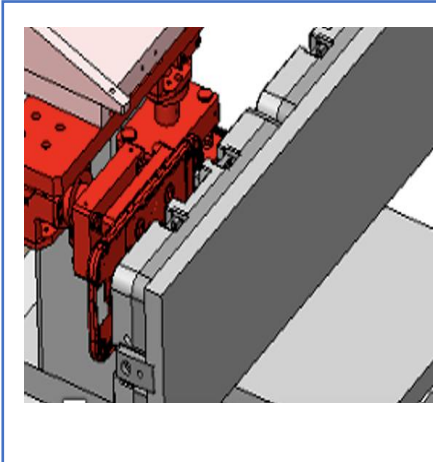


Monza | Snap Assy: Module Process Flow

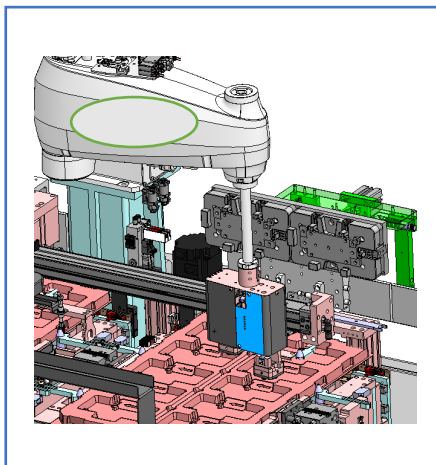
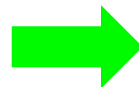
➡ HSG flow
➡ NG flow



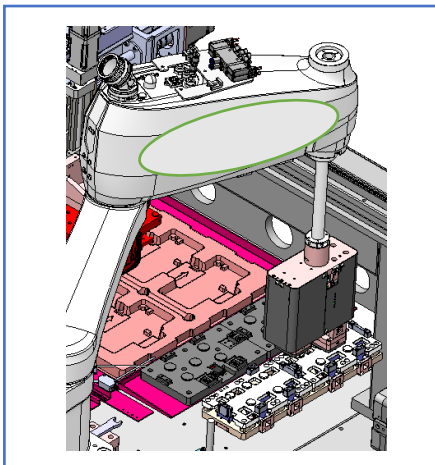
Fixture move to unloading position



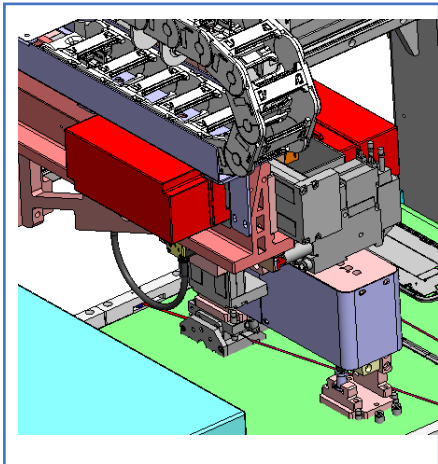
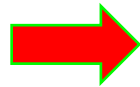
Robot pick up HSG from fixture



Robot unload HSG to tray

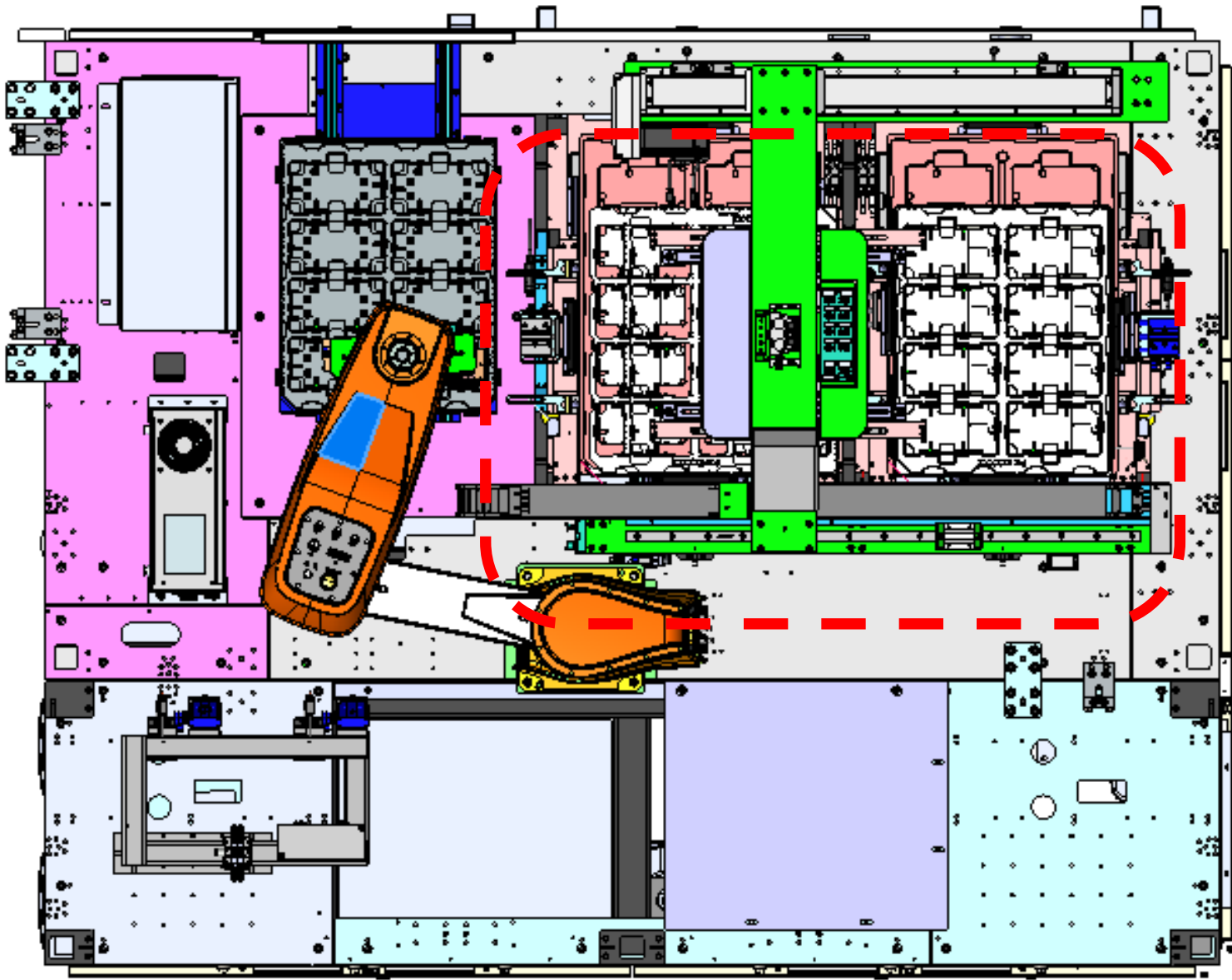


Robot unload HSG to buffer fixture

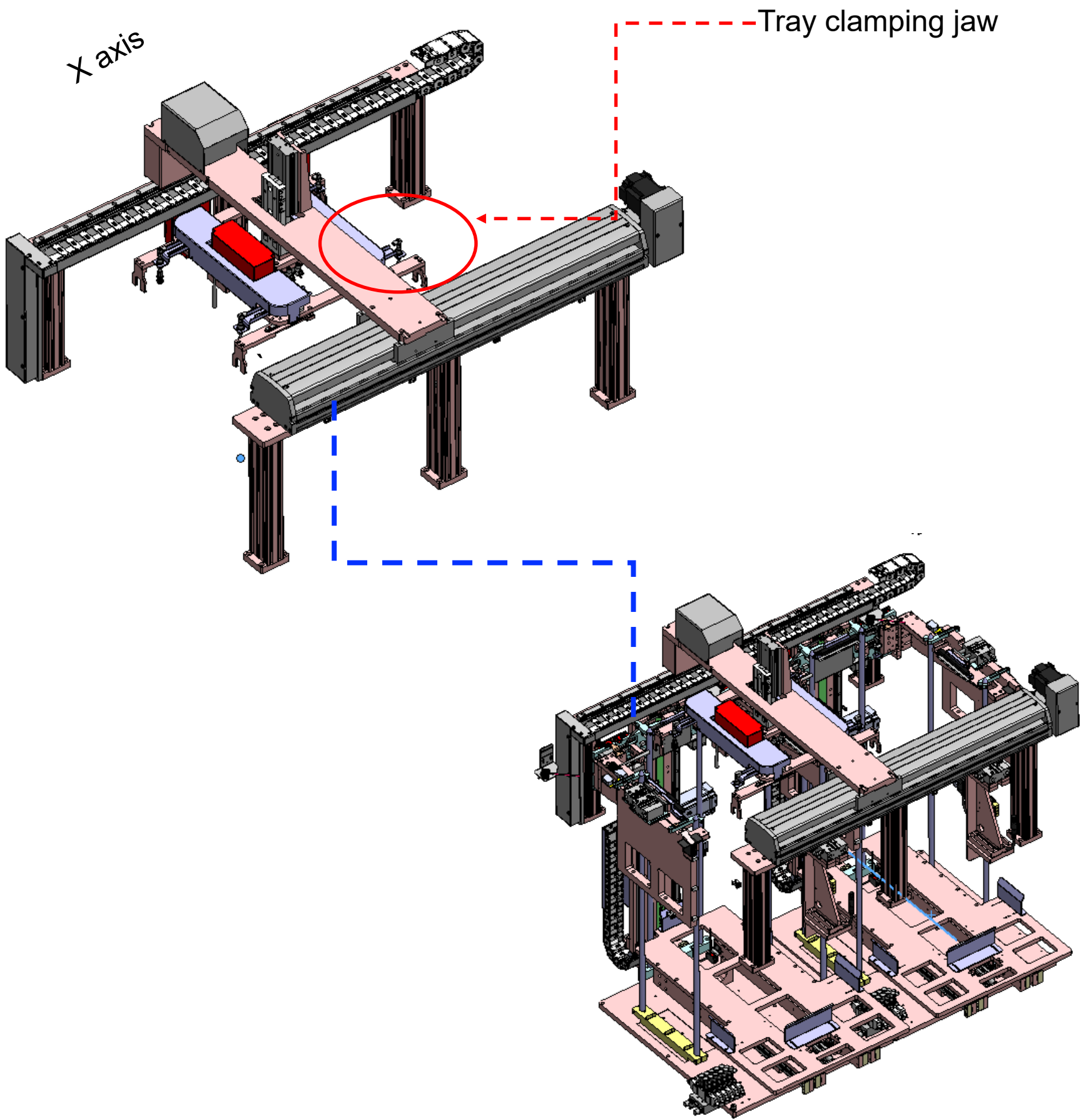


Module unload HSG to NG tray

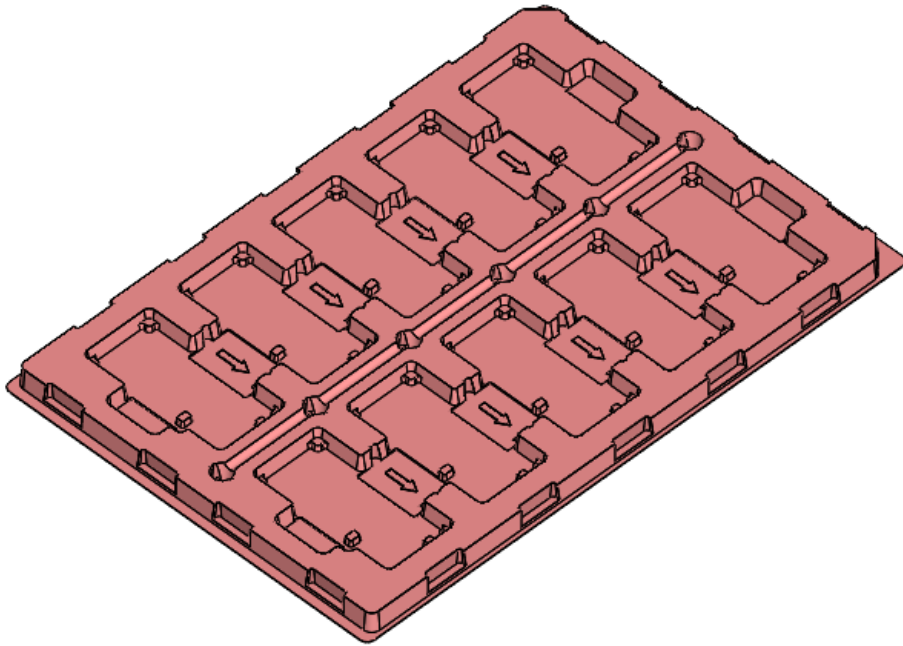
Monza | Snap Assy: Module Details – Storage Module

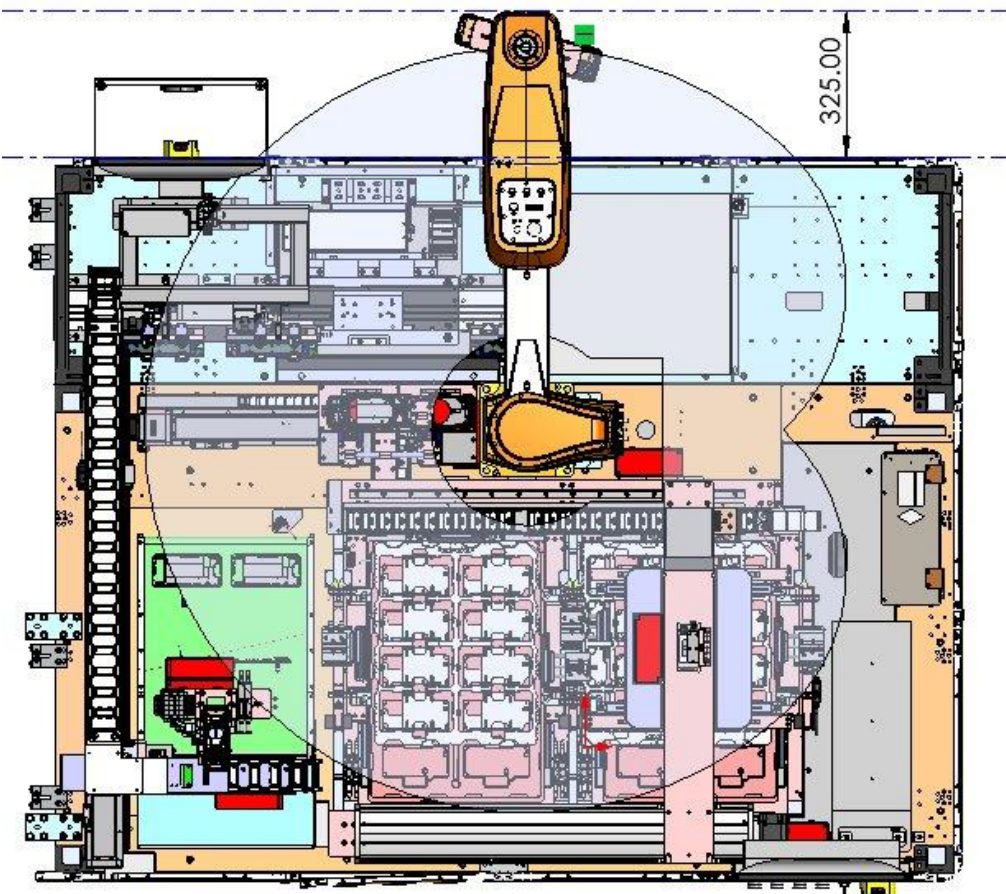
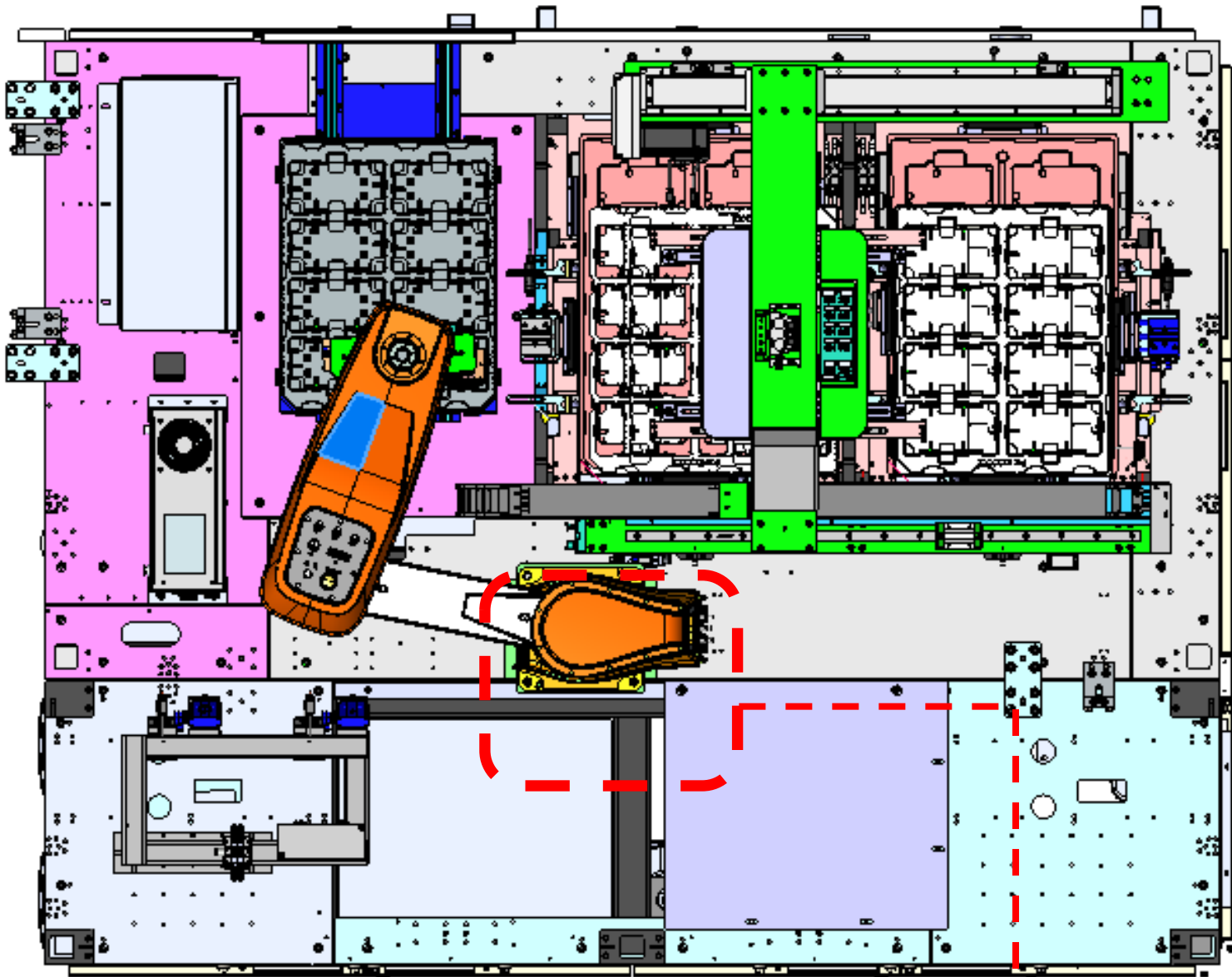


No.	Name	Model	Qty	Brand	Remark
1	X axis module	GTH8-L20-600-BC-P40-C4P5	1	Toyo	Stroke : 600mm Repeatability : $\pm 0.02\text{mm}$



Maximum tray size: 580*390
Minimum tray size: 420*350





No.	Name	Model	Qty	Brand	Remark
1	4-axis robot	KR 13 R850 Z200-2 CS	1	KUKA	

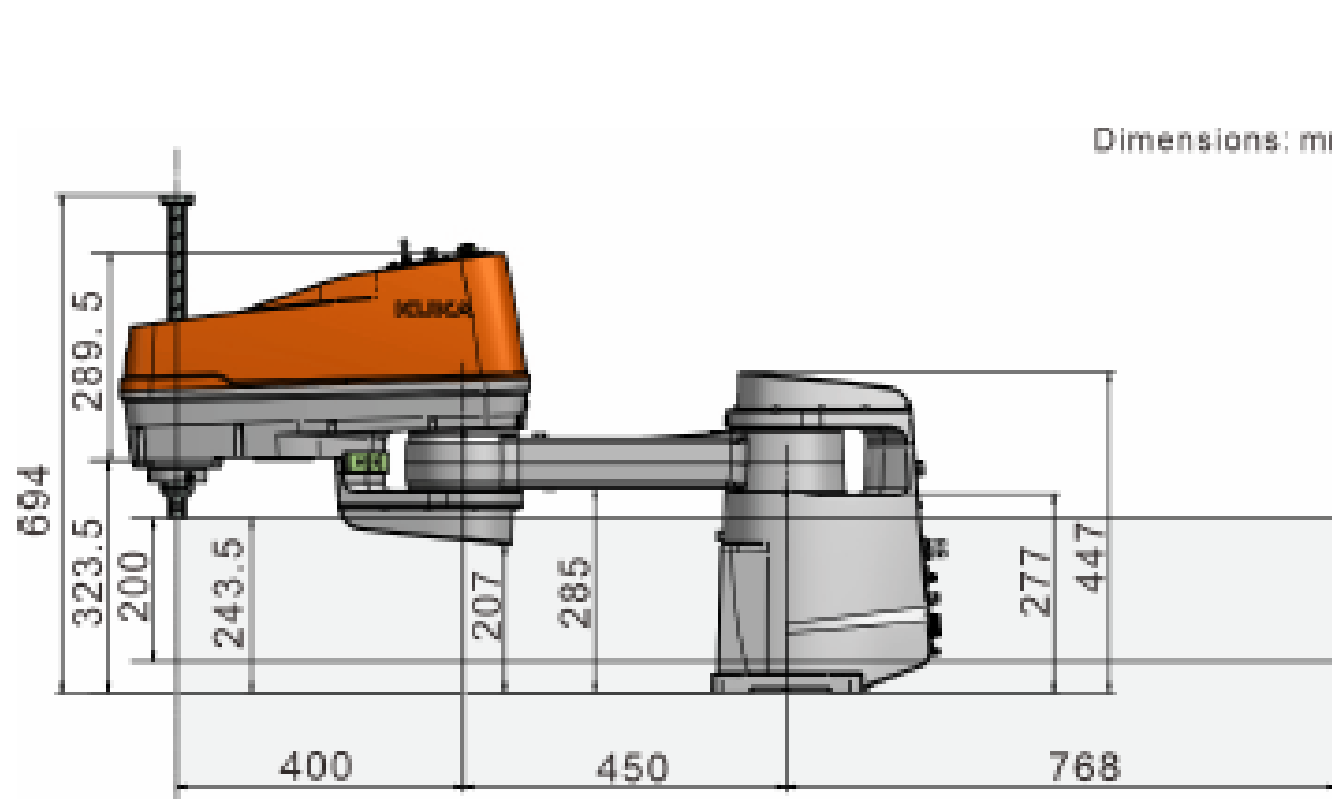


Fig. 4-65: KR 13 R850 Z200-2 CS, working envelope, side view

Designation	KR 13 R850 Z200-2 CS
Number of axes	4
Number of controlled axes	4
Volume of working envelope	0.397 m³
Pose repeatability XY	± 0.025 mm
Pose repeatability Z	± 0.01 mm
Pose repeatability R	± 0.01 °
Weight	approx. 46.5 kg
Rated payload	6 kg
Maximum payload	13 kg
Maximum reach	850 mm
Protection rating (IEC 60529)	IP20
Sound level	< 80 dB (A)
Mounting position	Floor
Footprint	210 mm x 210 mm

MA KR 13 SCARA-2 CS V1 | Issued: 21.02.2025 www.kuka.com | 95/233

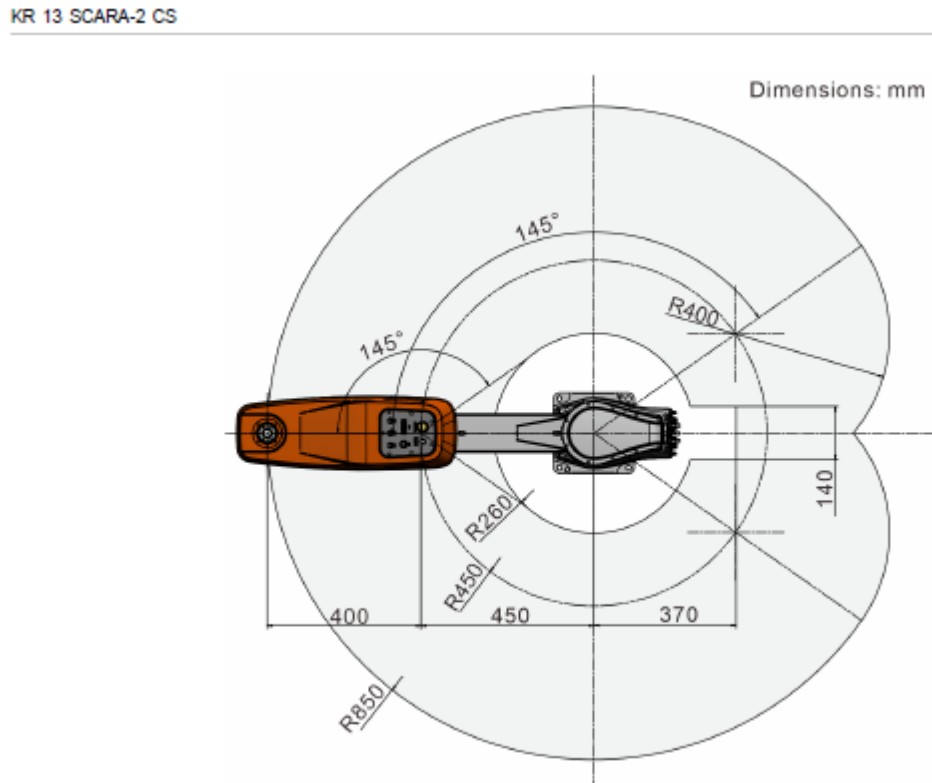


Fig. 4-66: KR 13 R850 Z200-2 CS, working envelope, top view

KR 13 SCARA-2 CS

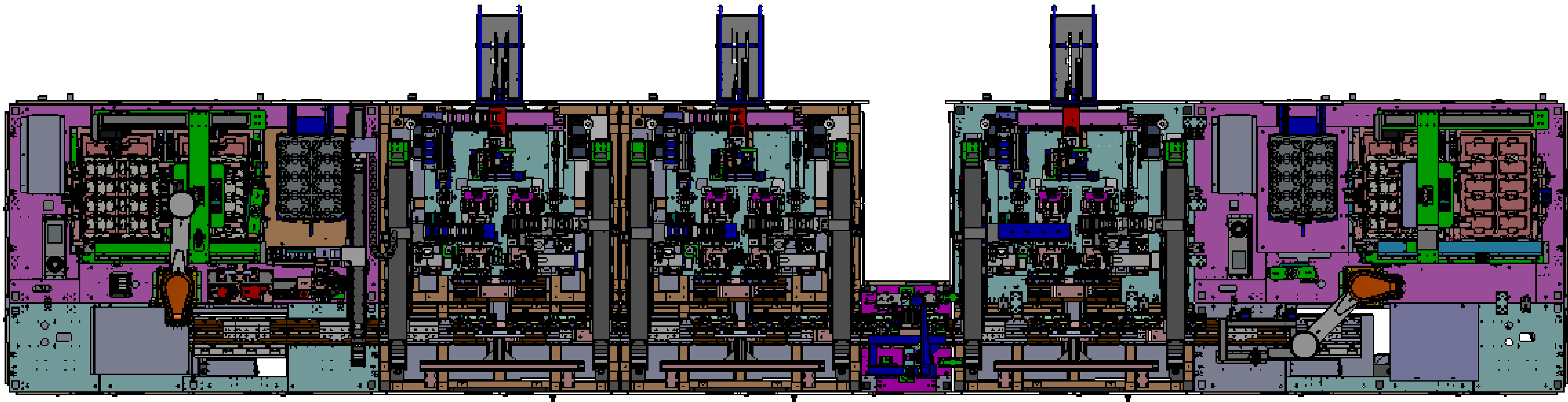
Designation	KR 13 R850 Z200-2 CS
Default color	KUKA Industrial Orange (RAL 2009); Signal white (RAL 9003)
Controller	KR CS Box-2
Time per cycle	0.44 s
Cleanroom class (ISO 14644-1)	-

Ambient conditions

Humidity class (EN 60204)	-
Classification of environmental conditions (EN 60721-3-3)	3K22
Ambient temperature	
During operation	0 °C to 45 °C (273 K to 318 K)
During storage/transportation	-25 °C to 60 °C (248 K to 333 K)

Note: When operating at low temperatures, it is necessary to reduce the speed and acceleration of the robot for preheating before normal use.

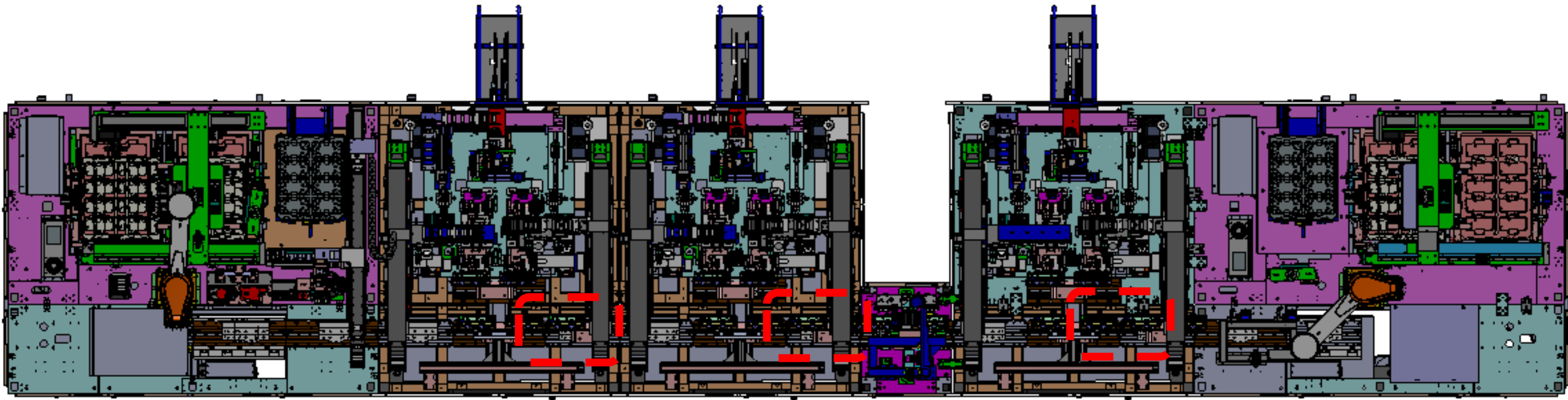
No.	Name	Model	Qty	Brand	Remark
1	Digital Microscope	Dino-lite AM412N -Portable	1	Dino-Lite	



- Usage of Digital Microscope for removable tuning, adjustment, debug camera.,
- Mounting positions will be Designed during Integration.,

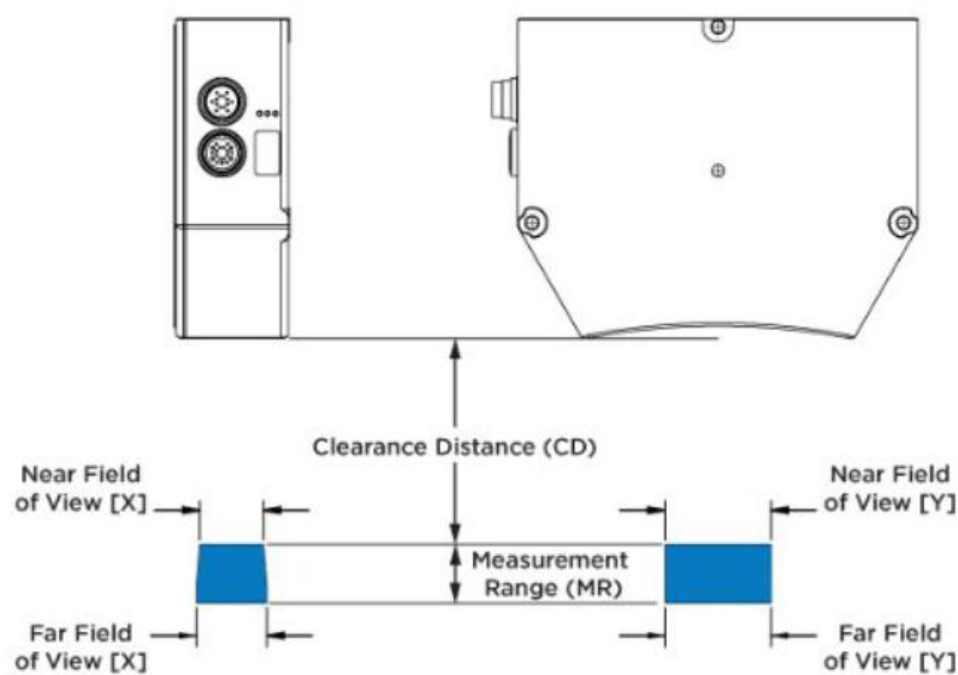
Monza | Snap Assy: Module Details – 3D snapshot sensor for final Inspection

No.	Name	Model	Qty	Brand	Remark
1	3D snapshot sensor	Gocator 3506	3	LMI	



- Usage of 3D Snapshot sensor for Final inspection to detect if any Screw / Snap Gap.

DATASHEET

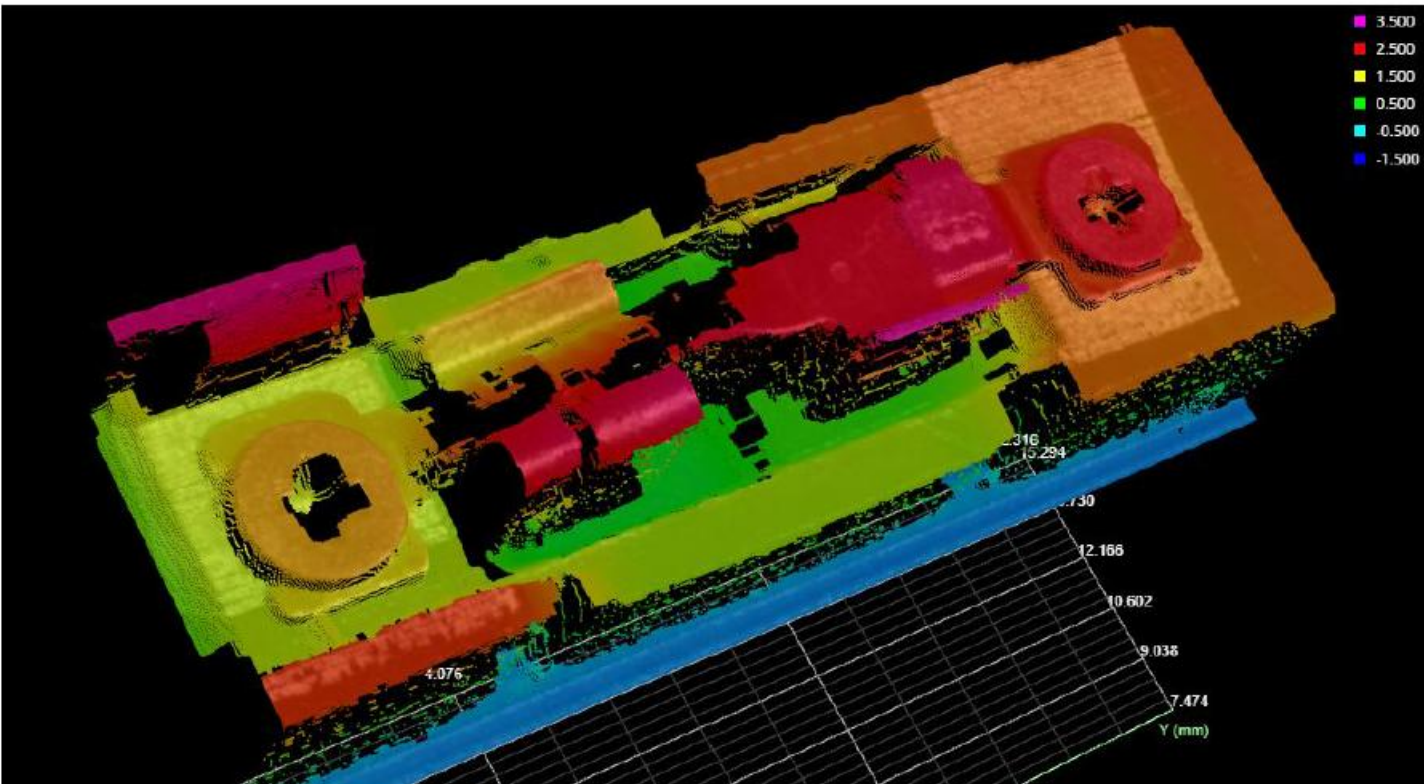


Model	3506
Resolution XY (μm)	20 - 25
VDI/VDE Accuracy (μm)	12(1)
Field of View (mm)	27 x 45 - 30 x 45
Clearance Distance (CD) (mm)	87
Measurement Range (mm)	25

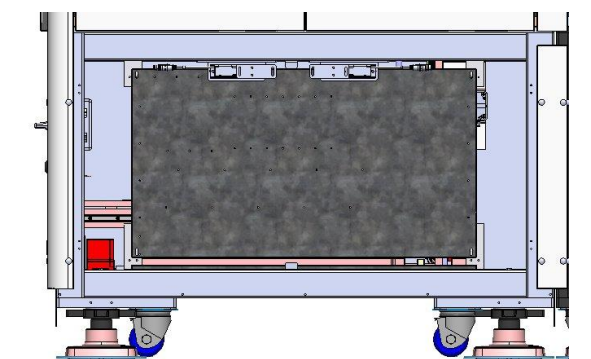
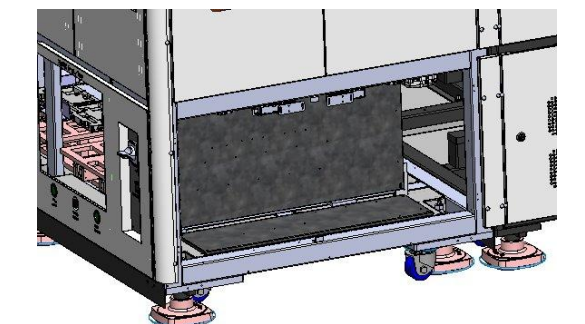
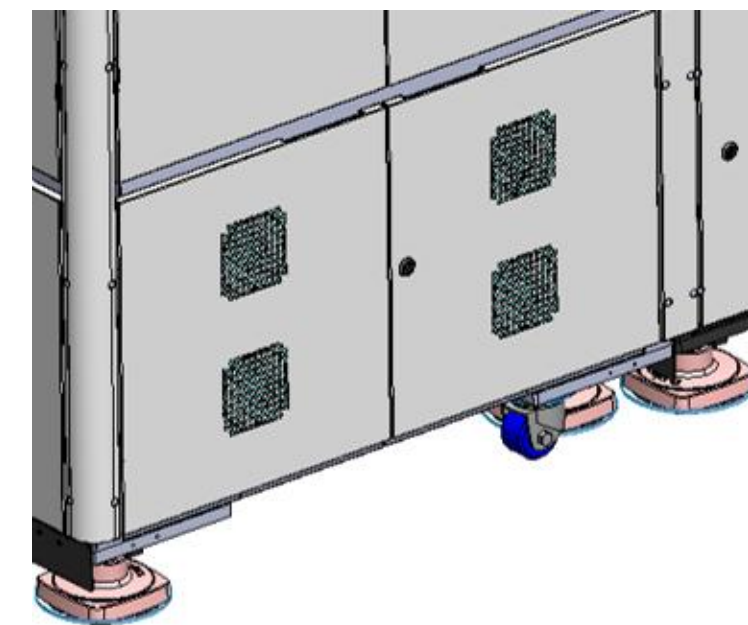
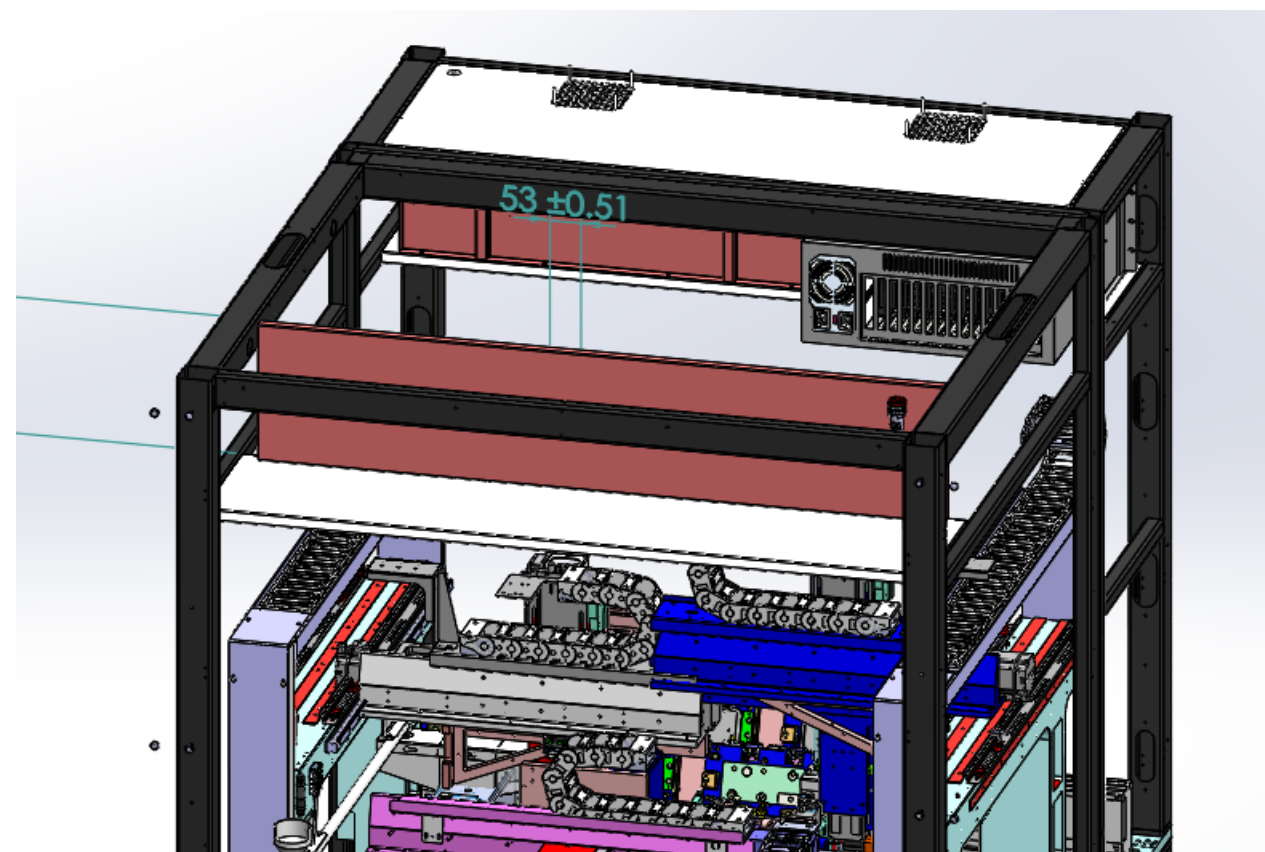
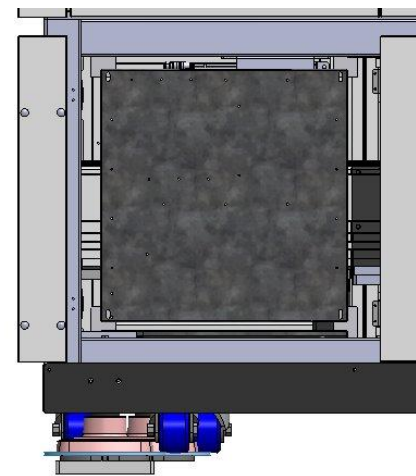
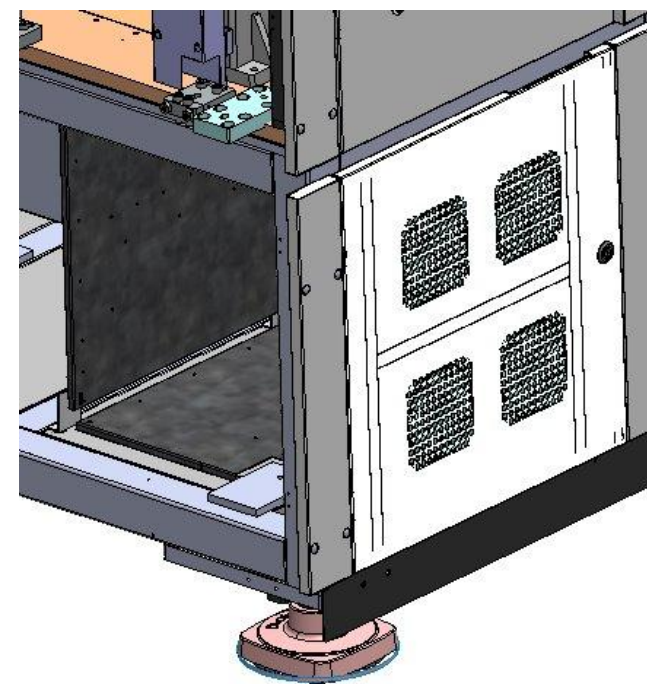
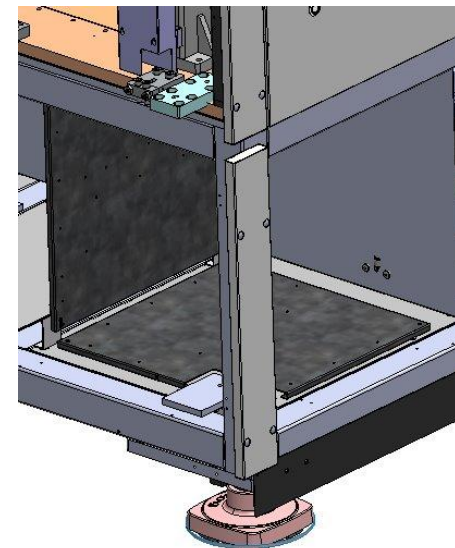
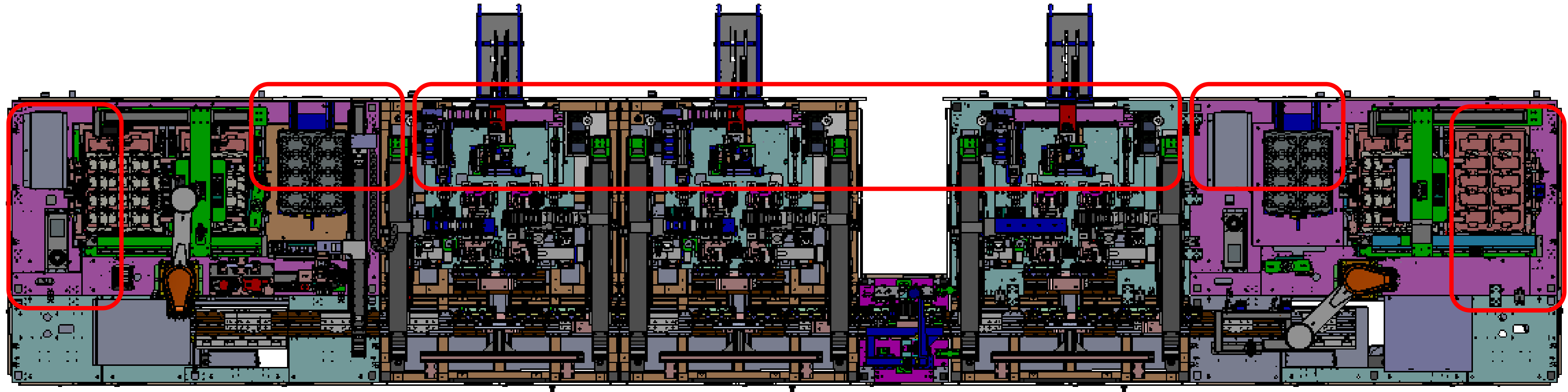
Factory Communication Protocols:



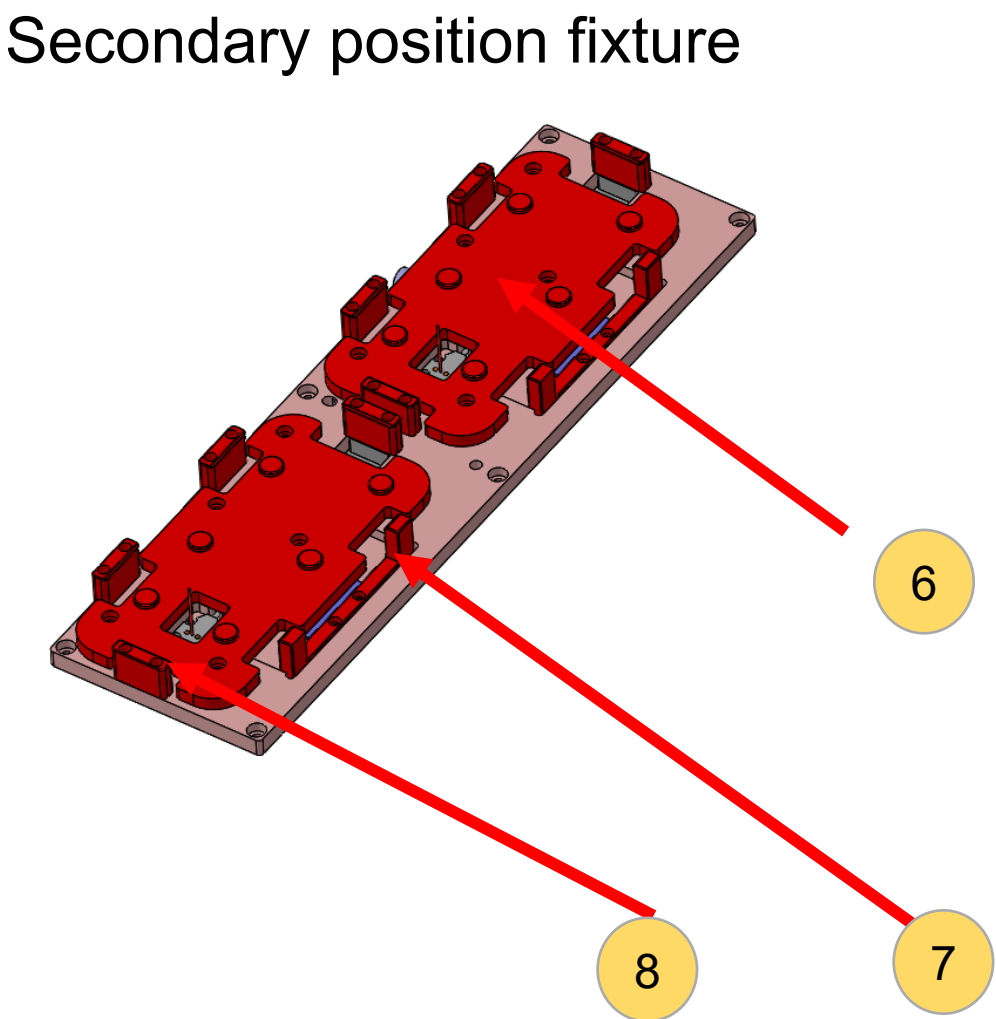
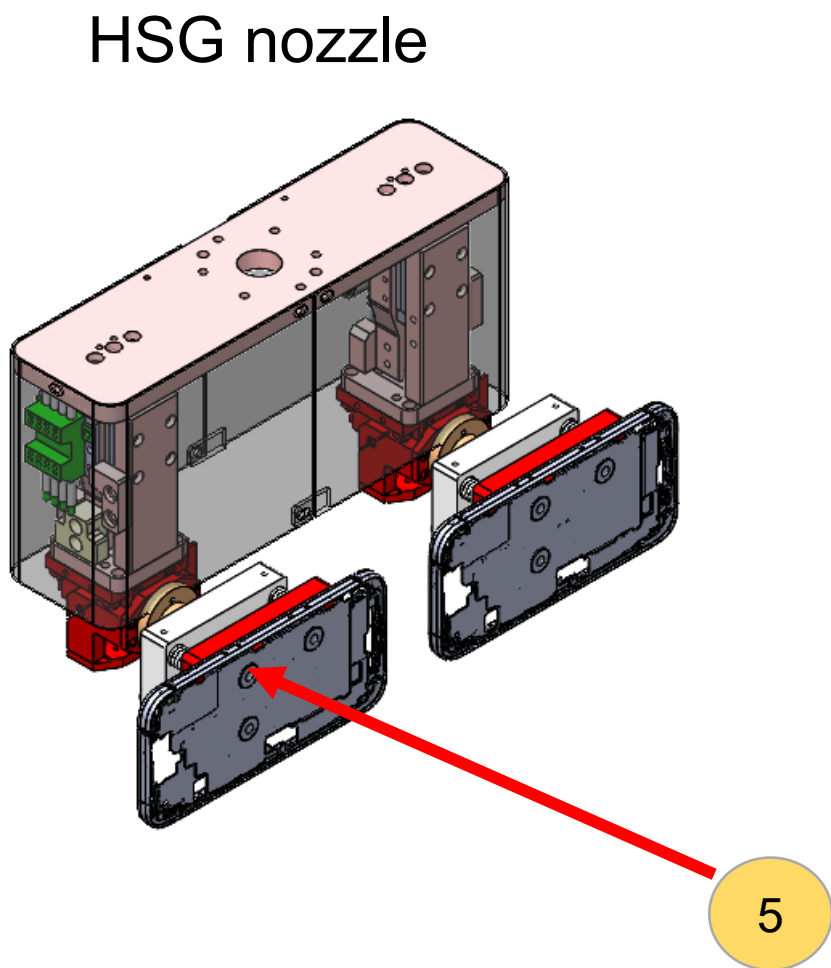
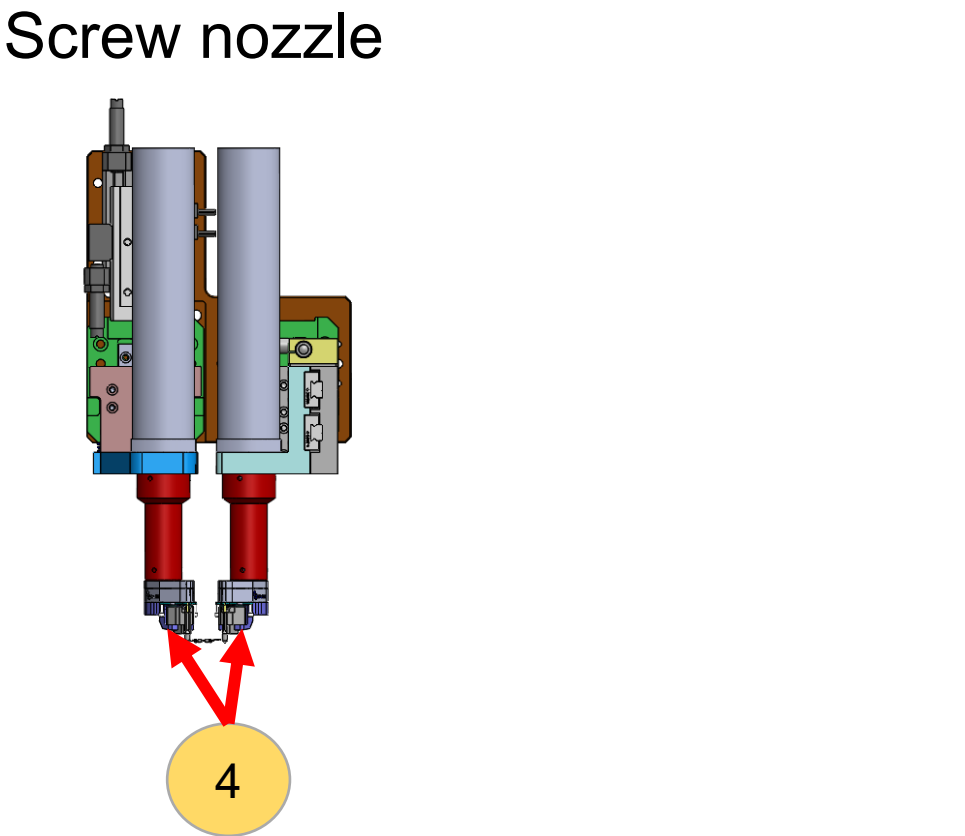
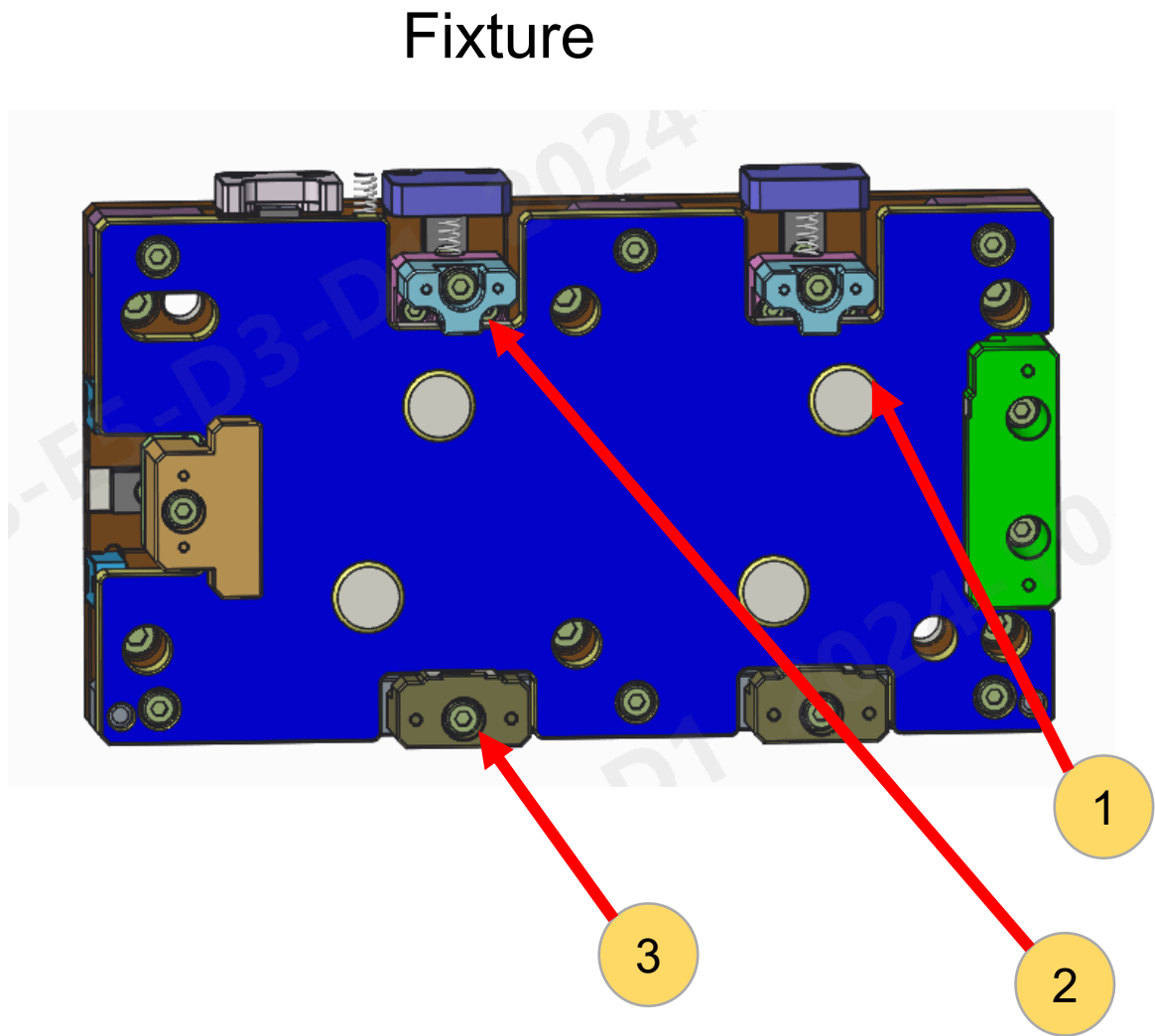
SCAN DATA OF SNAP WITH LMI SAMPLE



Monza | Snap Assy: Control panel mounting positions



Monza | Snap Assy: Module Details – DDS Description



Item	Description	QTY per assembly station	Material
1	Fixture body	4*2	PPS
2	Fixture Push block	2*2	PU
3	Fixture Datum block	4*2	PPS
4	Screw nozzle	4*2	Metal
5	HSG nozzle	3*2	Rubber
6	Fixture body	2*2	PPS
7	Fixture Push block	6*2	PU
8	Fixture Datum block	6*2	PPS

Thank you